



Review

Survey on deep learning-based weakly supervised salient object detection

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ABSTRACT

Salient Object Detection (SOD) in images (ISOD) and videos (VSOD) has always been an active field of research in computer vision. While fully supervised SOD algorithms have made significant progress over the years, they heavily rely on manual pixel-level annotations, which are time-consuming and labor-intensive. Weakly supervised learning paradigms have been explored in both image-based and video-based scenarios, leading to Weakly Supervised ISOD (WSISOD) and Weakly Supervised VSOD (WSVSOD) respectively. These algorithms aim to strike a balance between annotation cost and model performance by leveraging low-cost weak annotations as supervision. Based on the systematic analysis of research challenges in WSISOD and WSVSOD algorithms, this paper presents a comprehensive survey of recent advances in both deep learning-based WSISOD and WSVSOD algorithms. Furthermore, currently available benchmark ISOD and VSOD datasets were summarized, the evaluation metrics of the performance and the performance of existing ISOD and VSOD algorithms on different datasets were depicted. Finally, the existing problems and the future tendency of weakly supervised salient object detection research were discussed.

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1. Introduction

The major goal of Image Salient Object Detection (ISOD) is to detect the most prominent object in the image, whereas Video Salient Object Detection (VSOD) focuses on detecting the most attractive and motion-related objects in video (Cong, Song et al., 2023; Wang, Lai et al., 2022; Zhou, Porikli, Crandall, Van Gool and Wang, 2023). The ISOD task differs from the VSOD task in that, in addition to saliency cues, the motion characteristic plays a significant role in the video (Chen, Song, Peng, Wang, & Fang, 2022; Li, Chen, Li and Yu, 2019). Both of them are crucial for analyzing multimedia content intelligently and can be widely used in computer vision, such as image segmentation (Fan, Hou et al., 2018), image captioning (Fang et al., 2015), video compression (Hadizadeh & Bajic, 2014), video understanding (Wang, Shen, Lu, Hoi and Ling, 2021; Wang, Shen, Yang and Porikli, 2018; Wen et al., 2020), video tracking (Yao, Fu, Li, Zheng, & Ye, 2023; Zhou, Pei et al., 2021), video caption (Li, Tao, Li and Fu, 2019), person re-identification (Zhao, Ouyang, & Wang, 2013), video summarization (Evangelopoulos et al., 2009), semantic segmentation (Hou, Jiang, Wei, & Cheng, 2018), etc. However, human attention behavior is more complex in realistic scenarios, making SOD a challenging task (Koch & Ullman, 1985; Peterson, Kramer, & Irwin, 2004).

Deep learning has demonstrated exceptional performance in many vision tasks. Deep learning-based Salient Object Detection (SOD) algorithms have rapidly demonstrated superior performance over traditional algorithms, continuously pushing the boundaries of the field (Fan, Wang, Cheng, & Shen, 2019; Wang, Shen and Shao, 2018). These algorithms utilize advanced networks such as Convolutional Neural Networks (CNNs) (O'Shea & Nash, 2015), Recurrent Neural Networks (RNNs) (Zaremba, Sutskever, & Vinyals, 2015), and Transformers (Vaswani et al., 2017) to automatically extract features from data, accurately detect and track salient regions in images and videos, and significantly enhance detection precision and robustness. However, deep learning-based fully supervised ISOD and VSOD algorithms usually require large-scale manual pixel-level annotation datasets for training, which are time-consuming and labor-intensive. While Weakly Supervised Image Salient Object Detection (WSISOD) and Weakly Supervised Video Salient Object Detection (WSVSOD) algorithms only use weak annotations (such as category annotations, point annotations, scribble annotations, bounding box annotations, pseudo labels, etc.), which can achieve a trade-off between the annotation burden and the model performance, so WSISOD and WSVSOD have attracted increasing research attention in recent years.

In contrast to fully supervised learning, which relies on abundant annotated data, weakly supervised learning leverages limited or imprecise annotations to train models effectively. By using weak annotations such as category annotations or bounding boxes, weakly supervised algorithms can significantly reduce the annotation effort while maintaining a good balance in performance. On the other hand, unsupervised learning does not require any labeled data and instead focuses on learning patterns from the data itself. This paper mainly focuses on the algorithms of weakly supervised SOD and VSOD. The supervision information of the WSISOD and WSVSOD algorithms generally includes category (image-level) annotations, point annotations, scribble annotations, bounding box annotations and pseudo labels, whose visual examples are shown in Fig. 1. Compared to pixel-level

annotations, these coarse-grained annotations are easier to obtain. For category and point annotations, the complete lack of pixel-level details means that category labels cannot provide information about the position and appearance of objects, which are critical cues for salient object detection and help guide the model in distinguishing salient objects from the background. Consequently, saliency maps obtained through category-supervised algorithms often suffer from coarse boundaries and incomplete structures. Bounding box annotations can provide location information for locating salient objects. However, some salient objects tend to have irregular shapes, and the bounding box annotations often contain messy background information, leading to pixel-level objects' interference and inferior saliency prediction. The annotation area of scribble annotations is larger than that of point annotations, as they trace part of the object's structure, but smaller than bounding box annotations, which enclose the entire object. While they are sparse and low-cost, they also cannot convey sufficient information about object structure and details. Pseudo labels do not require manual annotation. Nevertheless, unlike point annotations and scribble annotations, the generation of pseudo labels is often uncontrollable, which can introduce inevitable noise. Additionally, hybrid labels can combine multiple types of annotations, such as category annotations and bounding box annotations, to leverage the advantages of different annotation types. In conclusion, while coarse-grained annotations offer a more accessible alternative to pixel-level annotations, each type of annotation comes with its own set of challenges and limitations. Understanding these nuances is crucial for effectively leveraging weakly supervised algorithms.

Compared with manual pixel-level annotations, weak labels are coarse, making the pipeline for WSISOD and WSVSOD algorithms more complex than fully supervised algorithms. The mainstream pipeline for WSISOD and WSVSOD algorithms is illustrated in Fig. 2. Firstly, the video and its corresponding coarse-grained annotations are input into the pseudo label generation module. This module transforms the coarse-grained annotations into pixel-level pseudo labels, which provide more detailed supervision information. This step is crucial as it enables the model to better learn the saliency cues of salient objects by offering finer-grained guidance. Next, these pixel-level pseudo labels are used to train a WSISOD/WSVSOD model. The trained model is then used to infer and generate even finer pixel-level pseudo labels. This iterative process helps in refining the saliency maps, thereby improving the overall performance of the model in detecting salient objects. By converting coarse annotations into detailed pseudo labels and iteratively refining them, the WSISOD/WSVSOD pipeline effectively bridges the gap between weak supervision and the need for precise pixel-level information. This approach not only leverages the advantages of weak labels but also mitigates their inherent limitations, leading to more accurate and reliable salient object detection.

Obviously, these weak annotations contain less detailed information compared to pixel-level annotations. They often lack fine edge and structural information, making it challenging for models to accurately identify and delineate salient objects. Therefore, the main challenge of WSISOD/WSVSOD lies in how to effectively learn accurate saliency cues from these coarse-grained annotations. This involves developing methods to bridge the gap between the coarse-grained annotations, such as category labels or bounding boxes, and the more precise pixel-level annotations required for high-quality salient object detection. Especially for WSVSOD, it faces several unique challenges that

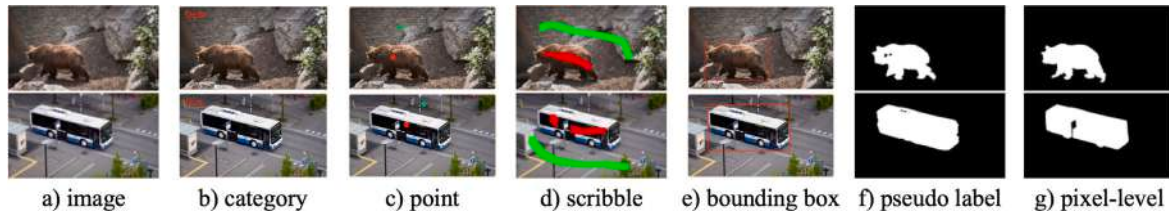


Fig. 1. Visual examples of different types of weak annotations. From left to right are: (a) images, (b) category annotations, (c) point annotations (foreground and background points are colored in red and green, respectively), (d) scribble annotations, (e) bounding box annotations, (f) pseudo labels, and (g) pixel-level annotations (ground truth).

distinguish it from WSISOD. First, videos inherently contain temporal information, requiring WSVSOD methods to ensure consistency in saliency predictions across frames—a challenge absent in WSISOD. Second, videos often include motion cues, such as object movement or camera motion, which WSVSOD methods must effectively leverage to improve saliency detection, whereas WSISOD focuses solely on spatial features within a single image. Third, while both WSVSOD and WSISOD rely on weak annotations (e.g., points, scribbles, or pseudo labels), WSVSOD faces additional complexity in propagating these annotations across frames to maintain temporal coherence. Finally, videos are computationally more demanding than images due to their temporal dimension, requiring WSVSOD methods to balance accuracy with efficiency—a consideration less critical in WSISOD.

WSVSOD fundamentally differs from WSISOD in several key aspects. First, WSVSOD explicitly considers temporal dynamics and frame-to-frame relationships, while WSISOD is limited to spatial features within a single image. Second, WSVSOD must propagate weak annotations across frames to ensure temporal consistency, whereas WSISOD only deals with annotations within a single image. Third, WSVSOD can leverage motion cues, such as optical flow, to enhance saliency detection, while WSISOD relies solely on static image features. These distinctions highlight the unique challenges and opportunities of extending weakly supervised techniques from images to videos.

Currently, there are numerous surveys on ISOD and VSOD available both domestically and internationally. For example, comprehensive surveys have been published on fully supervised ISOD (Borji, Cheng, Hou, Jiang, & Li, 2019; Gupta, Seal, Prasad, & Khanna, 2020; Wang, Lai et al., 2022), RGB-D SOD (CONG, ZHANG, XU, LIU, & ZHAO, 2022; Fu, He and Yang, 2022; Zhou, Fan, Cheng, Shen and Shao, 2021), deep learning-based VSOD (Wang, Zhang, Li, & Kpalma, 2020), light field SOD (Fu, Jiang et al., 2022), as well as fully and semi-supervised Video Object Segmentation (VOS) (Yao, Lin, Xia, Zhao, & Zhou, 2020; Zhou, Porikli et al., 2023) and fully/semi-supervised video saliency prediction and VSOD (Chen, Song, Song, Guo, & Jian, 2023). However, there is a lack of surveys on weakly supervised salient object detection algorithms. In contrast to the aforementioned works, this paper aims to bridge this gap by focusing on the WSISOD and WSVSOD algorithms. The contributions in this paper are summarized as follows:

- **Comprehensive Survey of WSISOD and WSVSOD Algorithms:** We conduct a systematic survey of WSISOD and WSVSOD algorithms, providing an in-depth analysis of the research challenges and recent advancements. This survey emphasizes how these algorithms leverage low-cost weak annotations to balance the trade-off between annotation burden and model performance.
- **Categorization and Taxonomy:** We categorize and summarize existing WSISOD and WSVSOD algorithms based on the type of annotation information and the designed strategies for different annotation types. This proposed taxonomy aims to help researchers gain a deeper understanding of the key features and differences among various WSISOD and WSVSOD models.
- **Benchmark Datasets and Evaluation Metrics:** We summarize several benchmark ISOD and VSOD datasets and outline the evaluation metrics used to assess algorithm performance. This includes a comparative analysis of the performance of existing ISOD

and VSOD algorithms across different datasets, highlighting their strengths and weaknesses.

- **Discussion of Current Issues and Future Directions:** We thoroughly discuss the existing problems within the weakly supervised SOD research field and propose potential future research directions. This discussion aims to inspire further investigations and improvements, offering valuable insights for advancing the state-of-the-art in weakly supervised SOD algorithms.

The roadmap of this paper is shown in Fig. 3. The rest of the paper is organized as follows. Section 2 summarizes the WSISOD and WSVSOD algorithms. In Section 3, several benchmark ISOD and VSOD datasets were summarized, evaluation metrics of the performance were depicted, and the performance of existing ISOD and VSOD algorithms was reported. Finally, the existing problems and the future tendency of WSISOD and WSVSOD task research were discussed in Section 4.

2. WSISOD and WSVSOD algorithms

As mentioned earlier, WSISOD and WSVSOD can achieve the balance between annotation cost and detector performance, attracting more and more researchers' attention. Based on the type of supervision information, WSISOD and WSVSOD algorithms can be broadly categorized into category annotations, point annotations, bounding box annotations, scribble annotations, pseudo labels, and hybrid labels.

Notably, unlike other types of annotations, category annotations provide only the presence or absence of certain object categories within a video frame/image without offering any pixel-level information. Since ISOD and VSOD are fundamentally pixel-level tasks, category annotations are difficult to directly provide pixel-level information for ISOD and VSOD networks. To obtain pixel-level details, category annotations are usually used to train a classification network to generate Class Activation Map (CAM) (Zhou, Khosla, Lapedriza, Oliva, & Torralba, 2016) as the pixel-level pseudo labels, which highlights the regions that are most relevant to the predicted class. Therefore, we classify Category-supervised WSISOD/WSVSOD algorithms under the category of pseudo label-supervised WSISOD/WSVSOD algorithms. In the following sections, we will delve deeper into these categories, exploring their specific characteristics, advantages, and the methodologies developed to leverage them effectively in WSISOD and WSVSOD algorithms.

2.1. Point-supervised WSISOD/WSVSOD algorithms

Point annotations can efficiently identify salient objects and backgrounds in images/videos at a very low annotation cost. However, due to the limited scope, models may struggle to fully learn the boundaries and detailed shape features of salient objects. This limitation can adversely affect the accuracy and robustness of the WSISOD and WSVSOD algorithms, particularly when dealing with complex scenes or diverse objects. To address this issue, researchers have employed various algorithms to enhance the model's ability to learn the boundaries of salient objects, striving to strike a balance between annotation cost and model performance.

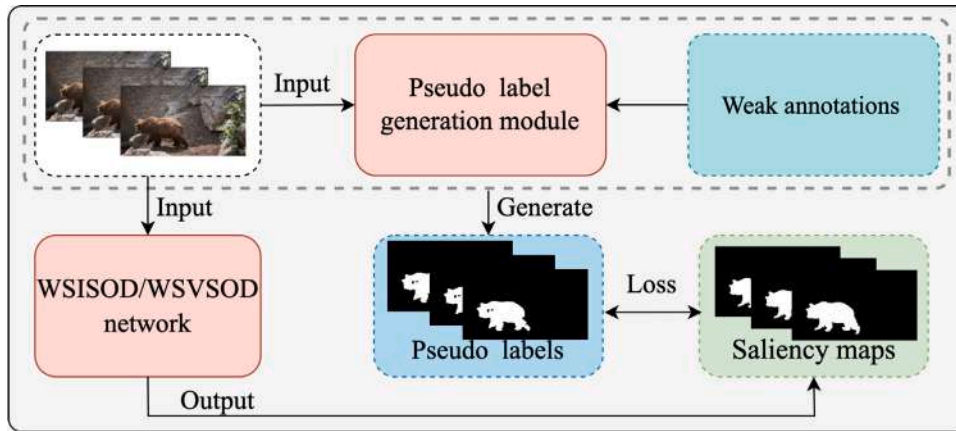


Fig. 2. The mainstream pipeline for WSISOD/WSVSOD algorithms.

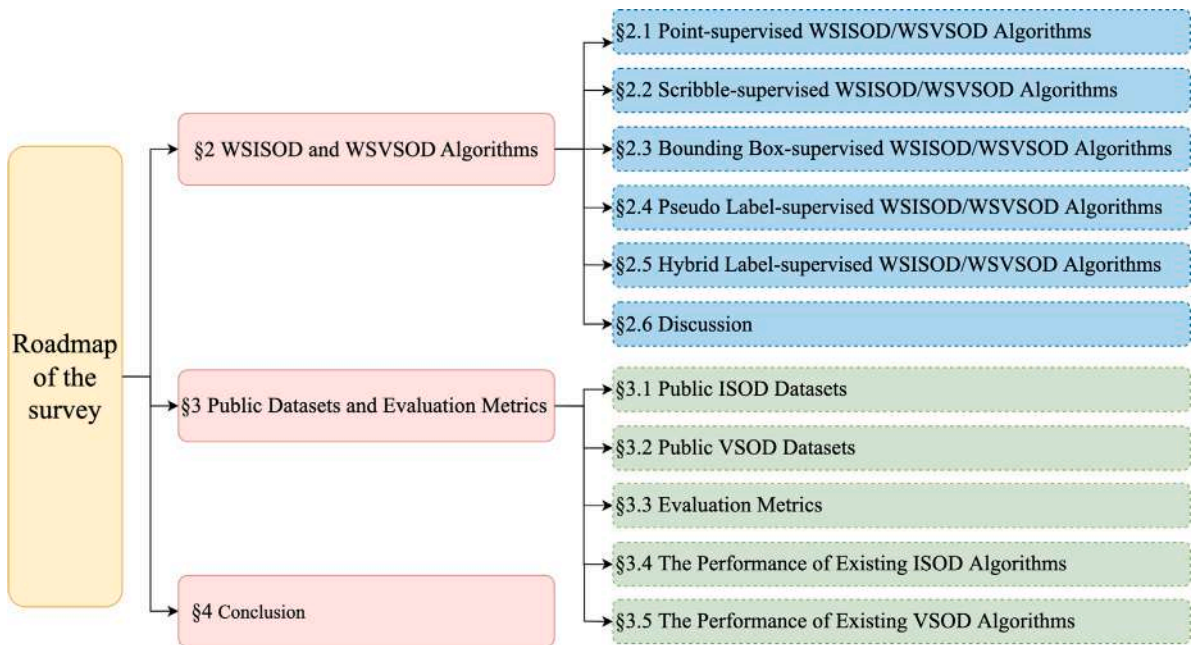


Fig. 3. The roadmap of this paper.

2.1.1. Point-supervised WSISOD algorithms

A WSISOD algorithm using point supervision is proposed by Gao, Zhang et al. (2022). They built a new point-supervised dataset (P-DUTS) by relabeling the DUTS (Wang et al., 2017) dataset. To infer the saliency map, they first design an adaptive masked flood filling algorithm to generate pseudo labels at first. Then, a Transformer-based point supervised salient object detection network was designed to produce the first round of saliency maps. However, due to the sparseness of the label, the weakly supervised model tends to degenerate into a general foreground detection model. To mitigate this issue, they introduced a Non-Salient Suppression (NSS) method, which optimized the erroneous saliency maps generated in the first round and used them for a second round of training, significantly improving the model's precision. Building on the work of Wu et al. (0000) attempted to prove a hypothesis: there exists a point annotation dataset where salient object detection models trained on it can achieve performance equivalent to those trained on a pixel-level annotated dataset. To prove the hypothesis, they proposed a novel Adversarial Trajectory-ensemble Active Learning (ATAL). Firstly, an adversarial attack triggering uncertainty was proposed to locate these uncertain pixels accurately. Then, the trajectory-ensemble uncertainty estimation method was proposed. Finally, the relationship-aware diversity sampling algorithm was

proposed to overcome while boosting performance. Their experimental results confirmed the hypothesis, demonstrating the potential of point supervision to achieve performance on par with fully supervised methods.

2.1.2. Point-supervised WSVSOD algorithms

While point-supervised methods have shown remarkable success in image salient object detection, their application to WSVSOD presents additional challenges. Video data introduces temporal dynamics and requires consistency across frames, which complicates the generation and utilization of point annotations. Despite these challenges, recent works have begun to explore point-supervised approaches in the video domain, building on the advancements made in image-based methods.

In 2022, Gao, Xing et al. (2022) extended their earlier work on point-supervised WSISOD to the video domain. They proposed a point-supervised VSOD algorithm that introduced a hybrid token attention module to fuse optical flow and frame information from orthogonal directions, enabling the extraction of short-term motion information. Additionally, they developed a long-term cross-frame attention module that leverages multi-frame tokens to assist the current frame in identifying salient objects. This work demonstrated the feasibility of

applying point supervision to video data while addressing the unique challenges of temporal consistency. Later, Ge et al. (2023) proposed Point-Teaching, which used a Hungarian-based point-matching method to generate pseudo labels at first, and then the multiple instance learning approach was proposed where the supervision information was the point annotations.

In addition, point annotations are widely used in tasks with higher labeling costs, such as nuclei segmentation (Inwan, Donggeun, & Kyunghyun, 2019; Qu, Wu, Huang, Yi, & Riedlinger, 2019), and cell segmentation (Zhao & Yin, 2021). In these tasks, precise pixel-level annotations often require specialized knowledge, substantial time, and professional personnel, making the labeling process costly and labor-intensive. Point annotations provide a lightweight and efficient alternative that can significantly reduce labeling costs. Although they may not provide the same level of detailed information as manual pixel-level annotations, they can still supply sufficient supervisory information for training models that achieve competitive performance in certain applications. By simplifying the annotation process and increasing efficiency, point annotations have become an ideal choice for large-scale data labeling in practical applications. This approach is particularly advantageous in fields like medical image processing, where there is a critical need to balance accuracy with cost-effectiveness.

2.2. Scribble-supervised WSISOD/WSVSOD algorithms

As a sparse and low-cost annotation method, scribble annotations can significantly alleviate the burden of pixel-level annotation in the video salient object detection task. Similar to point annotations, scribble annotations offer a more efficient and less resource-intensive alternative to traditional pixel-level labeling. By allowing annotators to mark key regions with simple strokes rather than meticulously outlining every pixel, it thereby reduces the time and effort for high-precision labeling. However, similar to point annotations, scribble annotations also fail to provide fine edge information for both the WSISOD task and the WSVSOD task. For the WSVSOD task, learning motion information from scribble annotations is also difficult.

2.2.1. Scribble-supervised WSISOD algorithms

In 2020, Zhang, Yu et al. (2020) were the first to explore the scribble-supervised WSISOD algorithm. Firstly, they relabeled an existing large-scale salient object detection dataset DUTS (Wang et al., 2017) with scribbles, namely S-DUTS. Then, in order to accurately locate the salient object boundary, an auxiliary edge detection branch is proposed to localize object edges. Given that scribble annotations cannot provide precise salient object boundaries, in the quest for a more precise boundary for salient objects based on scribble annotations, Yu, Zhang, Xiao, and EngGee (0000) proposed a local coherence loss. Unlike this method, Wang, Al-Huda, Peng and Tang (2023) focused on generating high-quality labels, they proposed a two-phase salient object detection framework to improve the performance of scribble annotation-supervised model. The scribble annotations were used as the initial supervised information, and high-quality pseudo labels were generated by mapping scribble annotations onto object contours. A deep neural network was then alternately used in training and prediction to refine the pseudo labels. In order to further expand the application of scribble supervision, Xu, Yu, Zhang, Zhu, and Wang (2022) proposed a scribble-supervised RGB-D SOD algorithm, which can explicitly exploit the complementary edge information from two modalities. They also designed a prediction consistency training fashion method to achieve more stable yet effective salient object predictions.

Many researchers attempted to optimize and improve the performance of WSISOD through network architecture and module design. Huang, Xiang, Chen and Dai (2022) present a novel Scribble-based Boundary-Aware Network (SBA-Net) for remote sensing salient object detection, which includes a Boundary-Aware Module (BAM) and a

Boundary Label Generation (BLG) module to learn the object boundary semantics and object structures, respectively. Xu, Liang, Liang and Chen (2024) proposed a self-consistent framework comprising a global integral branch and a boundary-aware branch. This framework guides the salient object detector to predict precise boundaries while accurately identifying salient objects. They later proposed a scribble-based visual representation-guided framework, which contains a task-related encoder, a general visual module, and an information integration module. These components aim to filter irrelevant information, provide general visual representations, and integrate various layers of task-related features. By carefully designing loss functions to constrain the learning process of neural networks, it is possible to achieve a more effective extraction of saliency cues (Xu, Liang, Gong, Liang, & Chen, 2023). Despite the effectiveness of fully convolutional networks (FCNs) (Wang, Shen, Shao, 2018) in fully supervised VSOD, studies (Luo, Li, Urtasun, & Zemel, 0000) have shown that even with a large number of convolutional layers, the effective receptive field of network units remains limited. This limitation is particularly critical in scribble-supervised salient object detection tasks, where the long-range dependencies between pixels play a crucial role in energy propagation, the transfer of energy from scribble-annotated pixels to unlabeled regions directly impacts the model's performance. Therefore, Liang, Yan, Xie, Zhu, and Wang (2024) proposed the Complementary Graph Reasoning Network (CGRNet), which aims to explore the impact of long-range dependence between pixels on energy propagation and model performance. They also introduced edge information and fusion strategy to assist the network learning to the fine edge. Li, Mao, Zhang and Dai (2024) explored the application of mutual information as a regularization technique within a weakly supervised framework. The aim is to enhance the robustness and accuracy of the model by maximizing the shared information between the input data and the predicted saliency maps. Wang, Zhang et al. (2024) leveraged the Transformer architecture to learn local-global representations, effectively integrating local features with broader contextual cues, which enhances the model's capability to identify salient objects in cluttered backgrounds. Han et al. (2024) proposed a scribble-based ultrasound image segmentation method named MambaEviScrib, which is a hybrid CNN-Mamba framework. Owing the Mamba's linear complexity (Gu & Dao, 2024; Zhu, Liao et al., 2024), they are usually used to replace the Transformer, and also introduced the Dempster-Shafer theory to enhance the stability of edge segmentation.

Notably, some weakly-supervised RGB-D (Li, Mao et al., 2024; Liu, Hayat, Yang, Peng and Lei, 2023) and RGB-T (Liu, Huang, Zhang, & Fang, 000) and optical remote sensing image (Yan et al., 2025) salient object detection tasks also use scribble annotations as supervision information. Liu et al. (000) proposed a novel framework that used scribble annotations to guide the RGB-T salient object detection. They employed a dual-stream architecture that integrates RGB and thermal information, leveraging the complementary characteristics of these two modalities to enhance the robustness of salient object detection in complex environments. By adopting a scribble-supervised mechanism, this framework effectively reduced the reliance on extensive pixel-level annotations, making it suitable for real-world scenarios where annotations are scarce. Li, Han et al. (2024) emphasized the necessity of achieving robust perception and precise segmentation in RGB-D data. They proposed a hierarchical segmentation approach that combined scribble annotations and depth information to improve detection accuracy. A multi-scale feature extraction strategy was implemented to capture details and contextual information, thereby enhancing segmentation performance. Yan et al. (2025) proposed a multi-source information fusion attention network consisting of a Boundary Detection Block (BDB), a Region Activation Block (RAB), and a Multi-source Information Fusion Block (MIFB). The BDB detects object boundaries using pseudo-boundary masks, and the RAB activates object regions with image-level labels. The MIFB decoder integrates these features to generate high-quality optical remote sensing saliency maps through scribble annotations.

2.2.2. Scribble-supervised WSVSOD algorithms

In 2021, Zhao, Zhang et al. (2021) proposed the first WSVSOD model based on relabeled scribble annotations. The proposed network consists of an appearance-motion fusion module and bidirectional ConvLSTM to achieve effective long-term temporal context modeling. In addition, they introduced a weak annotation boosting strategy, which elevated the model's performance by applying a new strategy for generating pseudo labels. Considering that scribble annotations can only provide very limited foreground/background information, Zhao, Cui, Shen and Liu (2024) tried to infer annotations that cover more complete salient objects and background regions through scribble annotations. Firstly, they proposed a label inference strategy, which assumes that pixels with similar colors and close positions should have consistent labels. Then, they trained the salient object detection model using inferred pseudo labels.

As previously discussed, VSOD networks face significant challenges in learning complete salient object structural information when relying solely on scribble annotations. To address this limitation, Wang, Yu et al. (2024) proposed a novel Cross-Frame Feature-Saliency Mutual Reinforcing (CFMR) training strategy. This approach introduces a cross-frame feature regularization head, which leverages both intra-frame and cross-frame deep feature consistency to regularize saliency predictions. By incorporating this auxiliary supervision, the CFMR strategy enhances the model's ability to capture structural information and improve the accuracy of salient object detection in video sequences.

It is well known that scribble-supervised algorithms often suffer from issues such as loss of detail and boundary information. To address this, Lu, Liang, Xu, and Liang (0000) proposed a background-foreground weight contrast method that incorporates label enhancement points, enabling the model to better learn the edges, details, and locations of salient objects. Additionally, they introduced a global structural localization framework to assist the model in achieving more accurate global localization through enhanced scribble annotations.

Similar to the VSOD task, the fully supervised VOS task also requires high-cost manual pixel-level labels, to this end, Zhou et al. (2024) focused on the scribble-supervised VOS task in the one-shot setting, in which the annotation of the object in the first frame is available during inference. Considering that the difficulties of scribble-supervised VOS tasks lie in two aspects: (1) It requires strong reasoning capability to accurately segment the objects. (2) It involves learning dense predictions from sparse scribble annotations, necessitating powerful learning capability. They proposed a Reliability-guided Hierarchical Memory Network (RHNet), which maintains a reliability-guided memory bank. They also proposed a scribble-supervised learning mechanism to facilitate the model learning for dense prediction. The experimental results demonstrate that the proposed method is promising. However, the proposed method still faces challenges in understanding the object details due to the lack of pixel-level supervision information.

Research on WSISOD primarily emphasizes image-level boundary learning and pseudo-label generation, whereas WSVSOD incorporates temporal context from videos, thereby enhancing the detection performance of salient objects. To address the sparsity issue of scribble annotations, numerous algorithms have introduced innovative techniques such as boundary-aware modules, cross-frame feature fusion, and consistency training strategies, which significantly improve model performance in complex scenarios. Scribble annotation is also widely used in many segmentation tasks, such as semantic segmentation (Lin, Dai, Jia, He, & Sun, 0000), volumetric image segmentation (Chen & Hong, 2022), and medical image segmentation (Zhao, Zhou et al., 2023).

2.3. Bounding box-supervised WSISOD/WSVSOD algorithms

In contrast to point and scribble annotations, which only mark portions of salient objects, bounding box annotations encompass the entire salient region by using minimal rectangular boxes. However,

this annotation method is not without challenges. Bounding boxes frequently capture not only the salient object but also the nearby background, producing labeling noise that is hard to differentiate. This noise is especially disruptive for pixel-level video salient object detection tasks, as it interferes with the algorithm's ability to accurately discern the most difficult part of the task—the boundaries of salient objects. The background data within the bounding boxes complicates the detection of object edges, negatively impacting the performance and precision of the algorithm.

Before being applied to the salient object detection (SOD) task, bounding box annotations were widely used in weakly supervised instance segmentation tasks. In 2019, Hsu, Hsu, Tsai, Lin, and Chuang (0000) addressed the challenge of uncertain figure-ground separation within bounding boxes by formulating the bounding box-supervised instance segmentation task as a Multiple Instance Learning (MIL) problem. Their approach involved generating positive and negative bags based on the sweeping lines of each bounding box. The MIL framework consists of two key components: the first estimates the foreground and background areas, while the second ensures the unity of the estimated object masks. This work laid the foundation for leveraging bounding box annotations in weakly supervised tasks, demonstrating their potential for complex visual understanding problems. Similarly, Lan et al. (2021) also designed a new framework where instance segmentation and semantic correspondence were jointly guided by a structured teacher in addition to the bounding box annotations. Cheng, Wang, Chen, Zhang, and Liu (2023) found that bounding box-supervised methods can produce some fine segmentation masks. Inspired by this, an end-to-end framework was proposed for weakly supervised instance segmentation, which leverages a sophisticated teacher to generate high-quality masks as pseudo labels. They also selected high-quality pseudo labels to train the network. Different from the above complex designs, Tian, Shen, Wang, and Chen (2021) showed significantly stronger performance with a simple design, whose core idea is to redesign the loss of learning masks in instance segmentation.

In 2021, Liu, Wang, Cao, Liang and Lau (2021) proposed a WSISOD algorithm that predicts pixel-level pseudo labels from just bounding boxes. Firstly, they took advantage of the unsupervised salient object detection algorithms to generate initial pixel-level pseudo labels and address the over/under prediction problems. Then, they designed a multi-task framework to learn the salient cues from the pseudo labels and bounding boxes to update the pseudo labels. Finally, the updated pseudo labels were used to train the salient object detection model. Building on this idea, He, Zhang, and Yu (2022) utilized the GrabCut algorithm (Rother, Kolmogorov, & Blake, 0000) to generate pseudo labels from bounding boxes, converting coarse bounding box annotations into finer-grained supervision. By fully exploiting both background and foreground information, they established a more accurate salient object detection model. Further advancing this line of research, Pei, Tang, Wang, Cheng, and Chen (2022) designed a cyclic global context salient instance segmentation network, which was supervised by the combination of salient regions and bounding boxes. They further designed a labeling updating scheme to iteratively update the weak labels.

Moreover, bounding box annotations are widely utilized in various pixel-level tasks, offering a balance between efficiency and annotation quality. Liang, Wang, Xu, Zhang, and Lau (2022) introduced a bounding box-supervised salient object detection algorithm for light fields, which generated pixel-level pseudo saliency maps by leveraging bounding-box annotations and a ground-truth hallucination method, thus reducing the need for expensive pixel-level labels. Chibane, Engelmann, Anh Tran, and Pons-Moll (2022) proposed Box2Mask, a 3D semantic instance segmentation algorithm, also based on bounding box supervision. Similarly, Lee, Yi, Shin, and Yoon (2021) utilized bounding boxes as weakly supervised information to achieve both semantic and instance segmentation. Furthermore, Fan et al. (2020) developed the CoSOD3k dataset, which is a high-quality resource for co-salient object detection, encompassing category, bounding box, object, and instance

levels. This dataset provides a solid foundation for training and validating weakly supervised models. Wei, Zhou, Cui, and Zhen (2024) proposed a weakly supervised point cloud salient object detection algorithm, which used cheap 3D bounding box annotations. To learn the noise-free supervision while mitigating shape biases inherent in coarse 3D bounding boxes, they introduced the mask-to-box transformation and a color consistency loss. Bounding box annotations have been applied in tasks such as point cloud instance segmentation (Liao et al., 2022), and semantic segmentation (Bonechi, Andreini, Bianchini, & Scarselli, 2018).

2.4. Pseudo label-supervised WSISOD/WSVSOD algorithms

CAM (Zhou et al., 2016) is generated from the classification network, which can roughly locate the object regions with the annotations of the image category, so it is increasingly used in various weakly supervised learning tasks, Li, Xie and Lin (2018), Wang et al. (2017), and Zeng et al. (2019). In addition to using CAM and traditional salient object detection algorithms (Nguyen et al., 0000; Zhang, Zhang, Dai, Harandi, & Hartley, 2018) to generate pseudo labels for training VSOD networks, many researchers have also explored the use of salient object detection models and Segment Anything Model (SAM) (Kirillov et al., 0000) to generate pseudo labels. The advantage of these approaches is that it does not require any manual annotation of video salient objects. Nevertheless, unlike point annotations and scribble annotations, the generation of pseudo labels is often uncontrollable, which can introduce inevitable noise. Additionally, pseudo labels may suffer from incomplete salient objects and missing edge information, which brings challenges to learning the complete object structure and precise boundary details.

2.4.1. Pseudo label-supervised WSISOD algorithms

Initially, researchers attempted to learn saliency cues from CAM directly. Wang et al. (2017) leveraged the observation that category annotations provide important cues of foreground salient objects, and developed a WSISOD algorithm. Firstly, the Foreground Inference Network (FIN) is jointly trained with a Fully Convolutional Network (FCN) for category annotation prediction. A global smooth pooling layer is proposed, enabling FCN to assign object category annotations to corresponding object regions, while FIN can capture all potential foreground regions with the predicted saliency maps. Then, FIN is fine-tuned with its predicted saliency maps as pseudo labels. Later, researchers attempted to improve the quality of pseudo labels. Hsu, Lin, and Chuang (2017) only used category annotations, indicating the presence or absence of the object in an image. Their framework is composed of two deep modules: a category classifier and a pixel-level pseudo label generator. In addition to the top-down guidance from category annotations, the pixel-level pseudo label generator is derived by also referring to other image information, including the background prior, area balance and spatial consensus, which greatly regularizes the training process and reduces the risk of overfitting. Wang, Liu et al. (2022) proposed a WSISOD algorithm based high-quality on only binary image labels. The fundamental concept is to create a dataset with images categorized into two classes: salient (containing prominent objects) and non-salient (lacking prominent objects). This binary classification serves as supervision to develop a salient object detector using existing pre-trained pseudo-label generation method. They introduce an object saliency map hallucinator that generates pseudo labels for the salient images based solely on these binary labels. These pseudo labels are then utilized to train the salient object detector. Zhou, Qiao, Yang, Lai, and Xie (2023c) used MoCo-v2 (Chen, Fan, Girshick and He, 2020) pretrained ResNet-50 (He, Zhang, Ren, & Sun, 2016) to generate pseudo labels and incorporated boundary-aware texture matching to refine the boundaries of salient objects.

Introducing other modal information is a promising scheme. Zhang, Zhang, Liu and Kong (2024) proposed a Category-aware Saliency Enhancement Learning (CSEL) method based on Contrastive Vision–Language

Pretraining (CLIP) (Radford et al., 2021), which performs image–text classification and pseudo label updating simultaneously. CSEL generates a category-aware saliency map evaluated by classification accuracy and uses quality confidence scores to update pseudo labels, allowing the two maps to mutually enhance each other and guide the learning of salient object detection network effectively. To reconcile the multimodal input signals for weakly supervised RGB-D salient object detection, Li et al. (2021) introduced semantic information and utilized spatial semantic modeling to generate more refined pseudo labels. Meanwhile, many researchers have improved model performance through enhancements in network architecture, training strategies, and other aspects. Li, Xie, Lin (2018) designed a multi-task network, which used the combination of a coarse CAM from the classification network and saliency maps generated with a traditional method as pixel-level annotations. Unlike the aforementioned algorithms that rely on a single label for training, Ma, Du, Zhuo and Li (2024) and Piao, Wang, Zhang, and Lu (2021) proposed a novel multiple label aggregation framework to address the limitations of single-label supervision. Their approach integrates comprehensive saliency cues from multiple pseudo labels, effectively mitigating the potential bias introduced by relying on a single label. By leveraging diverse sources of supervision, this framework enhances the robustness and accuracy of salient object detection, demonstrating the benefits of multi-label aggregation in weakly supervised scenarios. Specifically, Piao et al. (2021) adopt both pixel-wise and superpixel-wise algorithms on CAM for refinements to get multiple labels. The pixel-level refinement (Araslanov & Roth, 2020) and Conditional Random Field (CRF) (Krähenbühl & Koltun, 2011) were used. They proposed a Multi-Filter directive Network (MFNet), which includes a salient object detection network and multiple directive filters. The directive filter is used to extract more accurate saliency cues from the pseudo labels.

Additionally, considering the high amount of computation cost, Zhang, Huang and Zhu (2023) proposed a weakly supervised method with category annotation that only uses global average pooling. Firstly, they proposed a coarse object location network to roughly locate the object with category annotation. Secondly, they refined the coarse object location to generate pixel-level pseudo labels and proposed a quality check strategy to select high-quality pseudo labels. Given the considerable variation in the quality of CAM-based pseudo labels, Zhou, Chen, Yang, Xie and Lai (2023) came up with an Activation-to-Saliency (A2S) framework that effectively excavates saliency cues. They transformed an pre-trained pseudo-label generation network to aggregate multi-level features into a single activation map to obtain high-quality pseudo labels.

Given the significant variability in the quality of Class Activation Map (CAM)-based pseudo labels, Zhou, Chen et al. (2023) proposed an innovative Activation-to-Saliency (A2S) framework to effectively extract saliency cues. Their approach transforms a pre-trained pseudo-label generation network to aggregate multi-level features into a unified activation map, thereby generating high-quality pseudo labels. By addressing the limitations of traditional CAM-based methods, the A2S framework enhances the reliability and precision of weakly supervised salient object detection, demonstrating the potential of feature aggregation in improving pseudo label quality.

To improve the transferability and applicability, researchers attempted to design a unified framework (Chen, Wei, Guo and Gao, 2024; Hao et al., 2025; Su et al., 2024). Yuan, Liu, Gao, Dai, and Qin (2024a) proposed a progressive curriculum learning-based saliency distilling mechanism to extract saliency cues from a pre-trained salient object detection model. Following this, the authors employed a self-rectify pseudo label refinement mechanism to improve the quality of pseudo labels. Finally, they designed an adapter-tuning to transfer the acquired saliency knowledge. ResNet-50 (He et al., 2016) pre-trained by MoCo-v2 (Chen, Fan et al., 2020) along with several Squeeze-and-Excitation (SE) (Hu, Li, & Sun, 2018) modules, A2S (Zhou, Chen et al., 2023), and MIDD (Tu, Li, Li, Lang, & Tang, 2021) are employed as the pre-trained

salient object detection, saliency cues extraction, and saliency detector, respectively.

In addition to directly using weak annotations to generate pseudo labels, SAM can use weak annotations as a prompt to segment relatively fine-grained objects, thus compensating for the limitations of relying solely on weak annotations. Gao, Zhang, Yan, and Lu (2024) proposed a multi-scale and detail-enhanced SAM for SOD. They designed multi-scale and detail enhancement strategies, along with multi-level fusion techniques, to enable the network to learn detailed information about salient objects. The SAM possesses extraordinary segmentation and interactive prompt capabilities. Inspired by this, Liu, Deng et al. (2024) proposed a scribble-supervised salient object detection framework based on SAM, named SSFam. They are designed to handle multi-modal inputs such as RGB, RGB-D, RGB-T, and Visual-Depth-Thermal images. Modality-aware modules were developed to extract both modality-specific and modality-agnostic features, which are combined with SAM's powerful segmentation abilities. Additionally, a custom siamese decoder bridges the gap between training with scribble annotations and testing without prompts, allowing the model to achieve top performance across various modality combinations, nearing the results of fully supervised algorithms. Wang, Zhang and Shi (2023) combined CAM and the foundation model SAM, proposed a novel framework for weakly supervised change detection, namely CS-WSCDNet, which can achieve pixel-level results by training on samples with category labels. The framework is built upon the localization capability of CAM and the powerful zero-shot segmentation capability of SAM.

2.4.2. Pseudo label-supervised WSVSOD algorithms

Some approaches have attempted to leverage CAM for WSVSOD by integrating additional mechanisms to address the limitations of temporal information. Lu et al. (2020) proposed a unified, multi-granularity video object segmentation framework named MuG, which combined unsupervised saliency algorithms (Yang, Zhang, Lu, Ruan, & Yang, 2013) with CAM to enhance the discriminability of foreground and background representations within frames. Similarly, Ma, Du, Zhuo et al. (2024) employed a multiple pseudo label aggregation strategy, utilizing CAM and the Itti model (Itti, Koch, & Niebur, 1998) to generate diverse pseudo labels. Their Multiple Pseudo Label Aggregation Network (MPLA-Net) demonstrated the effectiveness of integrating multiple sources of saliency cues. These methods highlight the potential of combining category annotations with additional techniques to improve performance in WSVSOD tasks. Given the success of deep learning-based SOD tasks, Xu, Jiang, Liang et al. (2024) attempted to learn video saliency cues from SOD models. They utilized a progressive framework to locate and segment salient objects in videos without requiring video annotations. By leveraging knowledge from Image Salient Object Detection (ISOD) and introducing dynamic saliency, the method generates spatiotemporal location labels through a tracking mechanism across frames, enabling efficient knowledge transfer.

Although category annotations provide less information, especially the lack of inter-frame correlation information in the temporal dimension, these annotations are still widely used in various computer vision tasks due to their low annotation cost. For example, in the fields of RGB-T instance segmentation (Ma, Qian, Zhang, & Ma, 2020), image segmentation (Ghiasi, Gu, Cui, & Lin, 2022; Pinheiro & Collobert, 2015), semantic segmentation (Jing, Chen, & Tian, 2020; Wei et al., 2016), and object localization (Sun, Feng, Ma, & Liu, 2023), category annotations have demonstrated their cost-effectiveness and practicality. However, their limitations in providing temporal information continue to pose significant challenges for its application, prompting researchers to explore alternative weak supervision methods that better align with the dynamic nature of videos.

Recently, SAM is also used in the WSVSOD task, Zhang, Zhang, Wei, Dai and Zhu (2024) proposed STD-Net, which utilized the generated bounding boxes as prompts for SAM to produce segmentation results.

These results were subsequently used as pseudo labels to train the network for the video object segmentation task. In addition, SAM also has been widely used in many pixel-level computer vision tasks, such as medical segmentation (Gong et al., 2024; Huo, Tian, Zhou, Yu, & Li, 2024; Ma, He et al., 2024; Zhang, Deng and Lu, 2023), video super-resolution (Lu, Xiao, Bai, Xiong, & Wang, 2023), semantic segmentation (Kweon & Yoon, 0000; Zhang, Zhou et al., 2023), object detection (Chen, Wei et al., 2024; Qian, Lin, Chen, & Wang, 2024), vehicle re-identification (Wang, Wang, Min et al., 2024), and concealed object segmentation (He et al., 2023). However, due to the inherent differences between static images and video frames, the results generated by SAM, which is trained on image datasets, may not always be reliable (Xu, Jiang, Zhao et al., 2024). For example, He et al. (2023) used pseudo labels generated by SAM to train the concealed object segmentation network. However, the predicted salient objects are often incomplete, and the background is frequently misclassified as salient objects in many scenes. To address these limitations, Xu, Jiang, Zhao et al. (2024) proposed a multidimensional exploration strategy that integrates SAM with scribble annotations. Their approach introduces three key components: (1) a SAM-guided label enhancement method, (2) a SAM-driven spatiotemporal network, and (3) a global-aware loss function. This strategy effectively improves the model's ability to perceive global connections and accurately segment complete salient objects, overcoming the challenges of relying solely on scribble annotations.

While SAM has demonstrated impressive performance across various pixel-level tasks, it is not always perfect (Ji et al., 2024), and the SAM model is trained on images, so certain limitations arise when it is applied to video tasks. Consequently, Ravi et al. (2024) introduced an improved version, SAM2. This upgraded model boasts the ability to segment anything, whether in images or videos. Subsequently, researchers utilized SAM2 for the collection of the VOS dataset (Pan, Fang, Cong, Zhang, & Lu, 2024). And SAM2 is also used in surgical video segmentation (Shen, Ding, Shao, & Unberath, 2024; Yu et al., 2024), semantic segmentation (Zhang, Wang et al., 2024), camouflaged object detection (Tang & Li, 2024), as well as medical image/video segmentation tasks (Ma, Kim et al., 2024; Sengupta, Chakrabarty, & Soni, 2024; Xiong et al., 2024; Zhu, Qi and Wu, 2024). Some researchers also compared the performance of SAM2 and SAM in the medical image segmentation task (Sengupta et al., 2024).

In addition to SAM-based methods, many researchers have combined SAM and CLIP for pixel-level tasks. Wang, Vasu et al. (2024) integrated techniques of multitask learning, continual learning, and distillation to efficiently merge SAM and CLIP into a unified model named SAM-CLIP. This model not only reduces storage and compute costs but also enables zero-shot semantic segmentation. Li, Cao et al. (2024) summarized the limitations of using CLIP and SAM for anomaly segmentation tasks and proposed a targeted improvement scheme called Clip-SAM. Aleem et al. (2024) integrated SAM and CLIP into a unified framework for medical image segmentation. This integrated model, named SaLIP, effectively performs zero-shot organ segmentation. Koleilat, Asgariandehkordi, Rivaz, and Xiao (2024a, 2024b) proposed MedCLIP-SAM and MedCLIP-SAMv2, which integrates the CLIP and SAM, aimed at achieving universal text-driven medical image segmentation.

The pseudo labels generated by SAM and CLIP have proved valuable in their utilization, as they do not require any manual pixel-level annotation for salient objects. The WSISOD and WSVSOD algorithms trained using these pseudo labels can approach the performance of fully supervised algorithms. Moreover, this pseudo label generation method has found extensive utilization and demonstrated commendable performance in a plethora of other computer vision tasks.

2.5. Hybrid label-supervised WSISOD/WSVSOD algorithms

Both WSISOD and WSVSOD algorithms require capturing multi-level features from both global and local views while processing object

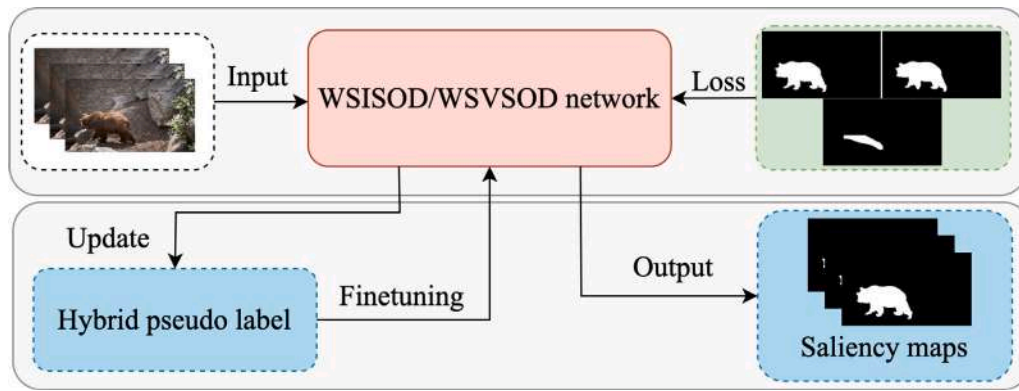


Fig. 4. The mainstream pipeline for hybrid label-based WSISOD and WSVSOD algorithms.

interiors and boundaries to obtain a clear salient map with sharp boundaries. Due to the limitations of weak annotations in salient object detection algorithms, combining hybrid labels and multi-source feature extraction presents a promising solution. This approach leverages diverse data sources and varying levels of annotation detail to improve feature representation and segmentation accuracy, addressing the challenges posed by incomplete or imprecise labels. The mainstream pipeline for a hybrid label supervised WSISOD and WSVSOD algorithm is shown in Fig. 4.

2.5.1. Hybrid label-supervised WSISOD algorithms

The use of a small amount of ground truth annotations combined with a large number of pseudo labels as supervisory information for WSISOD, resulting in a hybrid label scheme, is a common training approach. Tang et al. (2019) proposed a video salient object detection algorithm, which can learn a robust salient object detection model by using very limited manual pixel-level annotation data and a large amount of weakly labeled data that could be easily generated in a supervised approach. They designed a spatiotemporal cascade neural network architecture to learn the saliency cues. Similarly, Cong, Qin et al. (2023) proposed a hybrid label learning framework which ingeniously employs a small proportion of pixel-level labels and a larger majority of pseudo labels, generated through traditional approaches, to achieve salient object detection. They designed a complex, effective training strategy to address the imbalance between the pixel-level labels and pseudo labels, which includes an alternate iteration mechanism, group-wise incremental mechanism, and credibility verification mechanism. Wang, Zhang, Yu, and Han (2025) proposed an effective progressive expansion network (PENet) architecture, which used Semi-Supervised Learning Architecture (SSLA) to alleviate the cost of pixel-level annotation while ensuring detection performance. They designed two optimization strategies, including dynamic adjustment and active expansion, which were used to enable the model to continuously leverage pseudo labels to guide further exploration of unlabeled data and help the model focus on difficult-to-segment samples and acquire more useful supervised information, respectively.

Meanwhile, there are also many few-shot algorithms. Zhang, Tian and Han (2020) proposed an Adversarial-Paced Learning (APL)-based framework to facilitate few-cost salient object detection, which was based on a self-paced learning mechanism (Kumar, Packer, & Koller, 2010). The proposed algorithm can effectively approach the existing supervised deep salient object detection models with only 1k pixel-level annotated training images. Wu et al. (2022) proposed SODGAN, which was a novel approach for salient object detection that generates infinite high-quality image-mask pairs using Generative Adversarial Networks (GAN) (Andrew, Jeff, & Karen, 2018) based minimal labeled data, effectively replacing manual annotation pixel-level datasets for training SOD model. SODGAN's key contributions include a diffusion embedding network to address manifold mismatch, a few-shot saliency mask generator

to synthesize precise masks, and a quality-aware discriminator to filter high-quality pairs from noisy synthetic data, pioneering a new research paradigm for SOD.

Learning saliency cues from multiple types of weak annotations is also a common approach. Han et al. (2023) proposed a multi-source pseudo label learning method that is a two-stage framework. In the first stage, category annotations and bounding box annotations are used in the multi-source supervision to get CAM and salient bounding box, which are fused to obtain pixel-level pseudo labels. In the second stage, the generated pixel-level pseudo labels are used to train the Transformer network. Similarly, Wang, Wang, He et al. (2024) proposed a multi-source weakly supervised salient object detection framework, known as WBNet. They capitalized on scribble annotations, SAM-based pseudo labels prompted by point annotations and bounding boxes to learn the saliency cues. Many works are using a multi-source feature to achieve salient object detection (Li et al., 2022; Xu & Guo, 2023; Zhang et al., 2022). Liu, Xu, Su, Zhang and Li (2024) proposed a novel point-supervised segment anything model, which fine-tunes SAM using point annotations. They also considered the issue of noise in pseudo labels.

Numerous hybrid annotation fully supervised algorithms can provide insights for WSVSOD. Zhao et al. (2019) used both edges and pixel-level annotation as supervision information to address the issue of coarse object boundaries in existing salient object algorithms. Zhou, Shen et al. (2023) proposed an edge-guided recurrent positioning network, which consists of an edge extraction part and a feature fusion part. Building upon this foundational work, Gong et al. (2023) and Liu, Hou, and Cheng (2020) introduced skeleton information as additional supervision and demonstrated the fused features of edge and skeleton are essential in the salient object detection task. These ISOD methods provide new insights for weakly supervised methods. To obtain finer edge information, many weakly supervised methods also use edge detection as an auxiliary task.

Enhancing the learning strategy represents a promising approach for improving the performance of the model. Liu, Zhang, Wang, Ma et al. (2023) used multiple learning strategies to integrate weakly supervised, unsupervised and supervised learning for SOD task. The unsupervised learning module, weakly supervised learning module and supervised learning module are designed to generate coarse saliency features, produce relatively accurate saliency features based on scribble annotations, and process and refine information from the first two components. To obtain high-quality pseudo labels, Liu and Huang (2024) proposed a two-stage weakly supervised algorithm based on bounding-box annotation and SAM. In the first stage, bounding-box annotations are employed as prompts for SAM to generate initial pseudo labels, and reliable pseudo labels are subsequently selected to train the initial SOD model. Then, the selected high-quality pseudo labels are used to train the salient object detection network. In the second stage, the salient object detection model trained in the first stage is used to predict the

saliency map, and SAM is adopted to generate segmentation candidates to generate another pseudo label. These pseudo labels generated in two stages are used to train a refined salient object detection model.

2.5.2. Hybrid label-supervised WSVSOD algorithms

There are also some hybrid label schemes in WSVSOD. Yan et al. (2019) proposed a semi-supervised VSOD algorithm, which consists of a spatial refinement network and a spatiotemporal module. They used pixel-level pseudo labels and some ground truth to train the network, which can achieve performance comparable to fully supervised VSOD algorithms. As previously mentioned, the generation of pseudo labels is uncontrollable. To fully utilize the incorrect predictions in the pseudo labels to guide the network to generate more robust pseudo labels, Piao, Lu, Zhang, and Lu (2022a) proposed a semi-supervised VSOD algorithm, in which an uncertainty-guided pseudo label generator is designed to fully harnesses inter-frame information to ensure temporal consistency among pseudo labels. Further advancing hybrid label schemes, Zhao, Nan et al. (2023) proposed a novel annotation scheme that requires only the ground truth of the first frame and bounding boxes for subsequent frames. To fully exploit these labels, they designed a language-guided cross-frame segmentation method and a bi-level contrastive learning method, which enhance the model's ability to learn discriminative feature representations. Zhang, Cheng and Qi (2023) proposed SemiSAM, using extremely limited annotations to achieve medical image segmentation. SemiSAM leveraged the consistency of predictions between SAM and the semi-supervised segmentation model as an additional supervision signal to assist in the network learning.

Recent advances in few-shot salient object detection and video object segmentation methods have improved both efficiency and accuracy over time. The earliest approach, Caelles et al. (2017) introduced prototypical networks to perform one-shot VOS, using a single annotated frame to provide robust segmentation across video sequences. Building on this, Chanda and Wang (2019) proposed a method that transferred attention mechanisms from image tasks to video segmentation. By addressing the lack of annotated frames in videos, their work improved the generalization capability of few-shot VOS models. Subsequently, Meinhardt and Leal-Taixé (2020) proposed e-OSVOS, which decoupled object detection from segmentation using a modified Mask R-CNN and applies meta-learning (Hospedales, Antoniou, Micaelli, & Storkey, 2021) to optimize model initialization and learning rates, significantly reducing test-time computation. Zhao, Pang, Yang, Zhang and Lu (2021) proposed that a multi-source feature extraction network integrates global, boundary, and salient features through a multi-source global and local module, achieving superior performance. Recently, Chen, Zhao and Chan (2024) introduced a spatiotemporal auto-encoder that captures both spatial and temporal dynamics, enabling precise segmentation even with just one annotated frame. These works collectively tackle major challenges in one-shot VOS, offering increasingly sophisticated and efficient solutions over time.

In general, hybrid label-based WSISOD and WSVSOD algorithms typically build on weak annotations by incorporating a small amount of real pixel-level labels or additional information, such as edges and skeletons, as supervision. These algorithms can usually generate higher-quality pseudo labels for model training. The integration of these additional supervisory signals enables WSISOD and WSVSOD algorithms to generate saliency maps with enhanced precision, resulting in more finely boundaries of salient objects.

2.6. Discussion

The exceptional performance of deep learning-based algorithms is largely attributed to their strong dense representation learning ability (Shen et al., 2023), particularly excelling in handling intricate image details. However, in the context of ISOD and VSOD, using weak supervision still presents numerous challenges. This section summarized the state-of-the-art algorithms supervised by various types of supervisory

information, including point annotations, scribble annotations, bounding box annotations, pseudo labels, and hybrid labels, and analyzed the limitations and potential advantages of each approach.

Among the manually annotated supervisory information, category labels have the lowest cost but provide the weakest supervision. Category labels, point annotations, and scribble annotations offer limited supervisory information due to the lack of boundary and complete structural information, leading to models that struggle with global structure perception, resulting in relatively coarse boundaries. In contrast, bounding box annotations offer approximate location and contour information for salient objects, enriching the supervisory information. However, they still fall short in providing precise pixel-level annotations, especially in terms of object edges and fine details. While pseudo labels reduce the manual annotation workload, the generated labels may suffer from inaccurate salient objects and insufficient boundary information. Hybrid labels, by incorporating edge, skeleton, and partial pixel-level annotations, further enhance the model's capability to capture saliency cues, improving detection precision.

In the task of WSISOD and WSVSOD, the model's performance is closely tied to the level of weak supervision, with each type of annotation offering its own advantages and limitations. To bridge the gap between weak supervision and real pixel-level labels, researchers have proposed various algorithms, including pseudo label generation, boundary-aware modules, multiple pseudo label aggregation, and regularization, all aimed at maximizing the model's robustness and accuracy. Notably, by designing targeted strategies, such as introducing boundary-aware loss, improving pseudo label generation algorithms, and using hybrid labels, can effectively enhance its edge detection capability for salient objects, thereby improving its capability to capture salient object boundaries and finer details. Representative publications of WSISOD and WSVSOD algorithms are shown in Tables 1 and 2.

It can be seen that the publication information, backbone, supervision information (Sup.) and solution are summarized. In the tables, "T" denotes the label is generated by traditional methods, "SA" indicates scribble annotations, "PL" indicates pseudo labels, and "PA" indicates point annotations, "H" denotes hybrid labels, "/" indicates unavailable information. VGG16 is the very deep convolutional network proposed by Simonyan and Zisserman (2015). Simple Linear Iterative Clustering (SLIC) (Achanta et al., 2010) is an algorithm that performs preprocessing through superpixel clustering, while entropy-based probabilistic model (Leordeanu, Sukthankar, & Sminchisescu, 2012) captures saliency information using entropy and probabilistic models. The WSISOD and WSVSOD algorithms have been gaining increasing attention in recent years. Numerous researchers have made improvements in aspects such as network architecture, training strategies, loss functions, and pseudo label generation.

3. Public datasets and evaluation metrics

Performance evaluation is a crucial aspect of ISOD and VSOD algorithms. To ensure fairness, it is generally conducted on the same dataset. Firstly, this section provides a summary of five representative public ISOD datasets, and then the six benchmark public VSOD datasets are summarized. Secondly, the evaluation metrics are introduced. Subsequently, the existing ISOD and VSOD algorithms are summarized and analyzed from the perspectives of quantitative evaluation, qualitative comparison, and model complexity.

3.1. Public ISOD datasets

Five representative public ISOD datasets are widely used in the experiments, including PASCAL-S (Li, Hou, Koch, Rehg, & Yuille, 2014), HKU-IS (Li & Yu, 2015), ECSSD (Shi, Yan, Xu, & Jia, 2016), DUT-OMRON (Perazzi et al., 2016), DUTS (Ochs, Malik, & Brox, 2014). More datasets can be found in Ref. (Wang, Lai et al., 2022). Fig. 5 shows examples of images (including video frames and pixel-level 673

Table 1

Representative publications of WSISOD algorithms. “Sup.” denotes supervised information, “T” denotes the label is generated by traditional methods, “SA” indicates scribble annotations, “PL” indicates pseudo labels, “BB” indicates bounding box annotations, “H” denotes hybrid labels, “Cap” indicates image caption, “/” indicates unavailable information.

Algorithms	Year	Backbone	Sup.	Solution
SAG (Wang, Shen and Porikli, 2015)	CVPR'15	/	T	Geodesic measurement, spatial edges and temporal motion boundaries.
ASMO (Li, Xie, Lin, 2018)	AAAI'18	ResNet-101	PL	FCN, iteratively mining and correcting the labeling ambiguity on the saliency map.
MSW (Zeng et al., 2019)	CVPR'19	/	PL+Cap	A Multi-source, attention transfer loss, attention coherence loss.
WSSA (Zhang, Yu et al., 2020)	CVPR'20	VGG16	SA	An auxiliary edge detection task, edge-enhanced saliency prediction module, gated structure-aware loss.
MFNet (Piao et al., 2021)	ICCV'21	DenseNet-169	PL	A multi-filter directive network, multiple directive filters.
SBBs (Liu, Wang et al., 2021)	TIP'21	ResNet101	BB	Multi-task network, saliency prediction map fuse.
NSAL (Piao, Wu, Zhang, Jiang, & Lu, 2023)	TMM'23	DenseNet-169	PL	Adversarial learning, a noise-robust adversarial learning framework.
HSS (Cong, Qin et al., 2023)	TCSVT'23	ResNet-50	H	Three training strategies, including alternate iteration mechanism, group-wise incremental mechanism, and credibility verification mechanism.
A2S (Zhou, Chen et al., 2023)	TCSVT'23	ResNet-50	PL	Two-stage Activation-to-Saliency (A2S) framework, a self-rectification learning strategy.
A2SV2 (Zhou, Qiao, Yang, Lai, & Xie, 2023d)	CVPR'23	ResNet-50	PL	A confidence-aware saliency distilling strategy, boundary-aware texture matching strategy.
A2SV3 (Yuan, Liu, Gao, Dai, & Qin, 2024b)	IJCAI'24	/	PL	Unified framework, a progressive curriculum learning-based saliency distilling mechanism, a self-rectify pseudo label refinement mechanism.

Table 2

Representative publications of WSVSOD algorithms. “Sup.” denotes supervised information, “SLIC” denotes simple linear iterative clustering, “SA” indicates scribble annotations, “PA” indicates point labels, “PL” indicates pseudo labels, “/” indicates unavailable information.

Algorithms	Year	Backbone	Sup.	Solution
GF (Wang, Shen and Shao, 2015)	TIP'15	/	SLIC	Energy optimization, gradient flow field, spatiotemporal saliency energy.
SSOD (Zhang, Yu et al., 2020)	CVPR'20	VGG16	SA	An auxiliary edge detection task, structure-aware loss.
WS (Zhao, Zhang et al., 2021)	CVPR'21	ResNet-50	SA	Appearance-motion fusion module and bidirectional ConvLSTM based framework.
PVS (Gao, Xing et al., 2022)	MM'22	T2T-ViT_t-14	PA	Hybrid token attention module, a long-term cross-frame attention module.
MPLA-Net (Ma, Du, Zhuo et al., 2024)	TCSVT'24	ResNet-50	PL	Multiple pseudo label aggregation network, video frame quality evaluation, pseudo label consistency evaluation.
CFMR (Wang, Yu et al., 2024)	PR'24	ResNet-50	PL	A cross-frame feature-saliency mutual reinforcing training process.
UWS (Xu, Jiang, Liang et al., 2024)	TMM'24	ResNet-101	PL	Dynamic saliency and two-stream locating network.
MES (Xu, Jiang, Zhao et al., 2024)	TCSVT'24	ResNet-50	PL	SAM-driven spatiotemporal network and a SAM-based global-aware loss.

Table 3

Statistical information of the five public ISOD datasets. “PixL” indicates pixel-level annotation, “SA” indicates scribble annotations, “BB” indicates bounding box annotations, “/” indicates unavailable information.

Dataset	Year	Image	Train	PixL	SA	BB	Download link
PASCAL-S (Li et al., 2014)	2014	850	✓	/	/	/	http://cbl.gatech.edu/salobj/
HKU-IS (Li & Yu, 2015)	2015	4447	/	✓	/	/	https://i.cs.hku.hk/~gbli/deep_saliency.html
ECSSD (Shi et al., 2016)	2016	1000	/	✓	/	/	https://www.cse.cuhk.edu.hk/leojia/projects/hsaliency/
DUT-OMRON (Perazzi et al., 2016)	2016	5168	/	✓	/	✓	http://saliencydetection.net/dut-omron/
DUTS (Ochs et al., 2014)	2017	15 572	10 553	✓	✓	/	http://saliencydetection.net/duts/

annotations) from these datasets. The specific details for each dataset are provided as follows.

PASCAL-S dataset (Li et al., 2014) is built on the validation set of the PASCAL VOC 2010 segmentation challenge (Everingham, Van Gool, Williams, Winn, & Zisserman, 2010) in 2014, which includes 850 samples that are annotated by 8 subjects in eye-tracking tests.

HKU-IS dataset (Li & Yu, 2015) is proposed in 2015 and consists of 4447 images with many multiple-object scenes. The images were selected based on criteria such as having multiple disconnected salient objects, at least one salient object touching the image boundary, or a color contrast of less than 0.7.

Extension of Complex Scene Saliency Dataset (ECSSD) dataset (Shi et al., 2016) is the extension of Complex Scene Saliency Dataset (CSSD) (Yan, Xu, Shi, & Jia, 2013) dataset, which is proposed in 2013 and contains 200 images. ECSSD is proposed in 2016 and contains 1000 images of different sizes with obvious salient objects.

DUT-OMRON dataset (Perazzi et al., 2016) is proposed in 2016 and has 5168 images of complex backgrounds and diverse content. The dataset also includes bounding box and eye fixation annotations.

DUTS dataset (Wang et al., 2017) is proposed in 2017, containing 10,553 training images and 5019 test images. Since 2017, SOD models are typically trained on DUTS (Wang, Lai et al., 2022).

Table 3 provides detailed statistical information on five representative public ISOD datasets. The symbol “/” indicates the absence of

certain data points. “SA” denotes scribble annotations, “PixL” denotes pixel-level annotations, and “BB” denotes bounding box annotations. The data shows a clear trend over time: the size of ISOD datasets and the volume of annotations have steadily increased, with the content becoming more diverse and comprehensive.

3.2. Public VSOD datasets

For VSOD algorithms, six benchmark public VSOD datasets are widely used in the experiments, including SegV2 (Li, Kim, Humayun, Tsai, & Reh, 2013), FBMS (Ochs et al., 2014), ViSal (Wang, Shen, Shao, 2015), DAVIS (Perazzi et al., 2016), VOS (Li, Xia and Chen, 2018), DAVSOD (Fan et al., 2019). In addition, the DVSOD dataset (Li, Ji, Wang, Li, & Cheng, 2023) is the latest and more comprehensive dataset. For the sake of fairness, many authors divided the datasets into training set and testing set according to Gu et al. (2020), Li, Chen et al. (2019), and Zhang et al. (2021). Fig. 6 shows examples of images (including video frames and pixel-level annotations) from these datasets. The specific details for each dataset are provided as follows.

SegV2 (SegTrack_V2) dataset (Li et al., 2013) is an updated version of the georgia tech segmentation and tracking dataset (Tsai, Flagg, Nakazawa, & Reh, 2012), which is proposed in 2013 and contains 14 videos with about 1000 frames.

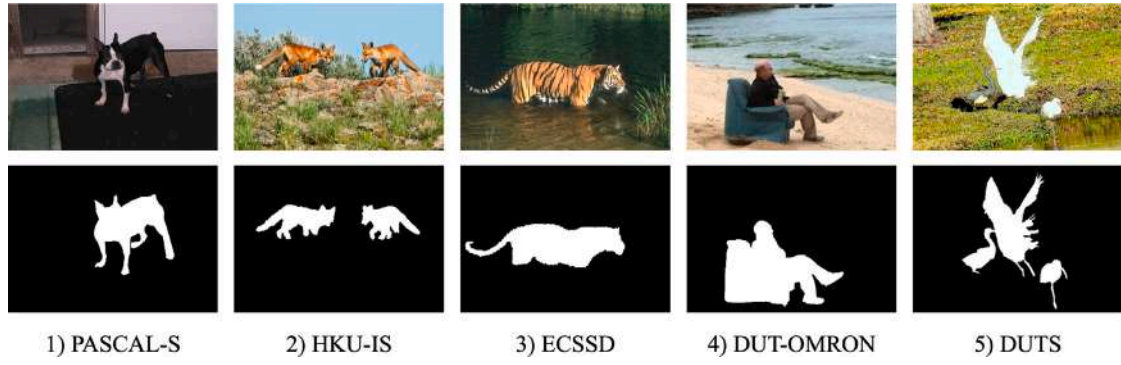


Fig. 5. Examples of images and pixel-level annotations from five ISOD datasets.

Table 4

Statistical information of the public VSOD datasets. “AnF” denotes the number of annotation frames, “PixL” indicates pixel-level annotation, “SA” indicates scribble annotations, “BB” indicates bounding box annotations, “/” indicates unavailable information.

Dataset	Year	Video	AnF	Train	PixL	SA	BB	Download link
SegV2 (Li et al., 2013)	2013	14	1065	/	✓	/	/	https://web.engr.oregonstate.edu/lif/SegTrack2/dataset.html
FBMS (Ochs et al., 2014)	2014	59	720	353	✓	/	/	https://lmb.informatik.uni-freiburg.de/resources/datasets
ViSal (Wang, Shen, Shao, 2015)	2015	17	193	/	✓	/	/	https://github.com/shenjianbing/ViSalDataset
DAVIS (Perazzi et al., 2016)	2016	50	3455	2079	✓	✓	/	https://davischallenge.org/davis2016/code.html
VOS (Li, Xia et al., 2018)	2018	200	7467	5927	✓	/	/	http://cvteam.net/projects/TIP18-VOS/VOS.html
DAVSOD (Fan et al., 2019)	2019	226	23 938	12 607	✓	✓	/	https://github.com/DengPingFan/DAVSOD
DVSOD (Li et al., 2023)	2023	237	7117	/	✓	✓	✓	https://github.com/DVSOD/DVSOD-DViSal

Freiburg-Berkeley Motion Segmentation (FBMS) dataset (Ochs et al., 2014) is proposed in 2014, which includes 59 video and 720 annotation frames.

ViSal dataset (Wang, Shen, Shao, 2015) is the first dataset dedicated to the VSOD task proposed in 2015, which consists of 17 videos containing complex color distributions, highly cluttered background, various object motion patterns, rapid topology changes and camera motion. Each video contains 30 to 100 frames, and all videos are manually annotated as the given classes.

Densely Annotated Video Segmentation (DAVIS) dataset (Perazzi et al., 2016) is proposed in 2016, which is a larger dataset. The dataset carefully addresses factors such as content diversity and data scale, enhancing its lifespan (Butler, Wulff, Stanley, & Black, 2012). The DAVIS dataset consists of 50 videos and 3455 annotated frames, all captured at 24 frames per second with full HD 1080p resolution.

VOS dataset (Li, Xia et al., 2018) is proposed in 2018, which consists of 200 videos. According to the complexity of the foreground, background, and motion, VOS is divided into two subsets, including VOS-E (easy subset, 97 videos) and VOS-N (normal subset, 103 videos).

Densely Annotated VSOD (DAVSOD) dataset (Fan et al., 2019) is proposed in 2019, whose stimuli comes from the dynamic eye-tracking dataset DHF1K (Wang, Shen, Guo, Cheng and Borji, 2018). The dataset encompasses a wide range of realistic scenes, varying object appearances and motion patterns, diverse object categories, and a broad spectrum of major challenges in dynamic environments. DAVSOD contains 226 videos with a total of 798 s and 23,938 annotation frames. DAVSOD is randomly divided into 90 training videos, 46 validation videos, and 90 test videos. According to the difficulty of the task, the 90 videos are further divided into 35 easy, 30 ordinary, and 25 difficult subsets.

RGB-D Video Salient Object Detection (DVSOD) dataset (Li et al., 2023) is proposed in 2023. This dataset features 237 diverse RGB-D videos alongside comprehensive annotations, including object and instance-level markings, as well as bounding boxes and scribbles. It is partitioned into training, validation, and test sets, which consist of 10 326 and 108 videos, respectively, with 3 560 200 and 3357 annotated frames each. The validation set is made up of 26 videos.

Table 4 provides detailed statistical information on six benchmark public VSOD datasets. The symbol “/” indicates the absence of certain

data points. “AnF” denotes the number of annotation frames, while “SA” denotes scribble annotations, and “PixL” denotes manual pixel-level annotations. The data reveals a clear trend over time: VSOD datasets have steadily increased in size and the volume of annotations, with content becoming more diverse and comprehensive. This expansion offers researchers a robust and reliable foundation for the development and refinement of VSOD algorithms, ensuring more accurate and generalizable results.

3.3. Evaluation metrics

Structure Measure (SM) (Fan, Cheng, Liu, Li, & Borji, 0000), Frequency Measure (FM) (Achanta, Hemami, Estrada, & Susstrunk, 2009), and Mean Absolute Error (MAE) (Perazzi, Krähenbühl, Pritch, & Hornung, 2012) are widely used metrics to evaluate the ISOD and VSOD algorithms. Enhanced-alignment measure (EM) (Fan, Gong et al., 2018) is more commonly used in ISOD algorithms. The closer the SM, FM, and EM values are to 1, the smaller the gap between the predicted saliency map and the ground truth, indicating better algorithm performance. Conversely, a smaller MAE value also represents better performance.

The formula for SM is as follows:

$$SM = \alpha * S_o + (1 - \alpha) * S_r, \quad (1)$$

where α is usually set to 0.5, S_o is object-aware similarity and S_r is region-aware similarity. S_o and S_r are calculated as follows:

$$S_o = \gamma * O_{FG} + (1 - \gamma) * O_{BG}, \quad S_r = \sum_{n=1}^N \omega_n * SSIM(n), \quad (2)$$

$$O_{FG} = \frac{2\bar{x}_{FG}}{(\bar{x}_{FG})^2 + 1 + 2\varphi * \sigma_{x_{FG}}}, \quad O_{BG} = \frac{2\bar{x}_{BG}}{(\bar{x}_{BG})^2 + 1 + 2\varphi * \sigma_{x_{BG}}}, \quad (3)$$

where γ is the ratio of foreground area in ground truth to image area. O_{FG} and O_{BG} represent the probability values of the saliency map and ground truth in foreground areas and background areas, respectively. x_{FG} and x_{BG} represent the probability values of the foreground region of the saliency map and ground truth, respectively. $\bar{x}_{FG}$ and $\bar{x}_{BG}$ denote the means of x_{FG} and x_{BG} , respectively. φ is the weight. $\sigma_{x_{FG}}$ and $\sigma_{x_{BG}}$ represent the standard deviation.

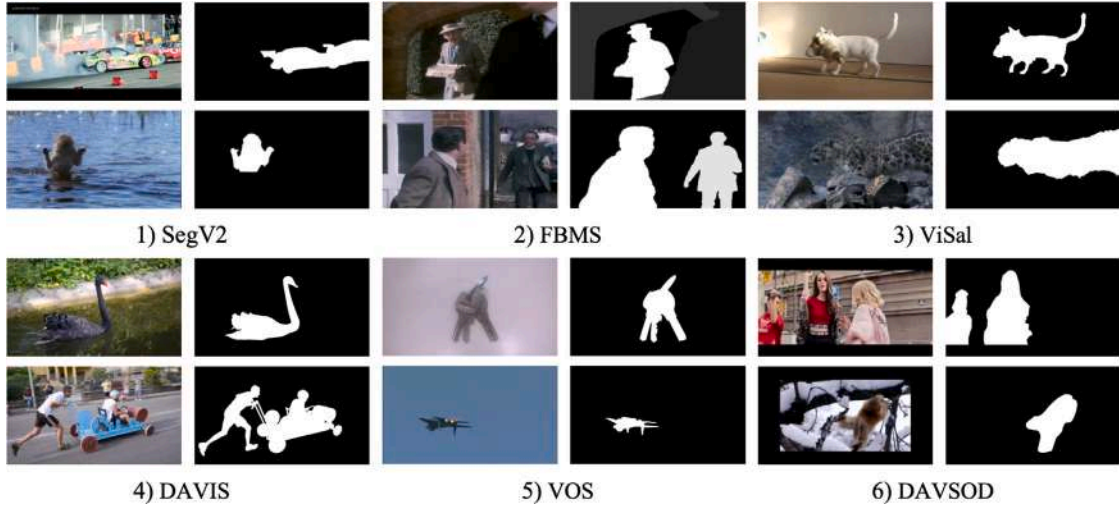


Fig. 6. Left to right: examples of video frames and pixel-level annotations from six VSOD datasets.

When calculating S_r , the saliency map and ground truth are divided into N blocks firstly. Then, the Structural Similarity Index Measure (SSIM) (Wang, Bovik, Sheikh, & Simoncelli, 2004) of each block is calculated. ω_n represents the ratio of the n th block to the foreground.

SSIM is a metric based on the human visual system, designed to evaluate image similarity by examining aspects such as luminance, contrast, and structural features. Compared to traditional pixel-based algorithms like MSE and Peak Signal to Noise Ratio (PSNR), SSIM aligns more closely with human perception, making it widely adopted in tasks like super-resolution reconstruction (Anwar, Khan, & Barnes, 2020; Nasrollahi & Moeslund, 2014), image quality assessment (Zhai & Min, 2020), video quality assessment (Chen, Wu, & Zhang, 2015), and image dehazing (Feng, 2020; Parihar, Gupta, Singodia, Singh, & Singh, 2020).

$$\text{SSIM}(x, y) = \frac{2\mu_x\mu_y + c_1}{\mu_x^2 + \mu_y^2 + c_1} \cdot \frac{2\sigma_x\sigma_y + c_2}{\sigma_x^2 + \sigma_y^2 + c_2} \cdot \frac{\sigma_{xy} + \frac{c_2}{2}}{\sigma_x\sigma_y + \frac{c_2}{2}} \quad (4)$$

$$= \frac{(2\mu_x\mu_y + c_1)(2\sigma_x\sigma_y + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)},$$

where the first, second, and third terms of Eq. (4) measure image luminance, contrast and structure similarity, respectively; μ_x , σ and σ_{xy} are local mean, variance, and covariance; x and y indicate two frames (saliency map and ground truth in VSOD task); c_1 and c_2 are small two stabilization constants.

The formula for FM is as follows:

$$FM = \frac{(1 + \beta^2) \times \text{Precision} \times \text{Recall}}{\beta^2 \times \text{Precision} + \text{Recall}}, \quad (5)$$

where β is set to 0.3 in general. Instead of plotting the whole FM curve, some algorithms only report maximal FM , or binarize the predicted saliency map by an adaptive threshold. Precision and recall are calculated as follows:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad (6)$$

where TP , TN , FP , and FN represent true-positive, true-negative, false-positive, and false-negative, respectively.

The formula for MAE is as follows:

$$MAE = \frac{1}{M} \sum_{m=1}^M |y^m - g^m|, \quad (7)$$

where y^m and g^m are m -th pixel in saliency map and ground truth, respectively.

EM is used to evaluate both local and global similarity between the predicted map and the ground-truth. The formula for EM is as follows:

$$EM = \frac{1}{W \times H} \sum_{x=1}^W \sum_{y=1}^H F_\phi(x, y) \quad (8)$$

where W and H represent width and height of the saliency maps respectively. F_ϕ denotes the enhanced alignment matrix.

While SM and FM are valuable metrics for assessing the performance of SOD models, they inherently favor approaches that successfully highlight salient regions. This bias can lead to models that neglect non-salient areas, which may be critical for a comprehensive understanding of visual content. MAE can remedy this, but it performs poorly when salient objects are small (Borji et al., 2019). To cope with it, existing algorithms usually employ multiple metrics to evaluate the performance.

3.4. The performance of existing ISOD algorithms

To gain a comprehensive understanding of the development of existing WSISOD algorithms, this section provides the performance of existing ISOD algorithms in recent years. These include seventeen fully supervised ISOD algorithms: UCF (Zhang, Wang, Lu, Wang, Yin, 2017)*, Amule (Zhang, Wang, Lu, Wang, Ruan, 2017)*, NLDF (Luo et al., 2017)*, RAS (Chen et al., 2018)*, PAGR (Zhang, Wang et al., 2018)*, BMPM (Zhang, Dai et al., 2018)*, DSS (Hou et al., 2019)*, AFNet (Feng et al., 2019)*, EGNet (Zhao et al., 2019)*, CPD (Wu et al., 2019)*, MINet (Pang et al., 2020)*, VST (Liu, Zhang et al., 2021)*, CII (Liu, Liu et al., 2021)*, DNA (Liu, Cheng et al., 2022)*, ICON (Zhuge et al., 2023)*, VSCode (Luo et al., 2024)*, SENet (Hao et al., 2025)*. Two unsupervised ISOD algorithms: SVF (Zhang, Han et al., 2017)†, C2S (Li, Yang et al., 2018)†, and seventeen WSISOD algorithms: SAG (Wang, Shen, Porikli, 2015), WSS (Wang et al., 2017), ASMO (Li, Xie, Lin, 2018), MSW (Zeng et al., 2019), WSSA (Zhang, Yu et al., 2020), MFNet (Piao et al., 2021), SBBs (Liu, Wang et al., 2021), OSG (Wang, Al-Huda et al., 2021), SHNet (Ma et al., 2022), PWSS (Gao, Zhang et al., 2022), NSAL (Piao et al., 2023), SLF (Liu, Zhang, Wang, Yang et al., 2023), HSS (Cong, Qin et al., 2023), A2S (Zhou, Chen et al., 2023), A2SV2 (Zhou et al., 2023d), A2SV3 (Yuan et al., 2024b), BWSS (Liu & Huang, 2024). The quantitative comparison, qualitative comparison and complexity comparison of these algorithms are reported as follows.

Quantitative comparison. Table 5 shows the quantitative comparison results in terms of SM, FM, EM, and MAE on five datasets. “F” and “U” indicate the fully supervised and unsupervised algorithms. “SA” denotes scribble annotations, “BB” denotes bounding box annotations,

Table 5

Performance comparison of the existing ISOD methods on various datasets. “F” and “U” denote fully supervised and unsupervised algorithms. “T” denotes the label is generated by traditional methods. “PL” indicates pseudo labels. “SA” indicates scribble annotations. “Cap” indicates image caption. “BB” denotes bounding box annotations. “H” denotes hybrid labels. “/” indicates unavailable information.

Methods	Pub.	Sup.	PASCAL-S				HKU-IS				ECSSD				DUT-OMRON				DUTS			
			SM ↑	FM ↑	EM ↑	MAE ↓	SM ↑	FM ↑	EM ↑	MAE ↓	SM ↑	FM ↑	EM ↑	MAE ↓	SM ↑	FM ↑	EM ↑	MAE ↓	SM ↑	FM ↑	EM ↑	MAE ↓
UCF (Zhang, Wang, Lu, Wang and Yin, 2017)*	ICCV'17	F	0.805	0.726	0.804	0.115	0.875	0.823	0.902	0.062	0.883	0.844	0.892	0.069	0.760	0.621	0.765	0.120	0.782	0.631	0.763	0.112
Amule (Zhang, Wang, Lu, Wang and Ruan, 2017)*	ICCV'17	F	0.818	0.757	0.802	0.100	0.886	0.841	0.912	0.051	0.894	0.868	0.901	0.059	0.781	0.647	0.779	0.098	0.804	0.678	0.794	0.085
NLDF (Luo et al., 2017)*	CVPR'17	F	0.805	0.769	0.839	0.098	0.887	0.874	0.929	0.048	0.875	0.878	0.91	0.063	0.77	0.684	0.816	0.08	/	/	/	/
RAS (Chen, Tan, Wang, & Hu, 2018)*	ECCV'18	F	0.799	0.777	0.836	0.101	0.887	0.871	0.929	0.045	0.893	0.889	0.914	0.056	0.814	0.713	0.846	0.062	0.839	0.751	0.861	0.059
PAGR (Zhang, Wang, Qi, Lu and Wang, 2018)*	CVPR'18	F	0.822	0.798	0.853	0.089	0.887	0.886	0.939	0.048	0.889	0.894	0.914	0.061	0.775	0.711	0.842	0.071	0.838	0.786	0.88	0.054
BMPM (Zhang, Dai, Lu, He and Wang, 2018)*	CVPR'18	F	0.845	0.758	0.842	0.074	0.907	0.871	0.937	0.039	0.911	0.868	0.914	0.045	0.809	0.692	0.837	0.064	0.862	0.745	0.86	0.049
DSS (Hou et al., 2019)*	TPAMI'19	F	0.798	0.801	0.847	0.093	0.878	0.902	0.934	0.040	0.882	0.904	0.912	0.052	0.790	0.740	0.842	0.063	/	/	/	/
AFNet (Feng, Lu, & Ding, 2019)*	CVPR'19	F	0.850	0.868	/	0.071	0.905	0.923	/	0.036	0.914	0.935	/	0.042	0.826	0.797	/	0.057	0.866	0.862	/	0.046
EGNet (Zhao et al., 2019)*	ICCV'19	F	0.852	0.817	0.854	0.074	0.918	0.902	0.949	0.031	0.925	0.920	0.927	0.037	0.841	0.755	0.868	0.053	0.887	0.815	0.891	0.039
CPD (Wu, Su, & Huang, 2019)*	CVPR'19	F	0.848	0.820	0.855	0.071	0.905	0.891	0.944	0.034	0.918	0.917	0.925	0.037	0.825	0.747	0.866	0.056	0.869	0.805	0.886	0.043
MINet (Pang, Zhao, Zhang, & Lu, 2020)*	CVPR'20	F	0.856	0.829	0.857	0.064	0.919	0.909	0.953	0.029	0.925	0.924	0.927	0.033	0.833	0.755	0.865	0.056	0.884	0.828	0.898	0.037
VST (Liu, Zhang, Wan, Shao and Han, 2021)*	ICCV'21	F	0.872	0.829	0.844	0.061	0.928	0.900	0.953	0.029	0.932	0.920	0.918	0.033	0.850	0.756	0.861	0.058	0.896	0.818	0.892	0.037
CII (Liu, Liu, Peng and Cheng, 2021)*	TIP'21	F	0.865	0.841	0.863	0.062	0.920	0.915	0.955	0.029	0.926	0.929	0.926	0.033	0.839	0.768	0.872	0.054	0.888	0.835	0.900	0.037
DIA (Liu, Cheng, Zhang, Nie and Wang, 2022)*	TCYBE'22	F	0.837	0.815	0.856	0.074	0.913	0.903	0.95	0.028	0.918	0.92	0.928	0.035	0.826	0.743	0.857	0.056	0.872	0.813	0.892	0.039
ICON (Zhuge et al., 2023)*	TPAMI'23	F	0.861	0.833	0.861	0.064	0.920	0.910	0.952	0.029	0.927	0.928	0.927	0.032	0.844	0.772	0.870	0.057	0.889	0.838	0.902	0.037
VSCode (Luo et al., 2024)*	CVPR'24	F	0.887	0.864	0.904	/	0.940	0.951	0.974	/	0.949	0.959	0.974	/	0.877	0.840	0.912	/	0.926	0.922	0.960	/
SENet (Hao et al., 2025)*	TIP'25	F	0.890	0.866	0.923	0.066	0.941	0.953	0.970	0.019	0.949	0.964	0.968	0.020	0.876	0.835	0.899	0.040	0.926	0.925	0.953	0.022
SVF (Zhang, Han and Zhang, 2017)†	ICCV'17	U	0.758	0.695	0.789	0.131	/	/	/	/	0.832	0.809	0.875	0.088	0.747	0.608	0.768	0.108	/	/	/	/
C2S (Li, Yang, Cheng, Liu and Shen, 2018)†	ECCV'18	U	0.826	0.754	0.838	0.087	0.873	0.839	0.919	0.051	0.882	0.853	0.906	0.059	0.780	0.664	0.817	0.079	0.817	0.710	0.841	0.066
SAG (Wang, Shen, Porikli, 2015)	CVPR'15	T	0.676	0.515	0.103	0.565	0.370	0.184	0.619	0.482	0.172	0.749	0.688	0.105	0.719	0.634	0.081	0.852	0.048	/	/	/
WSS (Wang et al., 2017)	CVPR'17	PL	0.744	0.715	0.791	0.139	0.822	0.821	0.896	0.079	0.811	0.823	0.869	0.104	0.725	0.603	0.768	0.109	0.748	0.654	0.795	0.1
ASMO (Li, Xie, Lin, 2018)	AAAI'18	PL	0.717	0.693	0.772	0.149	0.804	0.806	0.878	0.086	0.802	0.798	0.853	0.11	0.752	0.622	0.776	0.101	0.697	0.614	0.772	0.116
MSW (Zeng et al., 2019)	CVPR'19	PL+Cap	0.768	0.713	0.79	0.133	0.818	0.814	0.895	0.084	0.827	0.84	0.884	0.096	0.756	0.609	0.763	0.109	0.759	0.684	0.814	0.091
BMSA (Zhang, Yu et al., 2020)	CVPR'20	SA	0.797	0.774	0.837	0.092	0.865	0.86	0.927	0.047	0.865	0.87	0.901	0.059	0.785	0.703	0.84	0.068	0.804	0.742	0.857	0.062
MFNet (Piao et al., 2021)	ICCV'21	PL	0.782	0.746	0.818	0.112	0.852	0.839	0.917	0.058	0.837	0.844	0.877	0.084	0.726	0.621	0.783	0.098	0.778	0.692	0.83	0.079
SBBs (Liu, Wang et al., 2021)	TIP'21	BB	-	-	-	-	0.854	0.843	0.92	0.056	0.851	0.855	0.894	0.072	0.776	0.695	0.835	0.074	0.789	0.722	0.851	0.073
OSG (Wang, Al-Huda and Peng, 2021)	ISKE'21	SA	0.798	0.792	0.830	0.090	0.848	0.878	0.926	0.050	0.866	0.893	0.900	0.058	0.777	0.728	0.847	0.068	0.786	0.763	0.856	0.063
SHNet (Ma et al., 2022)	ArX'22	SA	0.697	0.699	0.732	0.147	0.806	0.777	0.883	0.073	0.750	0.758	0.817	0.126	0.704	0.641	0.763	0.094	0.733	0.637	0.797	0.094
PHWS (Gao, Zhang et al., 2022)	ArX'22	PA	0.853	0.866	/	0.065	0.902	0.923	/	0.032	0.910	0.936	/	0.039	0.824	0.809	/	0.036	0.853	0.858	/	0.045
NSAL (Piao et al., 2023)	TMM'23	PL	0.767	0.756	0.823	0.110	0.854	0.864	0.923	0.051	0.834	0.856	0.883	0.078	0.745	0.648	0.801	0.088	0.782	0.730	0.849	0.073
SLF (Liu, Zhang, Wang, Yang et al., 2023)	EAAI'23	SA	0.794	0.789	0.841	0.087	0.851	0.857	0.923	0.050	0.864	0.878	0.908	0.056	0.778	0.708	0.844	0.066	0.793	0.750	0.865	0.060
HSS (Cong, Qin et al., 2023)	TCSTV'23	H	0.828	0.755	/	0.069	0.887	0.892	/	0.038	0.886	0.899	/	0.051	/	/	/	/	0.837	0.807	/	0.050
A2S (Zhou, Chen et al., 2023)	TCSTV'23	PL	/	0.790	0.838	0.106	/	0.887	0.937	0.042	/	0.888	0.911	0.064	/	0.719	0.841	0.069	/	0.750	0.860	0.065
A2SV2 (Zhou et al., 2023d)	CVPR'23	PL	/	0.803	0.848	0.099	/	0.891	0.939	0.041	/	0.902	0.923	0.056	/	0.731	0.851	0.065	/	0.767	0.871	0.061
A2SV3 (Yuan et al., 2024b)	LCAI'24	PL	/	0.844	0.899	0.069	/	0.908	0.954	0.033	/	0.923	0.951	0.038	/	0.759	0.868	0.062	/	0.816	0.906	0.047
BWSS (Liu & Huang, 2024)	ERA'24	BB+PL	/	/	/	/	/	0.932	/	0.031	/	0.945	/	0.035	/	0.790	/	0.055	/	0.889	/	0.036

and “Cap” denotes the image caption. “PL” denotes the pseudo labels. For fair comparison, the data is directly cited from the Ref. (Hao et al., 2025; Liu, Zhang, Wang, Yang et al., 2023) and the original papers. It can be seen that there has been a significant improvement in weakly supervised algorithms over the years, demonstrating their growing advantages in handling tasks with greater diversity and complexity, especially in scenarios with limited labeled data.

Qualitative comparison. Fig. 7 presents a qualitative comparison of four representative images selected from the ECSSD dataset, comparing algorithms including four fully supervised algorithms, two unsupervised algorithms, and nine weakly supervised algorithms. It can be observed from the figure that different algorithms exhibit significant differences in performance on the task of ISOD, particularly in terms of precision in object extraction, detail preservation, and handling of background interference. Fully supervised algorithms perform best in most scenarios. For instance, MINet and EGNet can precisely extract the boundaries of salient objects, especially in complex backgrounds, where they effectively separate the objects from the background, ensuring the integrity and details of the objects. Even in more complex scenes (such as images of animals or people), their performance remains outstanding. In contrast, unsupervised algorithms perform relatively poorly. For example, SVF and C2S fail to effectively extract objects from the background when processing complex images, resulting in less accurate object extraction, particularly with noticeable flaws in handling details and boundaries. Weakly supervised algorithms, on the other hand, show potential in some simple scenarios, such as HSS and A2S, which can extract the target objects. However, in complex scenes, these algorithms generally do not perform as well as supervised ones, with blurrier boundaries and less refined handling of object details. In conclusion, although fully supervised algorithms still have an advantage in most scenarios, weakly supervised algorithms have begun to narrow the gap with them in some cases. The research of ISOD still needs to further optimize weakly supervised algorithms to obtain performance closer to that of fully supervised algorithms without relying on large amounts of labeled data.

Complexity comparison. Table 6 shows the complexity comparison results of eleven algorithms. Floating Point Operations (FLOPs) (Hunger, 2007), Parameters, and Frames Per Second (FPS) are used to

evaluate the computational complexity of the VSOD algorithms. These model complexity metrics are also widely applied in other computer vision tasks, such as object detection (Ma, Du, Gao, Zhuo and Wang, 2024; Ma, Zhuo, Li, Zhang, & Zhang, 2023b), pedestrian detection (Ma, Zhuo, Li, Zhang, & Zhang, 2023a), classification, and segmentation. It can be seen that, over time, with the continuous advancements in GPU hardware, the inference speed of models has significantly improved. This not only allows for faster real-time processing but also enables the deployment of more complex and deeper models without compromising efficiency. As computational power increases, it becomes possible to achieve better trade-offs between accuracy and speed, making these models more applicable in various real-world scenarios where both high precision and quick responses are crucial.

3.5. The performance of existing VSOD algorithms

To gain a comprehensive understanding of the development of existing WSVSOD algorithms, this section provides the performance of existing VSOD algorithms in recent years. These include sixteen fully supervised VSOD algorithms: FCN (Wang, Shen, Shao, 2018)*, FGRN (Li, Xie, Wei, Wang and Lin, 2018)*, PDB (Song, Wang, Zhao, Shen, & Lam, 2018)*, SPD (Li, Li, Chen, Hao, & Qin, 2020)*, SSAV (Fan et al., 2019)*, PCSA (Gu et al., 2020)*, TENet (Ren, Han, Yang, Guoqiang, & He, 2020)*, MGA (Li, Chen et al., 2019)*, MAM (Zhao, Liang et al., 2024)*, LSTI (Chen, Wang, Peng, Zhang and Qin, 2020)*, Fie-net (Wang, Sun et al., 2024)*, DCFNet (Zhang et al., 2021)*, and FSNet (Ji et al., 2021)*, CoSTormer (Liu, Nan, Zhao, Yao and Han, 2024)*, BMST-Net (Sharma, Singh, Kumain, & Kumar, 2025)*, RealFlow (Cho, Lee, Lee, & Lee, 2024)*. Two hybrid label-supervised VSOD algorithms: RCR (Yan et al., 2019)†, UGPL (Piao, Lu, Zhang, & Lu, 2022b)†, and nine WSVSOD algorithms: GF (Wang, Shen, Shao, 2015), SAG (Wang, Shen, Porikli, 2015), SSOD (Zhang, Yu et al., 2020), WS (Zhao, Zhang et al., 2021), PVS (Gao, Xing et al., 2022), MPLA-Net (Ma, Du, Zhuo et al., 2024), CFMR (Wang, Yu et al., 2024), UWS (Xu, Jiang, Liang et al., 2024), and MES (Xu, Jiang, Zhao et al., 2024). The quantitative comparison, qualitative comparison and complexity comparison of these algorithms are reported as follows.

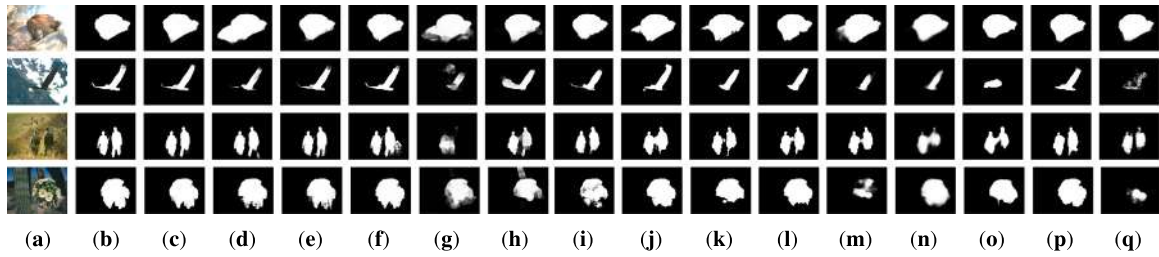


Fig. 7. Visual comparisons of the existing ISOD algorithms. (a)-(q) denote the Img, GT, MINNet (Pang et al., 2020), CPD (Wu et al., 2019), EGNNet (Zhao et al., 2019), VSCoDe (Luo et al., 2024), SVF (Zhang, Han et al., 2017)[†], C2S (Li, Yang et al., 2018)[†], HSS (Cong, Qin et al., 2023), A2S (Zhou, Chen et al., 2023), A2SV2 (Zhou et al., 2023d), A2SV3 (Yuan et al., 2024b), WSSA (Zhang, Yu et al., 2020), MFNet (Piao et al., 2021), NSAL (Piao et al., 2023), PWSS (Gao, Zhang et al., 2022) and SBBs (Liu, Wang et al., 2021), respectively.

Table 6

Complexity comparison results of the existing ISOD methods.

Methods	Years	Backbone	Platform	FLOPs(G)	Par.(M)	FPS
EGNet (Zhao et al., 2019)*	ICCV'19	ResNet-50	RTX 3090	244.14	111.66	20
CPD (Wu et al., 2019)*	CVPR'19	ResNet-50	RTX 3090	14.66	47.85	38
MINet (Pang et al., 2020)*	CVPR'20	ResNet-50	RTX 3090	87.03	162.38	38
VST (Liu, Zhang et al., 2021)*	ICCV'21	T2T-ViT t-14	RTX 3090	23.16	44.03	29
DNA (Liu, Cheng et al., 2022)*	TCYBE'22	NetNet-50	TITAN XP		20.06	25
C2S (Li, Yang et al., 2018) [†]	ECCV'18	VGG-16	GTX 1080Ti	/	/	33.33
WSSA (Zhang, Yu et al., 2020)	CVPR'20	VGG-16	RTX 3090	64.45	15.53	76
MFNet (Piao et al., 2021)	ICCV'21	DenseNet-169	RTX 3090	16.01	/	/
SHNet (Ma et al., 2022)	Apl'22	VGG-16	RTX 3090	110.64	20.71	25
SLF (Liu, Zhang, Wang, Yang et al., 2023)	EAAI'23	ResNet-50	RTX 3090	13.58	93.93	62
A2S (Zhou, Chen et al., 2023)	TCSVT'23	ResNet-50	GTX 1080Ti	/	28.4	/

Table 7

Performance comparison of the existing VSOD algorithms on various datasets. “F” denotes fully supervised algorithms. “H” denotes hybrid labels. “T” denotes the label is generated by traditional methods. “PL” indicates pseudo labels. “SA” indicates scribble annotations. “/” indicates unavailable information.

Algorithms	Pub.	Sup.	DAVIS			DAVSOD			VOS			ViSal			SegTrack_V2			FBMS		
			SM ↑	FM ↑	MAE ↓	SM ↑	FM ↑	MAE ↓	SM ↑	FM ↑	MAE ↓	SM ↑	FM ↑	MAE ↓	SM ↑	FM ↑	MAE ↓	SM ↑	FM ↑	MAE ↓
FCN (Wang, Shen, Shao, 2018)*	TIP'17	F	0.794	0.708	0.061	0.657	0.521	0.129	0.760	0.675	0.099	0.881	0.852	0.048	/	/	/	0.794	0.759	0.091
FRN (Li, Xie, Wei et al., 2018)*	CVPR'18	F	0.838	0.783	0.043	0.693	0.573	0.098	0.715	0.669	0.097	0.861	0.848	0.045	/	/	/	0.809	0.767	0.088
PDB (Song et al., 2019)*	ECCV'18	F	0.882	0.855	0.028	0.698	0.572	0.116	0.818	0.742	0.078	0.907	0.888	0.032	0.864	0.808	0.024	0.908	0.903	0.027
SPD (Li et al., 2020)*	TIP'19	F	0.783	0.892	0.763	0.773	0.685	0.072	/	/	/	/	/	/	/	/	/	0.851	0.821	0.064
SSAV (Fan et al., 2019)*	CVPR'19	F	0.892	0.861	0.028	0.724	0.603	0.092	0.819	0.742	0.073	0.943	0.939	0.020	0.851	0.801	0.023	0.879	0.865	0.04
MGA (Li, Chen et al., 2019)*	ICCV'19	F	0.912	0.892	0.022	0.751	0.656	0.081	0.792	0.735	0.075	0.941	0.94	0.016	0.88	0.829	0.027	0.908	0.903	0.027
PCSA (Gu et al., 2020)*	AAAI'20	F	0.902	0.880	0.022	0.741	0.655	0.086	0.827	0.747	0.065	0.946	0.94	0.017	0.865	0.81	0.025	0.868	0.837	0.04
TENet (Ren et al., 2020)*	ECCV'20	F	0.905	0.881	0.017	0.779	0.697	0.070	0.854	0.781	0.052	0.949	0.949	0.012	0.868	0.810	0.025	0.916	0.915	0.024
LSIT (Chen, Wang et al., 2020)*	TIP'20	F	0.876	0.890	0.022	0.750	0.654	0.081	/	/	/	0.922	0.943	0.027	0.891	0.835	0.028	0.816	0.795	0.084
DCFNet (Zhang et al., 2021)*	ICCV'21	F	0.914	0.900	0.016	0.741	0.660	0.074	0.846	0.791	0.06	0.952	0.953	0.010	0.883	0.839	0.015	/	/	/
FSNet (Li et al., 2021)*	ICCV'21	F	0.920	0.907	0.020	0.773	0.685	0.072	/	/	/	/	/	/	/	/	/	/	/	/
TCRN (Huang, Tian, Su, Lin and Chun-Wei, 2022)*	PRL'22	F	0.916	0.898	0.016	/	/	/	0.867	0.823	0.049	0.946	0.943	0.014	0.904	0.865	0.014	0.896	0.889	0.032
RealFlow (Cho et al., 2024)*	Ars'24	F	0.945	0.939	0.010	0.803	0.732	0.066	/	/	/	0.962	0.966	0.010	/	/	/	0.926	0.906	0.028
MAM (Zhao, Liang et al., 2024)*	TIP'24	F	0.897	0.877	0.020	0.777	0.708	0.065	/	/	/	0.947	0.948	0.012	0.886	0.85	0.014	0.894	0.883	0.032
Fie-net (Wang, Sun et al., 2024)*	SIVP'24	F	0.910	0.894	0.017	0.744	0.632	0.065	/	/	/	0.938	0.938	0.013	0.907	0.893	0.013	/	/	/
CoFormer (Liu, Nan et al., 2024)*	TNNLS'24	F	0.926	0.916	0.013	0.783	0.697	0.068	/	/	/	/	/	/	0.888	0.833	0.015	/	/	/
BMST-Net (Sharma et al., 2025)*	SIVP'25	F	0.929	0.939	0.012	0.798	0.724	0.032	0.893	0.837	0.019	0.973	0.961	0.007	0.908	0.893	0.014	/	/	/
RCR (Yan et al., 2019) [†]	ICCV'19	H	0.866	0.848	0.027	/	/	/	/	/	/	0.922	0.906	0.026	0.842	0.781	0.035	0.872	0.859	0.053
UGPL (Piao et al., 2022b) [†]	NeurIPS'22	H	0.914	0.900	0.019	/	/	/	/	/	/	0.930	0.926	0.019	0.844	0.778	0.027	0.899	0.892	0.027
GF (Wang, Shen, Shao, 2015)	TIP'15	T	0.688	0.569	0.100	0.553	0.334	0.167	0.615	0.506	0.162	0.757	0.683	0.107	0.699	0.592	0.091	0.651	0.571	0.16
SAG (Wang, Shen, Portikl, 2015)	CVPR'15	T	0.676	0.515	0.103	0.565	0.370	0.184	0.619	0.482	0.172	0.749	0.688	0.105	0.719	0.634	0.081	0.659	0.564	0.161
SSOD (Zhang, Yu et al., 2020)	CVPR'20	SA	0.795	0.734	0.044	0.672	0.556	0.101	0.682	0.648	0.106	0.853	0.831	0.038	0.733	0.664	0.039	0.747	0.727	0.083
WS (Zhao, Zhang et al., 2021)	CVPR'21	SA	0.846	0.793	0.038	0.705	0.605	0.103	0.765	0.702	0.089	0.883	0.875	0.035	0.819	0.762	0.033	0.778	0.786	0.072
PVS (Gao, Xing et al., 2022)	MM'22	PA	0.808	0.754	0.04	0.718	0.622	0.094	0.796	0.739	0.074	0.878	0.858	0.028	0.834	0.875	0.03	0.812	0.794	0.055
MPLA-Net (Ma, Du, Zhao et al., 2024)	TCSVT'23	PL	0.855	0.814	0.033	0.723	0.658	0.091	0.768	0.721	0.084	0.89	0.881	0.033	0.836	0.778	0.032	/	/	/
UWS (Xu, Jiang, Liang et al., 2024)	TMM'24	PL	0.869	0.844	0.041	0.744	0.659	0.085	0.811	0.729	0.074	0.954	0.958	0.011	/	/	/	0.873	0.862	0.042
CFMR (Wang, Yu et al., 2024)	PR'24	SA	0.867	0.828	0.028	0.737	0.646	0.085	0.768	0.696	0.088	0.883	0.869	0.035	0.848	0.802	0.022	0.798	0.802	0.065
MES (Xu, Jiang, Zhao et al., 2024)	TCSVT'24	SA	0.873	0.883	0.028	0.770	0.682	0.071	/	/	/	0.926	0.921	0.016	0.884	0.841	0.014	0.872	0.842	0.036

Quantitative comparison. Table 7 shows the quantitative comparison results in terms of SM, FM, and MAE on six datasets. It can be seen that a significant improvement in weakly supervised algorithms over the years, demonstrating their growing advantages in handling tasks with greater diversity and complexity, especially in scenarios with limited labeled data. Among these algorithms, MES (Xu, Jiang, Zhao et al., 2024) and UWS (Xu, Jiang, Liang et al., 2024) stand out, with MES (Xu, Jiang, Zhao et al., 2024) achieving comparable or even superior results to some supervised algorithms on certain datasets, and UWS (Xu, Jiang, Liang et al., 2024) attaining high performance across several datasets. Although the performance of WSVSOD algorithms is generally lower than supervised algorithms, they are narrowing the gap, and some have even outperformed earlier supervised models. This

progress highlights the potential and practicality of weakly supervised algorithms, which are not only becoming more competitive with recent supervised algorithms but also significantly reduce the reliance on large-scale pixel-level annotation datasets. “T” denotes the label is generated by traditional methods. “F” indicates the fully supervised algorithms. “H” indicates hybrid labels. “SA” denotes scribble annotations. “PL” denotes the pseudo labels.

Qualitative comparison. Fig. 8 shows the qualitative comparison of four representative video frames selected from the DAVIS dataset, and the comparison algorithms include nine supervised algorithms and six weakly supervised algorithms. It can be intuitively seen from the visual results that the performance of WSVSOD algorithms is gradually improving, and the edges and structure of salient objects are becoming

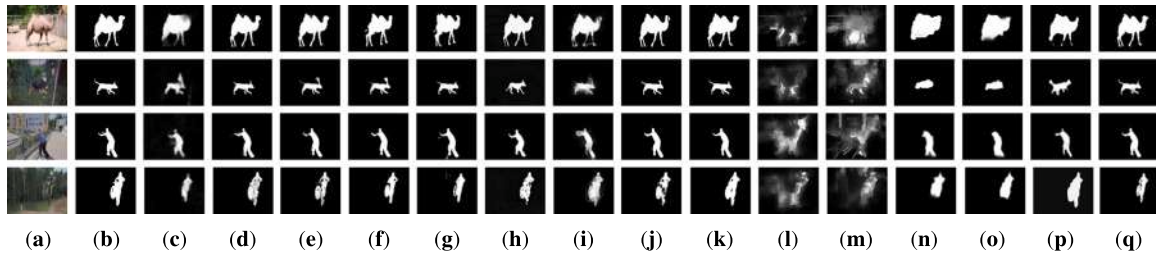


Fig. 8. Visual comparisons of the existing VSOD algorithms. (a)-(q) denote Img, GT, FCN (Wang, Shen, Shao, 2018)*, SSAB (Fan et al., 2019)*, PDB (Song et al., 2018)*, PCSA (Gu et al., 2020)*, TENet (Ren et al., 2020)*, MGA (Li, Chen et al., 2019)*, LSTI (Chen, Wang et al., 2020)*, DCFNet (Zhang et al., 2021)*, FSNet (Ji et al., 2021)*, GF (Wang, Shen, Shao, 2015), SAG (Wang, Shen, Porikli, 2015), SSOD (Zhang, Yu et al., 2020), WS (Zhao, Zhang et al., 2021), MPLA-Net (Ma, Du, Zhuo et al., 2024), and MTNet (Zhuge, Gu, Zhang, Qi, & Lu, 2024), respectively.

Table 8

Complexity comparison results of the existing VSOD algorithms.

Algorithms	Years	Backbone	Platform	FLOPs (G)	Par. (M)	FPS
FCN (Wang, Shen, Shao, 2018)*	TIP'18	VGG-16	TITAN X	/	/	2
PDB (Song et al., 2018)*	ECCV'18	ResNet-50	RTX 3090	/	/	20
SSAB (Fan et al., 2019)*	CVPR'19	ResNet-50	TITAN X	/	236.2	20
MGA (Li, Chen et al., 2019)*	ICCV'19	ResNet-50&101	GTX 1080Ti	2265.7	254	14
SPD (Li et al., 2020)*	TMM'20	Self-built CNN	GTX 1070	/	/	0.38
LSTI (Chen, Wang et al., 2020)*	TIP'20	VGG-16	RTX 3090	/	/	0.71
DCFNet (Zhang et al., 2021)*	ICCV'21	ResNet-101	GTX 2080Ti	/	/	28
FSNet (Ji et al., 2021)*	ICCV'21	ResNet-101	RTX 3090	/	/	12.5
RealFlow (Cho et al., 2024)	Arx'24	MiTb2 (Xie et al., 2021)	RTX 2080Ti	/	55.7	29.5
WS (Zhao, Zhang et al., 2021)	CVPR'21	ResNet-50	RTX 2080Ti	6.19	34.5	28.6
MPLA-Net (Ma, Du, Zhuo et al., 2024)	TCSVT'23	ResNet-50	RTX 2080Ti	9.788	34.38	39.5
MTNet (Zhuge et al., 2024)	TNNLS'24	ConvNext (Liu, Mao et al., 2022)	RTX 2080Ti	134.2	35.1	43.4
UWS (Xu, Jiang, Liang et al., 2024)	TMM'24	ResNet	TITAN XP	/	/	4

more and more refined. The supervised algorithms still outperform the weakly supervised ones in most cases, especially in handling complex scenes and background distractions. However, some weakly supervised algorithms have started to close the gap, delivering competitive results with more targeted solution designs. This indicates that the development of WSVSOD is heading in a promising direction, and there is potential for further optimization to achieve results closer to those of fully supervised algorithms.

Complexity comparison. Table 8 shows the complexity comparison results of eleven algorithms. The FPS of Zhao, Zhang et al. (2021) is obtained from the original paper, while the FLOPs and parameters are computed based on the publicly available source code. The data of other algorithms are cited from the Ref. (Chen et al., 2021; Su et al., 2024) and the original papers. When analyzing the complexity of VSOD, we can combine the results of ISOD, which are significantly different in terms of computing complexity, parameter quantity and real-time performance. ISOD typically involves feature extraction from single frames, resulting in relatively lower computational complexity. Representative algorithms like EGN and MINet, despite employing complex network structures (e.g., ResNet-50), have smaller computational complexity and parameter counts compared to video tasks, usually with FLOPs ranging from tens to hundreds of GFLOPs. Their processing speeds are also faster, with FPS maintained at high levels, such as WSSA achieving an FPS of 76, making them suitable for real-time applications. However, VSOD requires consideration of temporal sequence information, necessitating not only the processing of features in each frame but also the analysis of dynamic changes between frames. This significantly increases the computational complexity of video algorithms. For example, MGA has FLOPs as high as 2265.7 GFLOPs and a parameter count of 254M, far exceeding those of most image algorithms. The additional computational burden in video tasks mainly stems from temporal dimension processing, leading to greater computational resource demands and storage overhead. Moreover, the FPS of VSOD is generally lower than that of image algorithms. Although some video algorithms like MPLA-Net and MTNet perform relatively

well in terms of FPS, overall, video algorithms tend to have slower inference speeds due to the need to process more temporal information, with their average inference speed being significantly lower than that of image algorithms. Therefore, while video salient object detection can provide richer temporal information, thereby improving accuracy, its bottlenecks in real-time performance and computational resource consumption pose challenges for real-time tasks. With the advancement of hardware performance, complex video algorithms have gained better support, but finding an appropriate balance among accuracy, computational complexity, and real-time performance remains a significant challenge in the field of salient object detection.

The growing scale of datasets and the increasing availability of computational resources have provided a robust foundation for the design and validation of algorithms. From the comparison of the three aspects mentioned above, it is evident that the overall performance of WSVSOD algorithms has steadily improved in recent years, even surpassing some early fully-supervised approaches. This underscores the emerging potential of weakly supervised techniques in this domain, suggesting that further significant advancements may be achieved in the future. However, the performance of WSVSOD algorithms is still constrained by issues such as sparse annotations and the quality of pseudo labels, which leads to a performance gap between WSVSOD and fully supervised VSOD algorithms.

4. Conclusion

In this paper, we provide, to the best of our knowledge, the first comprehensive survey of the deep learning-based WSISOD and WSVSOD algorithms. WSISOD and WSVSOD algorithms, several public ISOD and VSOD datasets, evaluation metrics of the performance, and the quantitative comparison, qualitative comparison and complexity comparison of existing WSISOD and WSVSOD algorithms were depicted in the previous sections. The existing problems and the future tendency of WSISOD and WSVSOD task research are discussed in this section. As mentioned earlier, time-consuming and labor-intensive pixel-level

manual annotation brings profound challenges for fully supervised ISOD and VSOD tasks. Thus, many researchers have been employed economical weak annotations to achieve it. Despite significant advancements, there remains a considerable gap between weak annotations and pixel-level annotations. Weak annotations inherently possess limitations, such as coarse object edges and a lack of structural information. Thus, how to effectively learn salient cues from weak annotations to achieve performance comparable to that of pixel-level annotations has been a prominent research direction. The remaining research gaps that require attention in this field include the following:

Weak annotation datasets. While weak annotations are obviously more cost-effective than pixel-level annotations, they still demand considerable resources due to the large-scale datasets required for training deep neural networks. As shown in Table 4, the number of publicly available datasets with weak annotations for the WSISOD and WSVSOD task, as well as the richness of the annotation type, is highly limited. This scarcity poses a substantial challenge to exploring how various types of weak annotations affect model performance. Implementing new data annotation methods may offer an effective solution to this issue (Delatolas, Kalogeiton, & Papadopoulos, 2024).

The quality of labels. Generating pseudo labels through weakly annotations such as category annotations, point annotations, and scribble annotations, and then using a supervised network architecture to implement WSVSOD is a relatively common paradigm. However, due to the uncontrollable nature of pseudo label generation, the quality of pseudo labels varies greatly. Since this is a weakly supervised task that does not rely on real annotation information, it is currently difficult to measure the quality of pseudo labels, and the impact of their quality on model performance requires further exploration.

Fusion of multiple annotation. Single weak annotations may have prejudiced characteristics. Therefore, employing advanced technologies such as SAM, SAM2 to generate multiple annotations is an implementable strategy. It is crucial to explore fusion methods for extracting and utilizing saliency cues from these diverse annotations, which can significantly enhance the precision and robustness of the WSISOD and WSVSOD task.

Thorough exploitation of multimodal information. In real-world scenarios, the regions of interest for users are influenced by various modalities of data, including images, text, and audio. Effectively harnessing the salient cues present within this multimodal information to enhance detection accuracy represents a promising approach in the field.

Model generalization. Current WSISOD and WSVSOD algorithms predominantly rely on training and testing within specific datasets. While these algorithms may achieve satisfactory performance in controlled scenarios, they often struggle when applied to the diverse and dynamic environments of the real world, especially when diverse and continuously generated new videos present additional challenges. Therefore, future research may focus on enhancing the generalization capability of WSISOD and WSVSOD algorithms, using new technical means such as prompt learning (Luo et al., 2024) and SAM, enabling them to not only perform well on the training datasets but also adapt to various application scenarios and complex situations, thereby better meeting the demands of real-world challenges.

Inference speed of the model. In practical applications such as healthcare, autonomous driving, intelligent surveillance, and military operations, weakly supervised video salient object detection holds great potential. These scenarios place stringent demands on model inference speed and accuracy. By incorporating advanced techniques such as Vmamba (Liu, Tian et al., 2024), network pruning (Molchanov, Mallya, Tyree, Frosio, & Kautz, 2019), knowledge distillation (Li, Xu et al., 2023), and parallel processing (Yang, Zhu, & Pu, 2008), etc., it is feasible to design models that achieve high accuracy while maintaining fast inference speeds. These optimizations are crucial for meeting the rigorous performance requirements of real-time, mission-critical tasks in these fields.

Task extension. In addition to the traditional video salient object detection task, there are several emerging research avenues that focus on salient object detection diverse data modalities. For example, the exploration of RGB-T and RGB-D data (Bao et al., 2024; Li et al., 2023), saliency detection in point cloud data (Wei et al., 2024). Moreover, traditional ISOD and VSOD algorithms struggle to distinguish the relative importance of multiple objects and the relationships between them. This limitation can result in errors and reduced accuracy in downstream tasks that require an understanding of object importance. A new paradigm of salient object ranking has been proposed, which aims to prioritize objects based on their “importance order”, offering a more refined approach to object saliency in complex scenes (Fang et al., 2021; Qiao et al., 2024; Song, Li, Wu, Song, & Chen, 2024; Wang, Yan, Han, & Sun, 2019). These emerging directions present both theoretical challenges and practical opportunities in the field of salient object detection.

Large Language Models (LLMs) for salient object detection. Integrating LLMs into salient object detection presents both challenges and opportunities. On the one hand, bridging the gap between textual and visual modalities requires significant adaptation, and the computational cost of LLMs, combined with video processing demands, poses scalability issues. Additionally, translating LLM-generated textual descriptions into precise weak annotations is challenging due to language ambiguity and video complexity, while domain-specific adaptations may be needed for specialized applications. On the other hand, LLMs offer exciting opportunities, such as generating richer weak annotations (e.g., semantic labels or contextual cues), enabling cross-modal learning to improve saliency predictions, and automating the annotation process for large-scale datasets. Furthermore, LLMs can enhance model explainability by providing textual justifications for predictions, which is crucial in many applications.

CRedit authorship contribution statement

Lina Du: Investigation, Writing – original draft, review & editing. **Chunjie Ma:** Software, Investigation, Methodology, review & editing, Funding acquisition. **Junhao Chen:** Investigation, review & editing. **Huimin Zheng:** Investigation, review & editing. **Xiushan Nie:** Review, Resources, Supervision, Funding acquisition. **Zan Gao:** Resources, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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Climate Change Initiative Extension (CCI+) Phase 2
New Essential Climate Variables (NEW ECVS)
High Resolution Land Cover ECV (HR_LandCover_cci)

Product Validation and Algorithm Selection Report
(PVASR)

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Changelog

Issue	Changes	Date
1.0	First issue	21/01/2025
1.1	Updated based on RIDs	21/02/2025

Detailed Change Record

Issue	RID	Description of discrepancy	Sections	Change
1	ESA-01	GlobCover Land Cover map 300m 2019, to which map are you referring to? GlobCover or C3S MRLC 2019?	Table 5, Table 6, Table 7	The reference to the 'GlobCover Land Cover map 300m 2019' is incorrect. The validation dataset was extracted from the CCI Medium Resolution Land Cover (MRLC) Map 300m 2019. Tables and text in the document have been updated accordingly.
2	ESA-02	GlobCover validation points are from 2008, this dataset has been updated to 2019?	3.2.1.1 and 3.2.1.2.1	
3	ESA-03	Please remove the space from the link after "download/" http://maps.elie.ucl.ac.be/CCI/viewer/download/ESA_CCI-LC-Ph2-PUGv2_2.0.pdf	Reference [11]	The document is updated accordingly.
4	ESA-04	For the comparison maybe it is better to use the same scale for the values on the Y axis for both, the weekly and monthly plots.	Fig. 37 and 38	The document is updated accordingly.
5	ESA-05	Did you investigate the moisture index? $NDMI = (NIR - SWIR) / (NIR + SWIR)$. Maybe it could provide some additional information on the LC	3.4.2	We will consider NDMI for the following experiments.

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1 Introduction

1.1 Executive summary

By the end of the first year, the EOS team conducted extensive comparative testing and performance analyses to evaluate the effectiveness of different algorithms for specific components of the processing chain designed to produce high-resolution land cover (HR LC) products. The results of these experiments, along with recommendations for the best-performing techniques, are detailed in section 3. The analysis is ongoing, as the team continues to assess various approaches to select the final algorithms that will ensure optimal performance for both static and historical land cover (LC) and land cover change (LCC) products.

1.2 Purpose and scope

The Product Validation and Algorithm Selection Report (PVASR) provides an in-depth overview of the comparative tasks conducted to assess the best-performing algorithms and techniques for inclusion in the classification blocks of the overall processing chain. The current version presents the activities carried out during the first year, with a particular focus on the classification of optical and SAR imagery, decision fusion and change detection. Key areas of emphasis include:

1. Testing optical pre-processing and evaluating its performance in terms of accuracy, computational efficiency and composite quality.
2. Testing classifiers and evaluating their performance in terms of accuracy, computational efficiency, and adaptability for model/code modifications to meet specific requirements and implementation needs.
3. Exploring methods for creating reliable training datasets from existing products, which may be sub-optimal in terms of spatial resolution (coarse to medium) and legend detail (less comprehensive compared to HR LC products as outlined in the ATBD).
4. Assessing sets of multitemporal features used as inputs for classifiers.
5. Evaluating multisensory decision fusion methods in terms of both accuracy and computational time.
6. Evaluating optical composite generation to enhance detection accuracy and analyzing different feature spaces to ensure reliable change detection maps.

1.3 Applicable documents

Ref. Title, Issue/Rev, Date, ID

- [AD1] CCI HR Technical Proposal phase 1
- [AD2] CCI HR Technical Proposal phase 2
- [AD3] CCI Extension (CCI+) Phase 1 – New ECVs – Statement of Work, v1.3, 22/08/2017, ESA-CCI-PRGM-EOPS-SW-17-0032
- [AD4] CCI_HRLC_Ph2-D1.1_URD, latest version
- [AD5] CCI_HRLC_Ph1-D2.2_ATDB, latest version
- [AD6] CCI_HRLC_Ph2-D2.2_ATDB, latest version

1.4 Reference documents

Ref. Title, Issue/Rev, Date, ID

- [RD1] The Global Climate Observing System: Implementation Needs, 01/10/2016, GCOS200

1.5 Acronyms and abbreviations

3D-FCN	3-Dimensional - Fully Convolutional Network
AC	Atmospheric Correction
ATBD	Algorithm Theoretical Basis Document
BSI	Bare Soil Index

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CCI+	Climate Change Initiative Extension
dB	Decibel
DEM	Digital Elevation Model
DL	Deep Learning
ER	Ecoregion
GEE	Google Earth Engine
GRD	Ground Range Detected
HR	High Resolution
HRLC10	CCI High Resolution Land Cover Map at 10m resolution of 2019
HRLC30	CCI High Resolution Land Cover Map at 30m resolution from 1990 onwards every 5 years
HRLCC30	CCI High Resolution Land Cover Change Map at 30m resolution from 1990 onwards
IW	Interferometric Wide Swath
L-5/7/8/9	Landsat-5/7/8/9
LC	Land Cover
LCC	Land Cover Change
MOLCA	Map Of LC Agreement
MRLC	Medium Resolution Land Cover
NDVI	Normalized Difference Vegetation Index
NDWI	Normalized Difference Water Index
OA	Overall Accuracy
PA	Producer Accuracy
PCC	Post Classification Comparison
RF	Random Forest
S1/2	Sentinel-1/2
SAR	Synthetic Aperture Radar
SCL	Sen2Cor Scene Classification Layer
SITS	Satellite Image Time Series
SNAP	Sentinel Application Platform
SR	Surface Reflectance
UEXT	Urban EXTent
VH	Vertical-Horizontal polarization
VV	Vertical-Vertical polarization

2 Selection procedure

The overall procedure for the selection of best performing algorithms and methods is performed according to a three-step procedure. The algorithms presented in the Technical Proposals [AD1] [AD2] and ATBD [AD6] are considered for the comparisons together with a set of proposed solutions for each task such as generating training samples and building multitemporal features. The evaluation-selection procedure is devised in such a way that the selected algorithms/techniques are the most suitable to satisfy project requirements.

The three steps of the procedure are the following:

- **Step 1: Qualitative pre-screening of algorithms**

A pre-screening of the algorithms and methods from a State-of-the-art pool of competitors is carried out in order to identify the most relevant methodologies with respect to the project objectives. This preliminary analysis is driven by the selection criteria described in Section 2.1. In this first step, a high-level qualitative evaluation of these criteria is conducted in order to identify techniques that clearly cannot reach a satisfactory ranking on several categories of parameters. These techniques are discarded and not considered in the next steps. Algorithms and methods that passed the pre-screening are reported in the Technical Proposal [AD2] and more in detail in the ATBD [AD6]. In this report only the methods that passed the pre-screening are considered explicitly.

- **Step 2: Quantitative evaluation of algorithms**

Algorithms that pass the pre-screening in step 1 are analyzed in greater detail with a quantitative evaluation. This analysis is based on different parameters, ranging from a scientific and technical analysis to possible impacts on the application and users. For each investigated item (algorithm, method, technique, etc.) details on the quantitative evaluation of the comparison activities can be found in a dedicated section of this document.

- **Step 3: Final decision**

According to the analysis carried out for each individual comparison task, a **final decision** is taken according to the best performer and its relevance with respect to project objectives. Final decision is reported.

It is worth noting that the pre-processing algorithms are not included in the evaluation and ranking procedure because we expect to import in the project basic pre-processing chains already developed for both multispectral and SAR data.

2.1 Criteria

In this section the criteria adopted for evaluating the relevance of methods and algorithms with respect to project requirements are listed. Up to seven categories of parameters are considered divided in different issues.

1. **Scientific Background and Technical Soundness** – The scientific validity of the algorithms and of the methodologies on which the algorithms are based is considered as an important parameter. The rationale is that selected algorithms should be based on a solid theoretical background that guarantees the accuracy of its results also at an operational level. The guidelines for rating are as follows:
 - The methodology is solid;
 - The methodology is technical convincing;
 - The methodology is at the state-of-the-art;
 - The methodology is published in high quality journals;
 - The methodology is included in several other scientific publications or project technical reports.
2. **Robustness and Generality** – In order to obtain a reasonable estimation for the robustness and generality of the investigated algorithms, different parameters are considered, such as:
 - The method is suitable to be used with different kinds of images (e.g., S2, Landsat, SAR, etc.);
 - The method shows high performance on different images (Sentinel, Landsat, etc.) and over the three test areas as described in URD [AD4];
 - There are software implementations or examples for the implementation available;
 - The algorithm can be used in combination with other methodologies.

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3. **Novelty** – An appropriate candidate algorithm should have been published or reported for the first time relatively recently in the literature. It is not required that algorithms are completely innovative; the novelty may consist in both combining well established methodologies or applying well-known techniques in a novel way. As a main guideline, a tested method should be already applied in literature to solve existing problems.
4. **Operational Requirements** – The expected operational requirements (in terms of computational complexity, time effort, cost, etc.) for the final implementation of an algorithm/technique are evaluated. Although no actual constraints are fixed on the algorithm computational complexity, the most optimized implementations available in literature are preferred. Other crucial aspects are:
 - The algorithm is prone to architectural modifications;
 - The processing time scaling is likely to be linear with image size;
 - The hardware and disk-storage requirements are appropriate.
 Algorithm/method consistency with project requirements is also extremely relevant, following guidelines from GCOS [RD1] and SoW [AD3][AD3] :
 - Algorithms and methodologies must be effective for high resolution images (e.g., optical data at 10-30m).
 - Documented accuracy must be within the boundaries imposed by GCOS (see[RD1]) and as reported in SoW [AD3].
5. **Accuracy** – An algorithm is positively evaluated if able to provide a high absolute accuracy in all test areas, especially keeping into account the different climatological conditions and possibly different data availability conditions. Accordingly, the following guidelines are used for evaluating accuracy characteristics:
 - Accuracy/uncertainty to be in line with GCOS [RD1] requirements as reported in SoW [AD3].
 - The algorithm matches the end-user (climatologist and other users from the community) requirements;
 - For unsupervised tasks the accuracy should not depend on the availability/quality of prior information.
 - For supervised tasks the accuracy should be robust to the availability/quality of prior information.
6. **Level of Automation** – From an operational point of view, it is mandatory that an algorithm runs in a completely automatic way. Algorithms requiring any amount of manual work, strong interaction with the final users are negatively evaluated.
7. **Specific End-users Requirements** – From an operational point of view, capability of an algorithm to satisfy and meet possible end-user requirements is another important parameter of evaluation. The main guidelines for driving this ranking are:
 - The algorithm is robust to the use in several climatological regions;
 - The algorithm can be reasonably included in an operational procedure.

2.2 Evaluation

The evaluation procedure of each comparative task aimed at deciding on a specific algorithm/technique is carried out by considering all criteria listed before. To each reported activity, a thorough discussion is given regarding how these criteria are weighted in the overall evaluation, which aspects are given strong emphasis and which ones are considered less relevant. The evaluation activity provides answers about best performing algorithms/techniques that are included in the processing chain of the current version of HR LC products.

3 Algorithms and procedures (year 1)

3.1 Optical data processing

The optical processing chain is designed to primarily work with images at 10/30m resolution, producing outputs at the same resolution. It leverages multitemporal, multispectral data from recent years, including Sentinel-2 (S2) and Landsat-8/9 (L-8/9), alongside legacy data from Landsat-5/7/8 (L-5/7/8). The pre-processing methodology follows the same logical framework as Phase 1, with planned adjustments aimed at enhancing both

the quality of the output (e.g., optical composites) and the computational efficiency of the process.

In the classification stage, Phase 1 faced challenges including: (i) limited availability of photo-interpreted data suitable for mapping large areas; (ii) constraints related to input features; and (iii) the need to optimize the classification algorithm for efficiency. Phase 2 focuses on refining the classification pipeline to address these challenges.

The pre-processing phase addresses radiometric and geometric distortions specific to sensors and platforms, as well as harmonization tasks. Radiometric corrections are applied to mitigate issues such as variations in illumination, viewing geometry, atmospheric conditions, and sensor-specific noise or response variations. These factors depend on the sensor, platform, and acquisition conditions. A major challenge with optical imagery is cloud coverage, which requires targeted processing to accurately identify cloud and cloud shadow pixels, potentially incorporating restoration techniques to recover spectral information for occluded areas. Figure 1 shows the main blocks of the optical pre-processing chain.

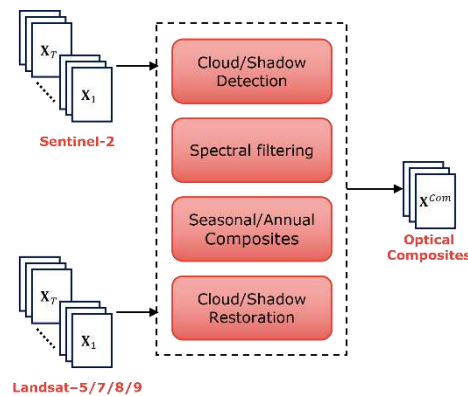


Figure 1. Optical pre-processing chain.

Figure 2 illustrates the optical data processing chain used to produce both static and historical high-resolution land cover (HRLC) maps. The workflow involves pre-processing the images to create optical composites, integrating these with ancillary data (e.g., Copernicus Digital Elevation Model [DEM]) to extract features for classification. The classifiers are trained on available training data points and subsequently generate pixel-wise class-posterior probabilities. These probabilities feed into the decision fusion processing chain, resulting in the final land cover products.

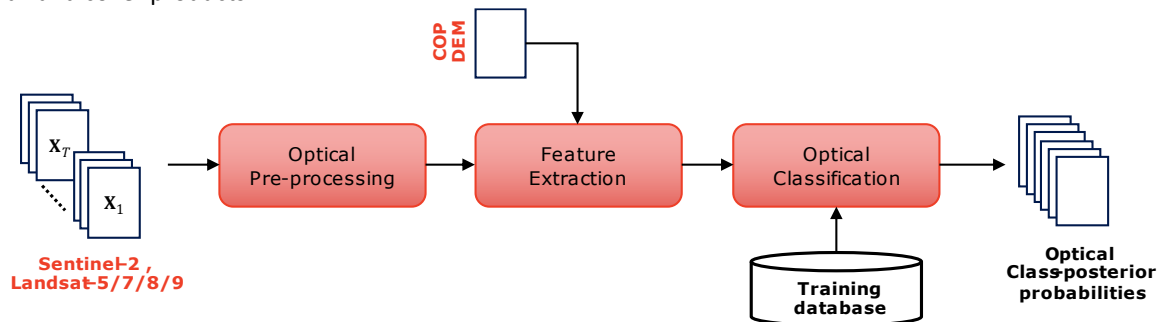


Figure 2. Optical data processing chain for the prototype production of both the static and the historical HRLC maps obtained by classifying the time series of HR optical data.

The following sections provide the results of the activities on the optical processing chain during year 1 of the project. The results here presented are focused on three main activities:

1. Cloud detection: validation of the S2 Sen2Cor Scene Classification Layer (SCL) Cloud and Cloud shadow mask enhancement;
2. Composite generation: code optimization and alternative compositing strategies;
3. Optical classification: preliminary evaluation of deep learning architectures for HRLC mapping with optical Satellite Image Time Series (SITS).

3.1.1 Cloud and Cloud shadow Detection

The improvements foreseen for the pre-processing chain of optical data require both quality and speed in the computation. This last requirement is essential for reducing costs and production times, especially considering the scaling of the production to large areas. For this reason, during Phase 1, Surface Reflectance (SR) products

were not generated as part of the processing chain, and instead they were retained from the providers, thus relying on the providers Atmospheric Correction (AC) algorithms. This allowed both to use consolidated AC strategies and to save processing time. On top of that, cloud detection is often coupled with AC during the generation of SR products. Therefore, clouds and cloud shadows masks are usually provided alongside the SR products. The availability of precomputed cloud and cloud shadows masks means that we have the possibility of saving computation time by utilizing the available masks. During Phase 1, Fmask [1] cloud and cloud shadows masks, provided with the SR Landsat products, have shown to be effective in the cloud detection in Landsat imagery with satisfying performance levels [AD5]. Therefore, it has been used for masking clouds in the historical processing chain for HRLC30 products. On the other hand, Sen2Cor has shown some limitations in the cloud detection capabilities with S2 SR products during Phase 1. Therefore, we developed a cloud and cloud shadow enhancement module. This approach is being re-evaluated in Phase 2, considering the recent improvements in the Sen2Cor cloud detection module. Indeed, the new Collection 1 of reprocessed S2 products (the data that will be used in the Phase 2) made available by ESA consider also upgraded cloud detection strategies that leverage the parallax effect of S2 MSI sensors.

Therefore, we are considering whether to use the new Collection 1 masks as they are or whether to apply the Phase 1 cloud enhancement module to them. The selection will weigh the performance improvements against the necessity of additional processing time, which would undermine the production time for HRLC10 products. Indeed, the cloud and cloud shadow enhancement module require the computation of a seasonal background image, used to better discriminate clear sky observations from clouds. This means that the cloud detection of each acquisition relies on the time series of observations of the current season, which is computationally expensive (similar to the composite generation). Therefore, this increase in computation time requires a sufficient improvement in cloud detection performance to be acceptable.

In this analysis, we compare the recent Sen2Cor and the Ph1 Cloud Enhancement module using the multi-temporal global benchmark dataset CloudSEN12+ [2], [3], for cloud and cloud shadow detection with S2. This dataset provides 49,250 S2 image patches (IPs) with different annotation types: (i) 10,000 IPs with high-quality pixel-level annotation, (ii) 10,000 IPs with scribble annotation, and (iii) 29,250 unlabelled IPs. The labelling phase was conducted by 14 domain experts using a supervised active learning system. A rigorous four-step quality control was designed to guarantee high quality in the manual annotation phase. Furthermore, CloudSEN12+ ensures that for the same geographical location, users can obtain multiple IPs with different cloud coverage: cloud-free (0%), almost-clear (0–25%), low-cloudy (25–45%), mid-cloudy (45–65%), and cloudy (>65%), which ensures scene variability in the temporal domain. Therefore, CloudSEN12+ provides a reliable benchmark for precisely evaluating different cloud detection algorithms. For this experiment, we consider only IPs of 2000×2000 pixels for which high quality annotations are available. Each pixel is labelled as “Clear”, “Thick Cloud”, “Thin Cloud”, or “Cloud Shadow”. The main difference between thick and thin clouds is that thin clouds are semi-transparent, while thick clouds are opaque and highly reflective in the visible bands. However, we don’t make a difference between thick and thin clouds, thus they are merged into the same class “Cloud” in our analysis. While CloudSEN12+ is a very large dataset, it does not contain all the adjacent acquisitions of each labelled scene, making it difficult to use with the Phase 1 Sen2Cor enhancement module, which requires the computation of a seasonal background by aggregating all the acquisition of the same season as the target acquisition. Therefore, we selected a subset of CloudSEN12+. Specifically, we selected three S2 tiles for which we have multiples IPs during the same season and a fair representation of each class:

- 18NWL with 4 IPs in Winter 2019;
- 21HUA with 3 IPs in Spring 2019;
- 36SWH with 4 IPs in Winter 2017.

For each of these tiles, we collected all the S2 L2A acquisitions of the corresponding season, and computed the enhanced masks. Figure 3 shows the visual comparison of the cloud and cloud shadow masks obtained by Sen2Cor and the enhancement module. From the images, it is clear that Sen2Cor overestimates clouds, and the enhancement module further accentuates this phenomenon. The quantitative results in Table 1 confirm the observations, reporting the User’s Accuracy (UA), Producer’s Accuracy (PA) and F1 score against the CloudSEN12+ subset. Indeed, the enhancement module reduces the omission errors at the cost of highly increasing the commission errors of the Cloud category over the Clear category. Cloud shadow performance remain unchanged.

Given the limited improvements provided by the enhancement module applied to the new Sen2Cor masks, the decision is to remove the enhancement step from the processing chain, reducing the overall computation time.

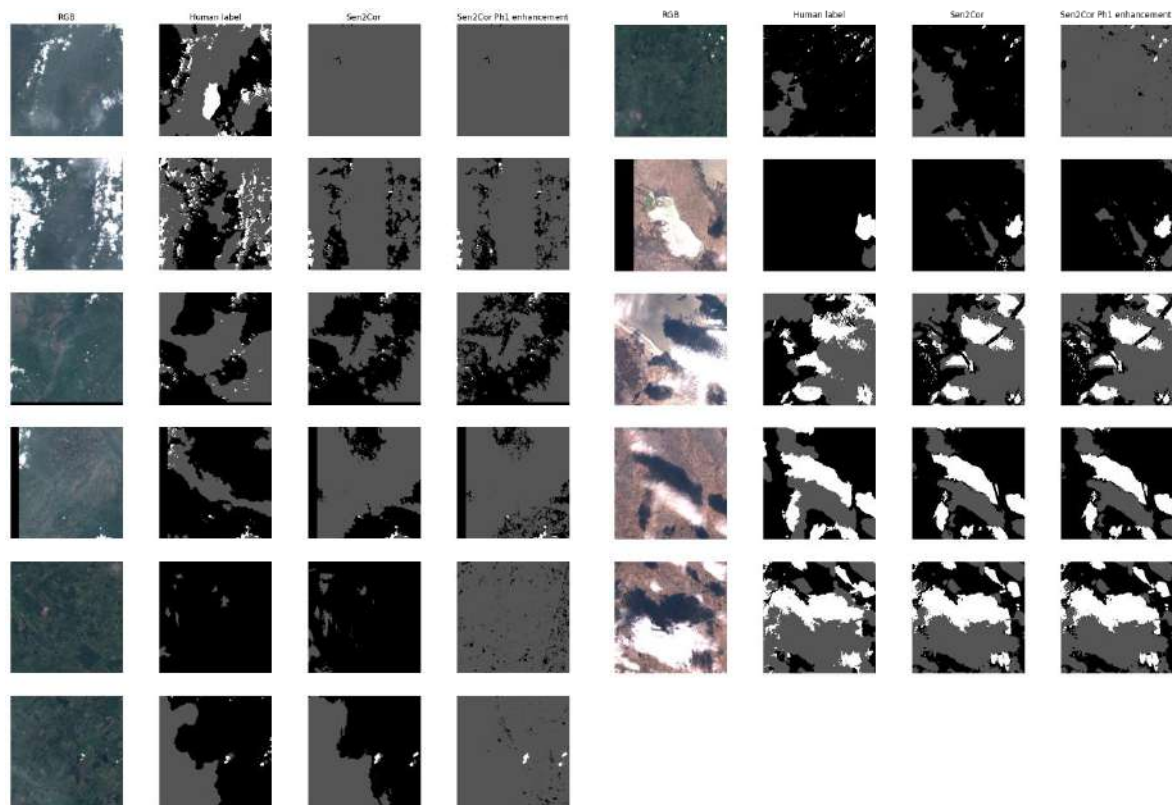


Figure 3. Comparison of the expert annotation with the Sen2Cor and Ph1 Improved Sen2Cor cloud and cloud shadow masks. (black = clear, gray = cloud, white = cloud shadow)

Table 1 Quantitative accuracy metrics from the benchmark analysis on the CloudSEN12+ dataset.

	Sen2Cor			Ph1 Sen2Cor Enhancement		
	UA	PA	F1	UA	PA	F1
Clear	0.91	0.85	0.88	0.86	0.44	0.58
Cloud	0.68	0.84	0.75	0.38	0.85	0.52
Cloud Shadow	0.91	0.75	0.82	0.91	0.75	0.82

3.1.2 Composite Generation

The composite generation is the most expensive step in the optical pre-processing chain. The activities of the first year focused on the improvement of the processing time while maintaining or improving the composite quality. During Phase 1, optical composites were generated by using a processor written in Python, using GDAL, numpy and basic multiprocessing capabilities. The approach is here compared to two different alternatives: FORCE and an improved Python processor adopting more advanced libraries such as xarray and dask.

FORCE has the advantage of being compiled, thus it allows generally fast processing time. Moreover, FORCE is developed with a focus on time series analysis: it implements the concepts of data cube and allows for parallel processing of long Satellite Image Time Series (SITS). However, it has some restrictions, as it requires the data to be saved on disk in a specific format, achievable only by processing Top-of-Atmosphere (TOA) data directly with FORCE AC, which adds an additional step to the processing chain. Also, FORCE does not provide a strategy for generating optical composites exactly as they were designed during Phase 1. Nevertheless, workaround strategies are under analysis both to avoid the AC processing of FORCE and to use instead the providers SR products, and to generate products equivalent to Phase 1 monthly/seasonal composites by exploiting specific parametrizations of FORCE time series analysis (TSA) tool.

The improved Python processor has been designed by keeping in mind two main aspects: the advantage of modelling data cubes for processing SITS data and the capability of scaling the processor in a distributed scenario. The improved Python processor implements two approaches to composite generation: band-wise median aggregation and medoid approach to most-representative-image selection (see ATBD [AD6]). The advantage of using the medoid approach is the generation of a composite whose spectral signature actually matches the spectral signature of on the observations that has been aggregated. This allows to have more consistent

representation of the spectral signature of the land cover. Indeed, the band-wise median aggregation creates a composite whose reflectance values can come from different acquisitions, thus distorting the spectral signature of the land cover by mixing the spectral signature of potentially very different acquisitions. In the current implementation, the medoid is computed for each pixel as the image that minimizes the Euclidean distance in the spectral domain from the median composite.

The test area considered for the analysis is the S2 tile 21KUQ in the Amazonia area. From this area, we collected a time series of 73 S2 acquisitions over the year 2023, with acquisitions every 5 days. For the experiment, we considered both monthly and bimonthly composites:

- Legacy Phase 1 processor: test only with 12 bimonthly composites (pixel-wise median of each band of all the acquisition in a given month $\pm$ 15 days);
- Improved Python processor: test both monthly and bimonthly composites;
- FORCE: a custom parametrisation of TSA is used to generate monthly composites.

Note that FORCE does not only compute the median for each band by month, but it estimates a cloud-free SITS by cloud masking and interpolation beforehand. First, all clouds, cloud shadows and defective pixels are discarded. Then, the acquisitions are reprojected on a regular time grid by interpolation, generating a full time series of multispectral acquisitions. Only after this, the pixel-wise and band-wise median is computed by month. This approach effectively fills in missing data in the SITS, which is partly equivalent to Phase 1 cloud/shadow restoration step. However, the Phase 1 approach does this after the composite generation. Therefore, FORCE is able to better exploit the temporal information in the original SITS to fill in the missing data. In order to efficiently interpolate the time series, FORCE uses temporal convolutions with Radial Basis Function (RBF) kernels [4]. This ensures fast processing exploiting low-level implementations of the convolution operation. In the experiments, we consider an interpolation step (defining the spacing in the temporal grid) of 14 days, using three RBF kernel with sigma values of 14, 28, and 42 days. The sigma values define the width of the gaussian bell of each filter. Using different filters allows to more precisely follow the SITS when frequent acquisitions are available with the smaller kernels, whereas it allows for interpolating more distant time steps with larger kernels when few acquisition area available. The three filters are combined by weighting them by the acquisition's density within each filter. Since gaussian kernels have in principle infinite width, a cut-off density of 0.95 is set, reducing the effective width of each kernel. The maximum width allowed for each filter is also limited to $\pm$ 54 days around the centre of the kernel. After interpolation, the FORCE folding capabilities is used to generate temporal aggregates on a monthly basis. FORCE does not allow bimonthly folds, thus we only tested monthly composites. Nonetheless, the interpolation capabilities of FORCE actually address the same issue addressed by the use of bimonthly composites over monthly composites, i.e., reduced data availability. Therefore, FORCE monthly composites are actually comparable to both the monthly and bimonthly composites generated by both the legacy Phase 1 processor and the improved processor.

Table 2. Processing time of the different processors for composite generation

	12 Monthly composites	12 Bimonthly composites
Legacy Phase 1 processor	–	$\geq 3h$
Upgraded Python processor (median)	33m	1h 26m
Upgraded Python processor (medoid)	1h 45m	–
FORCE (interpolation + median)	33m (+1h 18m L2 gen)	–

Table 2 reports the processing times of the different processors. These results were achieved on a workstation with 64GB of RAM (2x32GB DDR4 3200MHz CL16 non-ECC), a 8C/16T Ryzen 7 5800X CPU, and the data was stored on a M.2 NVMe v1.4 SSD Samsung 980. From these results, we can observe a clear 2x improvement in processing time over the legacy processors by the upgraded Python processor. If we consider the interpolation capabilities of FORCE, we achieve a ~6x improvement over the legacy processor. Considering monthly composites, the processing time of the upgraded processor with median aggregation and FORCE are similar. However, there are few aspects to keep into consideration. On one hand, FORCE performs both interpolation and median aggregation taking the same time as the upgraded processor, which performs only the median aggregation operation. On the other hand, FORCE needs the Level-2 imagery to be already stored following the specific FORCE requirements for tiling, projections, and file formats. Currently, the only way to achieve this is to use the Level-2 generation capabilities of FORCE, which adds an additional step to the processing (~1h 18m for generating all the Level-2 images of the 73 S2 acquisition considered). This makes the use of the upgraded processor more appealing. Approaches for avoiding this time consuming step are under analysis. If we can utilize the pre-computed Level-2 data from the providers, FORCE would become the most efficient approach to

composite generation, also including an interpolation step before temporal aggregation, which further reduce gaps in the data. In the case this is not possible, the upgraded Python processor will be updated to integrate the interpolation procedure before computing the median composite, thus aligning it with FORCE. The processing time of the upgraded processor with medoid aggregation takes 3x the median aggregation. This large increase in processing undermines the applicability of the medoid operator. The results for computing the bimonthly composites with the medoid operator are missing due to the high memory requirements of this approach. However, we can expect worse processing time than the legacy processor.

Figure 4 shows some examples of optical composites for January (Figure 4(a)-(c)-(e)-(g)) and February (Figure 4(b)-(d)-(f)-(h)) 2023 in tile 21KUQ. While January is populated with mostly clear-sky observations for the whole tile, February consists of almost completely clouded acquisitions, with the exception of a single acquisition (Feb. 5th 2023) with sparse cloud cover. This challenge provides a good benchmark for understanding the most proper composite generation approach. First, we can observe that the January composites in Figure 4(a)-(c)-(e)-(g) are all very similar and of good quality. This is expected given the good number of cloud-free observations for all the pixels. Instead, the February composites in Figure 4(b)-(d)-(f)-(h) show some differences. First, Figure 4(b)-(d), generated by monthly median and medoid composites, respectively, show some gaps in the scene (the black regions). This is expected given the availability of a single acquisition with low cloud cover (Feb. 5th 2023), whereas all the other acquisitions have a cloud cover > 90%. However, the other acquisitions still contribute to the final composite for the few available cloud-free pixels (from the computed cloud masks). Note that the cloud masks in these scenes are underestimating the cloud coverage, thus cloudy pixels are actually used in the composites. In these areas of the scene, we can observe a difference in the median and medoid composites (otherwise the same). In particular, the medoid composite seems noisier than the median (Figure 4(b)-(d)), as it tends to select the cloudy pixels, failing in filtering the available observations. If we consider the bimonthly composite of February 2023 (Figure 4(f)), we can see improvements related to absence of gaps in the scene. Thanks to the temporal window ranging from January 15th to March 15th, the bimonthly composite is able to exploit the cleaner observations of January and February to fill the gaps. However, we can observe artifacts due to the large temporal distance in the selected pixel values, which reflects the gaps due to clouds. The FORCE monthly composite, instead, is the one producing the cleanest February composite, despite using a median aggregation by month. This is possible thanks to the interpolation computed before aggregation, which not only allowed to avoid gaps in the scene by combining January and March acquisitions but also produced an artifact-free composite.

In conclusion, we improved the efficiency of the composite generation step, reaching good processing times. Among the considered approaches, medoid is the most expensive with no composite quality improvements, whereas FORCE has shown to be both efficient and to produce high-quality composites. The key aspect to the improved quality of FORCE can be identified in the interpolation procedure performed before median aggregation. However, we need to take into account that FORCE requires the input L2 data to be consistent to its internal representation. Therefore, we are considering two directions:

- The alignment of the upgraded processor to the FORCE TSA pipeline;
- The introduction of converter that is able to satisfy FORCE internal requirement, such that to remove the necessity of performing AC, which adds unavoidable processing time.



(a)



(b)

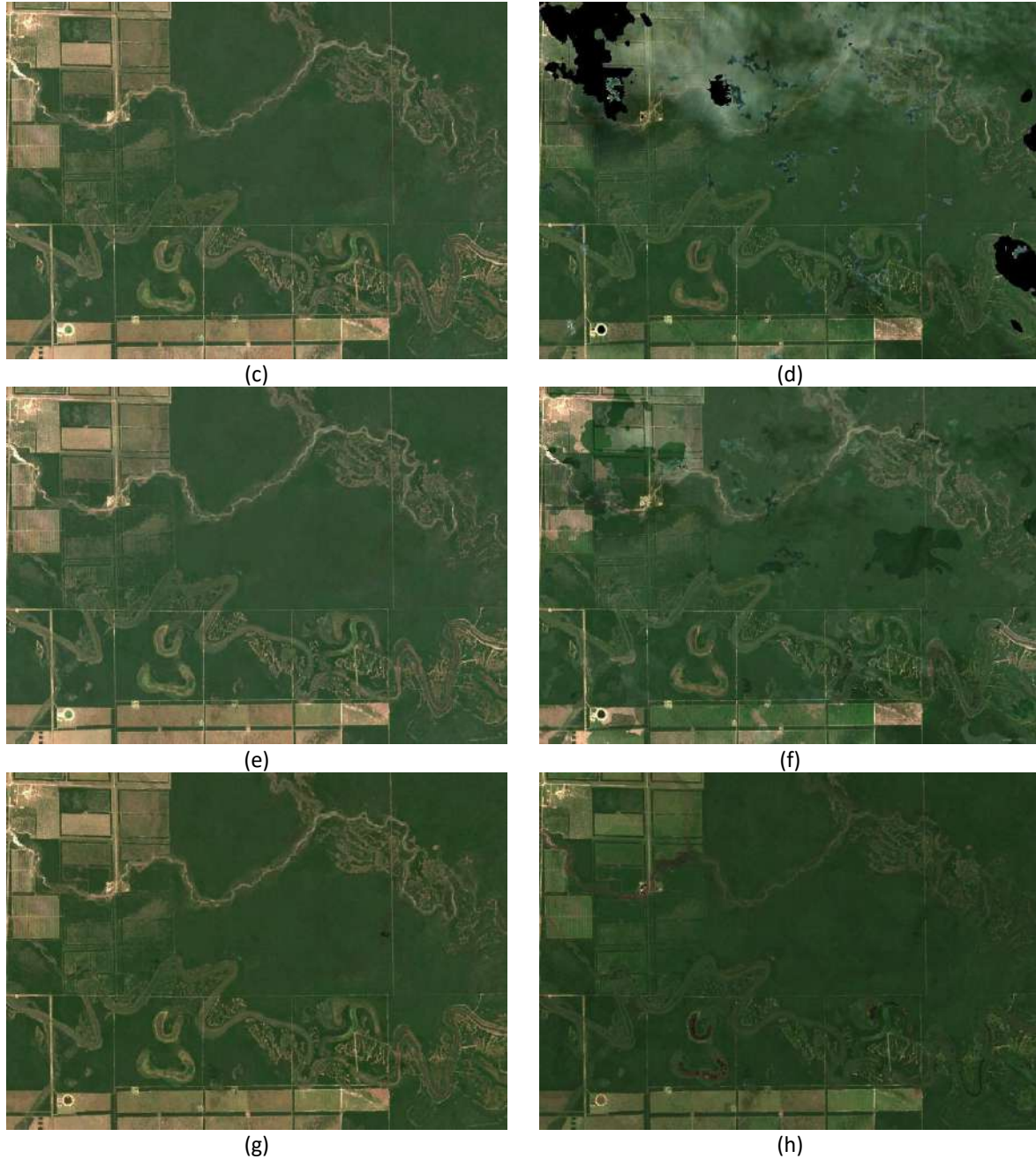


Figure 4. Examples of generated composites for S2 tile 21KUQ in 2019 for January [first column, i.e., (a) (c) (e) (g)] and February [second column, i.e., (b) (d) (f) (h)]. (a) (b) Monthly median composites. (c) (d) Monthly medoid composites. (e) (f) Bimonthly median composites. (g) (h) Monthly FORCE median composites. Black areas are pixels with not clear-sky observations in the considered period.

3.1.3 Optical Classification

During Phase 1, the optical classification was performed by using Support Vector Machines (SVMs). The main challenges in using SVMs were related to features selection and accurate pixel-wise class-posterior probabilities estimation. To solve both these problems in Phase 2, we consider deep learning strategies. Indeed, a deep learning architecture is able to learn a good feature representation starting from the raw input features, reducing time in tuning the optimal feature set, and it can directly estimate the pixel-wise class-posterior probabilities, instead of relying on external calibration approaches such as in the case of SVMs.

In our preliminary analysis, we compared three different architectures, all of which incorporate temporal information. To further examine the effectiveness of the methodologies in utilizing contextual information, we also included pixel-level networks. The considered architectures are the following:

1. Transformer: we employed the encoder architecture of a Transformer [5]. Positional encoding was applied to the SITS, focusing on classifying single pixels without incorporating contextual information.
2. Swin Transformer: we adapted a Vision Transformer (ViT), designed to capture both local and global

dependencies in the image. Instead of using global attention across the entire image like traditional ViTs, the Swin Transformer divides the image into non-overlapping windows, performing self-attention within each window. Our implementation considers an adaptation to the RS scenario of [6], initially designed for super-resolution of videos, here adapted to jointly model the spatial and temporal information of the SITS.

3. Convolutional Long Short-Term Memory (ConvLSTM): this hybrid model integrates convolutional neural networks (CNNs) with long short-term memory (LSTM) networks. It is tailored for spatio-temporal data, maintaining spatial information within individual time steps while capturing temporal dependencies across time steps [7].

Given the complexity of training deep network with the full HRLC legend on large areas, we performed a benchmarking exercise on restricted area in the Amazonia area. In order to provide a sufficiently large training dataset, we employed the HRLC10 map generated during Phase 1 as reference. Therefore, the accuracy metrics provide information on the level of agreement with the existing products. On one hand, a high level of agreement does not provide an actual accuracy metric. On the other hand, a high agreement means that the considered deep network agrees with the whole Phase 1 processing chain, including the optical-SAR decision fusion, spatial harmonization and post processing results, which improved the overall classification of the optical SVM classifier. Therefore, we consider this benchmark sufficient for comparing the different models and selecting the most suitable one for Phase 2. For our analysis, we selected the dataset corresponding to the MGRS S2 tile 22KGV in Brazil, chosen for its heterogeneous distribution of land cover classes, including tree cover (evergreen broadleaf, deciduous broadleaf, evergreen needleleaf), shrub (evergreen, deciduous), grassland, aquatic vegetation, cropland, bare areas, built-up areas, seasonal open water, and permanent open water.

We utilized all S2 images available for 2019, generating 12 monthly median composites. To ensure a balanced training dataset, we adopted a heterogeneity-based patch extraction approach. Specifically, we divided the S2 composite images into non-overlapping 5×5 pixel patches. For each patch, the dominant land cover class was identified. These patches were then randomly divided into training, validation, and test sets with a minimum requirement of N representative patches per class, where N was fixed at 1000.

For minority classes with fewer than N patches, 50% of available patches were allocated to the training set, 25% to the validation set, and 25% to the test set. This process resulted in a training dataset representing approximately 0.22% of the area. Validation set sizes were determined as 25% of the training samples per class, while the remaining patches in the tile, not included in the training or validation sets, comprised the test set. This methodology ensured a small but representative dataset for the full classification scheme.

To ensure the robustness of the evaluation, we conducted all tests five times, using distinct samplings for the training, validation, and test sets in each iteration. For reproducibility, we utilized predefined seed values for the random number generator (RNG). The evaluation metrics reported include Overall Accuracy (OA), mean F1 Score (mF1), and mean Intersection over Union (mIoU) on the test sets. Additionally, the standard deviation of these metrics was calculated as an unbiased estimator to provide insight into the consistency of the results. Table 3 shows the overall performance of the three deep architectures. In general, ConvLSTM and Transformer Encoder perform similarly, with no significant differences in overall performance metrics. On the other hand, Swin Transformer performs significantly better than the other, achieving the best performance overall.

Table 3. Comparison of the accuracies obtained by different architectures on the considered S2 tile. The best performance are in bold. The related standard deviation is reported in brackets.

	mF1 (%)	OA (%)	mIoU (%)
Transformer Encoder	60.80 (0.59)	80.21 (1.06)	48.66 (0.55)
Swin Transformer	65.42 (1.10)	83.53 (0.95)	53.14 (1.10)
ConvLSTM	61.81 (1.38)	81.65 (1.03)	49.73 (1.35)

In order to validate class-wise accuracies, we selected a random seed to initialize the networks' weights before training for all the considered architectures. For each class, we computed the User's Accuracy (UA), the Producers Accuracy (PA) and the F1 Score. Table 4 shows the accuracies metrics of the different architectures. Swin Transformer is again the best performing architecture in terms of F1 scores, achieving the best performance on most classes. However, Transformer Encoder is the one achieving the best PA scores on most classes. These results highlight that the best performance is achieved by architectures employing the attention mechanisms to model the temporal information (i.e., Transformer Encoder and Swin Transformer), and that the best results are achieved by combining it with the spatial context (Swin Transformer). Indeed, the proper modelling of the temporal information helps in identifying the phenology of different land covers, while the spatial context provides more control over commission errors on categories with spatial features such as shrubs and build-up areas.

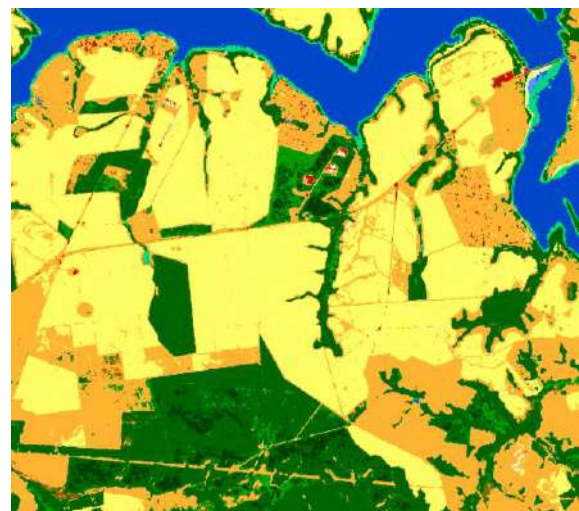
Table 4. Class-wise accuracy metrics obtained by different architectures on the considered S2 tile. The best performance are in bold.

	Transformer Encoder			ConvLSTM			Swin Transformer		
	UA	PA	F1	UA	PA	F1	UA	PA	F1
Tree cover evergreen broadleaf	91.18%	65.89%	76.50%	86.48%	74.50%	80.04%	85.33%	76.93%	80.91%
Tree cover evergreen needleleaf	46.95%	96.47%	63.16%	49.57%	91.90%	64.40%	54.53%	86.53%	66.90%
Tree cover deciduous broadleaf	43.66%	41.91%	42.77%	45.67%	44.54%	45.10%	43.38%	57.03%	49.28%
Shrub cover evergreen	25.10%	81.68%	38.40%	38.58%	56.37%	45.81%	35.62%	73.72%	48.04%
Shrub cover deciduous	5.70%	38.22%	9.92%	2.90%	37.48%	5.38%	5.10%	41.98%	9.09%
Grasslands	89.52%	89.79%	89.65%	89.72%	87.85%	88.77%	92.95%	87.05%	89.90%
Croplands	94.35%	90.95%	92.62%	91.66%	89.91%	90.77%	94.25%	90.88%	92.53%
Grassland vegetation aquatic	74.02%	74.29%	74.15%	72.18%	80.84%	76.27%	76.01%	84.24%	79.91%
Bare areas	23.30%	63.30%	34.06%	26.26%	55.74%	35.70%	27.75%	73.38%	40.27%
Built-up	42.49%	84.32%	56.51%	57.15%	78.33%	66.09%	66.20%	75.26%	70.44%
Open water seasonal	51.03%	81.18%	62.66%	57.19%	66.31%	61.41%	58.98%	74.52%	65.84%
Open water permanent	97.88%	98.40%	98.14%	98.89%	96.70%	97.78%	98.78%	97.22%	97.99%

Figure 5 shows a qualitative comparison of the predictions of the three deep architectures with the reference HRLC10 map. Notably, we can see that all the deep methods seem to better delineate the boundaries between the different land covers, especially for tree cover categories. Additionally, the deep learning methods seem to better capture the urban fabric, providing better results for the built-up class (especially the Transformer model). The Swin Transformer seems to provide noisy predictions in the example. However, the inference algorithm utilized with Swin Transformer was not optimized to perform proper semantic segmentation of the scene. Indeed, it process the image in a block-wise manner instead of using a rolling window approach. This leads to some checkboard artifacts, especially where the model is uncertain. Future analysis will include the qualitative results with proper inference algorithms, which will most likely improve the visual fidelity of the prediction maps of Swin Transformer.



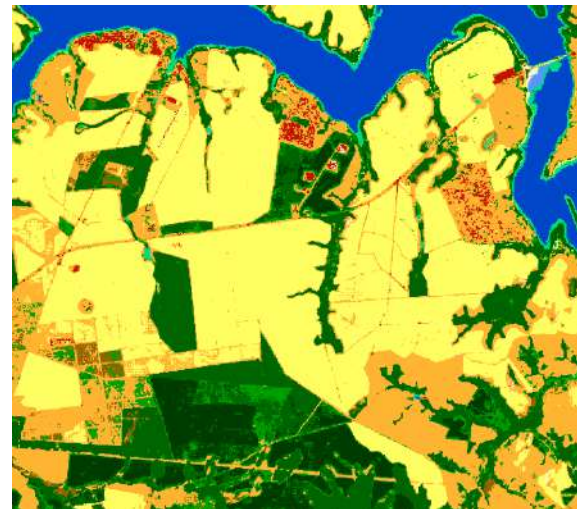
(a) January median composite



(b) HRLC10 map



(c) Transformer Encoder prediction



(d) Swin Transformer prediction



(e) ConvLSTM prediction

HRLC CLASSES	
CODE	DESCRIPTION
0	No data
10	Tree cover evergreen broadleaf
20	Tree cover evergreen needleleaf
30	Tree cover deciduous broadleaf
40	Tree cover deciduous needleleaf
50	Shrub cover evergreen
60	Shrub cover deciduous
70	Grasslands
80	Croplands
90	Woody vegetation aquatic or regularly flooded
100	Grassland vegetation aquatic or regularly flooded
110	Lichens and mosses
120	Bare areas
130	Built-up
140	Open water
141	Open water seasonal
142	Open water permanent
150	Permanent snow and/or ice

Figure 5. Qualitative comparison of the predictions from the different deep architectures considered. (a) RGB image of the January median composite of 2019. (b) Reference HRLC10 map. (c) Transformer Encoder prediction. (d) Swin Transformer prediction. (e) ConvLSTM prediction.

The preliminary analysis for the optical classification suggests that the use of deep learning architectures is promising, with results similar to the output of the full processing chain of HRLC10 (and in some cases qualitatively better). In particular, Swin Transformer and Transformer Encoder networks are promising options for improving the optical classification in Phase 2. Further analysis will focus on defining proper training strategies for defining deep networks able to operate at a large scale. This will include self-supervision approaches, as they have affirmed as the state-of-the-art approaches for model pre-training, and weak supervision approaches, in order to be able to augment the training dataset. Indeed, while the available photo interpreted points are fundamental for defining models that properly map the target land cover classes, the size of the training database is still scarce for training deep learning models.

3.2 SAR data processing

Figure 6 illustrates the processing workflow for generating the high-resolution (HR) land cover (LC) map through the classification of Sentinel-1 (S1) time series. Initially, the SAR images undergo preprocessing to convert the backscattered signal into σ_0 , expressed in decibels (dB).

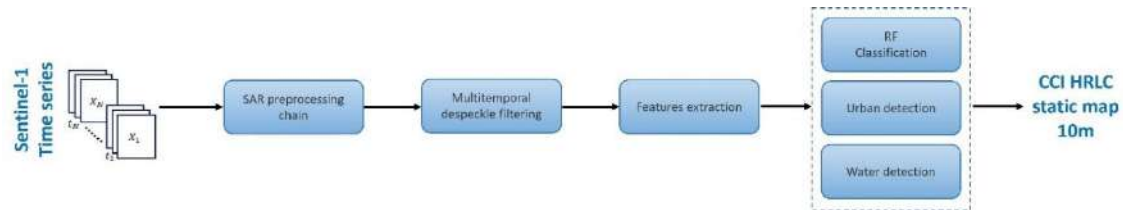


Figure 6. Block diagram of the SAR classification chain.

To determine the most suitable algorithm for the final SAR land cover classification process to be applied in Phase 2 of the project, a series of carefully designed experiments were conducted. These experiments aimed to assess the performance and accuracy of different approaches, with each outcome meticulously evaluated to ensure the optimal selection.

In more detail, the initial phase of testing employed the SAR classification chain, as illustrated in Figure 6, which had been developed during Phase 1 of the CCI+ project. This chain was used to generate preliminary classification outputs, providing an initial benchmark for performance. The Random Forest (RF) classifier was trained using the ground truth data collected by the team in Phase 1 through the hierarchical approach. Training points for the urban and water classes were excluded from the dataset.

However, recognizing the inherent limitations and challenges associated with traditional machine learning techniques for complex classification tasks, additional methods were considered. Specifically, alternative approaches leveraging Deep Learning (DL) techniques, Attention U-Net, Swin-Unet, and 3D FCN, were explored to address the limitations of the initial framework and potentially improve classification accuracy, particularly in dealing with the high-dimensional nature of SAR data. Figure 7 presents a simplified block diagram that outlines the key steps and processes involved in the system, providing a clear overview of the workflow and the interconnections between each stage of the procedure.

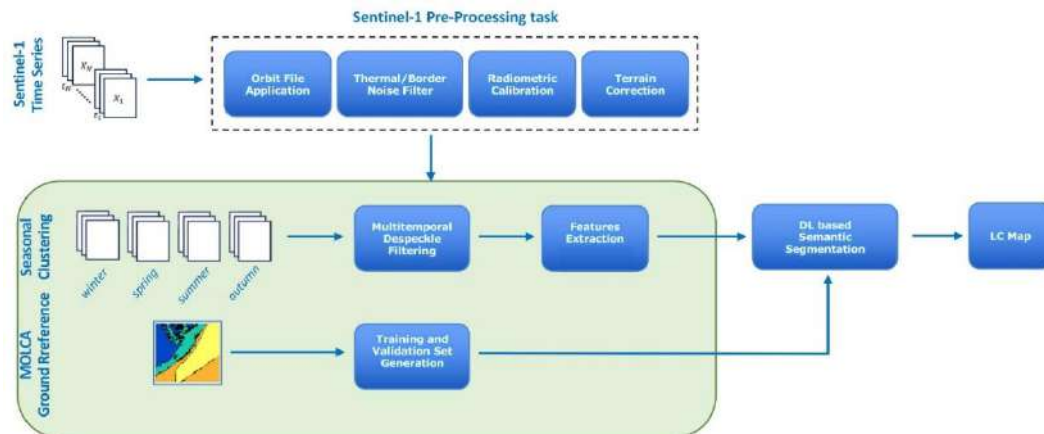


Figure 7. A simplified workflow illustrating the DL mapping procedure applied to SAR time series.

3.2.1 Data and Methods

A first test was carried out in the Amazonian region, specifically focusing on the 21KUQ tile, shown in Figure 8, which was selected as one of the benchmark areas during Phase 1 of the project.

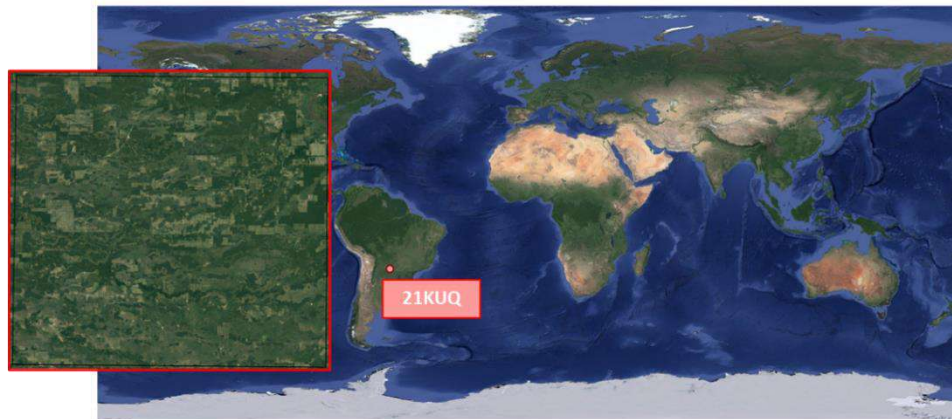


Figure 8. S2 tile 21KUQ in Amazonia, selected to test the enhancement of the RF classification.

This tile is particularly significant due to its diverse land cover types and ecological importance, making it an ideal test case for assessing the classification methods.

For this test region, the performance of the classification procedure based on the use of the RF algorithm was specifically evaluated, considering an expansion of the feature set compared to what was implemented during Phase 1 of the project.

S1 Level-1 Ground Range Detected (GRD) products, acquired in Interferometric Wide Swath (IW) mode, were employed for the analysis. The dataset spans the entire year 2019, offering a comprehensive temporal representation of land cover dynamics. The preprocessing stage constitutes a foundational step in generating a time series specifically tailored for land cover map production. S1 products are first subjected to radiometric calibration to ensure consistency in backscatter values, followed by terrain correction to account for topographical distortions. These steps are executed using the Sentinel Application Platform (SNAP) software provided by ESA, ensuring standardised and reliable data preparation.

For the multitemporal analysis, the images are stratified into the four annual seasons, winter, spring, summer, and autumn, to capture seasonal variations in land cover. For each seasonal subset, a multitemporal despeckling filter is applied to reduce noise while preserving significant spatial and temporal details. This process results in a "super image" for each season, representing an aggregated and noise-reduced view of the tile.

From each super image, a set of spatial features is extracted, including statistical metrics such as Lee, Min, Max, Max-Min, Mean, and Median, which provide valuable information about texture and spatial variability. These spatial features, combined with the super images themselves, form the inputs for training and executing a RF classification.

3.2.1.1 Random Forest classifier

Following Phase 1, a RF classifier was applied in this area. Just to quickly recall it, the RF algorithm has become a cornerstone in land cover mapping due to its robustness and adaptability, particularly when dealing with complex datasets such as those derived from S1 time series.

The classical RF approach, used as a baseline in Phase 1, relied on 28 features derived from four seasonal composites of S1 VH polarisation data collected in 2019. These features captured a range of spatial and statistical characteristics, including local edge enhancement, minimum and maximum values, the difference between them, as well as mean and median measures. While effective in providing a basic understanding of land cover patterns, this classical method was limited in its ability to distinguish subtle variations in the landscape. The need for more detailed classification motivated the development of an expanded feature set.

This initial test investigates potential improvements to the Phase 1 RF classification framework as per the previous paragraphs by incorporating an expanded feature set and implementing advanced preprocessing techniques, following the methodology outlined in [8]. These innovations aim to address the limitations of traditional approaches and provide more accurate and detailed maps, contributing to a better understanding of land cover dynamics in challenging environments like the Amazonian region. Specifically, the enhanced approach introduced data from both VH and VV polarisations, broadening the scope of information captured by the model. S1 images were grouped into 24-day intervals, and multitemporal despeckling was applied to reduce noise. For each period, an arithmetic mean composite was generated, adding 15 new features to the original set. This process resulted in a comprehensive 43-feature set, significantly increasing the algorithm's potential to discern finer details in land cover. These additional features offered improved temporal resolution and better

representation of surface dynamics, particularly in areas with rapid environmental changes. A simplified block scheme is depicted in Figure 9.

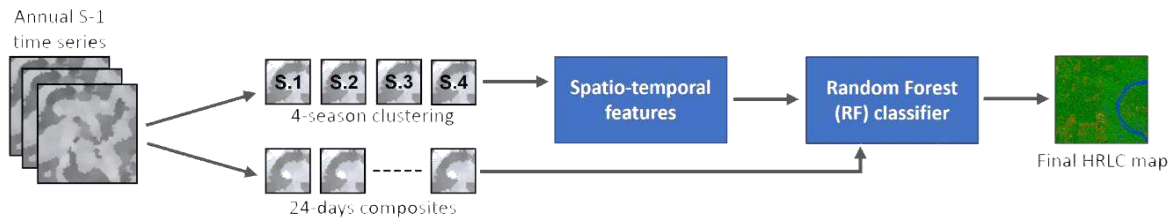


Figure 9. A simplified block diagram illustrating the land cover mapping procedure, highlighting enhancements to the feature pool.

Another critical aspect of the study was the application of an angular-based radiometric slope correction. This preprocessing step, inspired by the method proposed by Vollrath et al [9], addressed terrain-induced distortions in S1 images. These distortions, often present in rugged terrains, can skew classification results by altering the apparent radiometric values of the imagery. By correcting for these angular effects, the study sought to enhance the reliability of the data, especially in regions like the Amazon, where topographic variability can significantly impact radiometric measurements.

As mentioned, this first experimental setup was centred on the Amazonian region, specifically the S2 21KUQ tile. This area presented a diverse range of land cover types, making it an ideal testbed for the proposed enhancements. The training data used consisted of points manually collected through a hierarchical approach during Phase 1 within the region, covering a spectrum of land cover classes, including evergreen forests, grasslands, croplands, and bare soil. Figure 10 shows the spatial distribution of tile 21KUQ in the Amazon region, along with the corresponding legend indicating the number of points for each class.

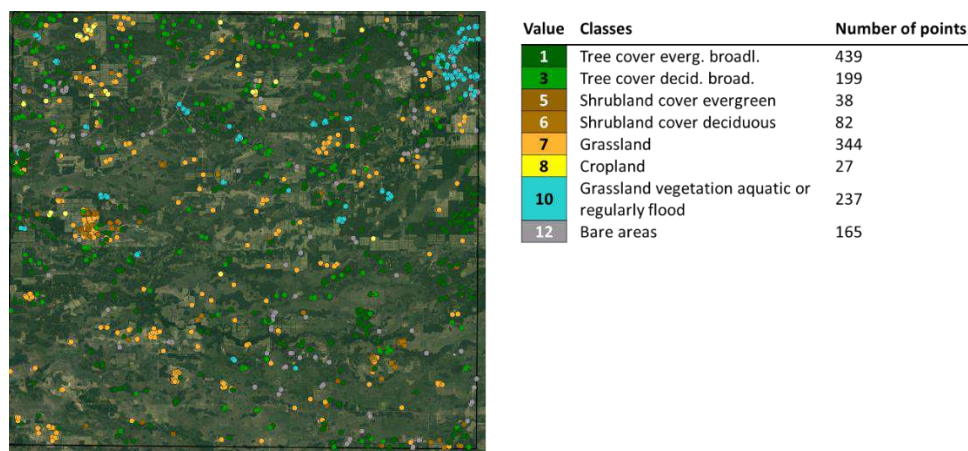


Figure 10. Spatial distribution and legend of the training points collected over the benchmark area 21KUQ in the Amazon region.

Three distinct classification experiments were conducted to assess the proposed enhancement under evaluation. The first experiment utilised the classical 28-feature set without any slope correction, serving as the control. The second incorporated the expanded 43-feature set, also without slope correction, to assess the impact of the additional features alone. The final experiment applied the expanded feature set with slope correction, examining the combined effect of the two enhancements. All classifications were implemented within Google Earth Engine, a platform that allowed efficient processing of large datasets.

Validation datasets for the quantitative assessment were sourced from well-established global products, including the Copernicus Global Land Cover (CGLC) at 100m 2019 [10], and the CCI Medium Resolution Land Cover (MRLC) Map 2019 [11], as well as a validation set provided by PoliMi and verified by UNITN, ensuring a robust comparison against external benchmarks. The validation points from existing global land cover products were randomly extracted to create a balanced and representative dataset for each land cover class. This approach ensures that the distribution of points adequately reflects the diversity of the classes, avoiding biases that could skew the classification accuracy assessment. Care was taken to select points from various geographic

regions within the study area, capturing variations in land cover types due to environmental and climatic factors. The balanced dataset enables a robust comparison across classes, providing a comprehensive evaluation of the classification model's performance.

Table 5. List of validation datasets used to evaluate the classification performance based on the RF approach.

Validation set description	Class	Number of Points
PoliMi/UNITN Validation Set	Tree cover (evergreen broadleaf)	658
	Tree cover (deciduous broadleaf)	338
	Grassland	308
	Cropland	137
	Built-up	250
CGLC Map 100m 2019	Tree cover (evergreen broadleaf)	1000
	Tree cover (deciduous broadleaf)	1000
	Shrubland (evergreen)	1000
	Grassland	1000
	Cropland	1000
	Grassland veg. aq. or reg. flooded	1000
	Bareland	198
	Built-up	1000
CCI MRLC Map 300m 2019	Open water (permanent)	1000
	Tree cover (evergreen broadleaf)	1000
	Tree cover (deciduous broadleaf)	1000
	Grassland	1000
	Cropland	1000
	Grassland veg. aq. or reg. flooded	1000
	Open water (permanent)	1000

3.2.1.2 Assessing the Impact of Enhancements on Random Forest Classification

The results of the improved RF provide significant insights into the effectiveness of the proposed enhancements. Qualitative assessments of the classification maps show that the expanded feature set improves the visual clarity and spatial detail of the outputs. Features such as roads and bare areas within the study region are more distinctly identified, highlighting the value of the additional temporal composites. However, the application of slope correction introduces unexpected challenges. While it aimed to standardise radiometric values, it also increased noise in the classification process, particularly in homogeneous regions, leading to less consistent maps, as highlighted in Figure 11.

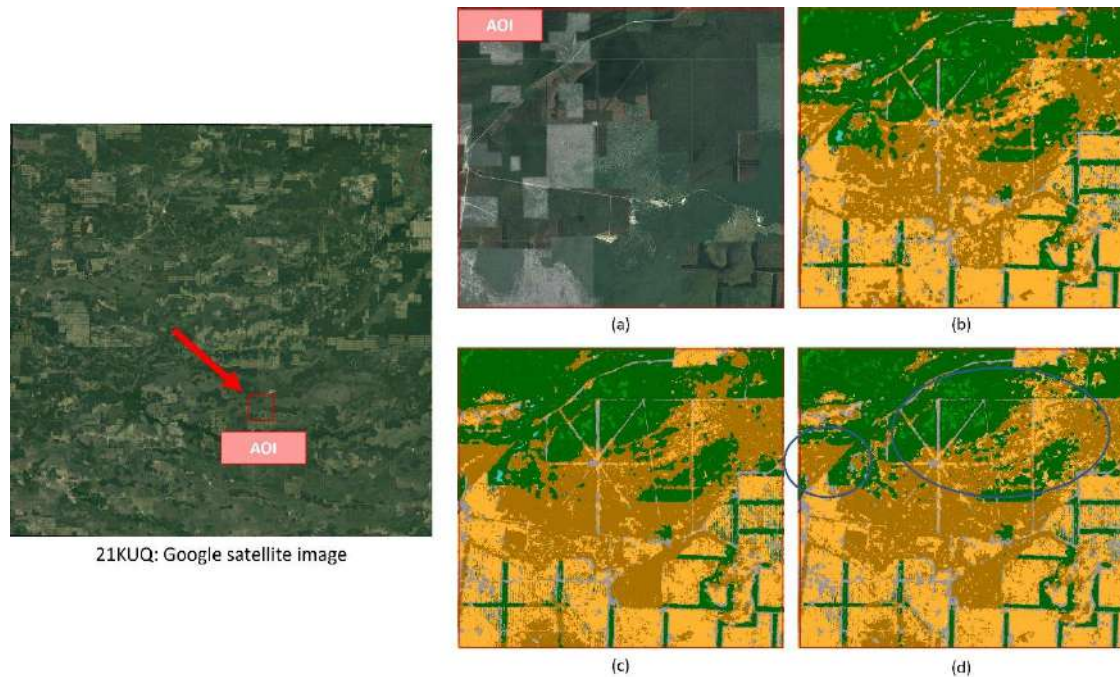


Figure 11. Visual comparison of the 21KUQ AOI tile between: (a) Google satellite image, (b) the RF map produced by the 'classical' approach, (c) S1 RF map with an extended feature set and no slope correction, and (d) S1 RF map with an extended feature set and slope correction.

A visual assessment in Figure 11 suggests that extending the feature space enhances classification performance, making spatial details more apparent. For instance, within the AOI, routes are more distinctly identified and classified as bare land. However, the slope correction appears ineffective in resolving class recognition confusion, resulting in a classification map that appears noisier. Notable examples of this issue are highlighted in the blue areas.

Quantitative analyses supported these findings, with OA metrics indicating a clear improvement when the expanded feature set was used without slope correction. This approach achieved higher differentiation between land cover classes and better alignment with validation datasets. Conversely, the slope-corrected classifications exhibited lower OA, underscoring the need for further refinement of the correction algorithm to avoid introducing artefacts into the data.

Table 6 compares the performance of different RF classification experiments across three validation datasets. The results are expressed in terms of Producer's Accuracy (PA_x, percentage accuracy per class x, where x indicates the class number) and OA, overall classification accuracy across all classes).

Table 6. Comparison of Producer's Accuracy (PA) for individual classes and Overall Accuracy (OA) across three validation datasets using different Random Forest (RF) experimental approaches: Classical Approach, Features Expanded, and Features Expanded and Slope Corrected.

Validation set	RF experiment	PA ₁ (%)	PA ₂ (%)	PA ₃ (%)	PA ₄ (%)	PA ₅ (%)	PA ₆ (%)	PA ₇ (%)	PA ₈ (%)	PA ₉ (%)	PA ₁₀ (%)	PA ₁₁ (%)	OA (%)
PoliMi/UNITN Validation Set	Classical approach	56	75	-	-	51	87	-	-	-	-	-	54
	Features expandend	56	80	-	50	93	-	-	-	-	-	-	53
	Features expandend and Slope corrected	55	79	-	53	92	-	-	-	-	-	-	53
CGLC Map 100m 2019	Classical approach	42	37	17	-	29	40	25	8	-	-	91	29
	Features expandend	43	35	-	-	31	40	20	8	-	-	91	29
	Features expandend and Slope corrected	43	41	-	-	30	46	25	7	-	-	92	28
CCI MRLC Map 300m 2019	Classical approach	47	27	-	-	18	36	44	-	-	-	91	28
	Features expandend	47	27	-	-	18	31	40	-	-	-	91	27,6
	Features expandend and Slope corrected	47	22	-	-	18	42	48	-	-	-	91	27,5

*PA: Producer Accuracy; 1: Tree cover (evergreen broadleaf); 2: Tree cover (deciduous broadleaf); 3: Shrubland (evergreen); 4: Shrubland (deciduous); 5: Grassland; 6: Cropland; 7: Grassland vegetation aquatic or regularly flooded; 8: Bareland; 9: Built-up; 10: Open water (seasonal); 11: Open water (permanent).
Note: The symbol '-' indicates the absence of validation points for a specific class.

Concerning the PoliMi/UNITN Validation Set, expanding features improves class accuracy, especially for PA5 (51% → 93%) compared to the classical approach. The slope-corrected approach achieves similar accuracy for PA5

(92%). All approaches yield similar OA (53%-54%), suggesting that improvements in specific classes do not significantly affect global performance.

For the CGLC validation set, the slope-corrected approach slightly improves PA₆ (46%) compared to the classical approach (40%). Low accuracy is observed for classes such as PA₁ (42%-43%) and PA₂ (35%-41%). Incorporating slope corrections results in a marginal decrease in OA (28% vs. 29%).

For the CCI MRLC map, the slope-corrected approach enhances PA₇ (48%) compared to other methods. Other classes remain stable or decline slightly (e.g., PA₂ drops from 27% to 22%). The OA remains almost unchanged (27.5%-28%). In conclusion, expanding features or adding slope corrections improves the accuracy of certain classes (e.g., PA₅ and PA₇). However, OA remains relatively consistent across experimental approaches, indicating that these refinements do not translate into significant global improvements. Performance varies across validation datasets, with the PoliMi/UNITN dataset achieving the highest OA (53%-54%) compared to the other datasets (27.5%-29%).

3.2.1.2.1 RF Performance Assessment with Ecoregion Classification

To improve classification performance, the RF model was trained using training points manually collected by the EO team and organised by Ecoregions, as shown in Figure 12. This approach takes advantage of the ecological distinctions specific to each Ecoregion, allowing for a more tailored and accurate land cover classification. The model was applied across all three scenarios under consideration, ensuring comprehensive coverage of varying ecological contexts. The performance of this Ecoregion-based RF model was evaluated using the same three validation datasets employed in prior experiments. The Ecoregion (ER)-based training and classification process was carried out using Google Earth Engine (GEE), and the resulting classification maps are in Figure 13.

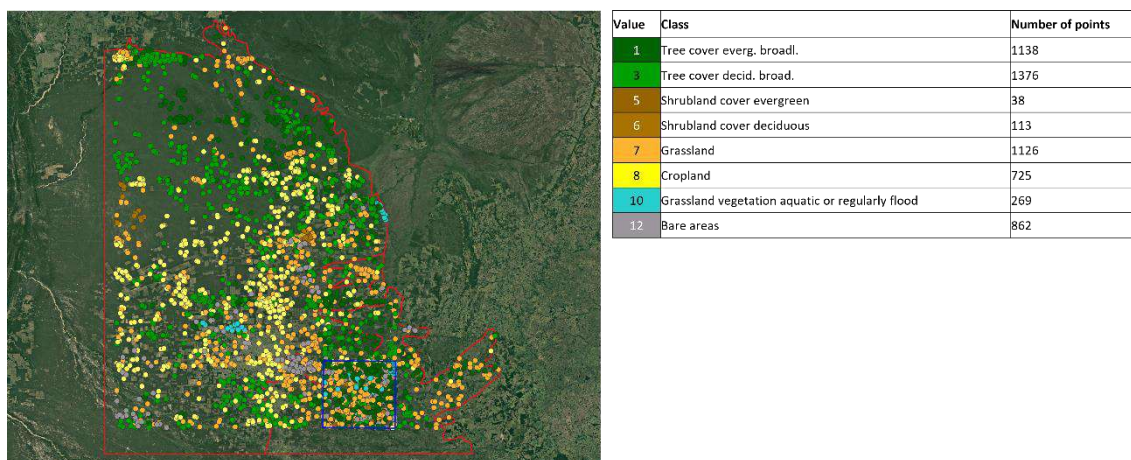


Figure 12. Spatial distribution and legend of the training points collected across the corresponding Ecoregions in the Amazon region.

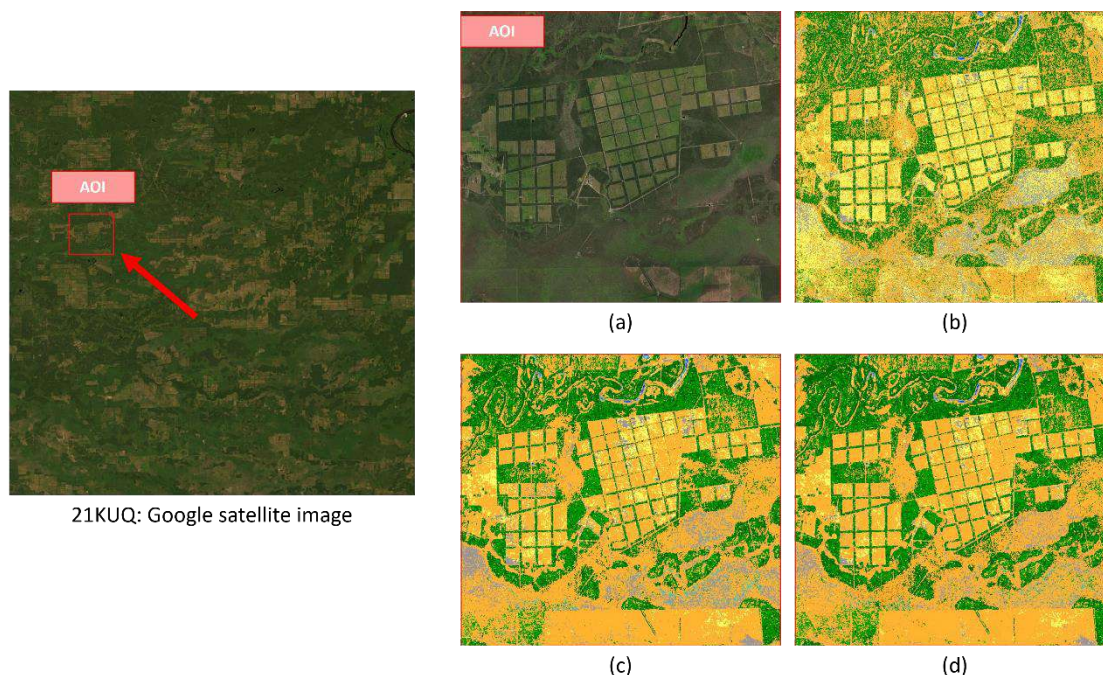


Figure 13. Visual comparison of the 21KUQ AOI tile between: (a) Google satellite image, (b) the RF map produced by the 'classical' approach ER-based, (c) the S1 RF map with an extended feature set and no slope correction ER-based, and (d) S1 RF map with an extended feature set and slope correction ER-based.

Furthermore, to mitigate residual misclassification errors and refine the output, a neighborhood reduction filter was applied as a post-processing step, with the resulting classification maps shown in Figure 14. This filter aims to smooth classification outputs by addressing isolated misclassified pixels, thereby enhancing spatial coherence and improving the overall map quality.

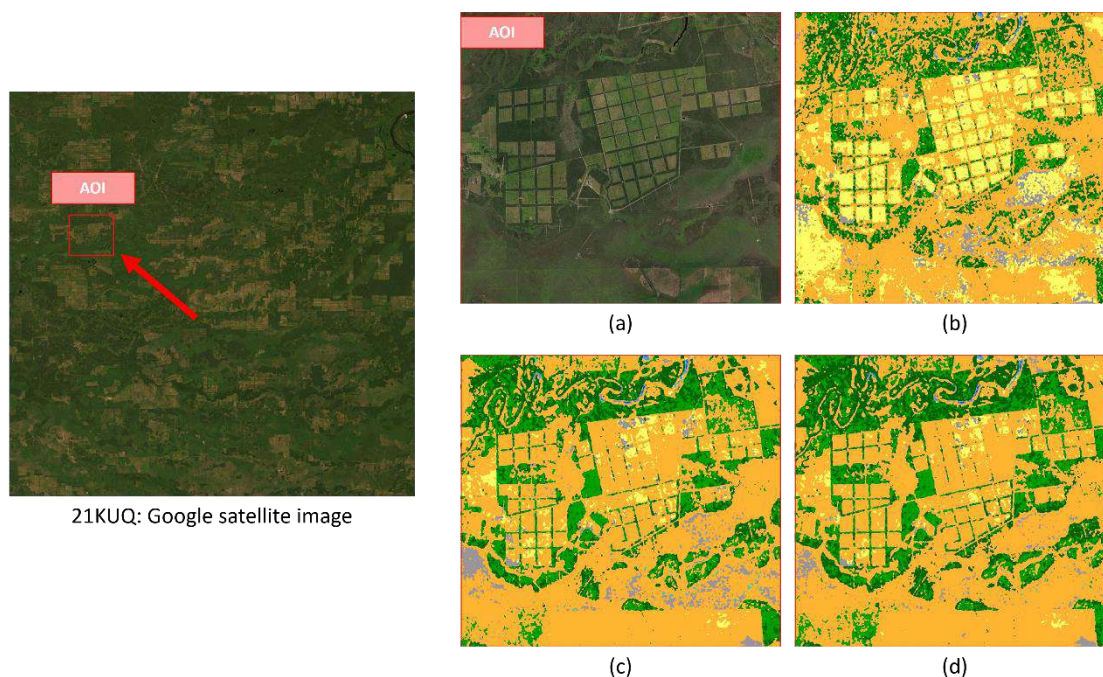


Figure 14. Visual comparison of the 21KUQ AOI tile between: (a) Google satellite image, (b) the RF map produced by the 'classical' approach ER-based, neighbourhood reduced (c) the S1 RF map with an extended feature set and no slope correction ER-based, neighbourhood reduced, and (d) S1 RF map with an extended feature set and slope correction ER-based, neighbourhood reduced.

This additional step is particularly effective in reducing noise in areas where class transitions are abrupt or where small misclassified patches might distort the final results. This technique significantly improved map homogeneity, reducing the 'salt and pepper' effect often observed in RF outputs. The filter proved particularly effective for the extended feature sets, further enhancing the usability of the classification results for practical applications.

Table 7 presents the results of classification performance for the different RF experiments, where training was based on Ecoregion (ER) data. The experiments were evaluated using the three validation sets used previously. The performance of the RF model is measured by PA for each class (PA₁, PA₂, PA₃, etc.) and OA.

Table 7. Comparison of Producer's Accuracy (PA) for individual classes and Overall Accuracy (OA) across three validation datasets using different Random Forest (RF) experimental approaches ER-based: Classical Approach, Features Expanded, and Features Expanded and Slope Corrected. The performances have been also compared with their corresponding neighborhood reduced filtering version.

Validation set	RF experiment	PA ₁ (%)	PA ₂ (%)	PA ₃ (%)	PA ₄ (%)	PA ₅ (%)	PA ₆ (%)	PA ₇ (%)	PA ₈ (%)	PA ₉ (%)	PA ₁₀ (%)	PA ₁₁ (%)	OA (%)
PoliMi/UNITN Validation Set	Classical approach ER-based	46	48	-	-	40	47	-	-	-	-	-	42,6
	Classical approach ER-based, neigh. reduced	52	55	-	-	68	39	-	-	-	-	-	51,1
	Features expandend ER-based	56	59	-	-	55	55	-	-	-	-	-	55,4
	Features expandend ER-based neigh. reduced	61	64	-	-	56	55	-	-	-	-	-	59
	Features expandend ER-based, Slope corrected	53	55	-	-	56	60	-	-	-	-	-	53
	Features expandend ER-based, Slope corrected, neigh. reduced	55	59	-	-	59	56	-	-	-	-	-	54,1
CGLC Map 100m 2019	Classical approach ER-based	36	26	5	-	18	33	12	3	-	-	91	24,6
	Classical approach ER-based, neigh. reduced	48	35	-	-	17	48	17	12	-	-	91	28
	Features expandend ER-based	40	26	-	-	21	34	15	6	-	-	91	25,8
	Features expandend ER-based neigh. reduced	47	29	-	-	20	36	15	11	-	-	91	26,6
	Features expandend ER-based, Slope corrected	37	27	-	-	27	35	15	7	-	-	92	25,9
	Features expandend ER-based, Slope corrected, neigh. reduced	47	29	-	-	20	36	15	11	-	-	91	26,6
CCI MRLC Map 300m 2019	Classical approach ER-based	40	22	-	-	23	33	28	-	-	-	91	27,9
	Classical approach ER-based, neigh. reduced	47	25	-	-	25	35	30	-	-	-	91	30,4
	Features expandend ER-based	46	29	-	-	27	43	36	-	-	-	91	31,9
	Features expandend ER-based neigh. reduced	51	32	-	-	26	54	36	-	-	-	91	32,9
	Features expandend ER-based, Slope corrected	44	28	-	-	27	36	36	-	-	-	91	31
	Features expandend ER-based, Slope corrected, neigh. reduced	50	33	-	-	27	50	36	-	-	-	91	33,2

*PA: Producer Accuracy; 1: Tree cover (evergreen broadleaf); 2: Tree cover (deciduous broadleaf); 3: Shrubland (evergreen); 4: Shrubland (deciduous); 5: Grassland; 6: Cropland; 7: Grassland vegetation aquatic or regularly flooded; 8: Bareland; 9: Built-up; 10: Open water (seasonal); 11: Open water (permanent).
Note: The symbol '-' indicates the absence of validation points for a specific class.

For the PoliMi/UNITN Validation Set, the highest OA (OA = 59%) is achieved by the "Features expanded ER-based neighbourhood reduced" experiment, indicating that adding feature expansion and applying the neighbourhood reduction filter leads to improved classification performance. The "Classical approach ER-based" experiment shows lower OA (OA = 42.6%), suggesting that using a classical approach without additional post-processing or feature expansion does not perform as well. "Features expanded ER-based" and "Slope corrected" methods provide relatively better results (OA = 55.4% and OA = 53%, respectively).

Using the CGLC set, the "Classical approach ER-based, neigh. reduced" experiment shows the best performance with an OA of 28%, followed by "Features expanded ER-based neigh. reduced" with an OA of 26.6%. Adding the neighbourhood reduction filter improves the classification, as seen in experiments like "Classical approach ER-based, neigh. reduced" (OA = 28%) and "Features expanded ER-based neigh. reduced" (OA = 26.6%). The "Features expanded ER-based" experiment without neighbourhood reduction or slope correction results in lower accuracy (OA = 25.8%).

With the CCI MRLC validation points, the best performance is observed in the "Features expanded ER-based neigh. Reduced" experiment, which achieves an OA of 32.9%, indicating that this combination of methods (feature expansion with the neighbourhood reduction filter) delivers the most accurate classification results. "Classical approach ER-based, neigh. reduced" (OA = 30.4%) and "Features expanded ER-based" (OA = 31.9%) also perform well, showing that adding neighbourhood reduction and expanding features consistently improves results. The lowest OA (OA = 27.9%) is obtained from the "Classical approach ER-based" experiment, underlining the importance of additional processing steps like feature expansion and neighbourhood reduction for improved accuracy.

The application of feature expansion and neighbourhood reduction significantly enhances classification accuracy, particularly when combined with Ecoregion-based training. The Slope corrected method provides some improvements in accuracy, but not as consistently as the neighbourhood reduction filter.

The Classical approach without feature expansion or post-processing methods yields the lowest accuracy, indicating that relying solely on basic RF techniques may not be sufficient for accurate classification across diverse land cover types. This analysis highlights the importance of data preprocessing and model refinement (e.g., feature expansion, slope correction, and neighbourhood reduction) in improving the performance of land cover

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classification using RF models.

Some classes consistently exhibit low PA values, which may indicate underlying issues related to the distribution of training samples or confusion between similar classes. These low PA values, coupled with equally low OA scores, suggest that the current RF approach may not be capturing the complexities of certain land cover types effectively. Such results point to the necessity of exploring alternative, more powerful methods for land cover classification. In particular, DL approaches, with their ability to model complex spatial relationships and learn hierarchical features directly from raw data, may offer significant improvements in performance. DL methods.

3.2.1.3 Deep Learning Models as an alternative to RF for LC Classification

The performance of the 'classical' RF classification, as used in Phase 1 and based on 28 features per season, was also evaluated in comparison with three advanced DL algorithms: Attention U-Net, Swin-Unet, and 3D Fully Convolutional Network (3D FCN).

For the training of the three DL algorithms the channel axis of the input tensors was utilised to encode the spatio-temporal information derived from the sequence of seasonal features extracted from the original SAR images. This approach allowed the networks to process the temporal dynamics effectively. The input tensors were structured as $B \times T \times W \times H$, where:

- B represents the batch size,
- T corresponds to the temporal dimension (seasonal features),
- W and H denote the width and height of the input images, respectively.

Training was conducted over 30 epochs, ensuring sufficient iterations for the models to converge. The remaining parameters were configured as described in reference [12]. Specifically:

- **Learning rate:** 10^{-4} , optimised for gradual and stable learning,
- **Batch size:** 1, chosen to efficiently handle the 28-feature temporal sequence while managing memory constraints.
- **Optimizer:** *Adam*, used for its effectiveness in handling sparse gradients and adaptive learning rates.
- **Loss function:** *Categorical Cross-Entropy*, applied to optimise for multiclass classification.

This configuration ensured the models could capture the intricate spatio-temporal relationships inherent in the seasonal SAR features, thereby enhancing their performance in classifying diverse land cover types.

The DL training set was extracted from the MOLCA dataset [13], which provides a comprehensive collection of labeled land cover data. This dataset includes various land cover classes with high spatial resolution, and it was specifically selected to ensure the availability of accurate training samples. The methodology for selecting and processing this training data is described in detail in the ATDB (Algorithm Technical Description Book) [AD6], where the steps for data extraction, sample refinement, and class assignment are outlined to ensure high-quality input for the DL model.

These algorithms were applied not only to the previously mentioned Amazonia tile, but to three static and geographically diverse areas representing the Amazon, Africa, and Siberia regions. These regions, selected during Phase 1 of the project for their distinct ecological and land cover characteristics, are illustrated in Figure 15.

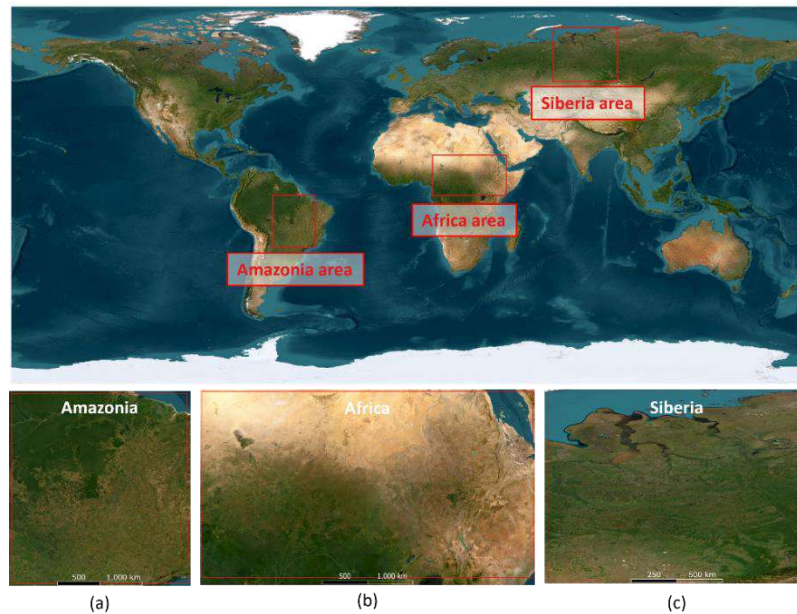


Figure 15. Views of the static areas identified in Phase 1: (a) Amazonia (62.1014° W, 23.5983° S to 42.9441° W, 0° N, WGS 84), (b) Africa (9.8986° E, 0.0885° S to 43.2908° E, 18.0891° N, WGS 84), and (c) Siberia (64.4361° E, 51.2789° N to 93.4017° E, 75.6847° N, WGS 84).

The performance of the classification models was rigorously compared across the three static areas., each represented by a subset of S2 tiles listed in Table 8 and shown in Figure 16.

Table 8. List of tiles selected to assess the DL classification performance for each static area: Amazonia (86 S2 tiles), Africa (103 S2 tiles), and Siberia (64 S2 tiles).

Amazonia	Africa	Siberia
20KQD	32NQF	41UPA
20KQV	32NRH	41VPE
20KRF	33NUA	41WPN
20LQH	33NUE	42UVD
20LQM	33NUG	42UWF
20LQR	33NVC	42VUP
20LRK	33NWJ	42VVH
20LRP	33NXA	42VVM
20MQD	33NXE	42VVR
20MQV	33NXG	42VWK
20MRB	33NYC	42VXP
20MRT	33PWN	42WVD
21KTS	33PWS	42WVV
21KVQ	33PXL	42WWB
21KVU	33PXQ	42WXT
21KWA	33PZN	42XWF
21KWS	33PZS	43UCU
21KYQ	33QXU	43UDA
21KYU	34NBF	43UFU
21LUC	34NBK	43VCC
21LUG	34NBP	43VCG
21LUL	34NCH	43VDE

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21LVE	34NCM	43VDL
21LVJ	34NEF	43VEG
21LXC	34NEK	43VFC
21LXG	34NEP	43VFJ
21LXL	34NFH	43WDQ
21LYE	34NFM	43WEU
21LYJ	34NHF	43WFN
21LZG	34NHK	43WFS
21LZL	34NHP	44ULF
21MUQ	34PCR	44UNF
21MUU	34PCV	44UPD
21MVN	34PEB	44VLK
21MVS	34PET	44VMJ
21MXQ	34PFR	44VMM
21MXU	34PFV	44VMR
21MYN	34PHB	44VPH
21MYS	34PHT	44VPP
21MZQ	34QCD	44WMV
22KBB	34QFD	44WNB
22KBF	35NLC	44WNE
22KCD	35NLG	44WPT
22KDV	35NNA	44XMF
22KEB	35NNE	45UUA
22KEF	35NNJ	45UWU
22KFD	35NPC	45UXA
22KGV	35NPG	45VUE
22KHF	35NRA	45VUG
22LBH	35NRE	45VVL
22LCK	35NRJ	45VWC
22LCP	35PLL	45VWG
22LDM	35PLQ	45VXE
22LEH	35PNN	45WVQ
22LER	35PNS	45WVU
22LFK	35PPL	45WWN
22LFP	35PPQ	45WWS
22LGM	35PRN	45WXQ
22MBD	35PRS	45WXU
22MCB	35QLU	45XVC
22MCT	35QPU	45XWA
22MED	36NUH	46VCP
22MEV	36NUM	46VCR
22MFB	36NVF	46XDH
22MFT	36NVK	
23KKS	36NVP	
23KLU	36NXH	
23KMA	36NXM	
23KMQ	36NYF	
23KNS	36NYK	
23LKC	36NYP	

23LKE	36PUR
23LKJ	36PUV
23LKL	36PWB
23LLG	36PWT
23LMJ	36PXR
23LNC	36PXV
23LNE	36PZB
23LNL	36PZT
23MKN	36QUD
23MKQ	36QXD
23MKS	37NBC
23MKU	37NBG
23MMN	37NCA
23MNQ	37NDE
23MNS	37NDJ
	37NEC
	37NEG
	37NFA
	37NGE
	37NGJ
	37PCL
	37PCQ
	37PDN
	37PDS
	37PFL
	37PFQ
	37PGN
	37PGS
	37QCU
	37QFU
	38NKF
	38NKM

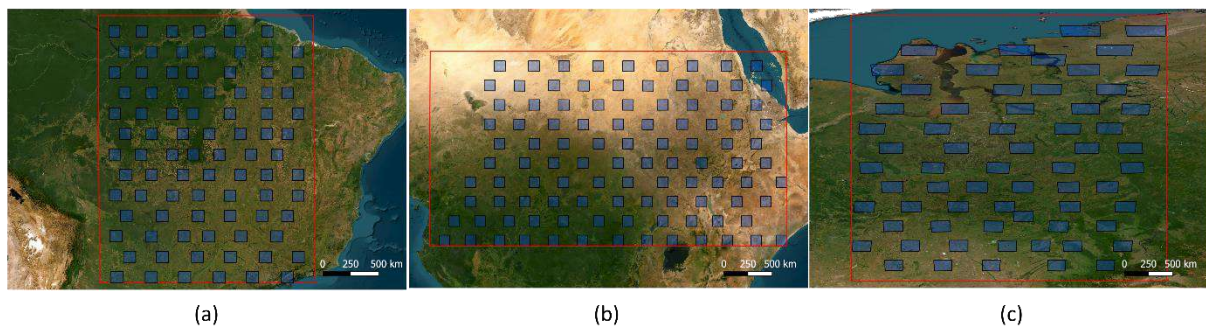


Figure 16. Visual representation of the spatial distribution of the tiles selected for (a) Amazonia, (b) Africa, and (c) Siberia, to assess the classification performance of the DL networks.

For the comparison, the S1 Level-1 GRD dataset from 2021 was utilised. The dataset was segmented into four subsequences corresponding to the seasons (winter, spring, summer, and autumn). This seasonal division provided a comprehensive temporal perspective for analysing land cover dynamics across diverse regions. The resulting dataset included:

- **Amazonia:** for 86 S2 tiles, a total of 5105 SAR images, distributed as 1264 for winter, 1310 for spring, 1338 for summer, and 1193 for autumn.
- **Africa:** A collection of 5827 SAR images for 103 S2 tiles, comprising 1470 for winter, 1480 for spring, 1654 for summer, and 1405 for autumn.
- **Siberia:** For 64 S2 tiles, a dataset of 3396 SAR images, with 768 for winter, 744 for spring, 984 for summer, and 900 for autumn.

All images were acquired in IW mode with VH polarisation and descending orbit configuration.

Each algorithm was evaluated in terms of its ability to accurately classify land cover types, with particular attention to how well each model handled the unique characteristics of the SAR data in these diverse regions. This evaluation aimed at determining the most effective DL model for the classification in the Phase 2 of the project, ensuring robust and reliable land cover maps across the selected global regions.

3.2.1.4 Comparative Performance Analysis of DL Models and RF for LC Classification

Table 9 displays the performance of different three DL models for the three regions: Amazonia, Africa, and Siberia. As a general statement, Swin-Unet consistently outperforms the benchmark CNN-based models (3D-FCN and Attention U-Net) and the RF model in terms of OA, kappa, and F1-score across all three regions. While RF remains robust in simpler scenarios, DL models outperform it by leveraging their ability to learn hierarchical and contextual features [14].

Table 9. Overall Accuracy (OA), Kappa Coefficient, F1-Score, and Producer Accuracy (PA) for the evaluated models across different regions.

Region	Model	O.A.	kappa	F1-Score	pa ₁	pa ₂	pa ₃	pa ₄	pa ₅	pa ₆	pa ₇	pa ₈
Amazonia	Random Forest	0.622	0.328	0.565	0.51	0.09	0.25	0.07	0.18	0	0.02	0.18
	3D-FCN	0.818	0.709	0.795	0.89	0.08	0.49	0.74	0.24	0	0.06	0.75
	Attention U-Net	0.843	0.760	0.832	0.91	0	0.61	0.6	0	0	0	0.96
	Swin-Unet	0.933	0.898	0.924	0.97	0.5	0.88	0.85	0.03	0	0.85	0.99
Africa	Random Forest	0.745	0.544	0.712	0.64	0.11	0.13	0.32	0.07	0.69	0	0.64
	3D-FCN	0.735	0.563	0.723	0.46	0.27	0.44	0.53	0.11	0.36	0.2	0.77
	Attention U-Net	0.738	0.596	0.751	0.66	0	0.55	0.18	0	0.02	0	0.89
	Swin-Unet	0.936	0.900	0.932	0.97	0.52	0.83	0.73	0.16	0.93	0.09	0.77
Siberia	Random Forest	0.677	0.362	0.634	0.31	0	0.34	0.18	0.06	0	0.18	0.42
	3D-FCN	0.854	0.760	0.842	0.87	0	0.58	0.75	0.21	0.16	0.01	0.87
	Attention U-Net	0.903	0.847	0.885	0.97	0	0.42	0.61	0.56	0.06	0.01	0.99
	Swin-Unet	0.974	0.959	0.974	0.99	0	0.86	0.95	0.79	0.75	0.92	0.98

*pa: Producer Accuracy; 1: Forest; 2: Shrubland; 3: Grassland; 4: Cropland; 5: Wetland; 6: Bareland; 7: Built-up; 8: Water.

A visual comparison of the results from the Swin-Unet model, including random S1 super images from the validation dataset, MOLCA reference data (Ground Truth), and the corresponding predictions, is illustrated in the Figure 17, Figure 18 and Figure 19, for Amazonia, Africa and Siberia, respectively.

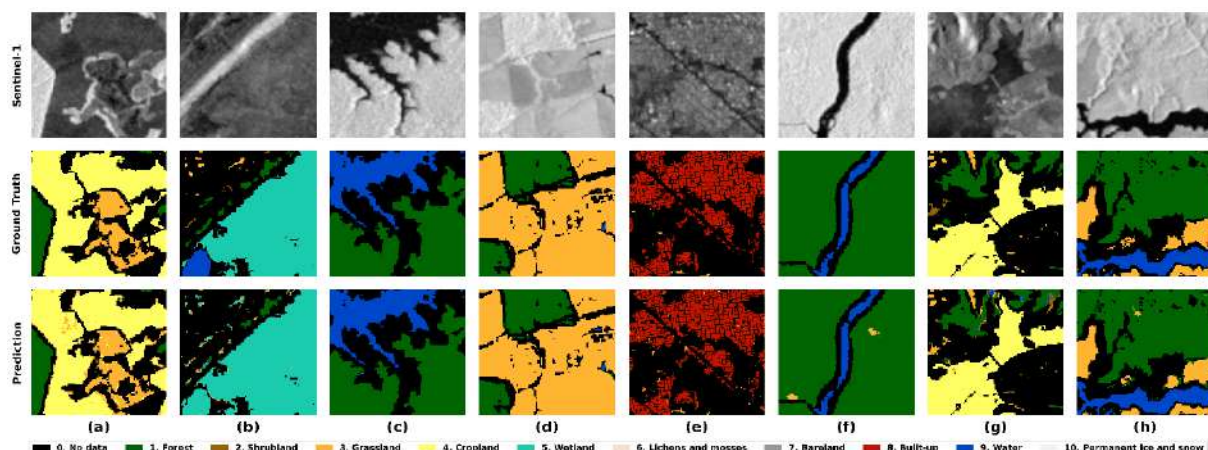


Figure 17. Visual comparison for the Amazonia region, showing Sentinel-1 super image data (top row), Ground Truth

(middle row), and predicted patches (bottom row).

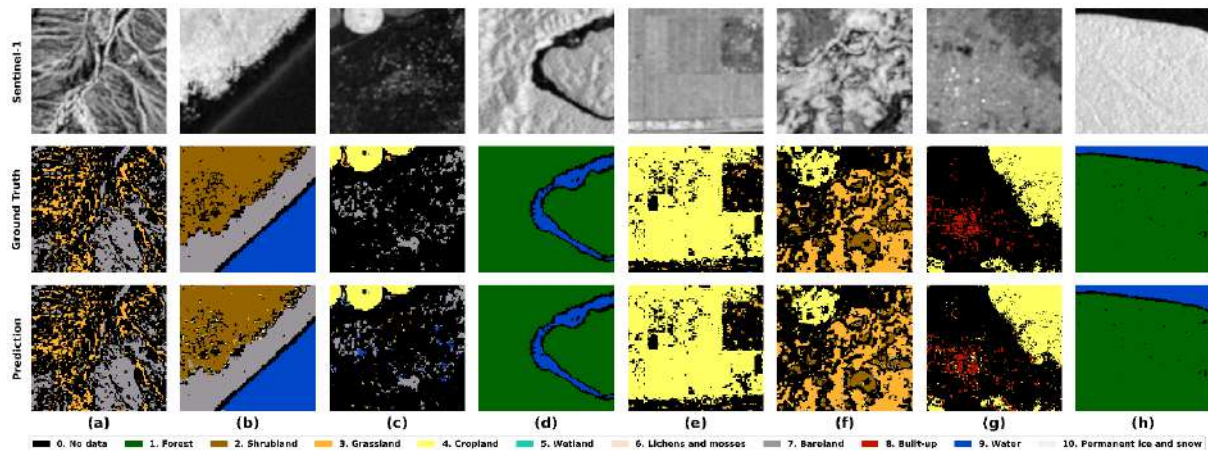


Figure 18. Visual comparison for the Africa region, showing Sentinel-1 super image data (top row), Ground Truth (middle row), and predicted patches (bottom row).

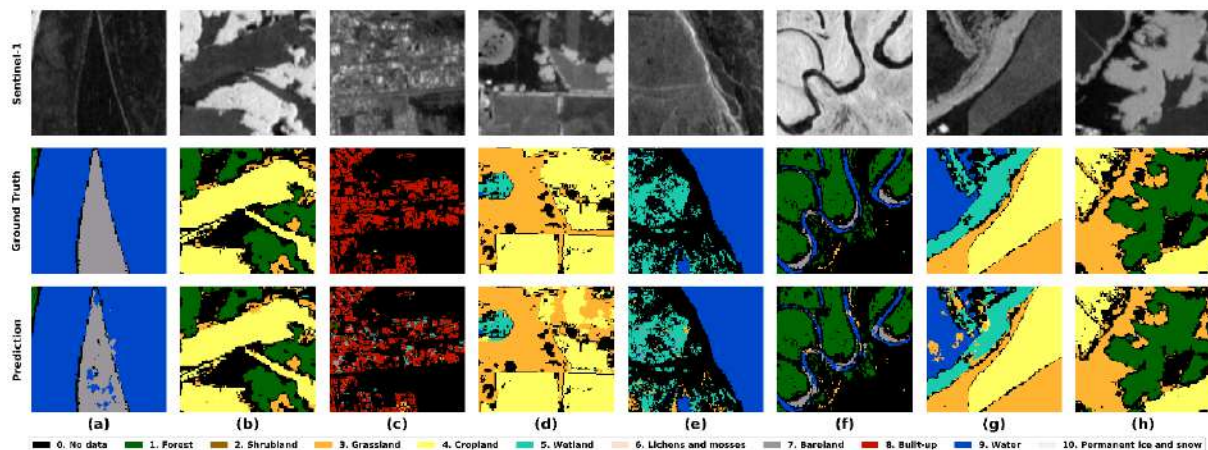


Figure 19. Visual comparison for the Siberia region, showing Sentinel-1 super image data (top row), Ground Truth (middle row), and predicted patches (bottom row).

In these regions, the Swin-Unet model achieves OAs of 93.3%, 93.6%, and 97.4%, respectively. The corresponding confusion matrices are presented in the Figure 20, with each diagonal element indicating the PA for the respective class. Notably, the results table indicates that the Forest, Grassland, Cropland, and Water classes are effectively extracted using the leading model across all three study areas.

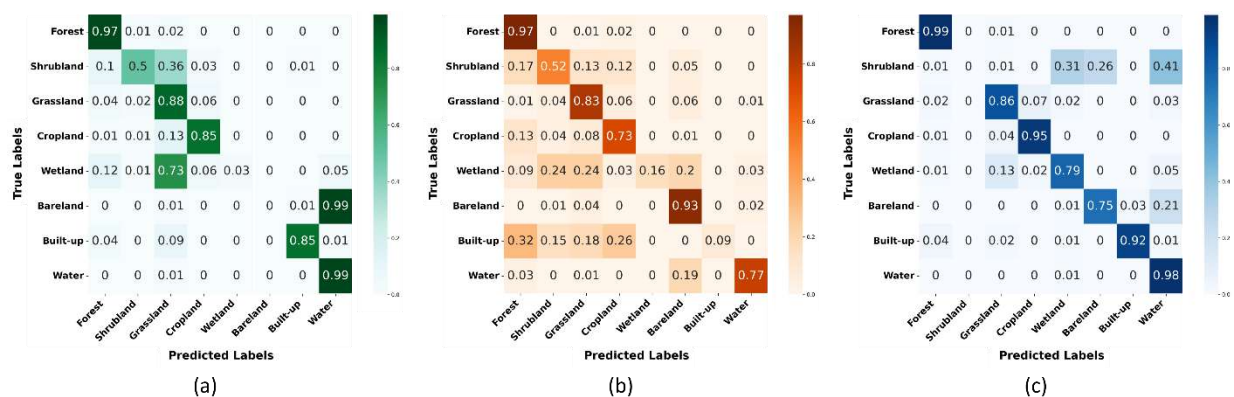


Figure 20. Normalized confusion matrices derived from the validation set for the regions: (a) Amazonia, (b) Africa and (c) Siberia.

The evaluation of the models on the African dataset indicates that CNN-based approaches may have difficulty differentiating Bareland from classes such as Water or Grassland within SAR imagery. In Africa, the 3D-FCN model scores 0.77 for Water, but only 0.36 for Bareland, 0.53 for Cropland, and 0.44 for Grassland. The Attention U-Net framework, on the other hand, achieves a PA of 0.89 for Water, 0.66 for Forest, and 0.55 for Grassland, but only 0.02 for Bareland. In contrast, the Swin-Unet achieves a final PA of 0.97 for Forest, 0.83 for Grassland, and 0.93 for Bareland, surpassing the other two models.

The improvement in recognizing Bareland with the Swin-Unet model is evident not only in Africa but also in the Siberian dataset. Despite representing only 1.04% of the available tiles, Bareland achieves a PA of 0.75 with this model, marking a substantial enhancement from the initial values of 0.16 and 0.06 recorded with the 3D-FCN and Attention U-Net, respectively. These outcomes suggest that the transformer model is capable of recognizing various LC classes and tends to make more balanced classifications compared to the other two CNN-based models, which generally show high PA only for specific classes like Water. This difference may stem from the Attention U-Net model's limited ability to generalize, which results in strong classification performance primarily for classes with easily identifiable spatial patterns or unique brightness values, such as Water or Forest. However, it struggles with more complex morphological relationships of specific LC classes like Bareland. Consequently, the Attention U-Net, which depends on locality-based attention mechanisms, may find it challenging to classify classes that share similar pixel value distributions. Regarding the 3D-FCN model, its three-dimensional logic-based structure does not provide additional benefits when the input data lacks a dense temporal sequence and instead consists of seasonal synthetic images, referred to as features. In this context, a 2D-CNN model, like the Attention U-Net, appears to be sufficient. In contrast, transformer-based models like Swin-Unet consider global pixel relationships, leading to a more accurate assessment of context and environment. The global attention mechanisms in these models can help recover this context, which CNNs cannot achieve due to the local nature of the convolutional kernel. Furthermore, the Built-up class showed significant improvement through the Swin-Unet model in Amazonia and Siberia with final values of 0.85 and 0.92. For the African region, the value for this specific class remains poor, likely due to the lower number of representative labels for the this class in this area (only 0.5%). Another contributing factor could be the small and fragmented nature of urban areas, particularly when compared to more spatially uniform LC classes like Forest, Grassland, or Cropland. In favor of these latter classes, the highest confusion is observed when predicting Built-up, indicating a significant challenge in visually distinguishing small urban areas from surrounding classes due to their reduced size and scattered appearance. For Wetland, a high PA of 0.79 is obtained in the Siberian region, while for the other two areas, the accuracy remains

For both Amazonia and Siberia, RF achieves lower OA than all three DL approaches. In Africa, RF reaches an OA of 0.745, slightly exceeding the OA of 3D-FCN and Attention U-Net. However, its Kappa coefficient (0.544) and F1-Score (0.712) remain lower than those of these DL models. The Swin-Unet achieves the highest metrics overall, with an OA of 0.936, a Kappa coefficient of 0.900, and an F1-Score of 0.932.

Finally, it is important to note that both the DL models and RF, show moderate performance in classifying the *Forest* and *Water* classes. RF records producer accuracies (PA) of 0.51 and 0.18 for these classes in Amazonia, 0.64 for both in Africa, and 0.31 and 0.41 in Siberia, respectively. However, RF struggles significantly with other classes. In Amazonia, it performs poorly for *Shrubland* (PA = 0.09), *Cropland* (PA = 0.07), *Bareland* (PA = 0), and *Built-up* (PA = 0.02). Similar issues arise in Africa for *Shrubland* (PA = 0.11), *Wetland* (PA = 0.07), and *Built-up* (PA = 0), as well as in Siberia for *Shrubland* (PA = 0), *Wetland* (PA = 0.06), and *Bareland* (PA = 0). These low PAs suggest that RF struggles with complex and highly variable LC classes.

3.2.1.4.1 DL Swin-Unet Performance Assessment with Ecoregion Classification

Additional experiments were conducted for the best DL model using ground truth data organized by ecoregions. Figure 21 shows the MOLCA training patches for the Amazon, Africa, and Siberia, divided by climatological zones. The legend includes a color map and numerical codes for each region.

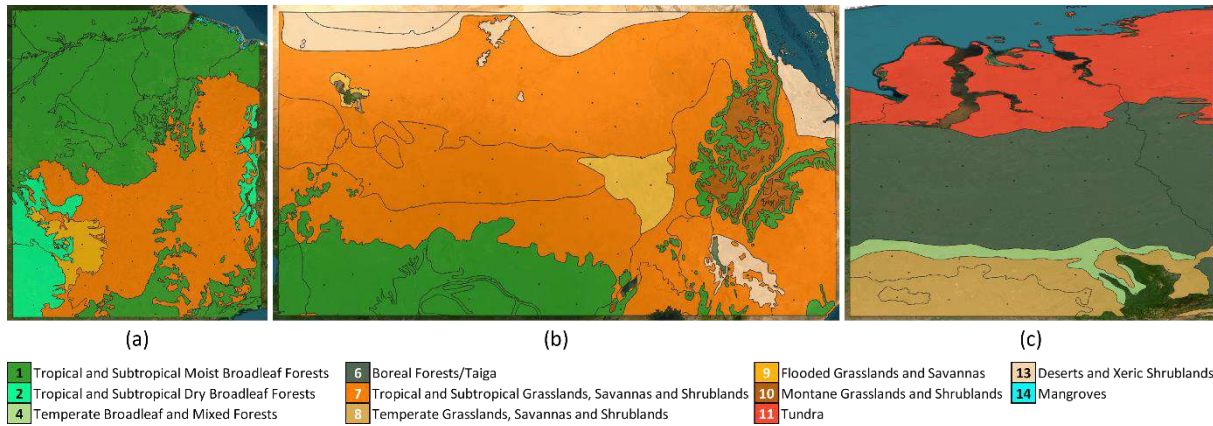


Figure 21. Division of the MOLCA patches in the training set based on ecoregions for (a) Amazonia, (b) Africa, and (c) Siberia.

The experiments were designed to incorporate the ecoregions within the three study areas. This additional analysis aims to evaluate the performance of the model when training and testing are done within the same ecoregion compared to when they are conducted across different ecoregions, with the goal of providing a deeper assessment. The results obtained using the Swin-Unet model are displayed in Figure 22, and are presented in terms of OA. Diagonal elements indicate cases where training and testing are done within the same ecoregion, while off-diagonal elements represent cross-validation across different ecoregions.

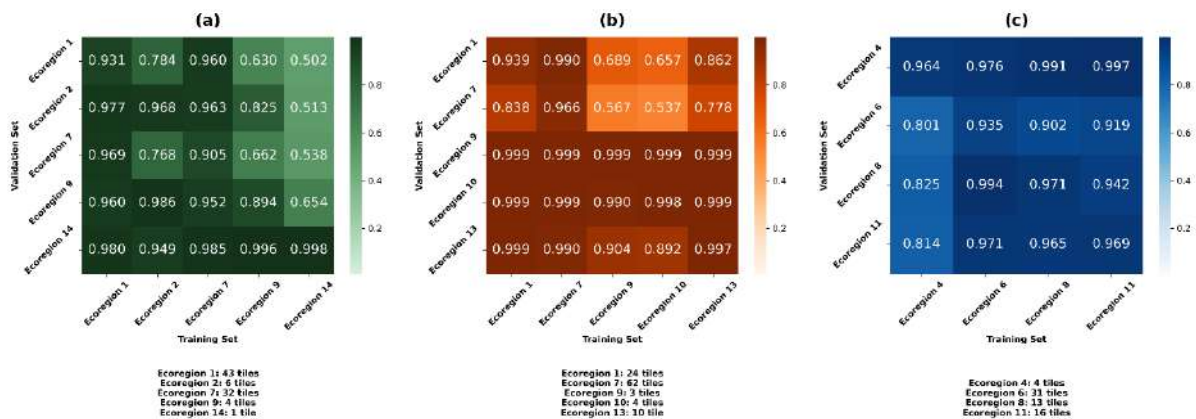


Figure 22. Overall Accuracy (OA) matrices derived from various training/test ecoregion pairings for (a) Amazonia, (b) Africa, and (c) Siberia.

In the Amazonia region in Figure 22(a), the OA is generally higher when both training and testing occur within the same ecoregion. However, a notable exception arises when Ecoregion 14 and Ecoregion 9 are used for training and tested on other ecoregions. The poorer performance in these cases is likely due to the limited number of tiles in Ecoregion 14 (1 tile) and Ecoregion 9 (4 tiles), restricting the diversity and quantity of training data. A similar trend is observed in the African region (Figure 22(b)), where performance is lower when training on Ecoregion 9 and Ecoregion 10, which contain only 3 and 4 tiles, respectively. This small dataset may be insufficient for the model to generalize well across more diverse ecoregions. In the Siberian region (Figure 22(c)), OA values remain high, surpassing 80% across all combinations. However, the lowest performance is noted when training on Ecoregion 4 and testing on other ecoregions, even though it contains just 4 tiles. Despite this, the model's accuracy remains relatively stable, demonstrating the strong generalization capability of the model, even when trained with limited data from other ecoregions.

3.2.1.4.2 Comparison with Conventional SAR Time Series Analysis

The following analysis presents a comparative evaluation classifications obtained in the static areas identified in Phase 1 for Amazonia, Africa, and Siberia either with the full time-series data or just the seasonal. The objective is to show that the approach based on seasonal SAR features surpasses the conventional method of using multiple temporal SAR images for each season.

For this experiment, twenty temporal images per tile were collected for 2021, i.e., five images per season. This

process generated datasets containing 1620 S1 images for Amazonia, 1860 for Africa, and 960 for Siberia. Strict selection criteria ensured that only tiles with complete spatial and temporal coverage were included. As a result, the final dataset was refined to 81 MOLCA tiles for Amazonia (reduced from 86), 93 for Africa (down from 103), and 48 for Siberia (from 64), preserving the data quality for all analyses. The seasonal distribution of the collected S1 time series is presented in Table 10.

Table 10. Seasonal count of S1 images from the 2021 time series for Amazonia, Africa and Siberia static area.

Region	Platform	Acquisition mode	Product format	Polarization	Resolution	Season*	Number of images
Amazonia	S1A/S1B	IW	GRDH	VH	10 m	Winter	405
	S1A/S1B	IW	GRDH	VH	10 m	Spring	405
	S1A/S1B	IW	GRDH	VH	10 m	Summer	405
	S1A/S1B	IW	GRDH	VH	10 m	Autumn	405
Africa	S1A/S1B	IW	GRDH	VH	10 m	Winter	465
	S1A/S1B	IW	GRDH	VH	10 m	Spring	465
	S1A/S1B	IW	GRDH	VH	10 m	Summer	465
	S1A/S1B	IW	GRDH	VH	10 m	Autumn	465
Siberia	S1B	IW	GRDH	VH	10 m	Winter	240
	S1B	IW	GRDH	VH	10 m	Spring	240
	S1B	IW	GRDH	VH	10 m	Summer	240
	S1B	IW	GRDH	VH	10 m	Autumn	240

*Winter: 01.01 to 03.31, Spring: 04.01 to 06.30, Summer: 07.01 to 09.30, Autumn: 10.01 to 12.31

The comparative study employed the Swin-Unet DL model, which demonstrated superior performance among the three analyzed DL approaches, alongside the RF classifier, ensuring a fair comparison by using identical training and validation sets. The results, reported in Table 9 and Table 11, reveal a significant reduction in accuracy when standard time-series images are used compared to the seasonal features series. These findings highlight the efficiency and reliability of the synthesized spatial information derived from the features, establishing it as a superior approach for multitemporal data analysis.

Table 11. Results for standard input time series for Amazonia, Africa and Siberia.

Region	Model	O.A.	kappa	F1-Score	pa ₁	pa ₂	pa ₃	pa ₄	pa ₅	pa ₆	pa ₇	pa ₈
Amazonia	Random Forest	0.612	0.309	0.553	0.49	0.09	0.21	0.08	0.18	0	0.02	0.16
	Swin-Unet	0.903	0.855	0.901	0.92	0.75	0.68	0.67	0.14	0	0.75	0.94
Africa	Random Forest	0.743	0.535	0.707	0.61	0.09	0.13	0.17	0.07	0.70	0	0.44
	Swin-Unet	0.872	0.795	0.854	0.95	0.03	0.68	0.16	0	0.89	0.27	0.77
Siberia	Random Forest	0.670	0.315	0.619	0.25	0.00	0.34	0.19	0.08	0	0.19	0.30
	Swin-Unet	0.896	0.817	0.893	0.93	0	0.44	0.73	0.65	0.57	0.28	0.79

*pa: Producer Accuracy; 1: Forest; 2: Shrubland; 3: Grassland; 4: Cropland; 5: Wetland; 6: Bareland; 7: Built-up; 8: Water.

Specifically, two significant drawbacks of the traditional time-series approach emerged:

- **Increased computational demand:** Handling numerous temporal images requires extensive storage and processing resources.
- **Data limitations:** Some regions lack sufficient temporal data consistency, hampering detailed studies.

The features series approach provides a significant benefit by condensing temporal dynamics into a single composite image, known as the super image. This image captures seasonal variations, offering a consistent representation while eliminating the need for extra spatial constraints. This method simplifies the workflow, enhancing coherence and reducing complexity.

3.2.1.5 Comparative Analysis for Water LC Class Recognition between the Swin-Unet DL Approach and SAR Water Detector

To assess the performance of water recognition by the dedicated detector selected in Phase 1 in comparison with the DL Swin-UNet approach, 10 benchmark areas were selected within the three static regions of interest. Each of these ten test areas were chosen to represent water bodies that are most commonly found in these regions, including rivers, streams, basins, lakes, seas, and other aquatic features. The tiles selected as

benchmarks for each area are listed in Table 12, and are shown in Figure 23, to offer a clear geographical context for each benchmark location.

Table 12. List of benchmark tiles selected for Amazonia, Africa, and Siberia to assess the performance of the Phase 1 water extractor against the DL Swin-Unet approach.

Amazonia	Africa	Siberia
21LXD	33PUK	41UPU
22KCA	33PVQ	42VVP
22KFB	33PYQ	42WWU
22LGK	34NBG	43WES
22LGP	35NKD	44UNF
22MFA	36NVG	44WNE
22MGD	36NZJ	45VWL
22MHT	36PVB	45WWR
23KMV	37NCG	45XVB
23MNT	37NDJ	46VEK

The selection ensures a diverse and comprehensive evaluation of the detection capabilities in varying geographical and environmental contexts. Each region presents unique challenges related to the scale and type of water bodies, enabling a detailed performance assessment.

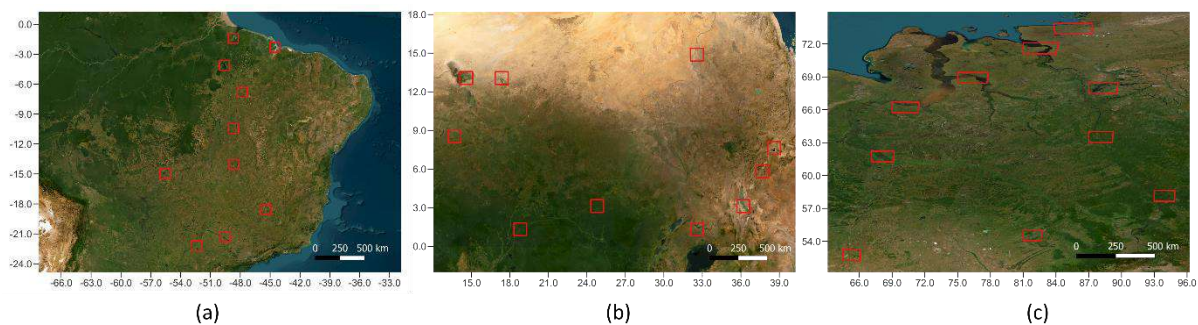


Figure 23. S2 tiles selected for evaluating water detection performance in (a) the Amazon, (b) Africa, and (c) Siberia. These tiles highlight regions with diverse hydrological features, providing a robust basis for performance comparison.

For each tile in each area, a representative patch of 549x549 pixels was selected through visual inspection of the optical imagery to identify potential water basins of interest. This approach allowed for a targeted evaluation by focusing on specific, clearly defined regions within the tiles, enabling a detailed understanding of the detector's robustness in real-world conditions, particularly in regions with diverse hydrological features.

For water extraction using the dedicated detector, the annual series of S1 data from 2021 was considered for each tile. The series was subsequently divided into monthly temporal subsets, for which the following temporal features were calculated: mean, minimum, maximum, and variance. These monthly feature sets served as inputs to the water extractor, which generates binary monthly water maps (where pixel values are 0 for non-water and 1 for water).

The water extraction is an unsupervised routine introduced in [15], utilizing a K-means clustering approach. This procedure differentiates areas with low backscatter and low variance along the temporal series (characteristic of water bodies) from other potential regions of interest. The final water map is derived from these monthly water maps, following the methodology described in the deliverable ATDB [AD1], particularly in the section titled "8.2.2.2 Water Dynamics Analysis: Seasonal vs. Permanent Water Identification."

The monthly aggregation of statistical measures such as mean and variance helps capture the temporal dynamics of water bodies, which may fluctuate due to seasonal changes or hydrological events. The reliance on K-means clustering makes this method adaptable to diverse regions without requiring labeled data, enhancing its applicability for large-scale studies. Concerning the final water map, by synthesizing monthly maps, the process distinguishes between seasonal and permanent water, providing a comprehensive view of water dynamics over the year.

To refine the final water map, a masking operation was applied to address desert areas or regions with strong

similarities to water due to their dominant backscattering mechanisms. Specifically, water areas identified by the detector that belong to desert or bare soil classes, according to coarser resolution land cover maps, such as the Copernicus Land Cover Map at 100 m spatial resolution, were excluded. This refinement step enhances the accuracy of the water map by reducing false positives, particularly in arid or semi-arid regions where surface characteristics can mimic water signatures in satellite imagery.

The classification of water bodies using the Swin-Unet approach is instead performed by utilizing the models trained in the previous experiments. Seasonal spatial feature sets were computed and used as input for the Swin-Unet DL framework. These features capture the temporal variability and spatial characteristics of water bodies, enabling the model to account for seasonal changes and better distinguish water from other land cover types.

To compare the water recognition performance of the two methodologies, the permanent and seasonal water classes from the water map generated by the dedicated extractor were merged into a single "water" class. This aggregation resulted in a binary map where pixel values of 0 represent "non-water" areas, and pixel values of 1 represent "water" areas. This simplification was necessary because the MOLCA dataset, used for training the DL approach, does not include seasonal information in its LC classes. Concerning the DL maps, regions encompassing various land surface types, such as forests, urban areas, agricultural fields, deserts, and other terrains that do not exhibit hydrological characteristics, such as low backscatter or high reflectivity, were grouped into the non-water land cover category. These areas are uniformly assigned a value of 0, distinctly separating them from water classes. This aggregation simplifies the classification process by consolidating diverse non-water types into a singular, cohesive category. Such an approach streamlines data analysis, enhancing the ability to compare and evaluate water-detection results against these non-water regions.

The classification results and the visual comparison are presented in Figure 24, Figure 25 and Figure 26, corresponding to the Amazonia, Africa, and Siberia, respectively. For each region, four representative S2 tiles were selected as examples. The water maps derived from the dedicated extractor and the DL Swin-Unet approach were evaluated by comparing them against two reference datasets: the ESRI optical basemap, which provides high-resolution visual reference imagery, and the super image calculated from the seasonal S1 temporal series, which aggregates backscatter data to highlight hydrological patterns over time.

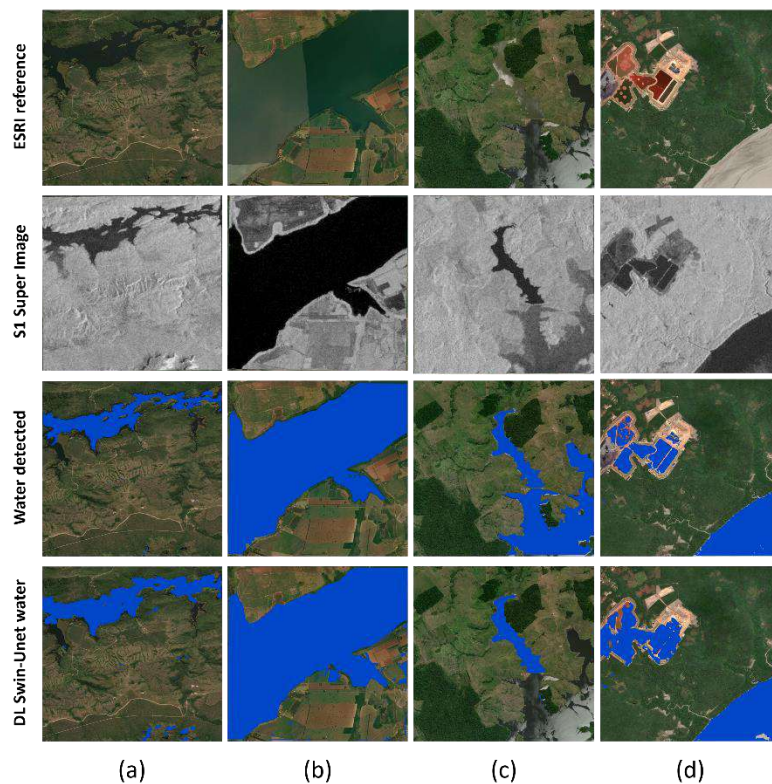


Figure 24. Visual assessment for the Amazonia region comparing (first row) the ESRI reference image, (second row) the S1 seasonal super image, (third row) the water map derived from the dedicated extractor, and (fourth row) the water land cover classes from the DL Swin-Unet classification. The comparison is presented for patches of 549x549 pixel size, corresponding to the following S2 tiles: (a) 21LXD, (b) 22KCA, (c) 22MFA, and (d) 23MNT.

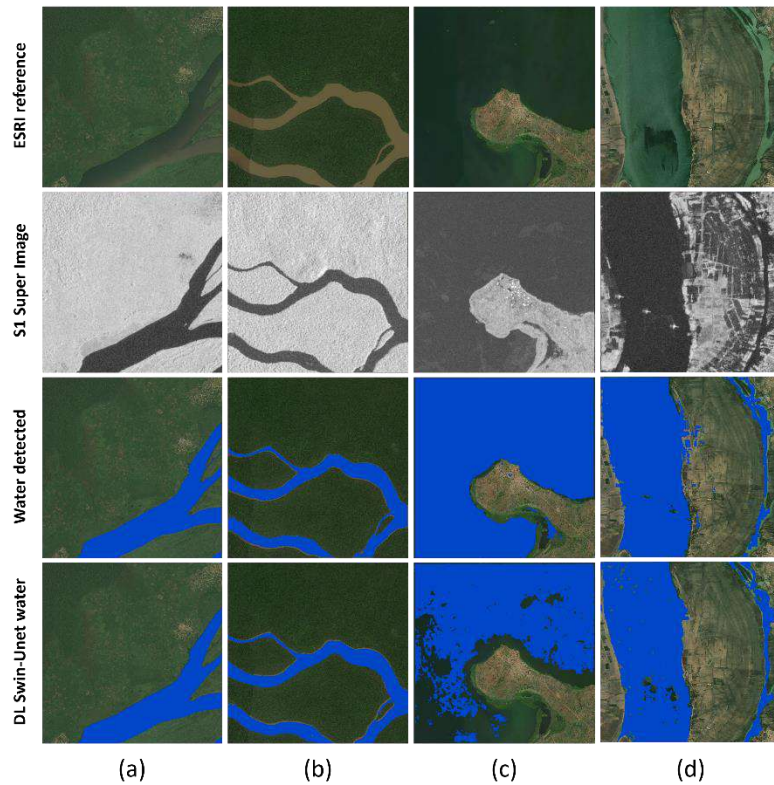


Figure 25. Visual assessment for the Africa region comparing (first row) the ESRI reference image, (second row) the S1 seasonal super image, (third row) the water map derived from the dedicated extractor, and (fourth row) the water land cover classes from the DL Swin-Unet classification. The comparison is presented for patches of 549x549 pixel size, corresponding to the following S2 tiles: (a) 34NBG, (b) 35NKD, (c) 36NVG, and (d) 36PVB.

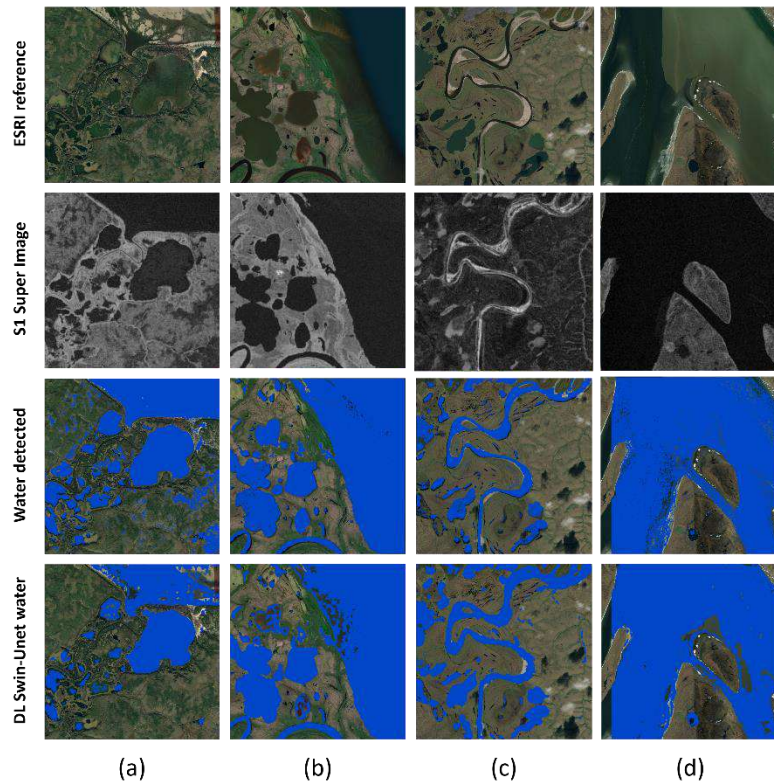


Figure 26. Visual assessment for the Siberia region comparing (first row) the ESRI reference image, (second row) the S1 seasonal super image, (third row) the water map derived from the dedicated extractor, and (fourth row) the water land cover classes from the DL Swin-Unet classification. The comparison is presented for patches of 549x549 pixel size, corresponding to the following S2 tiles: (a) 42WWU, (b) 43WES, (c) 44WNE, and (d) 45XVB.

This multi-faceted comparison highlights the differences in water detection performance, accounting for the strengths and limitations of the two methodologies. In the Amazonia region, for example, examining Figure 24 and column (a), it is evident that the DL approach can identify water bodies not detected by the dedicated extractor. Conversely, column (c) demonstrates a limitation of the Swin-Unet network: half of a lake within the examined patch is not classified as water, whereas the dedicated extractor's output map successfully identifies the entire lake. Similar observations can be made for the Africa and Siberia regions, as shown in the respective figures (Figure 25 and Figure 26). In these cases, the DL approach and the dedicated extractor exhibit complementary strengths, with the former excelling in identifying smaller or less prominent water features, and the latter showing greater reliability in capturing larger, more stable water bodies.

To complement the evaluation, a thorough quantitative analysis is introduced, providing a numerical framework to validate and expand upon the insights gained from the earlier qualitative assessment. Several metrics are considered, including OA Precision, Recall, F1-Score, and Producer Accuracy (pa) for both non-water (pa_0) and water classes (pa_1). These metrics provide complementary insights into different aspects of model performance:

- **OA** is used to assess the proportion of correctly classified pixels across the entire dataset, giving a general view of model effectiveness.
- **Precision** measures the accuracy of water detection, indicating the proportion of true water pixels among all those classified as water. This helps to understand how well false positives are minimized.
- **Recall** evaluates the ability of the model to identify all water pixels, reflecting how many true water features are correctly detected.
- **F1-Score** offers a balanced measure of Precision and Recall, particularly useful when there is an imbalance between water and non-water classes in the dataset.

These metrics were computed for each test area (Amazonia, Africa, Siberia) to allow both regional and aggregated comparisons between the water detection methodologies. The analysis, reported in Table 13, highlights not only the overall reliability of each method but also their strengths and weaknesses in capturing diverse hydrological patterns under varying environmental conditions. The ESA WorldCover map, with its 10-meter spatial resolution, was utilized as the validation dataset to assess the performance of the water detection approaches. This dataset provides a globally consistent, high-resolution land cover classification, enabling a reliable comparison between detected water bodies and the ground truth information.

Table 13. Overall Accuracy (OA), Precision, Recall, F1-Score, and Producer Accuracy (pa) for the evaluated approaches across different regions.

Region	Model	OA	Precision	Recall	F1-Score	pa_0	pa_1
Amazonia	Water extractor	0.96	0.97	0.96	0.96	98.06	90.40
	Swin-Unet DL	0.97	0.97	0.97	0.97	96.56	91.81
Africa	Water extractor	0.94	0.95	0.94	0.94	94.62	89.35
	Swin-Unet DL	0.80	0.90	0.80	0.80	99.47	63.65
Siberia	Water extractor	0.85	0.91	0.85	0.87	87.77	66.86
	Swin-Unet DL	0.92	0.95	0.92	0.93	93.31	90.51

* pa : Producer Accuracy; 0: Non-Water; 1: Water.

In Table 13, the performance in Amazonia shows that both the water extractor and the Swin-Unet DL model exhibit high accuracy. The OA values are 0.96% for the water extractor and 0.97% for Swin-Unet DL, indicating strong performance for both methods. The Precision, Recall, and F1-Score metrics are also very close, ranging from 0.96% to 0.97% for both models. For Producer Accuracy (pa), the performance for non-water areas (pa_0) is higher with the water extractor at 98.06%, compared to 96.56% for Swin-Unet DL. However, for the water class (pa_1), Swin-Unet DL outperforms the water extractor, achieving 91.81%, compared to 90.40% for the latter. This suggests that while both models perform similarly in most aspects, the water extractor has a slight edge in correctly identifying non-water areas, while Swin-Unet DL slightly excels in detecting water bodies in Amazonia. In Africa, as seen in Table 8, the water extractor outperforms Swin-Unet DL in several key metrics. The OA is significantly higher for the water extractor (0.94%) compared to Swin-Unet DL (0.80%). Similarly, the F1-Score of the water extractor (0.94) is considerably better than Swin-Unet DL (0.80%). This suggests that the water extractor is more effective in overall classification performance in Africa. Looking at the Producer Accuracy (pa) values, Swin-Unet DL performs better in detecting non-water areas (pa_0), with a value of 99.47%, compared to 94.62% for the water extractor. However, for water areas (pa_1), the water extractor significantly outperforms Swin-Unet DL, with a pa_1 of 89.35% compared to 63.65% for Swin-Unet DL. This disparity indicates that Swin-Unet DL struggles more with identifying water bodies in the complex and diverse African landscape, while the water extractor is more proficient in detecting water features in this region.

For the Siberia region, Swin-Unet DL shows better overall performance compared to the water extractor. The OA for Swin-Unet DL is 0.92%, surpassing the water extractor's OA of 0.85%. Similarly, the F1-Score for Swin-Unet DL (0.93%) outperforms the water extractor (0.87%). These metrics highlight Swin-Unet DL's stronger overall classification performance in Siberia. In terms of water class recognition, Swin-Unet DL also significantly outperforms the water extractor in terms of Producer Accuracy for water (pa_1), with a value of 90.51% compared to 66.86%. This indicates that Swin-Unet DL is more adept at recognizing water bodies in the Siberian landscape. However, for non-water class recognition (pa_0), the water extractor has a slight edge with a value of 87.77%, while Swin-Unet DL achieves 93.31%. This suggests that while Swin-Unet DL excels in identifying water bodies, the water extractor performs slightly better at distinguishing non-water areas. The corresponding confusion matrices for the models' performance in the Amazonia, Africa, and Siberia regions are presented in the Figure 27.

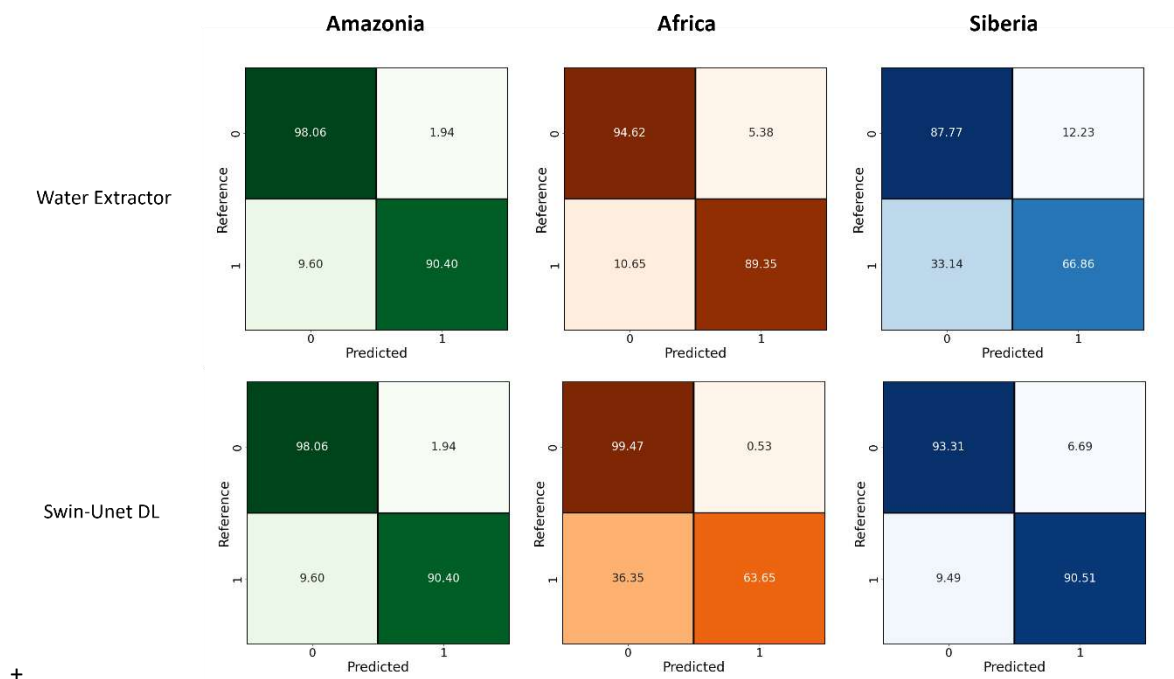


Figure 27. Confusion matrices for the (first row) water extractor and (second row) Swin-Unet DL approach across regions of (first column) Amazonia, (second column) Africa, and (third column) Siberia.

In conclusion, the water extractor typically offers a more consistent and balanced performance across all regions. It performs particularly well in Africa, where it demonstrates strong water detection capabilities, even within complex landscapes. In contrast, Swin-Unet DL shows more variability, excelling in Amazonia and Siberia, but struggling in Africa due to the region's complex land cover and seasonal dynamics. This indicates that while DL models such as Swin-Unet DL perform well in simpler or less dynamic environments, they may face challenges in regions with significant seasonal changes and diverse land cover types.

Furthermore, it is important to emphasize that the water extractor-based approach allows for the identification of seasonal variations in water bodies, unlike the DL network, which requires a reference dataset where the water land cover class is already separated based on seasonality. This makes the water extractor more feasible for use in the final SAR production chain, as it is capable of handling seasonal variations and land cover complexities more effectively than models like Swin-Unet DL. Given that the water extractor can operate independently of pre-labeled seasonal datasets, it offers a more flexible and robust solution, particularly in environments where seasonal dynamics play a crucial role in water detection. This adaptability makes it a practical choice for large-scale operational applications, where data consistency and continuous performance are key.

3.2.1.6 Comparative Analysis for Built-up LC Class Recognition between the Swin-Unet DL Approach and UEXT Detector

A similar analysis was conducted to compare the performance of built-up land class recognition using the Urban EXTent (UEXT) module and DL classification based on the Swin-Unet network. For this analysis, a test area shown in Figure 28, was identified in Siberia, specifically in the S2 45UVB tile, where open-pit mines were incorrectly

classified as built-up areas in the final product of Phase 1 of the CCI+ project. The aim is to assess whether the DL approach can resolve this issue and improve classification accuracy in such cases. Given the lack of an accurate urban reference dataset for this region, particularly one that is sufficiently distributed to cover the unique landscape features and urban patterns, only a visual comparison is presented in this section. This visual comparison helps identify the potential advantages of DL in distinguishing between built-up areas and non-urban land uses, such as open-pit mining, which may be misclassified due to their structural similarities to built-up regions. The analysis also highlights how well Swin-Unet can adapt to these complex, mixed landscapes compared to traditional methods like the UEXT module.

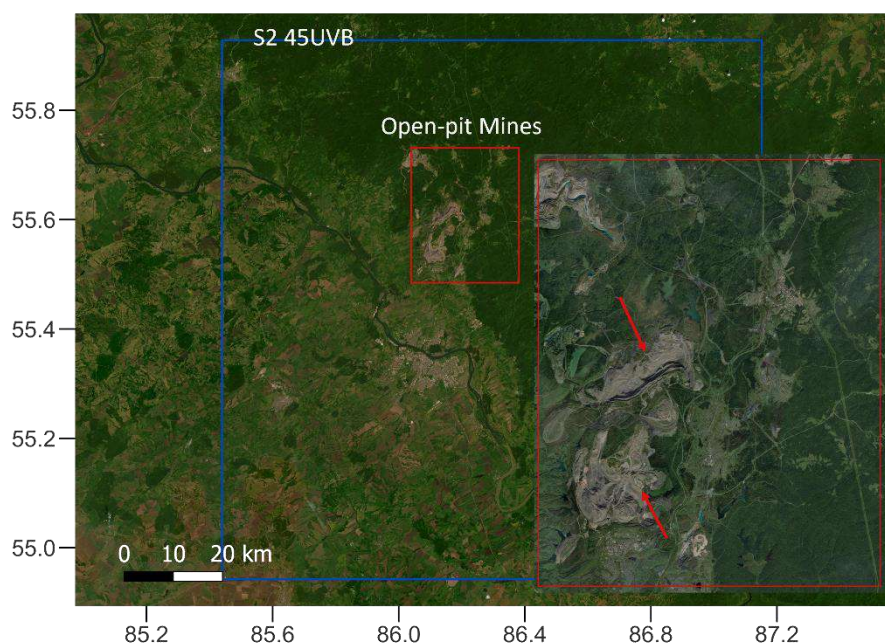


Figure 28- Open-pit mining site (in red) located in Siberia, within the S2 tile 45UVB (in blue), used for comparing built-up land class recognition performance between the UEXT module and the Swin-Unet DL classification approach.

The urban map was generated using the dedicated UEXT module, as described in the Phase 1 ATDB [AD1], leveraging the S1 time series for the year 2021. Similarly, for the same year, the LC map was derived using the DL Swin-Unet approach, from which the built-up land class of interest was extracted. Regarding the DL approach, the classification was executed by segmenting the input data into patches with dimensions of 549x549 pixels. To address potential inaccuracies at the patch boundaries, an overlay of 10% was applied between adjacent patches. This overlap strategy reduces edge effects, ensuring smoother transitions and improving classification coherence.

The generated patch-wise outputs were subsequently merged using a mosaicking process to create a continuous classification map covering the entire area of interest. This integration step not only consolidates the individual outputs but also addresses inconsistencies that may arise due to variations in local feature distributions. By combining overlapping regions, the approach enhances the overall spatial accuracy and ensures that all regions, including boundaries, are classified with minimal discrepancies.

Additionally, the methodology leverages the advantages of a patch-based workflow, such as scalability for large datasets and computational efficiency. This structured approach is particularly beneficial for extensive or heterogeneous regions, where variations in land cover types or topographic features might otherwise challenge the classification process. The resulting final map provides a detailed, high-resolution representation of the area, suitable for downstream analysis and validation.

The figure presents a performance comparison of these two results: one obtained via the UEXT module and the other using the DL approach. To ensure a comprehensive evaluation, the optical ESRI reference map and the S1 super image are also included in the analysis. The inclusion of these references provides additional context for assessing the consistency and accuracy of the built-up class identification.

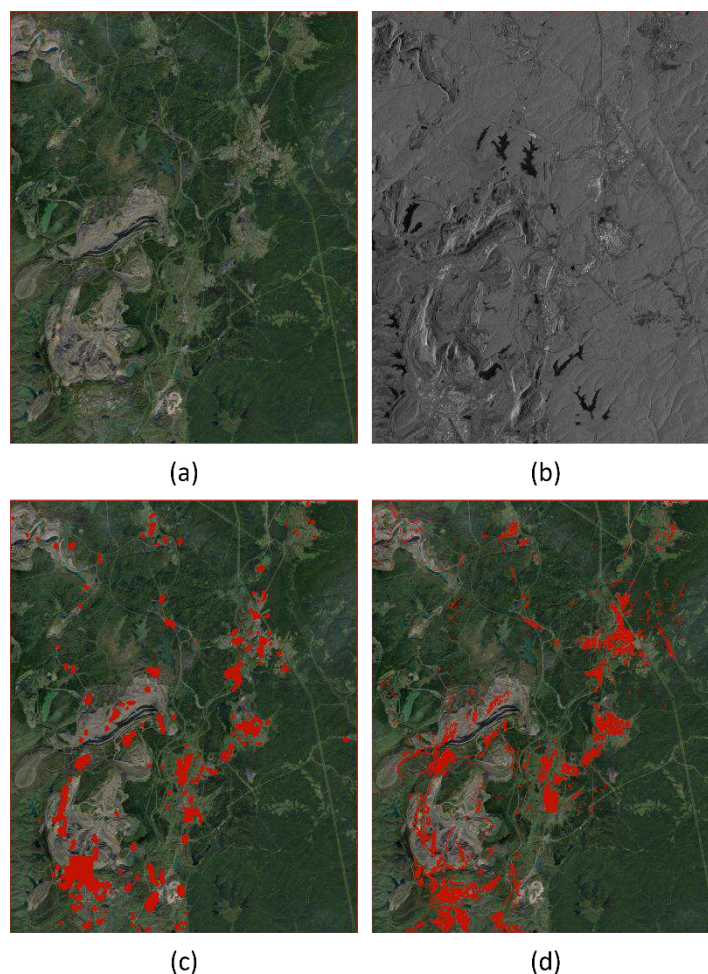


Figure 29. Visual assessment of an area of interest within the S2 tile 45UVB. The analysis was conducted to evaluate the performance of the UEXT detector and the Swin-Unet DL classification chain. The comparison includes: (a) the ESRI optical reference, (b) the 2021 S1 super image, (c) UEXT output, and (d) DL Swin-Unet classification network output.

Through visual inspection, the recognition of the built-up class by both methods appears to show comparable performance. Specifically, both the UEXT module output map and the Swin-Unet DL network's classification reveal misclassification of open-pit mines as "built-up" areas. This highlights the challenges presented by the spectral and spatial characteristics of open-pit mines, which closely resemble built-up areas in the feature space utilized by both methodologies.

This shows that there is no “free lunch” using DL approaches. To enhance the urban extraction performances for these models, a more accurate training dataset is required. This is what will be explored in the next step of the project, aiming at a reduction of open-mines misclassifications as urban areas, a limitation observed in both DL models and traditional methods like UEXT.

3.2.2 Final decision

The SAR classification chain identified for Phase 2 of the CCI+ project will involve the Swin-Unet DL model and the specialized water and urban area extractors. According to the previously described analysis, it combines innovative algorithms and robust validation processes to address the unique complexities of SAR data. The final framework represents a careful balance between DL model and specialized detectors, ensuring optimal results for diverse land cover applications.

The Swin-Unet DL network has been selected as the primary model for the SAR classification chain. Extensive testing demonstrated its superior performance in classifying complex and heterogeneous landscapes across regions such as Amazonia, Africa, and Siberia. The model's global attention mechanism enables precise feature extraction and contextual understanding, making it the most reliable and accurate solution for SAR-based land cover classification.

For the water class, the SAR classification chain will continue to use the dedicated water detector. This detector excels in capturing the seasonal dynamics of water bodies, a critical requirement for accurate hydrological

mapping. Its ability to differentiate between permanent and seasonal water ensures it remains the most effective tool for this purpose, particularly in regions with significant hydrological variability.

Regarding the built-up class, the choice of methodology will depend on future developments. While the Swin-Unet network has shown potential in detecting urban areas, challenges such as the misclassification of open-pit mines highlight the need for an enhanced training dataset. If improvements to the training set yield the desired results, the Swin-Unet could become the preferred choice for this class, given its advanced feature extraction capabilities. For the moment, however, the dedicated UEXT extractor remains a viable option, and there are plans to enhance its performance using spatial indicators to better address current limitations. This dual-path strategy ensures flexibility, allowing the project to adapt and adopt the most effective solution based on further testing and refinement.

In conclusion, the SAR classification chain for the HRLC ECV project aims at improving the performance of the Phase 1 processing chain. The adoption of Swin-Unet for general classification, the dedicated water detector for hydrological features, and a flexible approach for built-up areas demonstrates a commitment to achieving accuracy, reliability, and adaptability.

3.3 Decision Fusion

3.3.1 Decision fusion methods

3.3.1.1 Multi-temporal fusion method

In the context of historical land cover mapping where the availability of the data is different for each year, the classification process which is done independently in each historical year inevitably causes temporal inconsistencies in the classification product. To address this problem, a multi-temporal model called cascade model [16] was introduced in Phase 1. The idea behind that model was to propagate the information from the 2019 static map, whose generation benefits from more data availability, to other historical maps backward. The cascade model reduced incorrect land cover transitions significantly, but the use of the temporal information was limited because its processing considered only pairs of historical years i.e., 2019 coupled with each of other years. Taking into account this limitation, we adopt an approach based on the theory of Hidden Markov Models (HMMs) in Phase 2, to consider the information from posteriors coming from all of the years in which land cover map is produced. We refer to the ATBD for the general methodological formulation, and here, we only specify algorithmic details. Let T be the total number of observation times (years), ℓ_t be the class label of a generic pixel in time t ($t = 1, 2, \dots, T$), $\mathbf{x}_t$ be the corresponding feature vector (resulting from the optical and SAR observations), and $\mathbf{x} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T)$ be the vector collecting the observations of that pixel on *all* dates across the whole time series. The inference of the label at each time step $P(\ell_t|\mathbf{x})$ is accomplished using the forward-backward algorithm. Indeed, it is possible to prove that [17]:

$$P(\ell_t|\mathbf{x}) = \frac{\alpha(\ell_t)\beta(\ell_t)}{\sum_{\ell_t} \alpha(\ell_t)\beta(\ell_t)}.$$

Here, $\alpha(\ell_t)$ defines a forward step i.e., the joint probability of observing all feature vectors up to time t and the label at time t , evaluated through a sequential recursive procedure along the forward time direction:

$$\alpha(\ell_t) = \frac{P_F(\ell_t|\mathbf{x}_t)}{P(\ell_t)} \sum_{\ell_{t-1}} \alpha(\ell_{t-1}) P(\ell_t|\ell_{t-1}),$$

where $P_F(\ell_t|\mathbf{x}_t)$ (single-time pixelwise posterior) derives from the logarithmic opinion pool (LOGP) for the decision fusion of the posteriors received from the optical and SAR processing chains. $P(\ell_t)$ is the prior probability of the label, and $P(\ell_t|\ell_{t-1})$ is the transition probability stating the probability of one class in t changing to another class in $t - 1$.

In dual fashion, a backward recursive sequential procedure calculates the conditional probability $\beta(\ell_t)$ of all observations from time $t + 1$ up to T given the value of ℓ_t :

$$\beta(\ell_t) \equiv P(\mathbf{x}_{t+1}, \dots, \mathbf{x}_T|\ell_t),$$

$$\beta(\ell_t) = \sum_{\ell_{t+1}} \beta(\ell_{t+1}) \frac{P_F(\ell_{t+1}|\mathbf{x}_{t+1})}{P(\ell_{t+1})} P(\ell_{t+1}|\ell_t).$$

The calculation of $P(\ell_t|\mathbf{x})$ is done sequentially starting from 2019 backward to 1990. It is implied by the equations above that the algorithm formulated in Phase 2 enables the information propagation from the whole available years, because in each year, the forward procedure will include the information from the future years, and the backward process considers the information from the past years. Furthermore, the sequential inference approach allows the algorithm to run efficiently from the computational point of view.

3.3.1.2 Spatial fusion method

In Phase 1, a Markov Random Field (MRF) model [18] was introduced to consider the spatial context during the fusion process. This enabled the label regularization to be imposed on the fusion maps. As described in the latest Phase 2 ATBD [AD2], the energy function of an MRF model can be written as:

$$U(\mathbf{L}|\mathbf{X}) = - \sum_{s \in S} \alpha \log P_F(\ell_s|\mathbf{x}_s) + \sum_{\substack{s \in S \\ r \in \partial s}} V(\ell_s, \ell_r),$$

where $\mathbf{L}$ and $\mathbf{X}$ refer to the output label map to be generated and the input image data, respectively, S is the pixel lattice and s is a shorthand notation for a generic pixel location (i, k) . $\mathbf{x}_s$ collects the input optical and SAR data, and ℓ_s indicates the sample on pixel s of the random field $\{\ell_s\}_{s \in S}$ of class labels. $\partial s \subset S$ is the set of neighbouring pixels of pixel s , which can be typically in the form of four connected adjacent pixels (first order connectivity) or the surrounding eight pixels (second order connectivity). The energy function consists of: the estimated fused posterior probability $P_F(\ell_s|\mathbf{x}_s)$, and a positive weight α in its first term, which is associated with the likelihood of the class at the pixel level; as well as the second term $V(\ell_s, \ell_r)$ that is related to the spatial regularization process. While in Phase 1, we guided the degree of regularization through only one parameter γ , i.e., the weight of the pairwise potential in the energy function of the Potts MRF model:

$$V(\ell_s, \ell_r) = \gamma[1 - \delta(\ell_s, \ell_r)],$$

where $\delta(\cdot)$ is the Kronecker delta function, in Phase 2, we increase our control over the regularization by framing the approach within a more general family of probabilistic graphical models called Conditional Random Fields (CRFs). The extension of this spatial model includes two main points: Firstly, we change the one-for-all-class weight parameter γ for the label regularization to class-based weights through a function $\gamma(\ell_s, \ell_r)$:

$$V_{sr}(\ell_s, \ell_r|\mathbf{X}) = \gamma(\ell_s, \ell_r) [1 - \delta(\ell_s, \ell_r)].$$

This is to give us more freedom in compromising between regularization and detail preservation, considering that some classes may need stronger regularization while other classes should be preserved as they are. Secondly, we introduce a kernel function that measures the similarity associated with the feature vectors of the neighbouring pixels $\mathcal{K}(\mathbf{P}_s(\mathbf{X}), \mathbf{P}_r(\mathbf{X}))$, which makes the second term of the CRF energy function be written as:

$$V_{sr}(\ell_s, \ell_r|\mathbf{X}) = \gamma(\ell_s, \ell_r)[1 - \delta(\ell_s, \ell_r)]\mathcal{K}(\mathbf{P}_s(\mathbf{X}), \mathbf{P}_r(\mathbf{X})),$$

where $\mathbf{P}_s(\mathbf{X})$ is the vector collecting the fused posterior probabilities $P_F(\ell_s = \omega_k|\mathbf{x}_s)$, of all classes $\omega_k, k = 1, 2, \dots, \mathcal{C}$, where $\mathcal{C}$ is the number of classes. In the current formulation, we chose a radial basis function as the similarity kernel:

$$\mathcal{K}(\mathbf{P}_s(\mathbf{X}), \mathbf{P}_r(\mathbf{X})) = e^{-\varphi \|\mathbf{P}_s(\mathbf{X}) - \mathbf{P}_r(\mathbf{X})\|^2},$$

where φ is a positive parameter. This enables us to put restraints on the regularization process when the two neighbouring pixels are very different in terms of their posterior probabilities, while simultaneously pushing for more regularization over homogeneous image regions.

The optimization method of the CRF energy function is based on the Iterated Conditional Mode (ICM) algorithm because of the good tradeoff it usually favors between computational burden and accuracy [19]. ICM is applied iteratively and in each iteration, the label of each pixel is updated according to:

$$\ell_s \leftarrow \arg \min_{\omega_k} U(\ell_s = \omega_k | x_s).$$

The initialization makes use of the non-contextual map resulting from the pixelwise fused posteriors, i.e., $\arg \max_{\ell_s} P_F(\ell_s | x_s)$. From the implementation point of view, we aim to keep the efficient execution of the ICM algorithm during Phase 1 by keeping the strategy of formulating the ICM processing through convolution-like operations, which are designed to be feasible even when applied to the large-scale data of the CCI+ HRLC project. In this respect, the choice of ICM is confirmed as an appropriate tradeoff, as compared to more computationally intensive graph-theoretic energy minimization approaches.

3.3.2 Qualitative evaluation

3.3.2.1 Experimental Results – Multitemporal Fusion

The experiments on the multi-temporal model using HMM were done on representative tiles of each region, selected either from the benchmark tiles of the first production or because they were affected by temporal artifacts during the Phase 1. Here, we will present experimental results on tiles 20KPF, 42WWD, and 37PHL. Tile 20KPF (Amazon) is one of the tiles that showed temporal inconsistency that could not be solved even after applying the cascade model in Phase 1. The same holds true for tile 42WWD of Siberia, while tile 37PHL in Africa was drawn from the list of benchmark tiles for Phase 2. Moreover, for a representative analysis of the model performance, we also made sure that all classes defined in the CCI+ HRLC project exist in those tiles, except for the snow/ice which almost never shows inconsistency naturally. In the following, we show the results of the comparison between the cascade model from Phase 1 and the HMM-based algorithm of Phase 2. For a fair comparison, we applied the same spatial fusion process by MRF used in Phase 1 after both procedures. The production will make use of both approaches simultaneously, and the next subsection will focus on CRF-based results for spatial fusion, but here, we deemed more appropriate not to apply them jointly to disentangle their possible impacts.

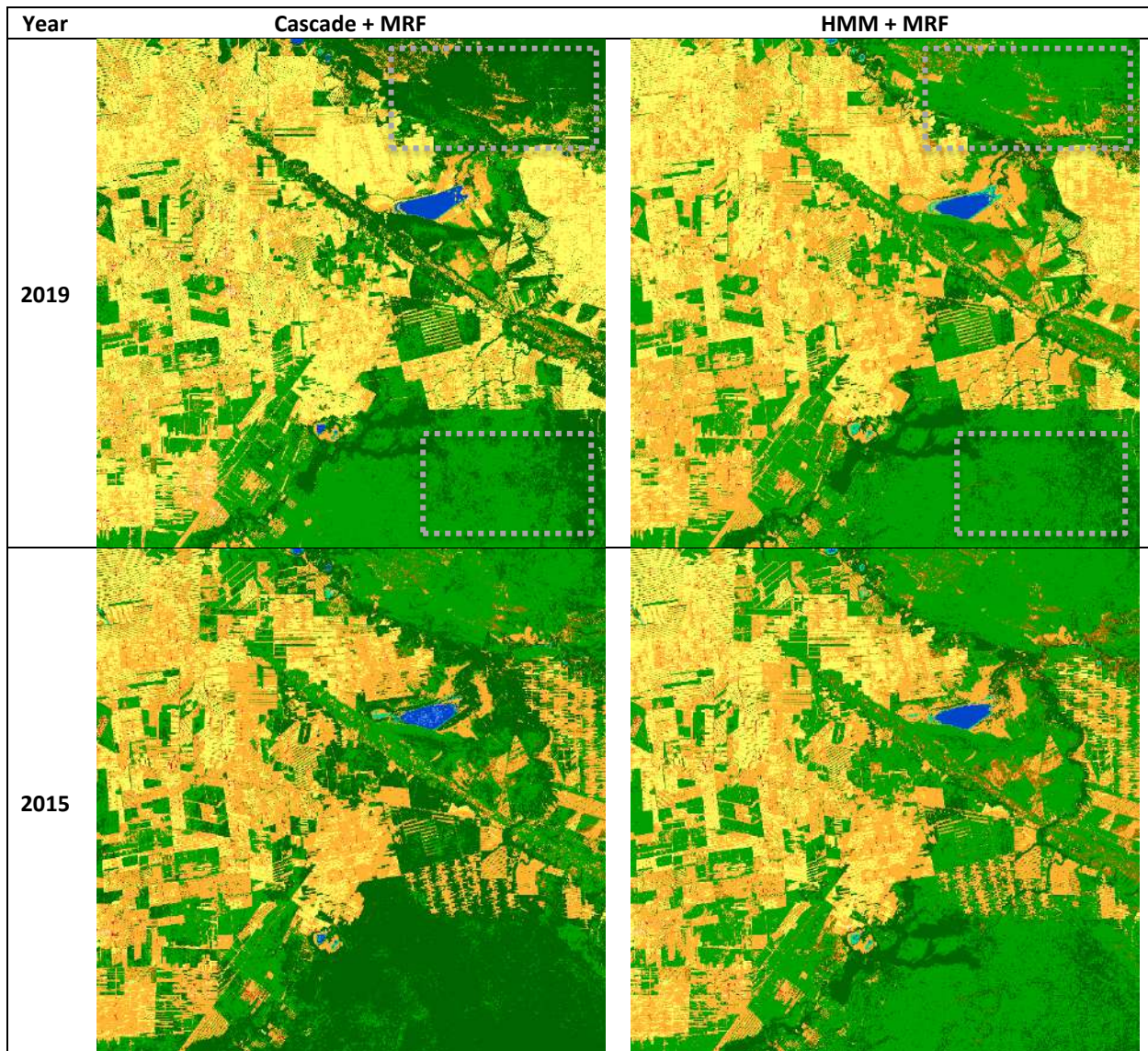
Figure 30 shows the comparison between the results of the cascade model and of HMM in the Amazonia area. The results using HMM provide more temporally consistent maps, especially in the areas marked by the grey rectangles placed on the 2019 maps. When the same areas are traced along the year, the cascade map produced inconsistencies between the evergreen broadleaf tree class (dark green) and the deciduous broadleaf tree (light green). For example, the bottom right rectangle on the cascade maps of 2019-2015-2010 shows an alternating pattern of deciduous-evergreen-deciduous, while the HMM maps indicate the pattern of deciduous with evergreen slowly disappearing as it goes backward. This artefact involving the evergreen class was indeed a major inconsistency problem pointed out during Phase 1. Furthermore, the information propagation by HMM is also beneficial in the case where the open water class (dark blue) disappears in the 2005 cascade map, while it is correctly revived in the 2005 HMM map because the class exists in every year except for that year. The favourable temporal behaviour can also be observed in the urban area (red) on the left side of the maps, where the HMM maps indicate more temporal consistency compared to the cascade maps, which for example exhibit an abrupt abundance in the 2010 instance.

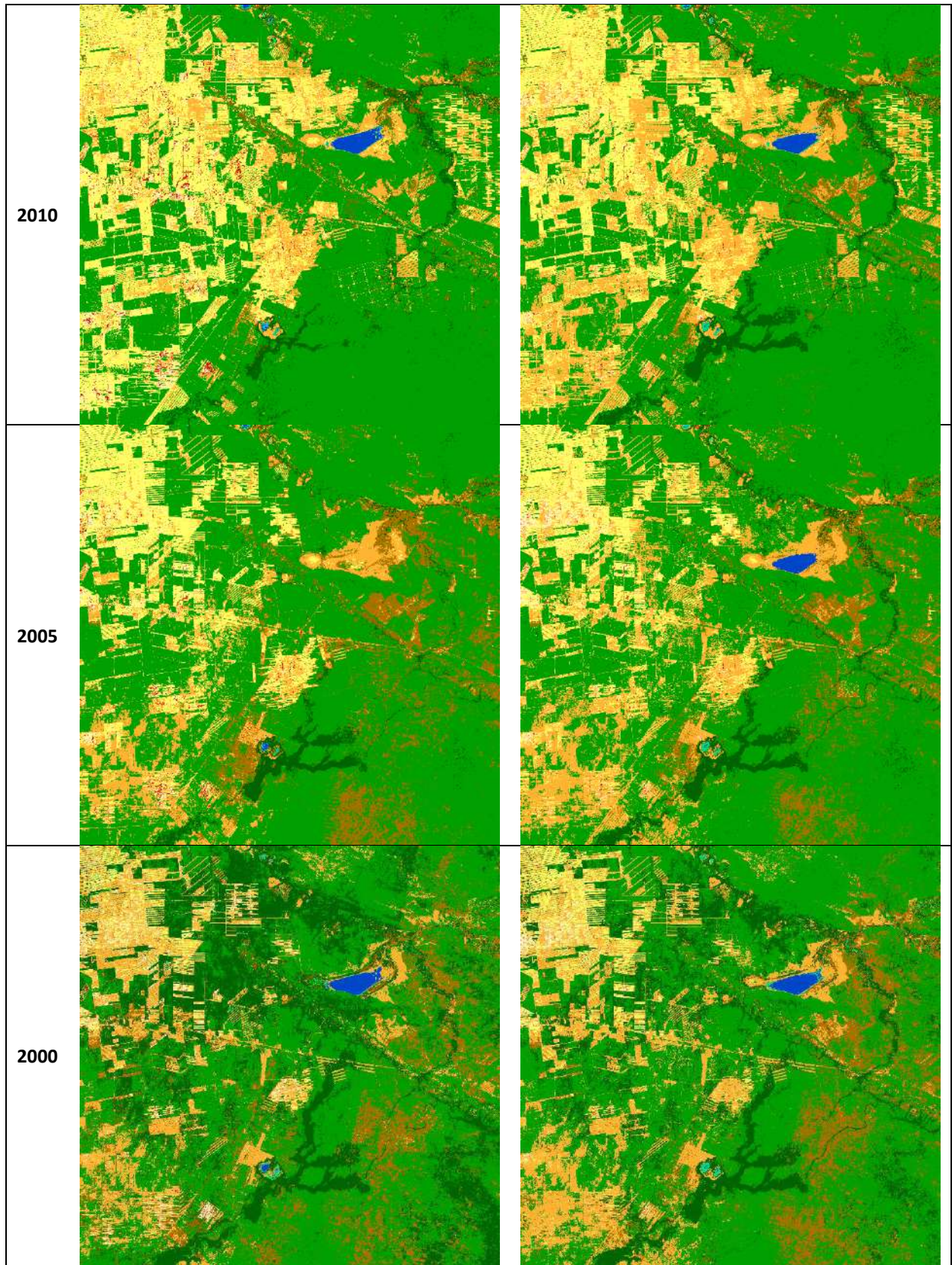
This consistent temporal behaviour is also confirmed in the Siberia area shown in Figure 31. For example, focusing on the grey rectangle area on the 2019 map, HMM exhibits more stability regarding the shrub class (brown) along the time, while the class appears less in the 2000 cascade map although it exists relatively abundantly in 2005 and 1995. In Africa's tile, the urban class (red), which is highly inconsistent and sometimes excessively abundant in the cascade maps shown in Figure 32, displays a more natural and progressive transformation along the time in the HMM results. The deciduous broadleaf tree class (green) in the region indicated by the grey rectangle on the 2019 maps of the same tile, which diminishes on the 2005 cascade maps, seems also to appear in a favourable regular pattern on the HMM maps.

The conducted experiments demonstrate that the multi-temporal model based on the developed HMM formulation is advantageous in favouring the consistency across the land cover maps over time, and in reducing unwanted temporal patterns that were observable from the maps of three or more consecutive years within the Phase 1. Those temporal pattern could not, even theoretically, be addressed by the cascade model because the information propagation only happens in this model between one pair of years. On the contrary, the HMM formulation proposed here addresses multitemporal fusion at the level of the whole time series, which allows taking into account the temporal information of the entire sequence in the labelling of each pixel.

On one hand, this is beneficial in decreasing the inconsistent pattern which is one of the main problems encountered during Phase 1. On the other hand, it is worth noticing that generally there is a risk of propagating information from maps that are less reliable (e.g., those with problems of lacking acquisition data), and of

correspondingly introducing noise in the classification results of the years which we have more confidence at. In the experiments conducted so far, this risk has not been observed. Nevertheless, in this regard, we are taking into account possible ways to further improve the model to minimize this risk. Moreover, in the next project period, we will also put our focus on tuning the transition probability matrix (TPM) which is one of the hyperparameters of the HMM that translates to the trade-off between the information propagation (which is associated to consistency) and the changes of classes in time. The goal is that the maps exhibit consistency over time but also do not end up censoring changes that actually happened.





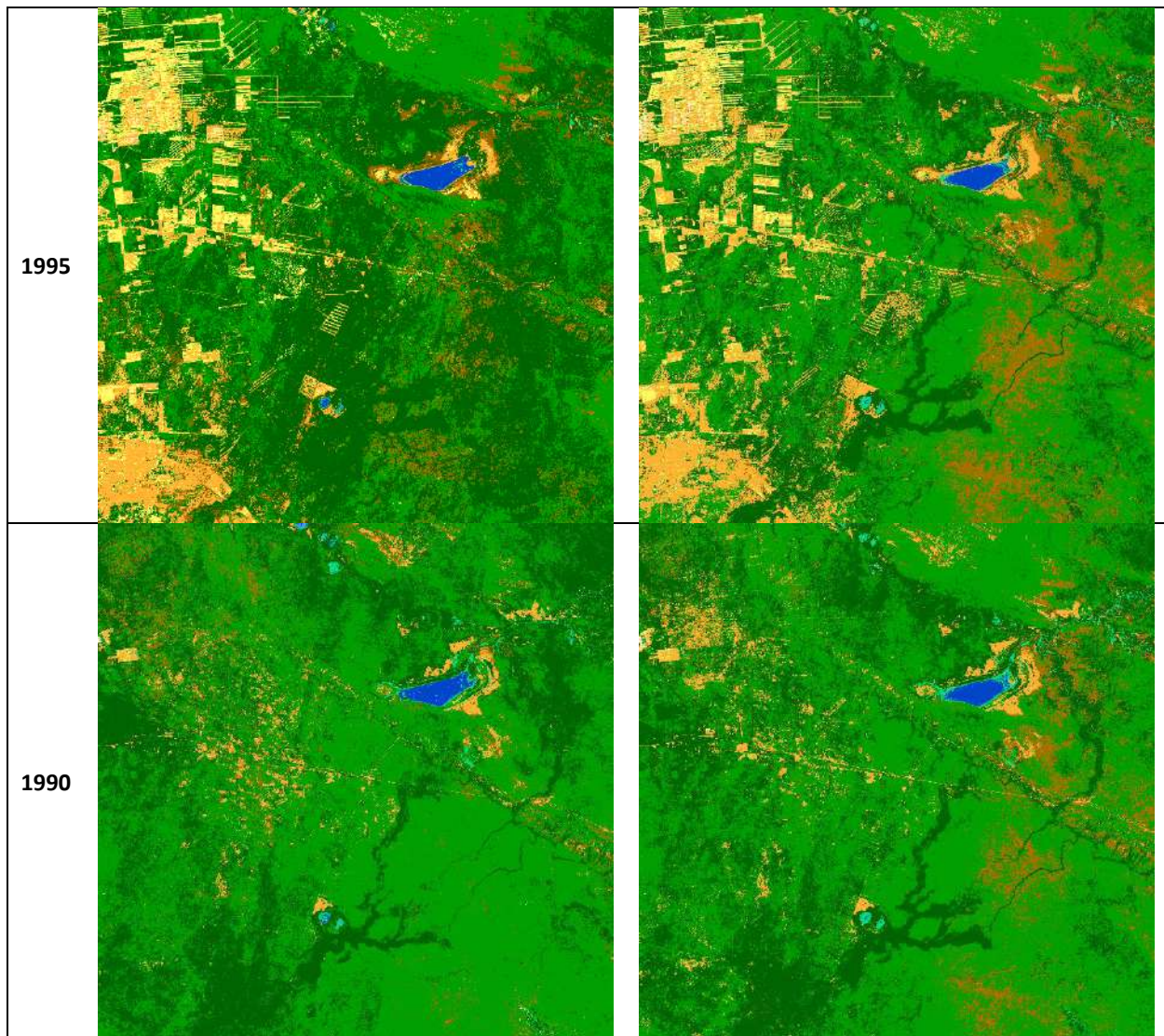
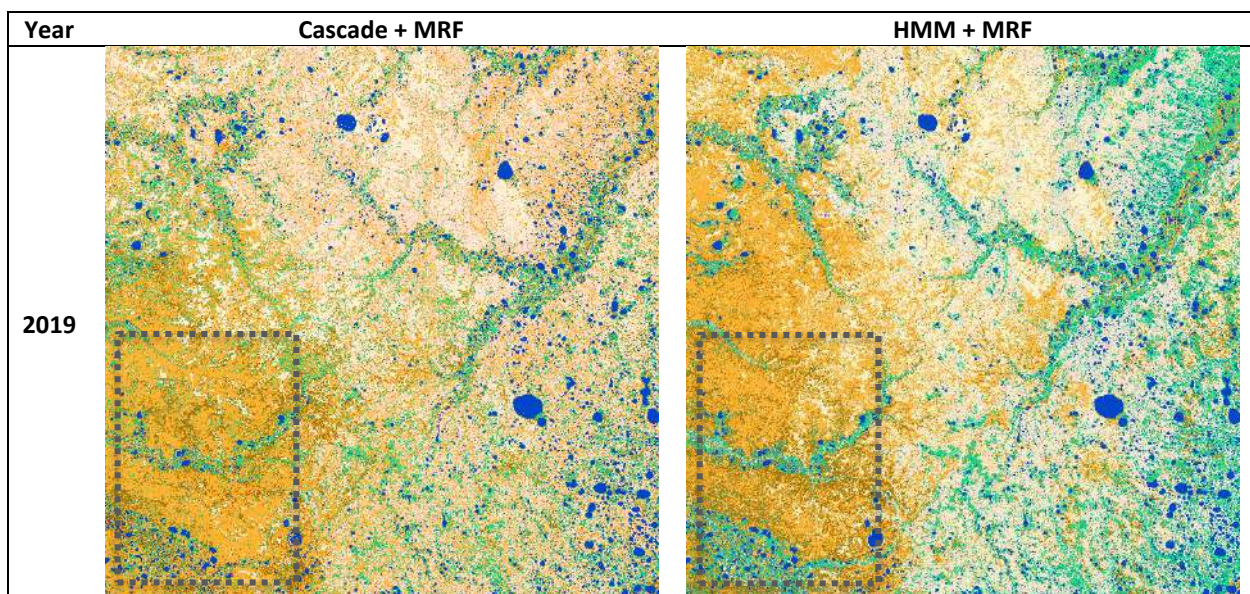
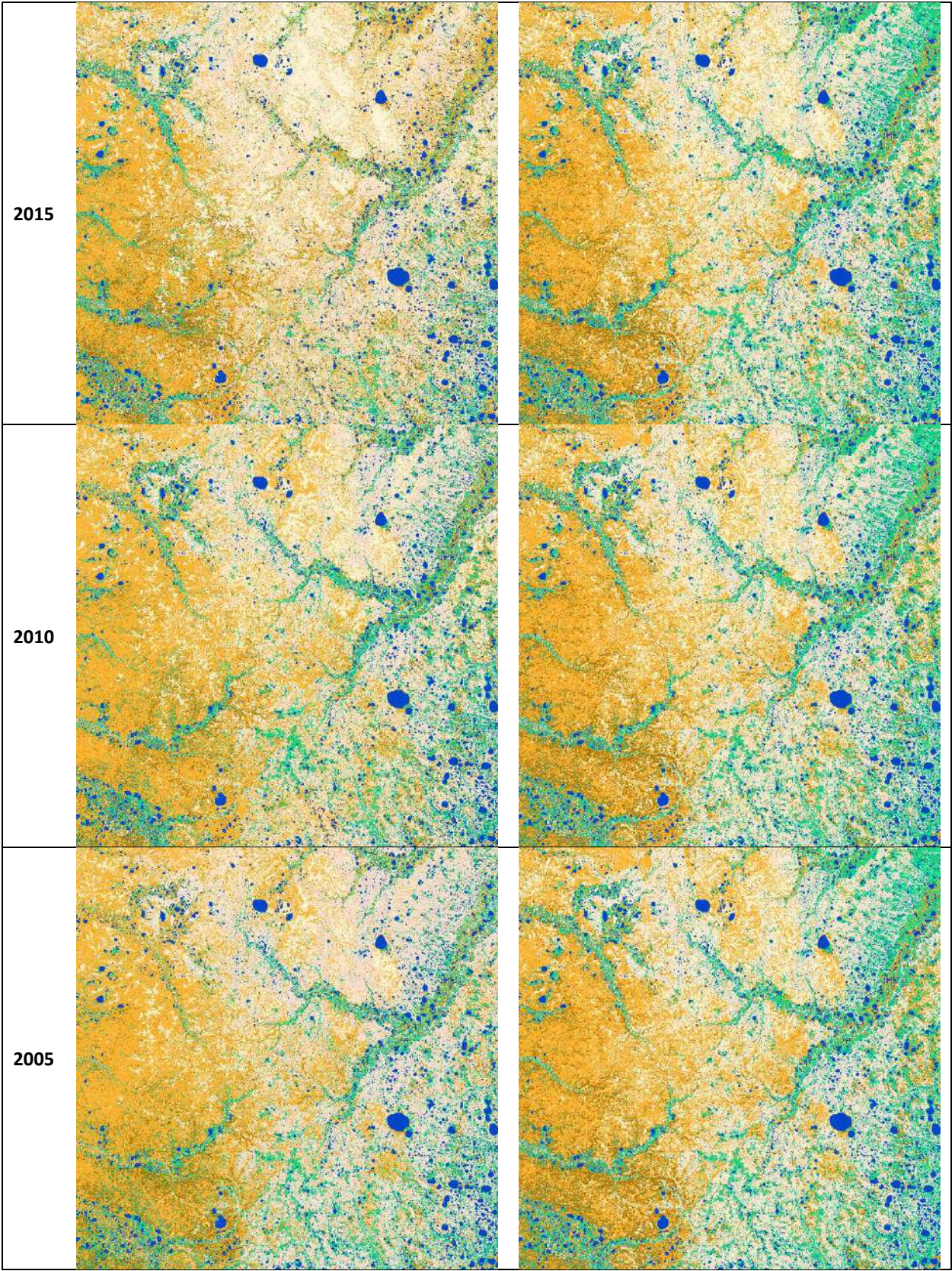


Figure 30. Comparison between the cascade model of Phase 1 (left) and the HMM of Phase 2 (right) for each historical year on tile 20KPF in Amazon





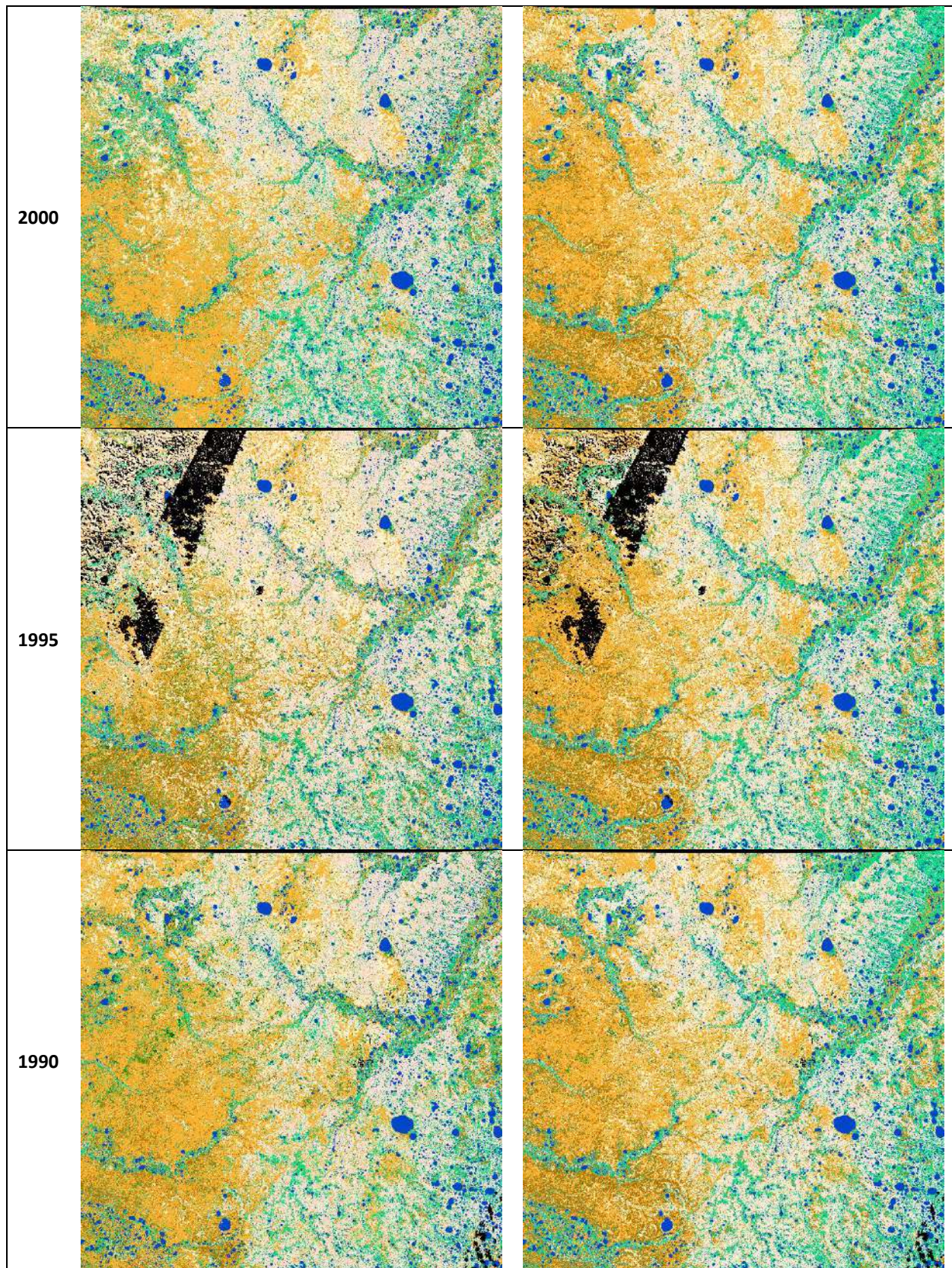
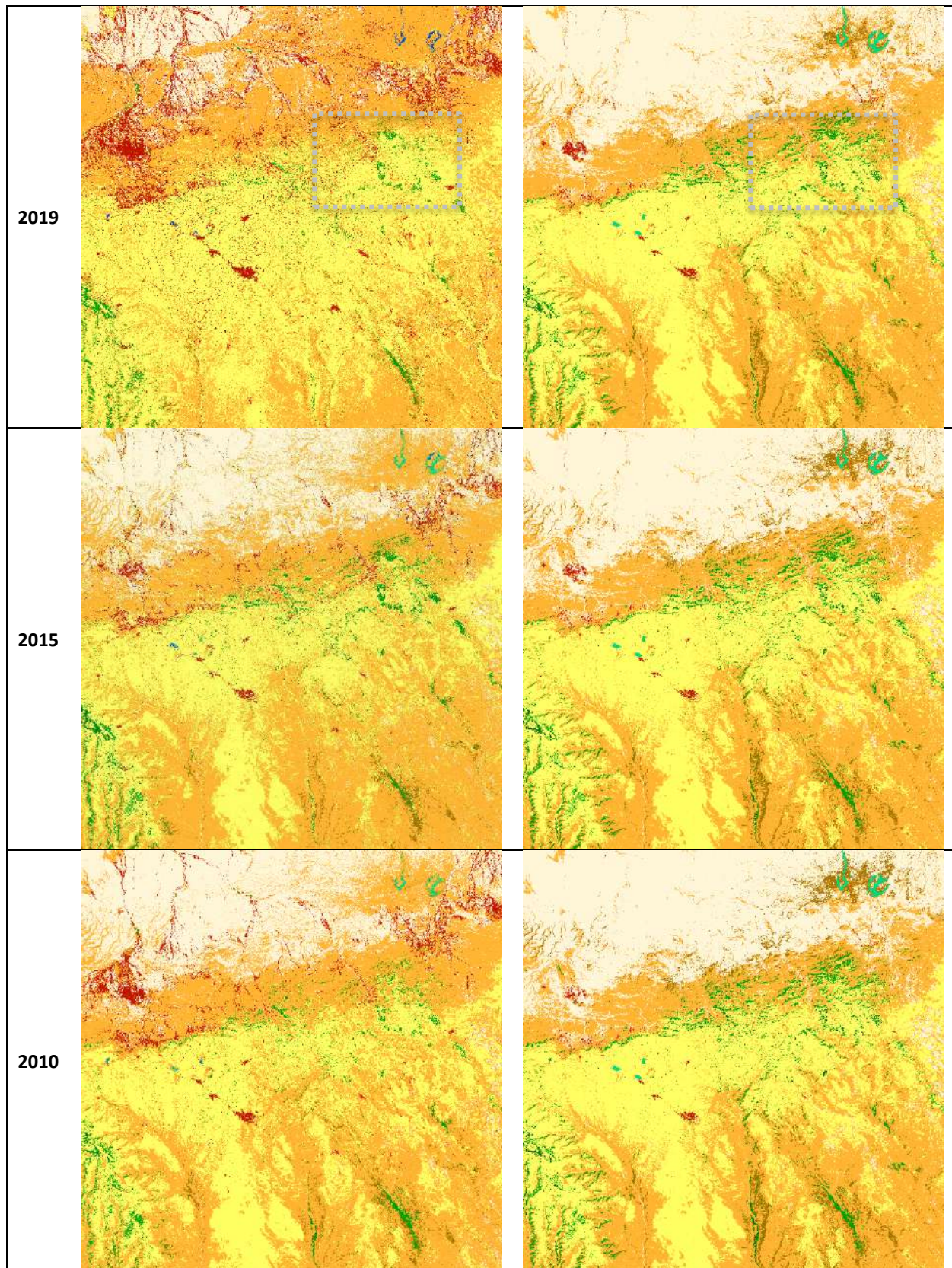
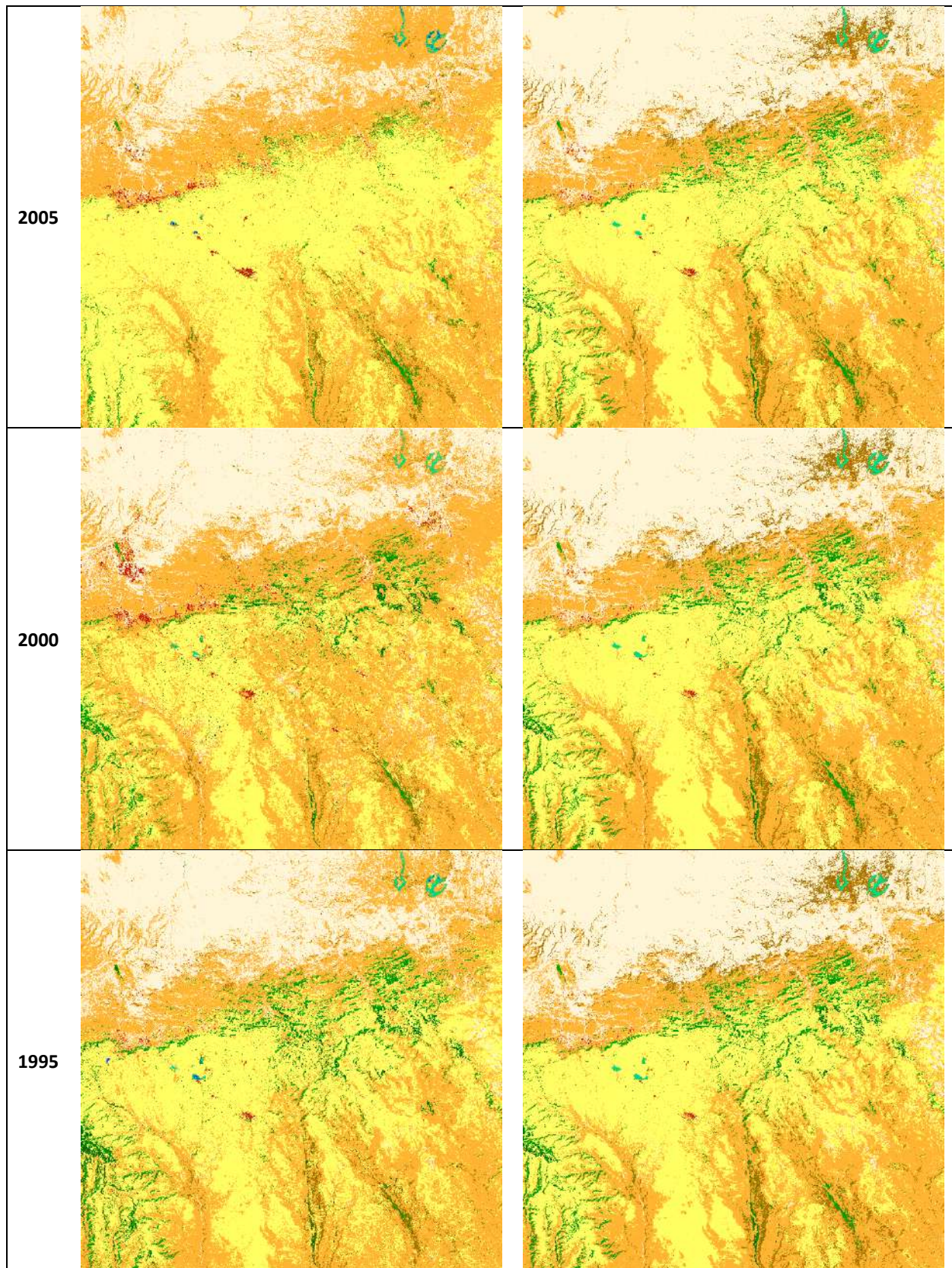


Figure 31. Comparison between the cascade model of Phase 1 (left) and the HMM of Phase 2 (right) for each historical year on tile 42WWD in Siberia. The black pixels in 1995 and 1990 correspond to areas with too limited data availability.

Year	Cascade + MRF	HMM + MRF
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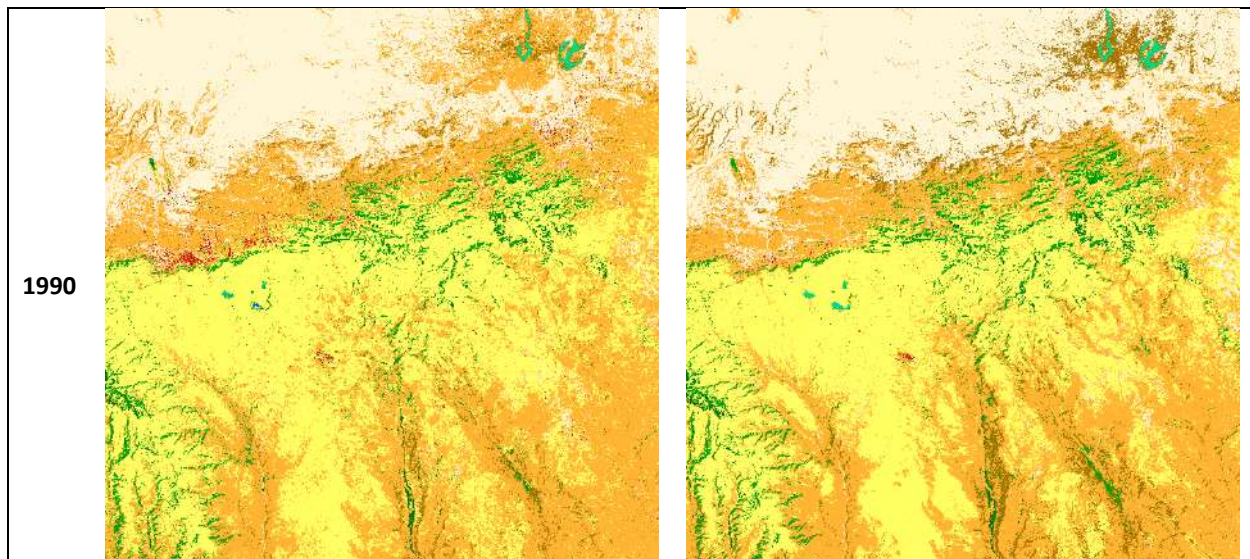


Figure 32. Comparison between the cascade model of Phase 1 (left) and the HMM of Phase 2 (right) for each historical year on tile 37PHL in Africa

3.3.2.2 Experimental Results – Spatial-Contextual Fusion

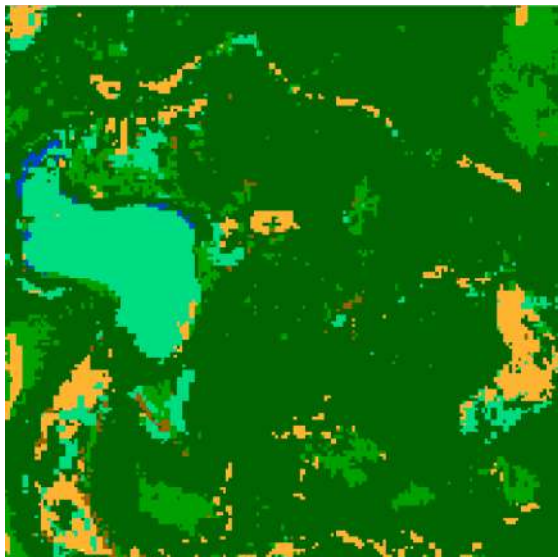
As mentioned in the methodology section above, the class-specific weights and a kernel function were introduced to give more flexibility in controlling the degree of spatial regularization. Preliminary class-specific weighting has been considered, but in general, the assignment of class-specific weights in the production will take into account the opinion from the Climate Team regarding which classes should be regularized more than other classes. Here, we will focus the discussion of the results especially on the behaviour allowed by the contrast-sensitive CRF model with its kernel function. The experiments were done on the case of 10-meter resolution static maps, and we tried several values of parameter φ in the Gaussian radial basis function kernel of the CRF model, to appreciate the changes in regularization behaviour. To illustrate this behaviour, we show in Figure 33, Figure 34, and Figure 35, spatial details of the resulting spatial-contextual classification maps. For visual comparison purposes, the non-contextual maps that would be obtained using only the optical pixelwise posteriors are also shown.

Figure 33 shows the zoom-in of an area in tile 20KPF in Amazon. First, let us observe how the regularization parameter in the MRF model, γ , affects the regularization of the pixels. When γ is set to be very small (b), the regularization on the image source (a) is very little, and as it is increased to a bigger value (c), the regularization becomes stronger, as it can be observed that some spatial details are gone. Interestingly, we can see that when γ is enlarged even further, the regularization effect does not take place in this specific area anymore. This is an expected behaviour when the best class has much higher posterior probability than the other remaining classes. Next, with the CRF model, we intentionally set γ to be very large to see a clearer effect of the contrast-sensitive CRF kernel. Theoretically, when φ is close to zero, we will see that the CRF behaves similarly to the MRF. This similarity can be seen in (e) and (f). With a fixed big value of γ , as the value of φ is risen (f, g, h, i), we can see that spatial details are gradually back, while the impact of noise remains limited and without reviving unwanted fragmentation that exists in (b). This effect is expected because the kernel acts, to a certain extent, as an inhibitor of the regularization but only when the probability of the neighbouring pixels are not too similar.

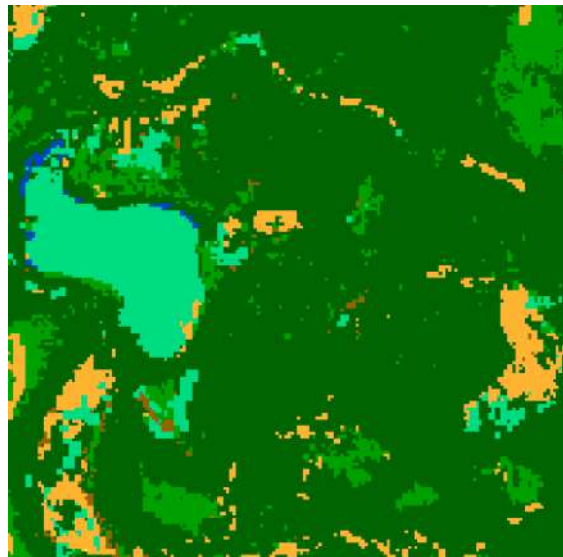
Figure 34 and Figure 35 illustrate the case in tile 37PGN in Africa and in tile 42WWD in Siberia, respectively. With the same γ parameter setting, these figures show the behaviour of the classification results as a function of the tuning of the φ parameter of the CRF model. While in the African tile, we can observe a behaviour similar to that described in the case of the Amazonian tile, it is interesting to see that, in the Siberian tile, the comeback of the details is not as apparent as in the other two tiles as the φ parameter increases. This is interpreted as due to the uncertainty in this area being higher than in the other two areas, i.e., the posterior probabilities of neighbouring pixels appear to be relatively similar. In turn, this implies that the increasing of the φ value brings back only spatial details that have dissimilarity to their neighbourhoods up to a certain point. Indeed, this is a desired behaviour since it confirms the capability of the proposed approach to take into account and propagate uncertainty from the input data (and their possible data availability issues) to the output land cover product.

These experimental results confirm that using the CRF model gives us improved control over the regularization level, compared to the MRF model. This yields an increased flexibility in taking into account the feedback and needs from the Climate Team. Indeed, the spatial regularization of the output map in the production will be tuned according to the indications from the Climate Team about the desired trade-off between preserving spatial

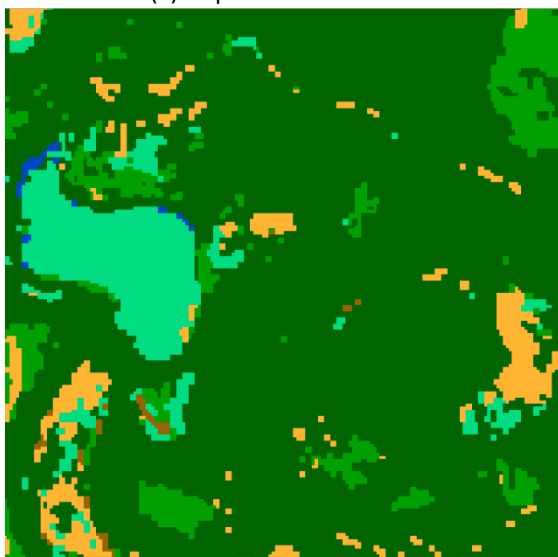
resolution and favouring spatial homogeneous labelling and about how this trade-off may change in the case of different classes.



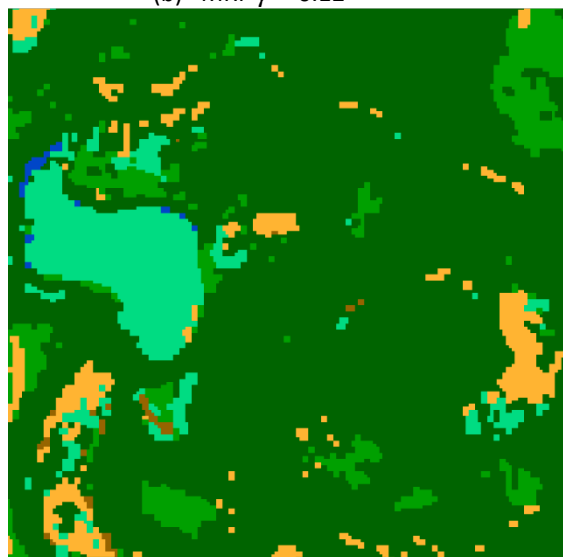
(a) Optical



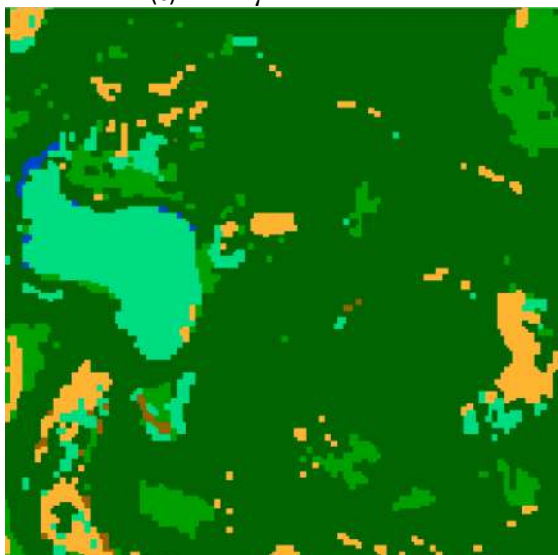
(b) MRF $\gamma = 0.12$



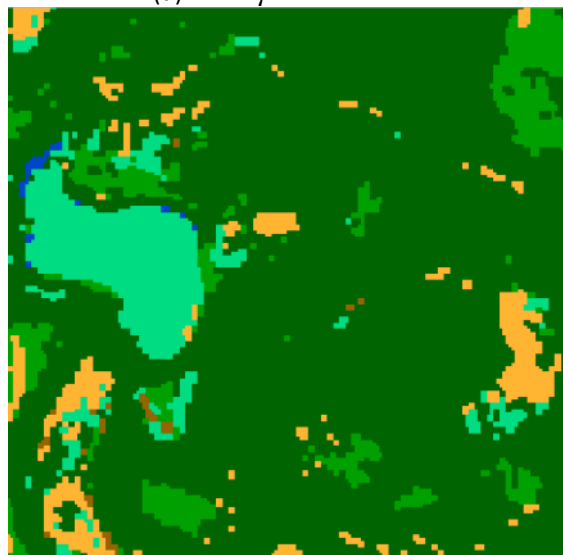
(c) MRF $\gamma = 1$



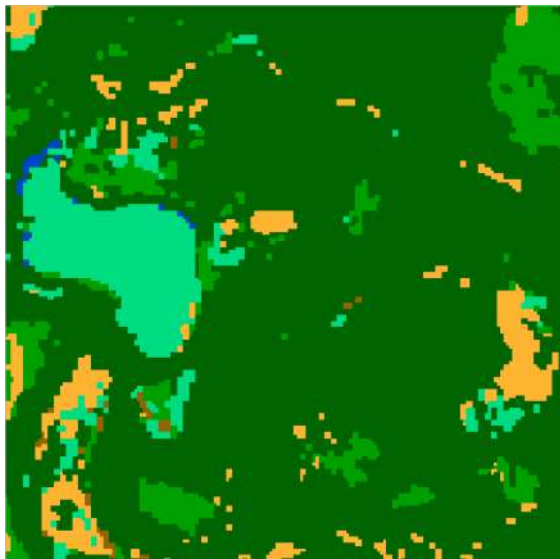
(d) MRF $\gamma = 2$



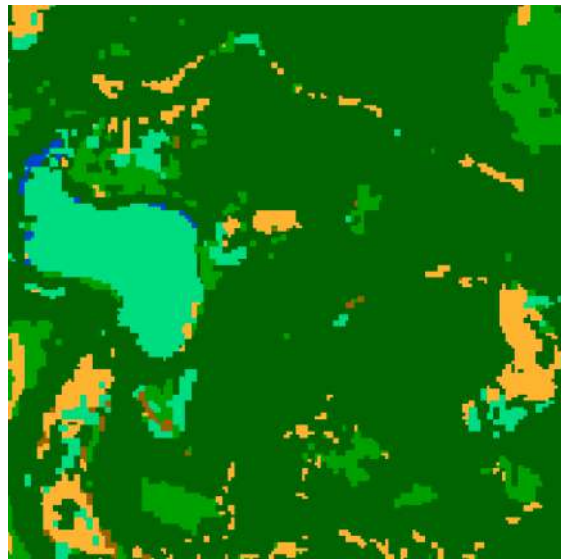
(e) MRF $\gamma = 50$



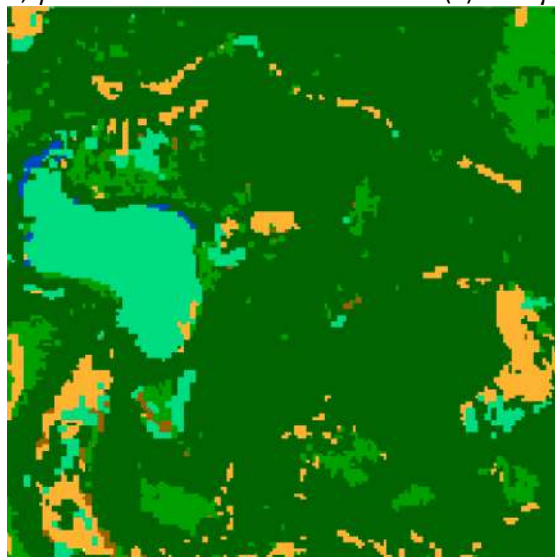
(f) CRF $\gamma = 50, \varphi = 0.5$



(g) CRF $\gamma = 50$, $\varphi = 2.5$

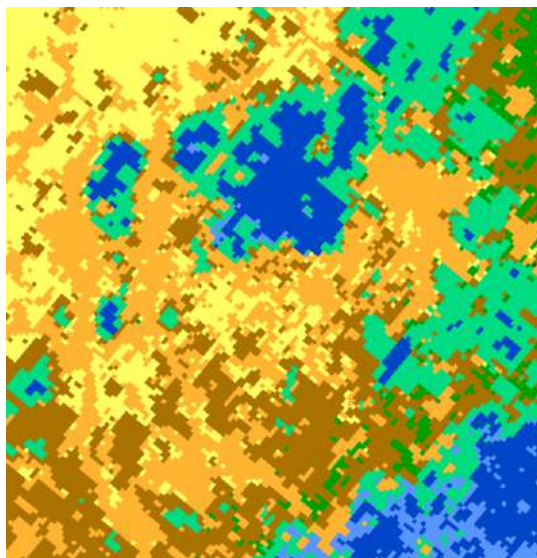


(h) CRF $\gamma = 50$, $\varphi = 5$

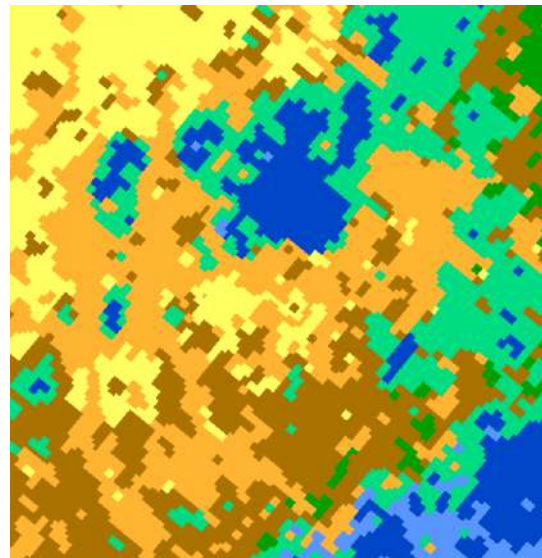


(i) CRF $\gamma = 50$, $\varphi = 10$

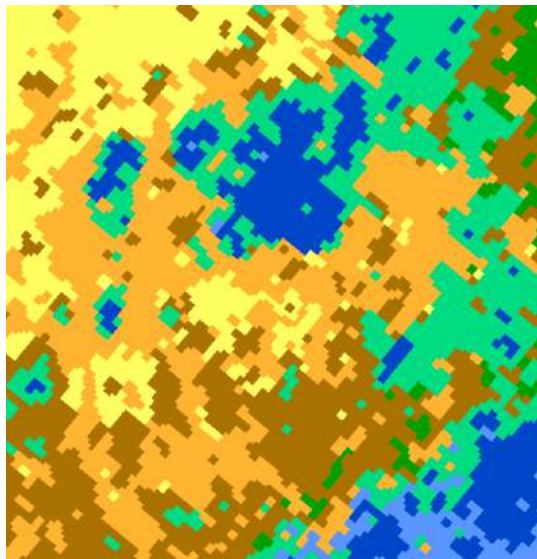
Figure 33. The comparison among (a) the map resulting from the original optical posteriors, (b,c,d,e) MRF with several regularization parameters, and (f,g,h,i) CRF with several kernel parameters



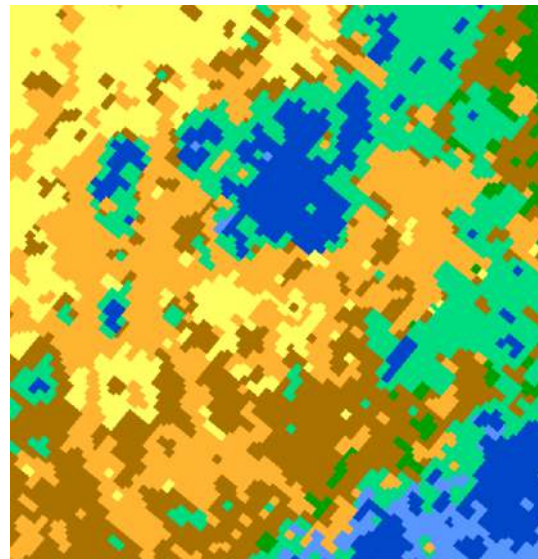
(a) Optical



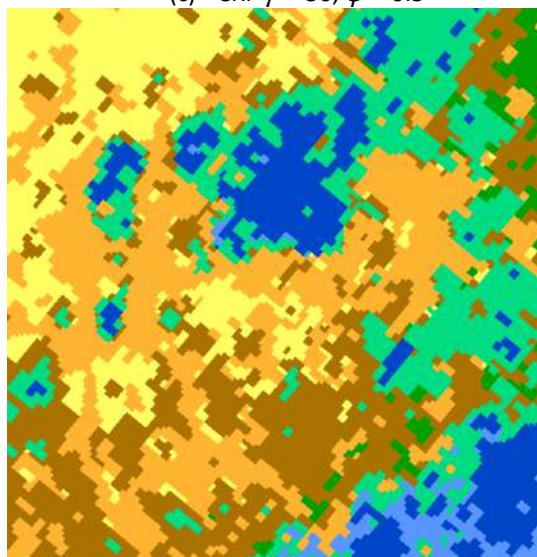
(b) MRF $\gamma = 50$



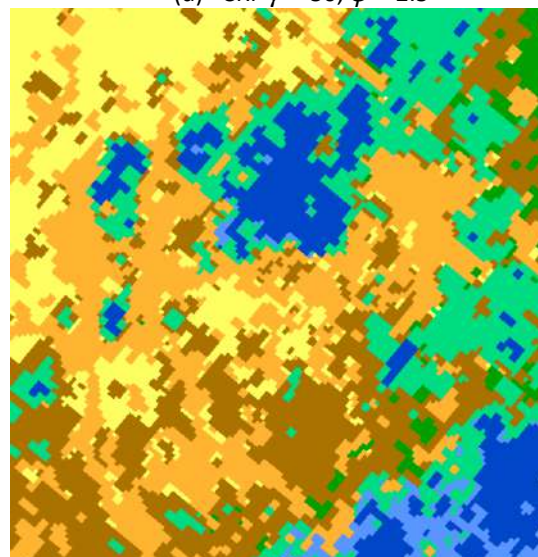
(c) CRF $\gamma = 50$, $\varphi = 0.5$



(d) CRF $\gamma = 50$, $\varphi = 2.5$

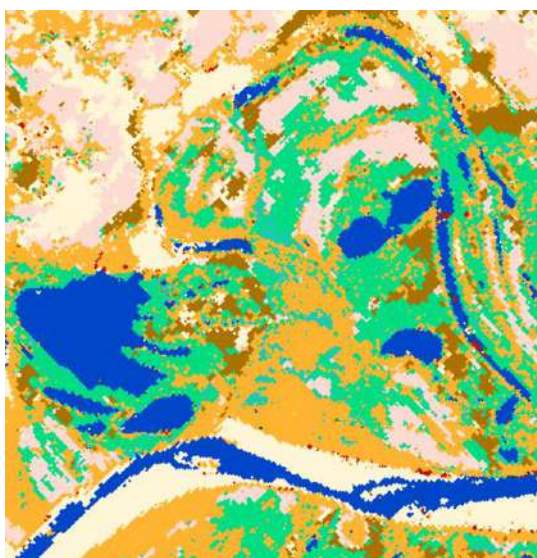


(e) CRF $\gamma = 50$, $\varphi = 5$

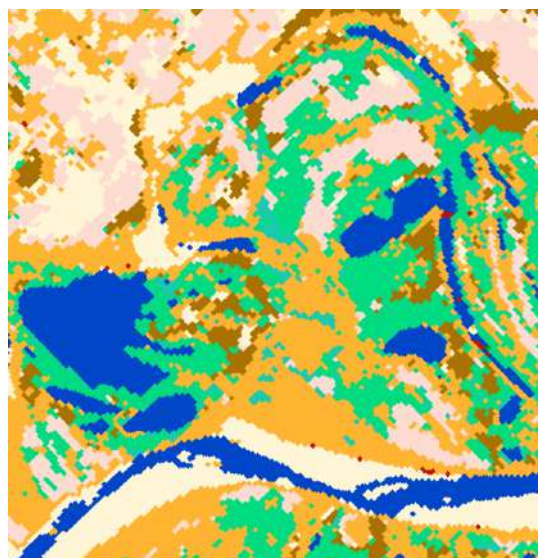


(f) CRF $\gamma = 50$, $\varphi = 10$

Figure 34. The comparison among (a) the map resulting from the original optical posteriors, (b) MRF with a big regularization parameter, and (c,d,e,f) CRF with several kernel parameters



(a) Optical



(b) MRF $\gamma = 50$

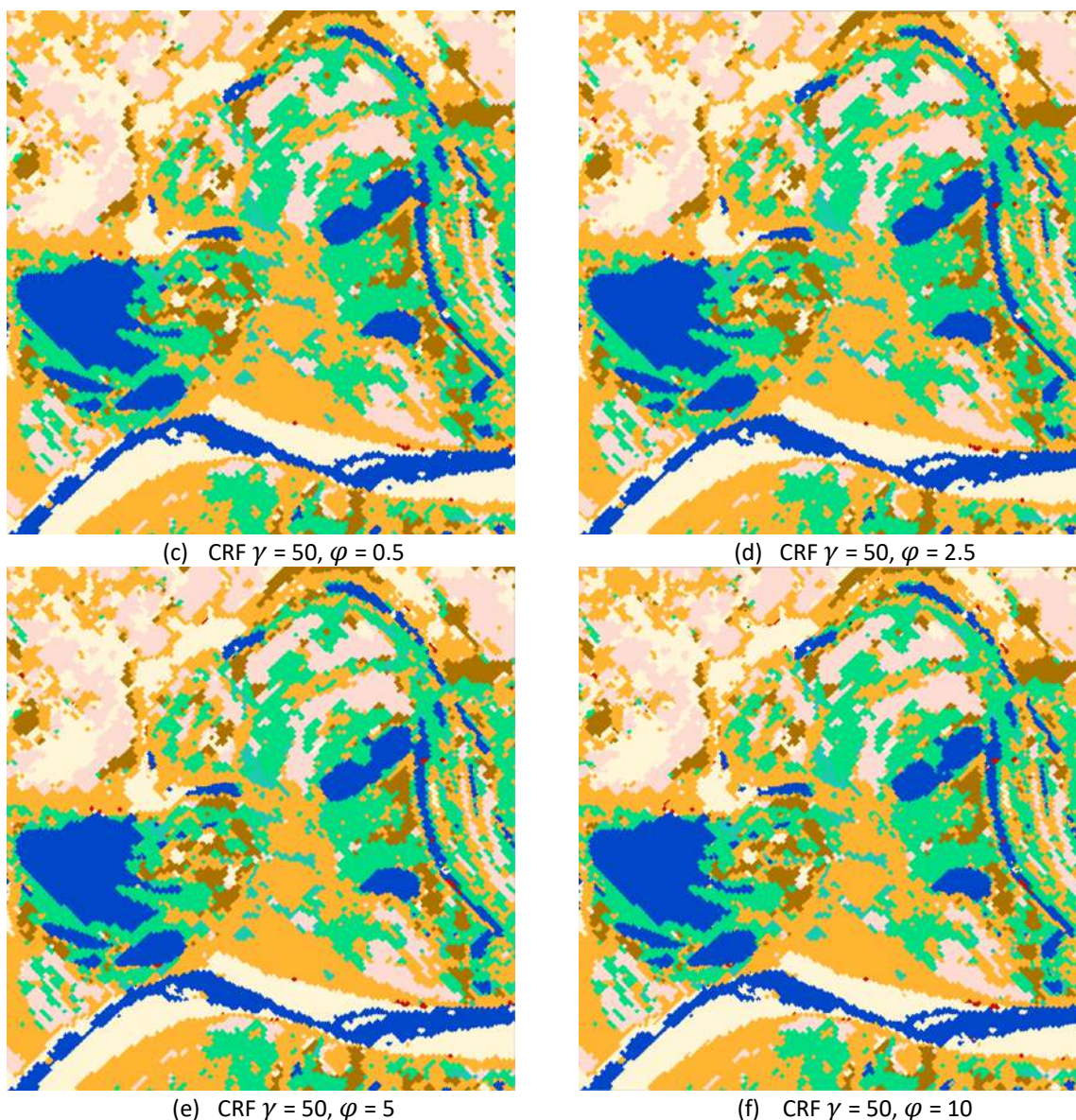


Figure 35. The comparison among (a) the map resulting from the original optical posteriors, (b) MRF with a big regularization parameter, and (c,d,e,f) CRF with several kernel parameters

3.3.3 Final decision

In the context of multi-temporal fusion, HMM has demonstrated to provide more consistent results temporally compared to the cascade model. Similarly, the CRF model gives enhanced flexibility in guarding the spatial regularization process than the MRF model can do and allows for stronger spatial smoothing while preserving edges and small-scale details. Therefore, HMM and CRF-ICM are identified as appropriate solutions towards the first production.

Further development of both models may regard the tuning of the transition probability matrix of HMM, and the evaluation of alternate kernel choices in the CRF energy function, which may consider introducing similarity operators between the two probability distributions $P_s(\mathbf{X})$ and $P_r(\mathbf{X})$ instead of calculating the distance between them within a radial basis function kernel. These possible variations will be considered also in relation to the point of view of computational cost during the production.

3.4 Landcover Change Detection

In phase 2, the analysis and change detection process expands to include new regions and incorporates S2 datasets. The CD processing chain continues to operate at the pixel level on a yearly basis from 1990 to 2024, with a focus on three subcontinental areas: Amazonia, Africa, and Siberia (see Figure 36). This phase introduces

several updates to the methodologies, accounting for the higher spatial resolution of S2 data, the challenge of harmonizing multi-sensor data across these diverse regions, and the need to incorporate advanced techniques for managing both temporal and spatial variability in land cover dynamics. The regional differences in land cover patterns, environmental conditions, and spatial distribution of vegetation types require careful adaptation of the methodology to ensure accurate change detection outcomes.

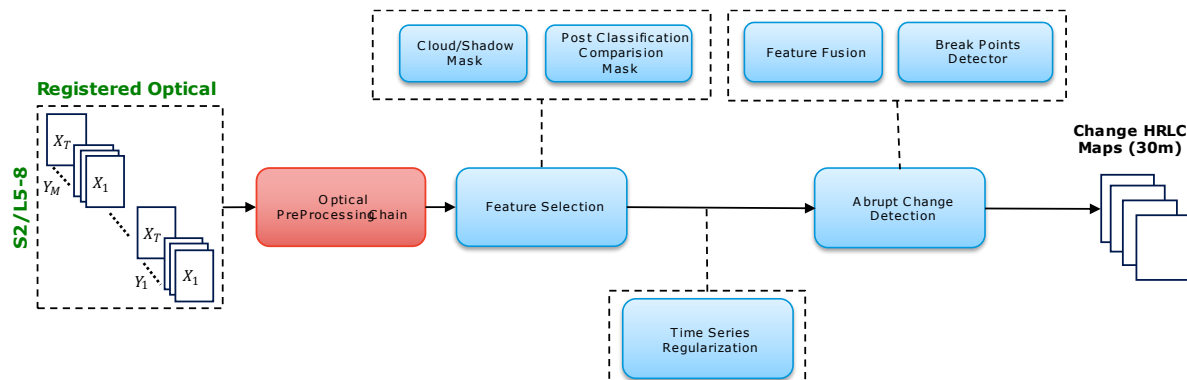


Figure 36. Multi-annual Multi-feature LC change detection.

To address these challenges, the preprocessing stages from optical data analysis were revisited, aiming to create an integrated and unified workflow for both land cover classification and land cover change detection. The goal is to streamline the preprocessing process across different data types (optical and radar), reducing the number of steps required while ensuring consistency and robustness in the data preparation pipeline. This approach is particularly important given the increasing complexity of data sources and the need for efficient processing across large datasets, such as those from S2, which provide more frequent, and higher-resolution imagery compared to other sources.

In terms of feature space design, an analysis was conducted to identify the most suitable features for the change detection task, considering the specific characteristics of the regions under study. The feature space will be regionally optimized, with distinct feature sets developed for Amazonia, Africa, and Siberia to capture the unique land cover dynamics of each area. This customization is essential for improving the accuracy of change detection. Furthermore, the integration of advanced techniques for managing spatio-temporal variability, such as those derived from machine learning models and time-series analysis, will allow for more accurate detection of abrupt changes and better handling of temporal variations.

3.4.1 Time Series Reconstruction

An analysis was conducted to evaluate the trend of the time series using the monthly composites generated in phase 1 for one of the tiles in Amazonia. The primary goal of this analysis was to assess whether the monthly composites, derived from the optical processing chain, could effectively capture the underlying trends of land cover changes over time without the need for time-consuming and computationally heavy time series reconstruction. The reconstruction process typically involves interpolating or filling in missing data points, which can be resource-intensive, especially when working with large datasets. By considering only one composite per month, the analysis aimed to test if this approach could maintain the temporal dynamics of the land cover changes and still yield accurate results.

The results of the analysis, displayed in the Figure 37 and Figure 38, show the comparisons between the weekly time series reconstruction from phase 1 and the monthly composites. This comparison indicates that the monthly composites are capable of handling the irregularities in the time series and adequately reflecting the temporal trends, even without the traditional reconstruction step. This suggests that it is possible to use monthly composites to capture the variation and dynamics in land cover changes, making the process more efficient. Consequently, this approach offers a promising alternative to time series reconstruction, especially when dealing with high-resolution S2 data in phase 2, as it reduces computational costs while still ensuring reliable breakpoint detection.

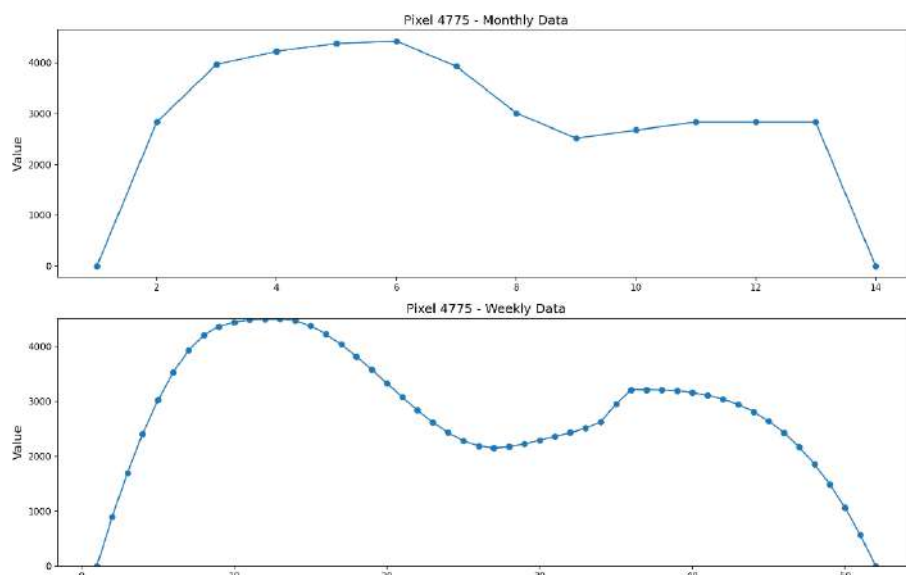


Figure 37. Comparison of time series data of the weekly time series reconstruction method and time series of the monthly composites for 2019.

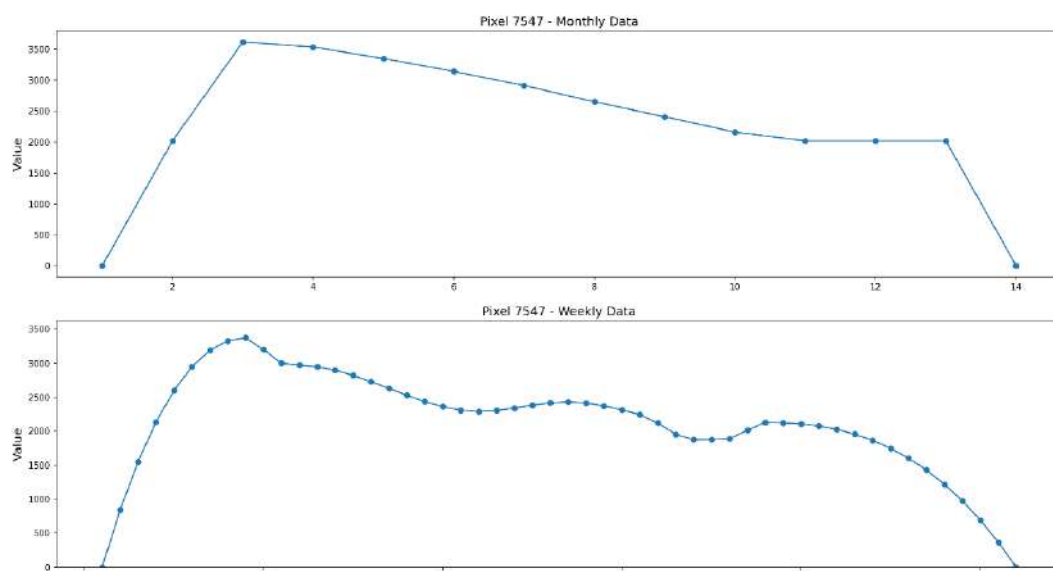


Figure 38. Comparison of time series data of the weekly time series reconstruction method and time series of the monthly composites for 2019.

3.4.2 Feature Selection

Another analysis was conducted to optimize feature selection for the developments in phase 2, focusing on several indices that are under evaluation for each of the areas of study. Each index provides valuable information about land cover, aiding in the analysis of changes. The analysis demonstrates that, depending on the specific area of study, certain indices may be more effective than others. In this context, temporal features are particularly useful for capturing how spectral information evolves over time, facilitating the identification of abrupt land cover changes. Time series analysis of indices such as Normalized Difference Vegetation Index (NDVI), Normalized Difference Water Index (NDWI), and Bare Soil Index (BSI) provides insights into patterns like deforestation, regrowth, and seasonal fluctuations in vegetation or water bodies. Additionally, feature selection will consider features extracted from the optical processing chain, such as deep learning-derived features and textural metrics like GLCM. Based on their relevance to change detection, these features will be incorporated into the feature selection and feature fusion process to enhance the accuracy and reliability of the analysis. So, for a vegetation area affected by deforestation, the indices NDVI, NDWI, and BSI were combined for further analysis. The fusion of NDVI and BSI was done using the following equation:

$$fused_{index} = \sqrt{\frac{NDVI^2 + NDWI^2 + BSI^2}{3}}$$

This equation assumes that all three indices (NDVI, NDWI, and BSI) are equally weighted. Different weights could be assigned to each index, by considering the modified equation below. For example, if you wanted to give more importance to NDVI, the equation might look like this:

$$fused_{index} = \sqrt{\frac{\omega_1 \cdot NDVI^2 + \omega_2 \cdot NDWI^2 + \omega_3 \cdot BSI^2}{\omega_1 + \omega_2 + \omega_3}}$$

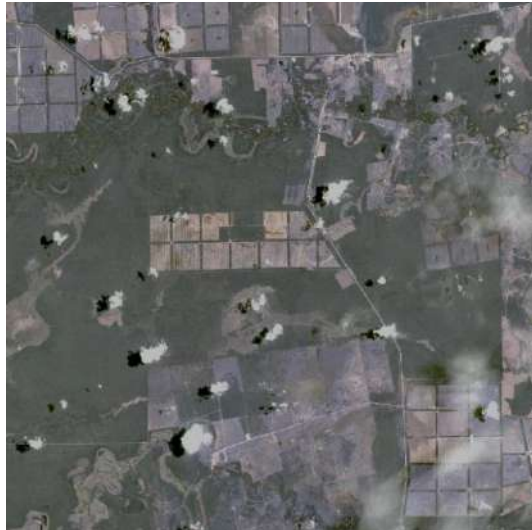
The weights for combining indices can be defined in several ways, depending on the specific application and what aspects of the indices are most important for the analysis. An analysis is currently ongoing to select an efficient and effective method for defining the weights using state-of-the-art methods [20], [21].

The outcome of this analysis, performed for the years 2018 and 2019 in the Amazonia area that experienced deforestation because of the crop construction, is shown in Figure 39. In this analysis, monthly composites were generated using high-resolution Sentinel-2 data and compared with the change map produced in Phase 1 using the Landsat dataset. The analysis was conducted without using the PCC mask, to assess the effectiveness of feature representations and the breakpoint detector in achieving accurate change detection. For the latter analysis, the PCC will be regenerated using Phase 2 products to align with the land cover (LC) maps and reduce computational burden.

By considering the sample S2 images provided in first row of the Figure 39, the changes that occurred from 2018 to 2019 reveal that the crop field has continued to be constructed, indicating ongoing agricultural development in the area. The change map generated using fused indices demonstrates following aspects:

- Improved change detection through fused indices: The change map generated using fused indices (NDVI, NDWI, and BSI) demonstrates enhanced change detection compared to the Phase 1 map. This improvement is due to the complementary information provided by these indices, which collectively increase the accuracy of identifying changes specific to the study area.
- Higher resolution insights with Sentinel-2 data: The use of high-resolution Sentinel-2 data at 10m resolution reveals finer details of the changes, offering a more precise representation of the landscape (see Figure 40). Notably, even the stripes within the crop field are distinctly detected, underscoring the capability of Sentinel-2 data to capture fine spatial details.
- Effective temporal change detection: The breakpoint detector effectively identified the year of change, showcasing its ability to handle high-resolution data for accurate temporal change detection.
- Efficient analysis without time series reconstruction: The analysis demonstrates that it is possible to bypass the computationally intensive time series reconstruction process by using Sentinel-2 composites. These composites provide sufficient information to capture temporal trends and detect changes accurately.

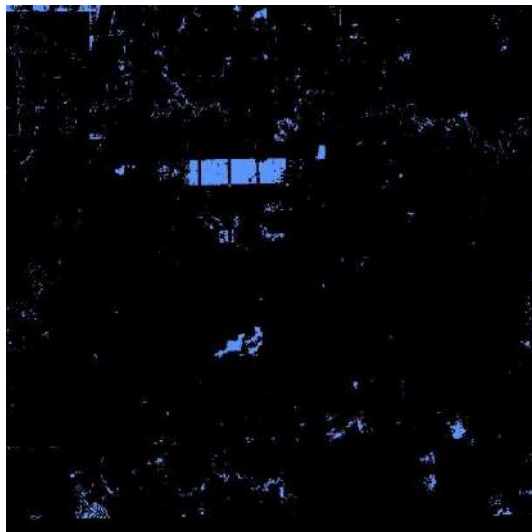
The updated approach enhances change detection by both in space and time while offering an efficient alternative to time series reconstruction, thereby reducing the computational burden on the processing chain. Furthermore, the analysis continues to focus on generating a comprehensive feature space for each area of interest, using the features generated from optical classification chain to further refine and enhance the results.



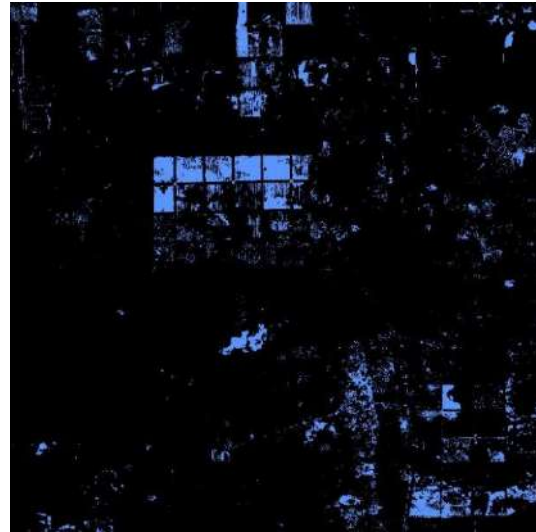
S2 - 2018



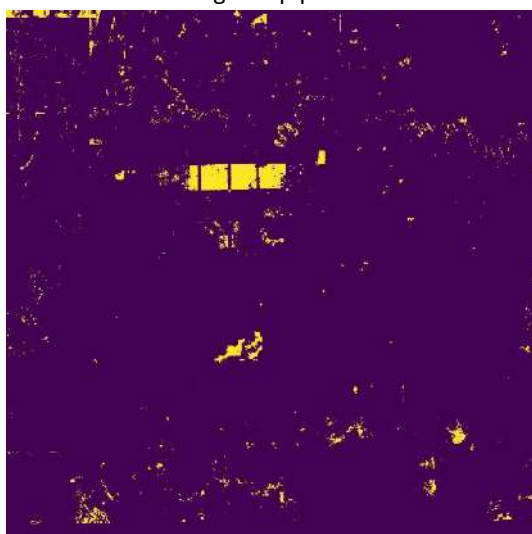
S2 - 2019



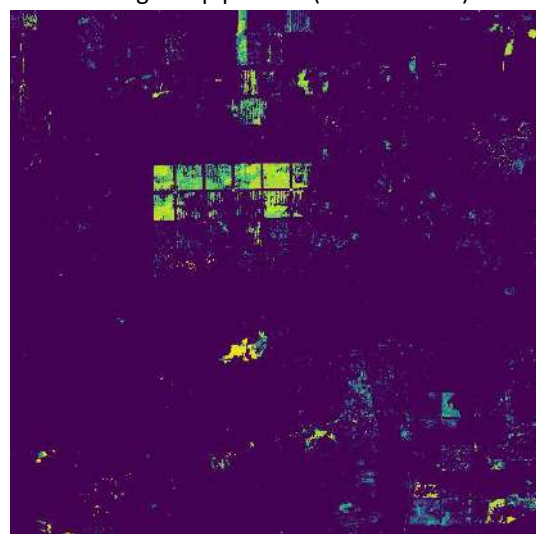
Change
Change map phase 1



No-change
Change map phase 2 (fused indices)



Probability of change phase1



Probability of change pahese 2 (fused indices)



Figure 39. A comparison of phase 1 change map and the change map using fused feature space.

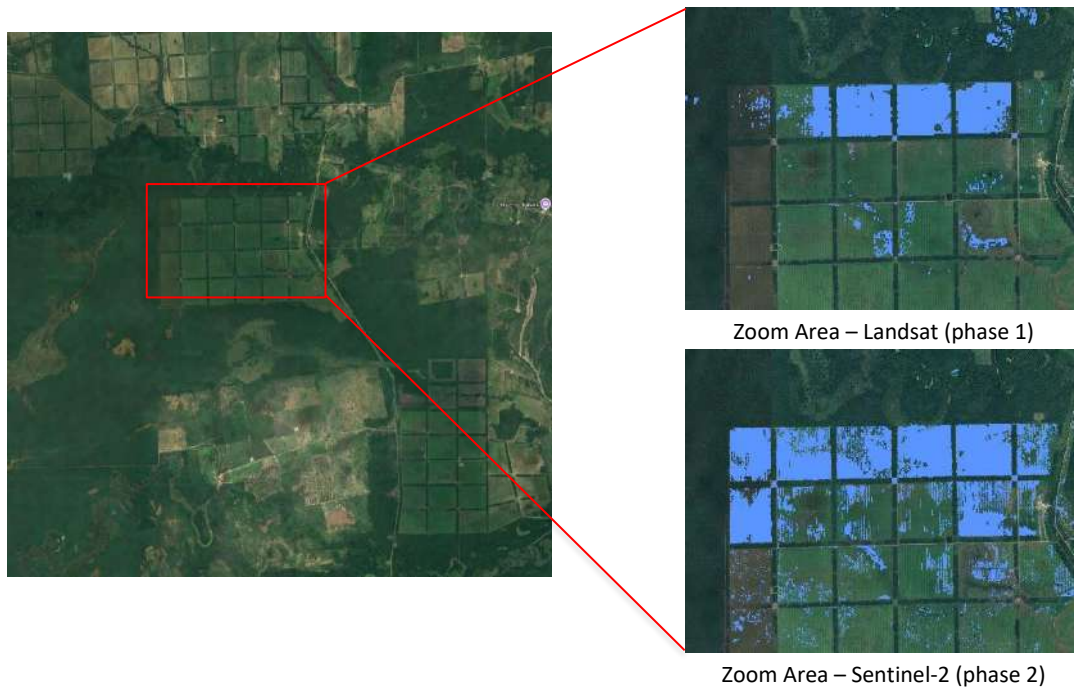


Figure 4041. The details of the generated change maps using Sentinel-2 fused features and phase1.

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Weakly Supervised Deep Learning in Radiology

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Conflicts of interest are listed at the end of this article.

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Deep learning (DL) is currently the standard artificial intelligence tool for computer-based image analysis in radiology. Traditionally, DL models have been trained with strongly supervised learning methods. These methods depend on reference standard labels, typically applied manually by experts. In contrast, weakly supervised learning is more scalable. Weak supervision comprises situations in which only a portion of the data are labeled (incomplete supervision), labels refer to a whole region or case as opposed to a precisely delineated image region (inexact supervision), or labels contain errors (inaccurate supervision). In many applications, weak labels are sufficient to train useful models. Thus, weakly supervised learning can unlock a large amount of otherwise unusable data for training DL models. One example of this is using large language models to automatically extract weak labels from free-text radiology reports. Here, we outline the key concepts in weakly supervised learning and provide an overview of applications in radiologic image analysis. With more fundamental and clinical translational work, weakly supervised learning could facilitate the uptake of DL in radiology and research workflows by enabling large-scale image analysis and advancing the development of new DL-based biomarkers.

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Deep learning (DL) uses multilayered artificial neural networks for the automatic analysis of images. DL has become the standard tool in computer-based analysis of radiologic imaging data. DL reaches expert-level accuracy in tasks such as object detection, semantic segmentation, and image classification in both two-dimensional and three-dimensional imaging data. Increasingly, DL is also used to extract quantitative features from images that can be used as image-based biomarkers for prognostic or predictive purposes, for example in oncology.

So far, DL has predominantly been used in research. However, we are currently in a transition phase in which the first such models are actively being used in clinical radiology to support radiologists. For example, having received regulatory approval, commercial products using artificial intelligence are available to aid in the routine task of detecting bone fractures on conventional radiographs. These products flag potential sites of fractures on radiographs, and the final diagnosis is left to the radiologist (1). Similar commercial products are available that flag potential findings such as effusion or pneumonic infiltrates on thoracic radiographs—again with the radiologist making the final diagnosis (2). In both of these scenarios, radiologists are still needed to finalize the report by employing their expertise to check the accuracy of the DL model's predictions. In addition, in situations in which two radiologists usually read the same examination, DL models could replace one of them. A landmark study in this regard is the randomized prospective trial by Lång et al (3) of mammography screening in approximately 80 000 women. In this study, Lång and colleagues

demonstrated that double reading in mammography screening can be replaced by a single reading by a human plus a reading by a DL model without decreasing the cancer detection rate.

Most of the DL models currently in use in clinical practice were trained in a supervised manner (Figs 1, 2). This method uses large data sets with carefully crafted reference standard labels. For example, using supervised learning to train a model to segment malignant lesions on screening mammograms requires a large database of mammograms in which radiologists have annotated the precise position of every malignant lesion. The advantages of this training paradigm are that it has been proven to work well and that DL model performance scales with the availability of annotated data (4). However, the major disadvantage is that the construction of such databases for every radiologic use case can be prohibitively expensive when one takes into account the time required to manually label each radiologic image and the substantial median salary of radiologists (5).

On the other end of the spectrum, unsupervised learning (Fig 1) relies on only the data, such as radiologic images, and does not use external labels. A key method of unsupervised learning is clustering. For instance, Nam et al (6) clustered features extracted from a model trained to predict survival based on preoperative CT images in patients with lung adenocarcinoma. CT imaging features served as a surrogate for histopathologic and metabolic profiles of adenocarcinoma that could be used for postoperative survival prediction.

Weakly supervised learning is a training paradigm situated between these two extremes that uses an incomplete,

Abbreviations

DL = deep learning, LLM = large language model



Summary

This review explores the transition of deep learning in radiology from laborious fully supervised methods to more scalable weakly supervised methods, emphasizing the potential and challenges for future research.

Essentials

- Weakly supervised learning enables scalability in radiology by using incomplete, inexact, or inaccurate labels, thus reducing or entirely removing the need for manual labeling.
- Vast data sets can be obtained automatically by using labels extracted from free-text radiology or pathology reports; these labels may be incomplete, inexact, or inaccurate.
- Weakly supervised learning accelerates the development of deep learning models for tasks such as diagnosis, prognostication, and segmentation.
- Researchers with access to large radiologic image databases and substantial computational resources are encouraged to train foundation models using self-supervised learning (ie, without labels generated by humans) and to publish these pretrained models.
- This sharing will enable researchers with limited resources to fine-tune these pretrained models using even small cohorts, thereby democratizing the field and fostering the exploration of rare conditions.

inexact, or inaccurately labeled training data set (Fig 1) (7). This approach allows for the inclusion of a much larger data set that requires less expert manual labeling compared with fully supervised learning. In this article, we describe weakly supervised learning systematically and present possible application scenarios in radiology. We then outline how large language models (LLMs) can help leverage the potential of weakly supervised learning.

Labels in Weakly Supervised DL

Supervised DL in radiology relies on pairing image data with labels. These labels can have varying qualities. In the best-case scenario, they are accurate and represent exactly what the DL model is tasked to learn. Going back to the earlier example, if a radiology model is tasked with identifying the location of malignant lesions on screening mammograms, an ideal training set would contain mammograms with associated labels that denote the image location of every biopsy-proven lesion without a single lesion being missed (blue circle in Fig 1). The creation of such labels often requires the work of highly skilled experts, typically with substantial salaries. Thus, the creation of large data sets can become prohibitively expensive (5). This is a problem since DL models have a strong tendency to improve with the availability of larger data sets (8). To counter these limitations, an alternative approach, weak supervision (7), has been introduced. Weak supervision comprises three types: incomplete supervision, inexact supervision, and inaccurate supervision (Table).

An incompletely labeled data set combines labeled data with unlabeled data (yellow circle in Fig 1). This is the realm of semi-supervised learning and active learning, which we regard as weakly supervised learning algorithms following the

definitions of Zhou (7). We discuss these approaches in depth later. Briefly, semi-supervised learning exploits unlabeled data in addition to labeled data, without human intervention. Active learning aims to reduce the amount of data that needs to be annotated by an expert.

Inexact labels are labels that are not a precise representation of the target. For example, instead of labeling the exact location of each malignant lesion on a mammogram, an inexact label might just denote the presence of such a lesion somewhere on the image (orange circle in Fig 1). This particular type of inexact label is referred to as a case label—that is, it is determined at the level of the case. Note that the definition of *inexact* depends on the context. For example, for a DL model tasked with flagging cases for the presence of malignant lesions without specifying the exact location, these case labels would be regarded as exact.

Finally, inaccurate labels are labels that contain errors. For example, an inexperienced labeler might denote a benign lesion as being malignant, or a labeler working at an accelerated pace might outline the boundaries of a lesion incorrectly (red circle in Fig 1). Inaccurate supervision uses data sets that include some inaccurate labels for these or other reasons.

The main takeaway is that the databases required to train a DL model with weakly supervised learning require less labeling by experts. Such databases require expert labels for only a minority of the images while the majority of the data can be annotated by nonexperts or by automated processing of existing radiology reports. Erroneous labels are explicitly allowed within this paradigm, with the idea that the sheer size of the database will lead the DL model to incorporate the correct concept, even in the presence of a few faulty labels.

Incomplete Supervision: Reducing the Number of Labels

Incomplete supervision involves training models with a combination of labeled and unlabeled data to reduce annotation efforts. Two important methodologies within incomplete supervision are semi-supervised learning and active learning.

Semi-supervised Learning

Semi-supervised learning uses a combination of labeled and unlabeled radiologic images. A primary type of semi-supervised learning employs self-supervised learning. In this approach, generalist DL models are created by performing a first training session—the so-called pretraining stage—on unlabeled data sets (9,10), such as unlabeled imaging data from CT, MRI, or radiography (Fig 3). These pretrained models are referred to as foundation models when they have been trained on vast data, typically through self-supervision, and are adaptable to a variety of downstream tasks (11,12). Self-supervised pretraining teaches the model to identify relevant features on images. For example, contrastive learning is based on the idea that most relevant image features remain unchanged by slightly altering the images (9,13,14). The model learns to cluster images that are altered versions of the same original image close together. Typical alterations are changes in the contrast or turning the image on its side. Another paradigm for self-supervised learning is

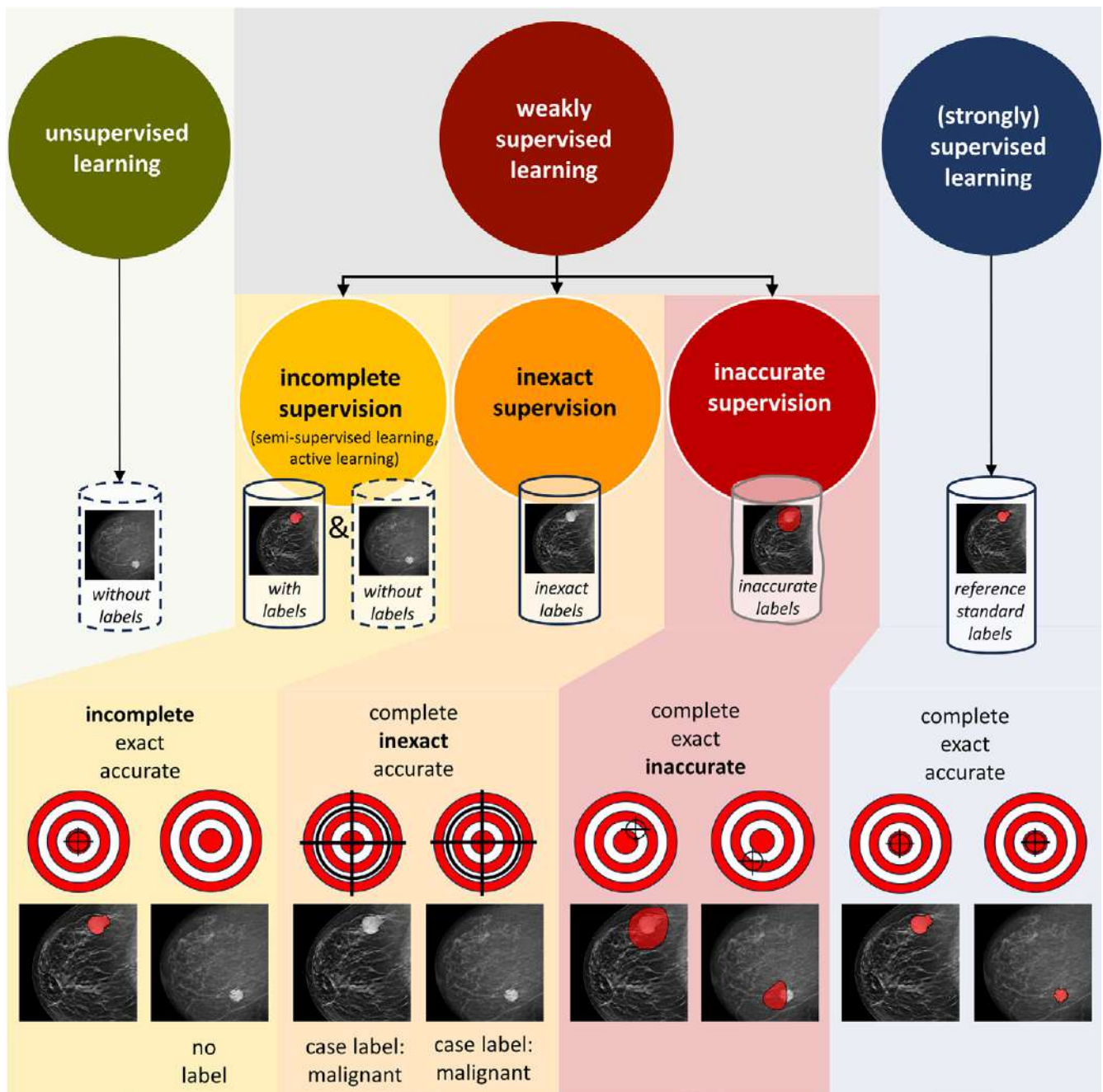


Figure 1: Diagram of different levels of supervision. Unsupervised learning uses no labels; weakly supervised learning uses incomplete, inexact, or inaccurate labels; and strongly supervised learning uses complete, exact, and accurate reference standard labels. The different types of labels are illustrated at the bottom for the example of segmenting malignant lesions on mammograms.

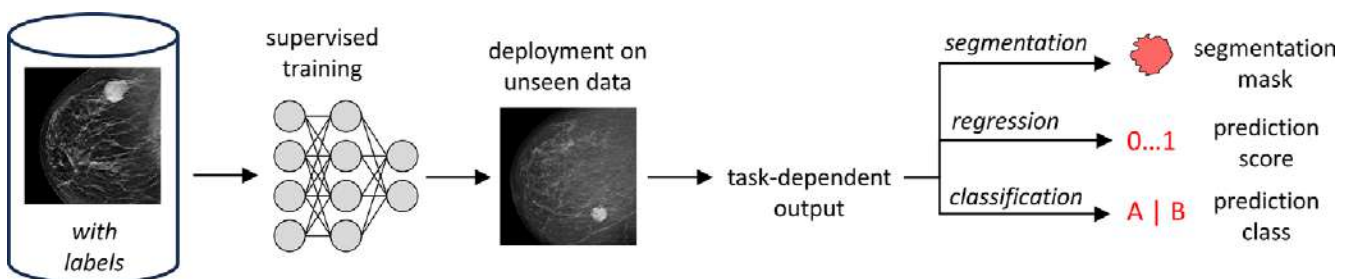


Figure 2: Diagram of supervised learning workflow. Supervised training uses labeled images to generate a model that typically generates segmentation masks, scores, or classes for unseen cases.

Overview of Types of Weakly Supervised Learning

Type	Description	Example
Incomplete supervision	Model trained with a mix of labeled and unlabeled data; subtypes are (a) semi-supervised learning, which uses labeled and unlabeled data for model training and (b) active learning, in which, from a pool of images, only the most informative images are annotated, for ideal performance while minimizing annotation cost	Semi-supervised learning: tumor segmentation model that is trained on MRI scans that contain segmentation labels and MRI scans that do not contain segmentation labels; active learning: tumor model for which only the most informative MRI scans from a large pool of unlabeled data are annotated
Inexact supervision	Model trained with labels that are not as exact as desired, such as case labels instead of segmentation labels	Tumor localization model that is not given the exact position of a tumor, but only the information that a tumor is present somewhere in the image
Inaccurate supervision	Model trained with a mix of data with correct labels and data with erroneous labels	Liver disease detection model that is trained with labels that were generated automatically from free-text radiology reports, some of which are inaccurate

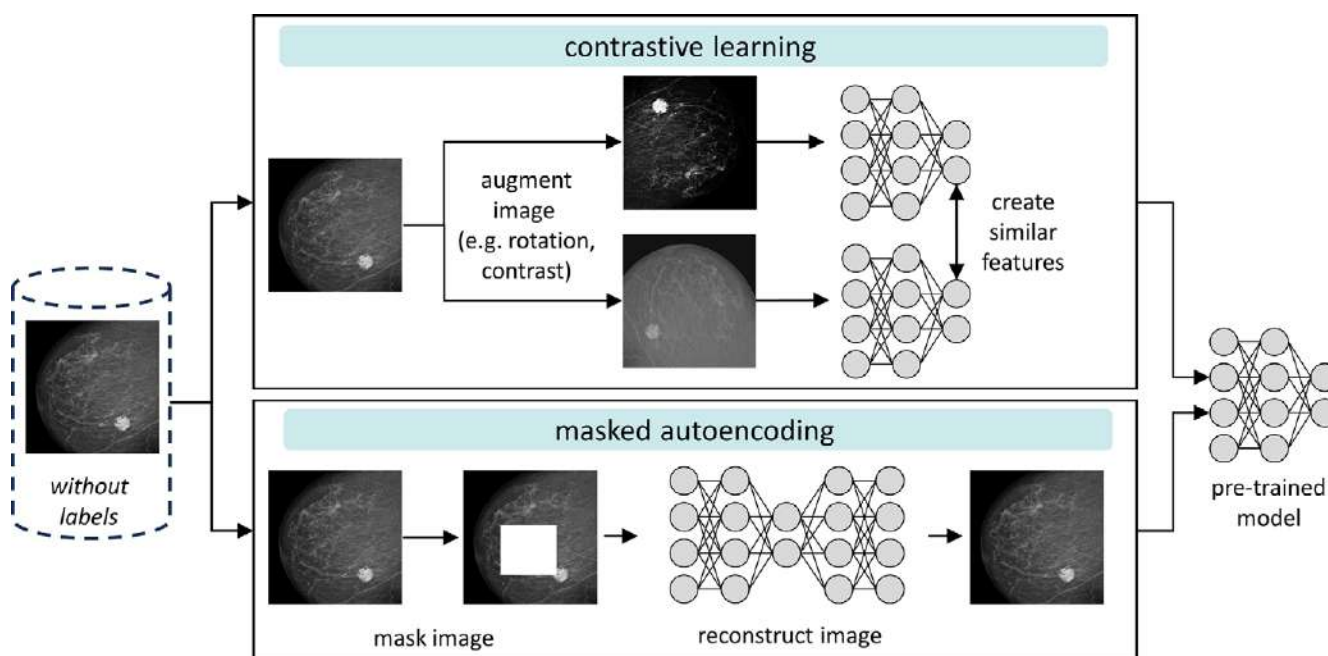


Figure 3: Diagram of self-supervised learning for model pretraining. Self-supervised learning trains the model without labels to extract relevant features from the images. Contrastive learning tasks the model to cluster images that are altered versions of the same original image close together. Masked autoencoding tasks the model to reconstruct the missing parts of the masked image.

masked autoencoding, where parts of images are deleted, and the model is tasked with reconstructing the missing parts (15).

These pretrained models can subsequently be trained with labeled data on various downstream tasks, such as disease diagnosis (14,16), prognosis (14,17), or segmentation (18,19). Self-supervised pretraining typically improves model performance on downstream tasks, in particular when only a few labeled images are available (14,18). One study that exemplifies this type of semi-supervised learning was conducted by Azizi et al (14). This study aimed at predicting atelectasis, consolidation, pulmonary edema, pleural effusion, and cardiomegaly from chest radiographs. To this end, the model was first pretrained with self-supervised contrastive learning using a large collection of chest radiographs. Then the model was fine-tuned using labeled chest radiographs. The authors reported that pretraining the model

with self-supervised learning improved model performance and that the pretrained model required substantially fewer labeled images to reach clinically applicable performance.

Another type of semi-supervised learning is pseudo-labeling, where only a fraction of the images are labeled and the remaining labels are inferred automatically (20). This training paradigm was used by Wang et al (21) for breast tumor segmentation on MRI scans. Instead of manually annotating all tumors, the model was trained using only 20 labeled MRI scans together with 180 unlabeled ones. The model inferred the missing labels and iteratively corrected them during training. Using this specialized training approach, the model reached a segmentation performance comparable to that of a model trained using fully labeled data, thereby greatly reducing the annotation workload for radiologists.

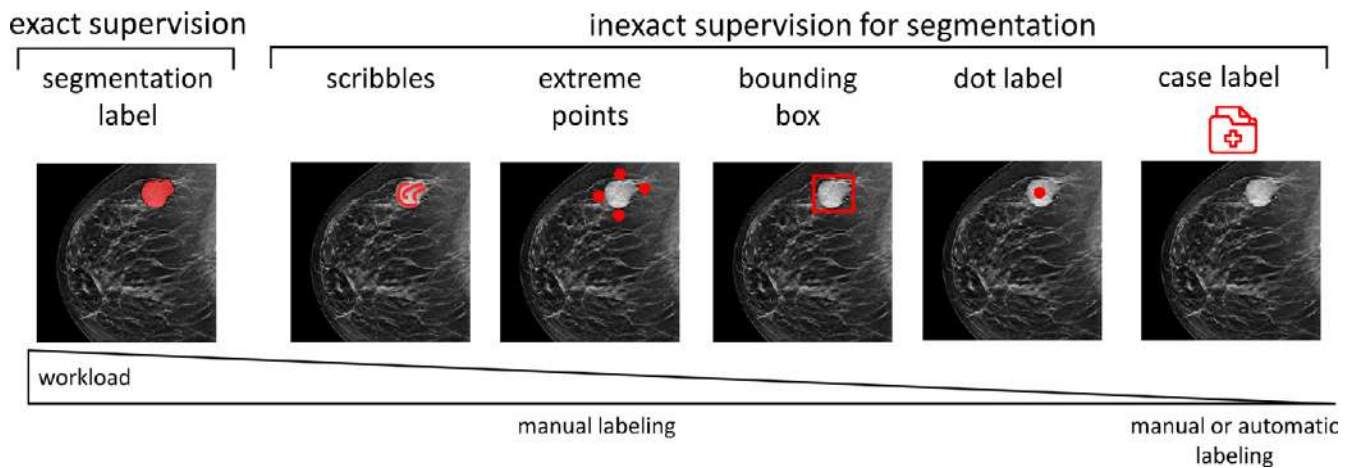


Figure 4: Diagram of exact and inexact labels for training segmentation models. Inexact supervision uses inexact labels (ie, labels that are not a precise representation of the target) for training segmentation models. Commonly used labels for radiologic images are illustrated for lesion segmentation on a mammogram.

A similar approach can be adopted when labels are inaccessible, for instance, when follow-up information is missing. For example, Guo et al (22) used incomplete data to train a model predicting prognosis in patients with extranodal natural killer/T-cell lymphoma, nasal type, based on pretreatment PET/CT imaging. The labeled data set with complete follow-up information was used to train an initial model, which subsequently inferred labels for the patients who had incomplete or missing follow-up information. Using both the existing labels together with the inferred labels for model training instead of the existing labels alone improved the downstream prognosis prediction.

Semi-supervised methods are also used in the context of image quality enhancement. For example, Desai et al (23) proposed a model for MRI reconstruction from undersampled acquisitions (undersampled acquisition speeds up the process but typically compromises image quality). Training the reconstruction model in a supervised fashion would require numerous fully sampled acquisitions. However, in clinical practice, often more undersampled scans are available than fully sampled ones. Desai et al used just one fully sampled and 13 undersampled acquisitions for training. Adding undersampled acquisitions to the training improved reconstruction. Evaluating the model's performance, two board-certified radiologists reported that the undersampled reconstructions produced by the model were comparable in quality to the fully sampled references for the analyzed knee MRI scans.

Active Learning

Active learning is a separate concept within incomplete supervision. Its fundamental idea is to select only specific images for labeling from the pool of unlabeled images—that is, those that are most likely to enhance the model's performance (24). This technique has been used to train a model to segment livers on CT scans (26), which in principle is not a hard task for DL. Segmentation algorithms have been around for several years now (25) and work well on three-dimensional images given sufficient amounts of training data (19). However, an algorithm that succeeds in simple cases but fails in cases

that are challenging and hard to segment due to unusual organ shapes or image artifacts would be of limited clinical use. It is therefore important that a DL model learns to deal with such challenging cases. This can be accomplished either by building an extensive training set that contains these outlier cases or by using weakly supervised learning, as shown by Marinelli et al (26) for a liver segmentation model. Instead of training the DL model in one go, the authors first trained it using a relatively small data set comprising 131 CT scans with segmentation labels outlining the liver. Then the model was tasked with segmenting unlabeled CT scans, and the authors identified particularly challenging cases, that is, ones where the model either strongly over- or underestimated liver volume. These challenging cases were then manually segmented by medical experts and added to the training set for retraining. This procedure, active learning, improved the quality of the model-generated liver segmentations.

Active learning was used in a similar way by Tallam et al (27), who investigated abdominal CT biomarkers derived from pancreas imaging. To fully automate this pipeline, automated segmentation of the pancreas was necessary. However, developing a reliable segmentation model proved difficult since small inaccuracies in the segmentation can, for example, strongly influence the automatically calculated fat content of the pancreas. The problem was addressed by iteratively developing a segmentation model, identifying challenging cases for it, manually segmenting these cases, and then training an improved model based on the enriched training set. In this particular study, this helped to establish more reliable biomarkers for diabetes type II prediction.

Inexact Supervision: Creating Labels Faster

For training segmentation models that precisely localize objects of interest such as organs or lesions, an effective method is to accelerate the labeling process itself. This is the realm of inexact supervision where, instead of precise pixel- or voxel-level segmentation labels, only coarse-grained labels are used for training (Fig 4). We divide the inexact training labels into two types: those typically obtained manually and case labels, which can also be obtained automatically.

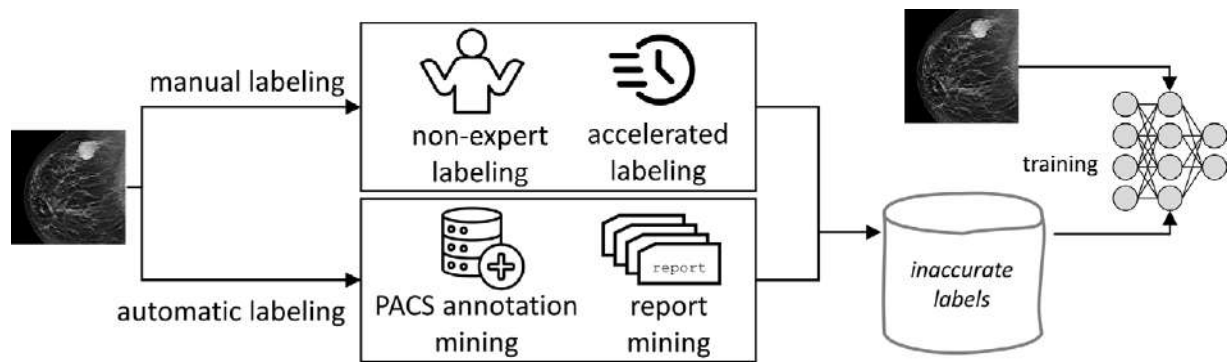


Figure 5: Diagram of inaccurate supervision. Inaccurate supervision uses data sets that contain some inaccurate labels (ie, labels that contain errors) for model training. Manually obtained labels may be inaccurate due to the labeler's inexperience or accelerated labeling. Automatically obtained inaccurate labels may be sourced from a picture archiving and communication system (PACS) or from free-text radiology reports.

Inexact Dot, Bounding Box, Extreme Point, and Scribble Labels

Manually obtained inexact labels, such as dot labels, bounding box labels, extreme point labels, and scribble labels, can be used to train segmentation models, particularly for organ or tumor segmentation. For instance, Kervadec et al (28) used dot labels to train a model to segment the prostate on MRI scans. Similarly, Wang et al (29) used three-dimensional bounding box labels to train a model to segment the prostate, bladder, and rectum on three-dimensional CT scans. Dorent et al (30) used extreme point labels to train a model to segment vestibular schwannomas on brain MRI scans. Liu et al (31) used scribble labels to train a model to segment COVID-19-infected regions on lung CT scans. Here, the authors chose scribbles over bounding boxes and points due to their ability to better capture the complex shapes of the affected regions in the lung. Dot labels, box annotations, extreme points, and scribbles can be obtained relatively quickly compared with precise voxel- or pixel-level labels; some studies have found a speedup of about 13–44-fold (29–31). Although some studies found that using these types of inexact labels instead of voxel- or pixel-level labels resulted in a slight reduction in segmentation performance (28–30,32,33), others reported similar performance between the two approaches (31,34).

Case Labels

Case labels can be obtained in the most time-saving and scalable manner, either through manual labeling or through automatic labeling by LLMs, as will be described later. Several technical articles have shown the feasibility of training a model to localize lesions on radiologic images using case labels only. In the work of Li et al (35), a model was trained to segment tumors on breast US images based on case labels indicating the presence or absence of tumors on the images. Initially, the model created a heatmap that coarsely localized a tumor. This initial segmentation map was later refined by only accepting proposed tumor regions in the fat and mammary gland, thereby improving segmentation performance. Similarly, other studies used case labels for tumor segmentation on brain MRI scans (36,37) or for pneumothorax localization on chest radiographs (38,39). It is important to note that the performance of segmentation models trained with case labels (36–39) is generally still worse than that

of segmentation models trained with exact voxel- or pixel-level labels (40,41). However, recent works (42–44) have narrowed the performance gap between training with inexact image-level labels and exact pixel-level labels for semantic segmentation on natural nonradiologic images, and this trend is likely to extend to radiology.

To improve segmentation performance, Nam et al (45) combined case labels with precise segmentation labels, both obtained manually. In this study, the authors aimed to train a DL model for the detection of malignant pulmonary nodules on chest radiographs and employed weakly supervised learning to reduce annotation efforts. Instead of segmenting each nodule in the full data set of 8625 radiographs, radiologists segmented nodules on only 3213 radiographs. The detection algorithm was then trained using the manual case labels for all 8625 radiographs and the pixel-level labels for the 3213 radiographs, representing a mix of inexact supervision and exact supervision. The authors mentioned that this method was not only cost-effective but also enabled the trained model to learn features of nodules that may be missed by radiologists. In a follow-up clinical study (46), the same research team showed that a model trained with a combination of inexact and exact supervision reduced reading errors when used as a second reader by detecting lung cancers that would otherwise have been overlooked.

Inaccurate Supervision: Toward Fully Automated Labeling

Another way to minimize the labeling workload is to use inaccurate supervision, where labels are used for model training that may contain some errors (Fig 5) (47). Radiologic images might be tagged with incorrect labels because of radiologist bias or lack of experience, hurried efforts during the labeling process, or automatic label generation.

One study that used this training paradigm was conducted by Liu et al (48). In this study, the authors proposed a training approach to handle noisy segmentation labels by correcting them during training. This approach was employed for the use case of segmenting spinal cord white matter and gray matter on MRI scans. To test the performance of this approach, the reference standard segmentation labels were artificially corrupted by enlarging, shrinking, or deforming them, to simulate

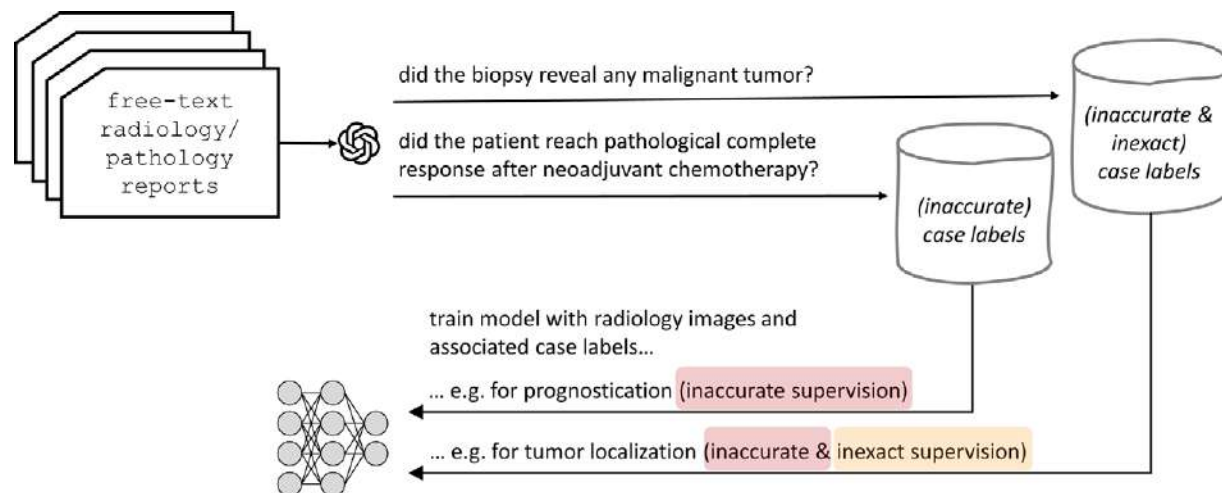


Figure 6: Diagram of using large language models to extract labels from reports. Case labels can be extracted from free-text reports to automatically create large data sets with potentially inaccurate and inexact labels for model training.

real-world segmentation labels of bad quality. These corrupted labels were then used to train the segmentation model. The experiments showed that even when half of the 341 reference standard segmentation labels used for training were heavily corrupted, the model segmented the spinal cord white matter and gray matter with similar performance compared with the model trained with the reference standard segmentation labels. If models are designed to be robust against a certain number of inaccurate labels, resources could be saved by speeding up the labeling process or partially delegating labeling to less experienced observers.

When creating a data set to train a DL model, it is highly desirable to harness data that are already available. In the study conducted by Swinburne et al (49), a model was trained to detect and localize tumors on brain MRI scans using labels extracted from the picture archiving and communication system of a cancer center. Typically, radiologists at this center marked tumor locations and sizes on MRI sections using two perpendicular lines. To harness this information, the MRI scans that had at least one line annotation on an axial section were automatically exported. From these line annotations, a two-dimensional bounding box was derived for each annotated tumor. Each two-dimensional bounding box was then extended to adjacent axial sections, creating a three-dimensional bounding box that approximated the tumor boundaries. These bounding box labels were inaccurate because the radiologists often did not annotate all tumors within each scan, and approximately 1.1% (about 118 of 10 623) of the extracted bounding boxes used for training marked nontumor regions. The authors reported that the model trained with all such available annotations effectively detected and localized tumors on brain MRI scans.

Another approach involves using information from reports, such as free-text radiology reports, that often contain unstructured data. Tushar et al (50) leveraged radiology reports to train a model to detect common abnormalities on CT scans, for example kidney atrophy or kidney lesions. To this end, labels indicating whether the investigated abnormalities were present were extracted from radiology reports using a rule-based language

processing technique. Then these computer-extracted labels were used for model training. This approach enabled model training without having to manually parse the reports. This approach represents inaccurate supervision because these extracted labels were sometimes incorrect. Specifically, the researchers estimated that 1% (about six of 589) of the extracted training labels indicating kidney atrophy were incorrect and 9% (about 130 of 1447) of the extracted training labels indicating kidney lesions were incorrect. Using these extracted inaccurate labels, the model was able to detect abnormalities such as kidney atrophy and kidney lesions. Of note, the next section discusses using LLMs to extract case labels, which generally achieves higher accuracy than using simpler rule-based methods.

LLMs as Enablers of Large-Scale Weak Supervision

Recent technical advances in natural language processing have considerably improved the capabilities of so-called LLMs over the past 2 years. Proprietary models such as OpenAI's ChatGPT (51) or Google's PaLM (52) have demonstrated an understanding of human-written text that now makes it possible to extract labels on a huge scale. Recent studies have shown that structured reports can now be generated from written prose radiology reports (53,54). We argue that this will lead to the availability of large radiologic data sets that can be used to develop DL models through weakly supervised learning (Fig 6).

As an illustrative example, consider the development a model for the detection and localization of malignant tumors on digital mammograms. By directly prompting the LLM with the question "Does the radiology report mention the presence of a malignant tumor?" or "Does the pathology report mention the patient responding to neoadjuvant chemotherapy?" and by applying this scheme to a hospital's database of reports, one could efficiently build a database of mammograms paired with labels. These labels could subsequently be used to train models to detect and localize malignant tumors on mammograms (55) or predict the therapy response of patients with breast cancer based on mammograms (56).

Inevitably, such labels will be lower-quality labels, in particular when using the smaller open-source LLMs that can be deployed on-site. These open-source models are often trained on a smaller data set and contain fewer parameters, so they tend to exhibit lower performance in processing information (57). This in turn leads to labels that can be erroneous (ie, inaccurate). The labels might also be inexact: From radiology reports, LLMs will be able to extract the information that breast cancer was present in the examination but will not be able to provide pixel-perfect locations of lesions. By definition, weakly supervised learning techniques are very well suited to deal with these situations (Table).

Future Directions

Looking forward, regarding incomplete supervision, we encourage research groups with access to large radiologic image databases and substantial computational resources to train foundation models through self-supervised learning. These pretrained foundation models should also be published, as has already been done by several groups (58–60), to enable researchers with limited resources to fine-tune these pretrained models for specific downstream tasks such as diagnosis, prognosis, and segmentation. Fine-tuning would be possible even with small cohorts, thereby democratizing the field and fostering the exploration of rare conditions. Apart from that, further studies should explore self-supervised learning techniques for various types of two-dimensional and three-dimensional radiologic images. These studies should specifically investigate optimal transformations for contrastive learning and modifications for reconstruction. This is important since many popular self-supervised learning techniques were not initially developed for processing medical images (13,15,58).

In addition, future work should focus on extracting structured case labels from free-text reports, for example, radiology or pathology reports. Specifically, future studies should systematically assess how accurately LLMs can extract case labels from free-text reports and under which conditions they fail. These assessments could include scenarios where LLMs handle non-English reports, encounter uncommon abbreviations, or deal with rare findings. Furthermore, it is important to validate the performance of models trained with these automatically generated, possibly inaccurate, case labels. Specifically, in addition to investigating models that predict abstract targets such as therapy response, researchers should also explore whether segmentation models, when trained with a vast amount of extracted case labels, can achieve performance comparable to that of models trained with precise voxel- or pixel-level segmentation labels.

Weakly supervised learning reduces human supervision in the development of DL models. However, this may make the generalizability of model performance and the certainty of the model outputs more opaque compared with supervised learning approaches. Consequently, it is crucial to validate weak supervision-trained models with accurate reference standard labels and to investigate what level of human supervision is necessary in the clinical use of these models.

Conclusion

The bottleneck for using deep learning (DL) models in radiology is often not the acquisition of a large number of images but rather obtaining the labels. Weakly supervised DL marks a shift from using manually obtained complete, exact, and accurate labels to leveraging data sets with incomplete, inexact, or inaccurate labels that require little or no additional manual annotation from experts. With this shift, it becomes feasible to leverage vast clinical databases to swiftly construct large and balanced data sets for model training. With careful validation of the trained models, weakly supervised learning could accelerate the development of DL models for clinical applications including diagnosis, prognosis, and segmentation.

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A Hierarchical Self-Consistent Regularization Approach to Satellite Image Time Series Classification

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Abstract—Deep learning has become increasingly important in remote sensing image classification due to its ability to extract semantic information from complex data. Classification tasks often include predefined label hierarchies that represent the semantic relationships among classes. However, these hierarchies are frequently overlooked, and most approaches focus only on fine-grained classification schemes. In this paper, we present a novel Semantics-Aware Hierarchical Consensus (SAHC) approach to learn hierarchical features and relationships by integrating hierarchy-specific classification heads within a deep network architecture, each specialized in different degrees of class granularity. The proposed approach employs trainable hierarchy matrices, which guide the network through the learning of the hierarchical structure in a self-consistent manner. Furthermore, we introduce a hierarchical consensus mechanism to ensure aligned probability distributions across different hierarchical levels. This mechanism acts as a weighted ensemble being able to effectively leverage the inherent structure of the hierarchical classification task. The proposed SAHC method is evaluated on two benchmark datasets with different degrees of hierarchical complexity on different tasks, considering varying spectral and spatial resolutions. Experimental results show both the effectiveness of the proposed approach in guiding network learning and the robustness of the hierarchical consensus for remote sensing image classification tasks. The codes will be released at <https://github.com/rslab-unitrento/sahc>.

Index Terms—Consensus, consistency, deep learning (DL), hierarchical, multi-granularity, semantics-aware, Sentinel-2, time series, multispectral, very high resolution, remote sensing

I. INTRODUCTION

IMAGE classification and semantic segmentation are central tasks in Remote Sensing (RS) image processing. In recent years, the proliferation of RS data and open distribution policies have significantly increased the availability of Land-Use and Land-Cover (LULC) products, along with their corresponding classification schemes [1]. These products play a critical role in various applications and are widely used by government agencies, municipalities, and ministries. However,

global and regional LULC maps often differ in terms of resolution and are affected by semantic inconsistencies in class definitions and production algorithms [2]. While Deep Learning (DL) methods have revolutionized LULC mapping by capturing complex spatial and spectral dependencies, standard approaches typically treat classification as a flat, single-level task. In practice, Land-Cover (LC) classes are naturally organized into hierarchical label structures that reflect intrinsic levels of semantic granularity. This multi-level hierarchy is a property of the physical landscape. For instance, within a single dataset, a specific target might be unambiguously identified at a coarse-grained level (e.g., forest), while its finer-grained classification (e.g., deciduous versus coniferous) remains uncertain due to inherent label ambiguity, complex scene compositions, or the physical limits of the sensor. Neglecting this inherent hierarchical structure discards valuable semantic information and can lead to suboptimal or inconsistent predictions, particularly when dealing with label ambiguity or complex RS scenes. An example of this hierarchical problem definition can be seen in Fig. 1.

In the literature, most of the works related to RS image classification focus on single-level approaches, discarding the hierarchical information contained in the data. Hierarchical classification schemes, such as the CORINE Land Cover (CLC) maps, have mostly been adopted in object-based image classification methods based on Very-High Resolution (VHR) satellite images [3], often considering a limited number of classes to simplify the classification task. To validate the hierarchical structure, few works analyzed the inter-level relationships, considering decision trees in an object-based classification scenarios [4], or validation approaches based on intercomparison of different maps [5]. Instead, the wide majority of works in the RS domain focus on single-level approaches, discarding the hierarchical information contained in the data. While these methods show promising results at individual hierarchical levels, they fail to incorporate the hierarchical structure of the data into the decision-making process, overlooking important information included in the source data.

Only recently, hierarchical and multi-granularity classification methods have emerged to address these challenges in RS. These methods leverage the hierarchical structure to encode semantic relationships among labels, enabling knowledge transfer across different levels of granularity. Recent studies have demonstrated the benefits of adaptive label granularity

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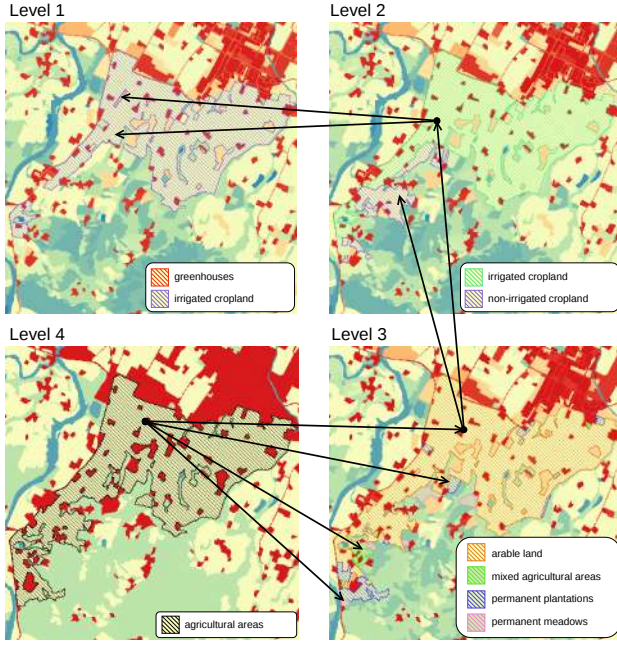


Fig. 1. Hierarchical classification task example. Level 4 represents the coarsest class definition, where each class is progressively broken down into its fine-grained representations.

selection to balance classification accuracy and specificity [6], as well as the integration of parent-child relationships and sibling mutual exclusivity to improve hierarchical representation learning [7]. Despite these advancements, integrating hierarchical information effectively into the learning process remains an open challenge. Many existing methods rely on complex, task-specific architectures or rigid rule-based post-processing steps, which limit their flexibility across different DL backbones and fail to explicitly enforce probability consistency across different hierarchical levels in a generalized manner.

To address these limitations, we propose a novel and modular Semantics-Aware Hierarchical Consensus method for learning hierarchical features and relationships. Our approach integrates hierarchy-specific classification heads within standard DL architectures, each specialized in different degrees of class granularity. By introducing trainable projection matrices and explicitly modeling the semantic relationships between these levels, the proposed method ensures consistent predictions across the hierarchy without relying on rigid aggregation rules.

The key contributions of this work can be summarized as follows:

- 1) We define a hierarchical approach to semantic segmentation and scene classification that integrates multi-level classification modules with DL backbones, along with trainable projection matrices to model hierarchical semantic relationships.
- 2) We propose a novel self-consistent consensus loss function that enforces consistency across different hierarchical levels, improving the semantic coherence of the

classification results.

- 3) The proposed method is general and can be applied to any DL backbone and hierarchical classification scheme, offering flexibility in defining architecture details.
- 4) We demonstrate the effectiveness of the proposed approach on two RS datasets that include images with varying resolutions and different complexities in their hierarchical relationships.

The remainder of this paper is organized as follows. Section II reviews related works on hierarchical semantic segmentation. Section III details the proposed hierarchical consensus approach. The datasets and experimental setup are described in Section IV while Section V presents and discusses the experimental results. Finally, Section VI concludes the paper and outlines future research directions.

II. RELATED WORKS

Image classification and semantic segmentation are widely studied tasks in RS image processing, with DL models achieving state-of-the-art performance across various domains. Standard approaches typically treat classification as a “flat”, single-granularity problem, predicting probabilities over a set of mutually exclusive classes without explicitly considering potential semantic relationships or hierarchical structures among them. However, many real-world RS datasets possess inherent taxonomies where classes can be naturally organized into hierarchies (e.g., standardized land-cover systems, target recognition taxonomies with super- and sub-categories). Ignoring this structure can lead to suboptimal performance, as standard loss functions treat all misclassifications equally. Without an explicit hierarchical representation, model fail to differentiate between errors involving closely related classes and those involving semantically distant ones.

Leveraging hierarchical label structures during model training and inference has emerged as a promising direction to improve accuracy and semantic understanding. Early work demonstrated that standard Convolutional Neural Networks (CNNs) implicitly capture some hierarchical information, with shallower layers grouping high-level categories and deeper layers specializing in finer distinctions [8]. Building on this, researchers have explored explicit methods to incorporate hierarchy. Typically, this is achieved by mapping the label hierarchy to specific network architectures or by introducing loss functions that handle the correlation between different granularity tasks [9]–[11]. For instance, several methods propose level-wise hierarchical cross-entropy losses based on conditional probabilities, or introduce custom features to minimize the divergence between aggregated predictions at different hierarchical levels, aiming to reduce the severity of misclassification. Bertinetto *et al.* [12] proposed a level-wise hierarchical cross-entropy loss based on conditional probabilities within the hierarchy and used mapping functions to embed class relationships. Similarly, Garg *et al.* [13] introduced the Hierarchy Aware Features (HAF), incorporating multiple losses, including minimizing the Jensen-Shannon Divergence (JSD) between aggregated predictions at different hierarchy levels and enforcing geometric consistency in the feature

space, aiming to reduce the severity of misclassifications. In [14], the authors proposed hierarchy and exclusion graphs to model semantic relations between two labels assigned to the same objects. They considered exclusions, overlaps, and subsumptions, and provided a detailed theoretical analysis of the structure of the hierarchical problem. Building on this work, a combinatorial loss is proposed by Chen *et al.* [15], which addresses the scenario where samples might be labelled at different levels of granularity. The loss is designed to maximize the marginal probability of the observed ground truth label by aggregating information from related labels in the hierarchy. The authors introduced a Hierarchical Residual Network (HRN) architecture, where features from parent levels act as residual connections to inform child-level features, explicitly facilitating hierarchical feature interaction. Following this work, Zhou *et al.* [16] proposed a multi-granularity archaeological dating method that depends on an explicit knowledge-guided relation graph, employing a focal-type probability classification loss to mine the relationship between specific object attributes and their broader eras. Their network leverages a bidirectional multi-granularity module with gradient truncated addition to explicitly fuse coarse- and fine-grained features. Similarly, Cai *et al.* [17] introduced HierSwin, a hierarchical vision transformer, which explicitly models the semantic correlations embedded within a predefined three-level hierarchical tree to gradually improve accuracy from fine- to coarse-grained classifications.

More recently, some works addressed the Multigranularity Classification (MGC) in RS to handle targets that are discernible only at specific variable semantic levels due to differences in spatial resolution, sensor modalities, and data quality. Chen *et al.* [6] introduced an accuracy-specificity curve to adaptively select task-dependent prediction thresholds. Their approach also employs a penalty term based on the global semantic distance across the label tree to adjust class logits. Addressing a similar challenge in ship classification, Chen *et al.* [7] proposed a hierarchical contrastive learning framework. By explicitly incorporating parent-child dependencies and mutual exclusivity among sibling categories, their method aligns class-specific representations with semantic relationships and enforces probability coherence across levels through a consistency loss.

Despite these significant advancements, many existing multigranularity and hierarchical learning methods still rely on the assumption that a predefined and accurate hierarchy is available, where no hierarchy violation is possible. This requires rigid post-processing rules, fixed decision thresholds, or highly specific non-modular architectures. Furthermore, many existing techniques require explicit guidance or architectural modifications tailored to a specific fixed hierarchy [18], potentially limiting their flexibility and ability to adapt if the underlying hierarchical understanding evolves or is uncertain. For example, some approaches decompose the problem into sequential decisions or binary classifications guided by the known structure [19], [20], rather than allowing the model to learn or refine hierarchical relationships implicitly from the data. The challenge remains to develop methods that can effectively exploit hierarchical semantics, even when the hierarchy

is imperfect, labels are provided at varying granularity levels, or needs to be partially inferred in a robust and data-driven manner.

III. PROPOSED METHODOLOGY

In this work, we address the above-mentioned challenges by proposing a novel deep learning framework for hierarchy-aware image classification. Our approach models the hierarchical semantic relationships within image data by integrating hierarchical loss components with a robust consensus mechanism. This allows the model to leverage potential hierarchical structures effectively while mitigating the impact of potential inaccuracies or inconsistencies in the predefined taxonomy. Specifically, our method aims to learn and exploit hierarchical relationships in a more implicit, self-consistent fashion within a hierarchy-aware architecture, reducing the reliance on rigid, externally defined structure.

A. Hierarchical Problem Formulation

Let us denote column vectors by lowercase bold letters (e.g., $\mathbf{v}$) and matrices by uppercase bold letters (e.g., $\mathbf{B}$). The i th component of a vector $\mathbf{v}$ is denoted by v_i , while the i th row and j th column of a matrix $\mathbf{B}$ are denoted by $\mathbf{B}_{i,\cdot}$ and $\mathbf{B}_{\cdot,j}$, respectively. Then, let us consider a user-defined hierarchy label tree with H levels, where the leaves represent the finest level of detail. We denote each hierarchical level as $h \in [1, 2, \dots, H]$, ranging from the coarsest level $h = 1$ to the finest one $h = H$. The root node $h = 0$ is not considered, as it represents the trivial case in which all classes are mapped into a single class. Let $\mathbf{x} \in \mathbb{R}^b$ be the input vector, where b is the number of input features. Its associated labels are $y^h \in \Omega^h$, where $\Omega^h = \{\omega_1^h, \dots, \omega_{|\Omega^h|}^h\}$ is the set of labels at hierarchical level h , with $|\Omega^h|$ number of labels. For every $h_1 < h_2$, we have that each fine-grained class $\omega_j^{h_2}$ is mapped to a coarse-grained class $\omega_i^{h_1}$, and $|\Omega^{h_1}| \leq |\Omega^{h_2}|$. We define this mapping relationship by $\omega_j^{h_2} \implies \omega_i^{h_1}$. Let $\mathcal{D} = \{(\mathbf{x}_n, Y_n) | n = 1, \dots, N\}$ be the training set, where for each input feature vector $\mathbf{x}_n$, we have an associated set $Y_n = \{y_n^1, \dots, y_n^h, \dots, y_n^H\}$, which denotes the ground truth labels at all H hierarchical levels, for a total number of N training samples. The labels in the dataset satisfy a nested label structure, where each class at a finer level is contained within a class at a coarser level.

B. Fully-Supervised Multi-Level Learning

The proposed methodology can be applied to any backbone network by simply introducing the required additional hierarchical classification modules. We refer to each of the hierarchical classification modules $\mathbf{g}^h : \mathbb{R}^d \rightarrow \mathbb{R}^{|\Omega^h|}$ on top of a given backbone network as the composition $\mathbf{g}^h \circ \mathbf{f} : \mathbb{R}^b \rightarrow \mathbb{R}^{|\Omega^h|}$, where $\mathbf{f} : \mathbb{R}^b \rightarrow \mathbb{R}^d$ is the backbone feature extractor. Fig. 2 provides a high-level overview of the considered multi-level classification approach.

Each classifier $\mathbf{g}^h$ estimates class-posterior logits. Let $\hat{\mathbf{p}}^h(\mathbf{x}) \in \mathbb{R}^{|\Omega^h|}$ be the vector of estimated class-posterior probabilities at level h given $\mathbf{x}$, i.e., $\hat{\mathbf{p}}^h(\mathbf{x}) =$

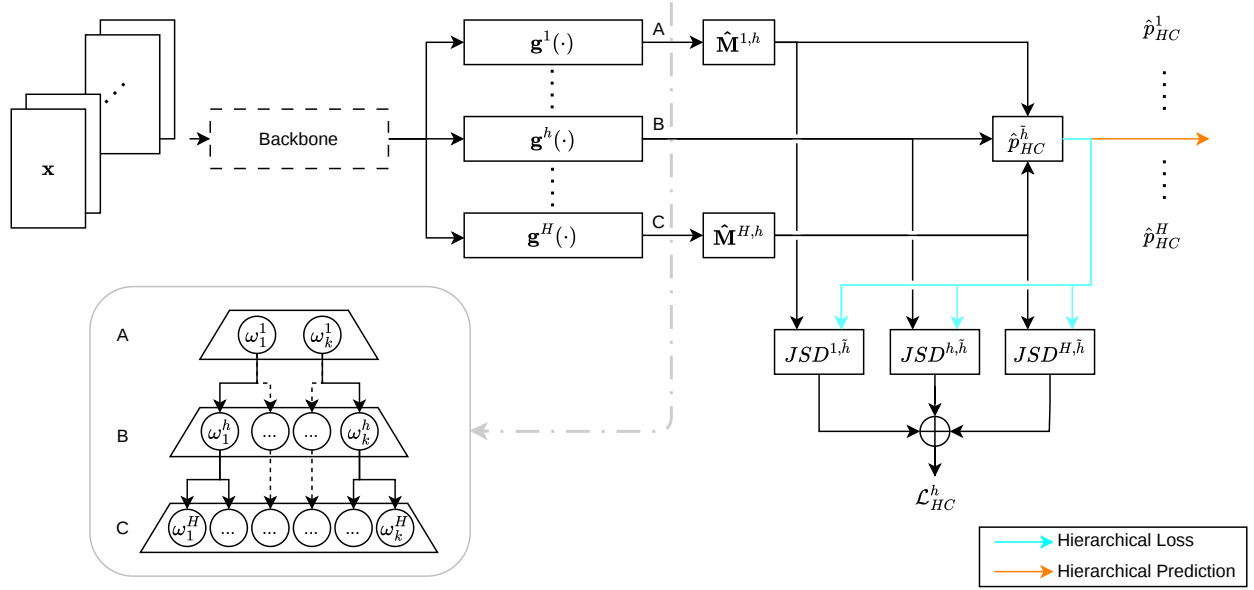


Fig. 2. Illustration of the proposed hierarchical architecture with the hierarchical consensus calculated on $h = \bar{h}$. Heads tailored to different granularity levels are reprojected on the target level and averaged to extract the consensus probability $\hat{p}_{HC}^{\bar{h}}$. Other than for prediction, this consensus probability is used to calculate the hierarchical self-consistency loss $\mathcal{L}_{HC}^{\bar{h}}$.

$[\hat{P}(\omega_1^h|\mathbf{x}), \dots, \hat{P}(\omega_{|\Omega^h|}^h|\mathbf{x})]^T$, where T refers to the transpose operator. Then, the class-posterior probabilities $\hat{\mathbf{p}}^h$ for the input sample $\mathbf{x}$ at level h can be retrieved as follows:

$$\hat{\mathbf{p}}^h(\mathbf{x}) = \sigma[\mathbf{g}^h(\mathbf{f}(\mathbf{x}))], \quad (1)$$

where $\sigma: \mathbb{R}^d \rightarrow [0, 1]^d$ is the *softmax* operator, with $\sigma_i[\cdot]$ computed as follows:

$$\sigma_i[\mathbf{g}] = \frac{\exp g_i}{\sum_j \exp g_j} \in [0, 1]. \quad (2)$$

The training of $\mathbf{g}^h$ is done by computing the categorical cross-entropy (CCE) loss as follows:

$$\begin{aligned} \mathcal{L}_{CCE}^h &= -\frac{1}{N} \sum_{n=1}^N \sum_{i=1}^{|\Omega^h|} \mathbb{1}[y_n^h = \omega_i^h] \log \hat{P}^h(\omega_i^h|\mathbf{x}_n) \\ &= -\frac{1}{N} \sum_{n=1}^N \sum_{i=1}^{|\Omega^h|} \mathbb{1}[y_n^h = \omega_i^h] \log \sigma_i[\mathbf{g}^h(\mathbf{f}(\mathbf{x}_n))], \end{aligned} \quad (3)$$

where $\mathbb{1}[\cdot]$ serves as an indicator function and takes a value of one when the argument is true, else zero. We apply the CCE loss function to all hierarchical levels, obtaining the total CCE loss $\mathcal{L}_{CCE}$ as the sum of the level-wise losses:

$$\mathcal{L}_{CCE} = \sum_{h=1}^H \lambda^h \mathcal{L}_{CCE}^h. \quad (4)$$

As it has also been noted in previous works [21], adopting several CCE loss functions for diverse classification levels may negatively affect performance at the most granular classification level. To mitigate this behavior, we introduced the weighting terms $\lambda = \{\lambda^1, \dots, \lambda^H\}$ to limit the impact of coarser levels, while still accounting for the additional losses.

C. Hierarchy Matrices

The CCE loss functions at each hierarchical level lead to independent classifications, which are not constrained to follow the hierarchy definition formulated in the problem statement. To impose the hierarchical constraints, additional consensus predictions and self-consistent penalty terms are introduced. These penalty terms are defined through hierarchy projection matrices designed to map the correct relationships between the different layers.

Let us consider for simplicity the case of $H = 2$ levels, with h_1 representing the coarse-grained labels and h_2 the fine-grained labels, i.e., $h_1 < h_2$. We refer to Ω^{h_1} as the set of the *macroclasses* (i.e., the coarse-grained classes) and to Ω^{h_2} as the set of the *microclasses* (i.e., the fine-grained classes). We can define the hierarchy indicator matrix $\mathbf{I}^{h_2, h_1} \in \{0, 1\}^{|\Omega^{h_2}| \times |\Omega^{h_1}|}$, which represents the *user-defined* hierarchical relationship between level h_2 and h_1 :

$$\mathbf{I}^{h_2, h_1} = \begin{bmatrix} I_{1,1} & I_{1,2} & \dots & I_{1,|\Omega^1|} \\ I_{2,1} & I_{2,2} & \dots & I_{2,|\Omega^1|} \\ \vdots & \vdots & \ddots & \vdots \\ I_{|\Omega^2|,1} & I_{|\Omega^2|,2} & \dots & I_{|\Omega^2|,|\Omega^1|} \end{bmatrix}. \quad (5)$$

This approach can be easily extended to a multi-level scenario, considering any given number H of hierarchical levels. In this case, the total number of hierarchy matrices that can be determined is $((H-1) \cdot H)/2$, as also mappings of distant relationship levels can be defined (e.g., from the root of the label tree to the leaves, skipping intermediate levels) if needed. A generalized example is illustrated in Fig. 3. In a strict hard-hierarchy, $I_{i,j}^{h_2, h_1}$ assumes a value of one when the microclass $\omega_i^{h_2}$ is contained in the macroclass $\omega_j^{h_1}$ through the

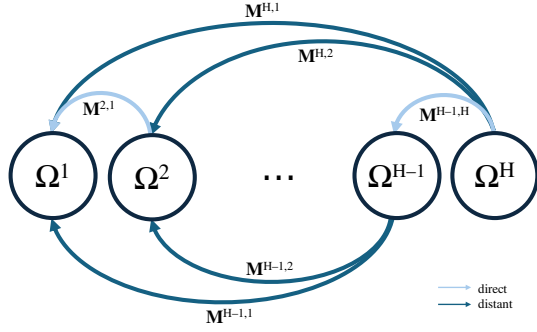


Fig. 3. Illustration of both direct and distant hierarchical mappings, from fine-grained classes to coarse-grained classes.

relationship $\omega_i^{h_2} \Rightarrow \omega_j^{h_1}$, and zero otherwise. However, this rigid formulation often fails to account for semantic overlaps in real-world hierarchies. Therefore, we distinguish between *user-defined* aggregations, established a priori for a given task, and *data-driven* aggregations, which are inferred directly to capture the data's true underlying organization. To combine the *user-defined* prior with a *data-driven* optimization, we parameterize the hierarchical relationships in the unconstrained logit space. Instead of strictly fixing the matrices to binary values or directly bounding them in $[0, 1]$, we define learnable parameter matrices $\mathbf{W}^{h_2, h_1}$. These matrices are initialized based on the *user-defined* indicator matrix $\mathbf{I}^{h_2, h_1}$ to provide a stronger prior:

$$W_{i,j}^{h_2, h_1} = \begin{cases} 0.5 + \epsilon & \text{if } I_{i,j}^{h_2, h_1} = 1 \\ -5.0 + \epsilon & \text{if } I_{i,j}^{h_2, h_1} = 0 \end{cases}, \quad (6)$$

where $\epsilon \sim \mathcal{N}(0, 0.01^2)$ is a small random noise added to break network symmetry during initial phases of optimization. The two selected values act as positive and negative biases towards valid connections and unmapped relationships. During the forward pass, these unconstrained parameters are dynamically mapped into valid hierarchy log joint distribution matrices $\mathbf{L}^{h_2, h_1} = \log \mathbf{J}^{h_2, h_1} \in \mathbb{R}^{|\Omega^{h_2}| \times |\Omega^{h_1}|}$ via a log-softmax operation on the flattened parameter matrices $\mathbf{W}^{h_2, h_1}$. Each term of the matrix $J_{i,j}^{h_2, h_1} = P(\omega_j^{h_1}, \omega_i^{h_2})$ models the probability of observing a relationship between microclass $\omega_i^{h_2}$ and macroclass $\omega_j^{h_1}$. This constraint relaxation allows us to better combine the *user-defined* aggregation with a *data-driven* centered architecture, by learning these matrices directly from the data while still reliably considering the initial *user-defined* mappings.

D. Self-Consistent Hierarchical Consensus Learning

Let us return to the general case of H hierarchical levels, considering projections from source levels h to a target hierarchical level $\tilde{h}$. Each hierarchical level acts independently of the others. To ensure that the network is *hierarchy-aware*, an additional constraint based on the prediction associated to the target hierarchical level and the projected predictions from the other levels is defined. In particular, we take inspiration from the disagreement-based active learning field [22] to define the disagreement between the predictions calculated at each hierarchical level. For each level $\tilde{h}$, we consider a committee

composed of all H levels. All predictions and their projected versions can be viewed as “soft” votes, which define each committee member’s confidence. Then, we introduce a self-consistent loss that penalizes the committee members’ disagreement with the overall committee consensus, thus guiding the learning of *data-driven* relationships in $\mathbf{J}^{h, \tilde{h}}$ and providing further supervision to the network based on the predictions at the different levels.

1) *Hierarchy-based Projected Predictions*: In order to generate the committee at each hierarchical level, the predictions first need to be projected to the other levels. For simplicity, let us consider the case of projection from source-level classes to target-level classes. Here we refer to estimated matrices as $\hat{\mathbf{J}}^{h, \tilde{h}} \in [0, 1]^{|\Omega^h| \times |\Omega^{\tilde{h}}|}$. The projected probability distribution $\hat{P}^{h \rightarrow \tilde{h}}(\omega_j^{\tilde{h}} | \mathbf{x})$ from source-level classes to target-level classes can be obtained from the probability distribution over the source-level classes $\hat{P}^h(\omega_i^h | \mathbf{x})$ and the hierarchy projection probabilities $\hat{P}(\omega_j^{\tilde{h}} | \omega_i^h)$ through the following relationship:

$$\hat{P}^{h \rightarrow \tilde{h}}(\omega_j^{\tilde{h}} | \mathbf{x}) = \sum_{i=1}^{|\Omega^h|} \hat{P}(\omega_j^{\tilde{h}} | \omega_i^h) \cdot \hat{P}^h(\omega_i^h | \mathbf{x}), \quad (7)$$

where the projection probabilities $\hat{P}(\omega_j^{\tilde{h}} | \omega_i^h)$ are directly derived by row-normalizing the estimated joint matrices $\hat{\mathbf{J}}^{h, \tilde{h}}$, i.e., we can define hierarchy projection matrices $\hat{\mathbf{M}}^{h, \tilde{h}} \in [0, 1]^{|\Omega^h| \times |\Omega^{\tilde{h}}|}$ such that $\hat{M}_{i,j}^{h, \tilde{h}} = \hat{P}(\omega_j^{\tilde{h}} | \omega_i^h)$. In general vector form, we can write the class-posterior probability vector $\hat{\mathbf{p}}^{h \rightarrow \tilde{h}}(\mathbf{x})$ projected from level h to level $\tilde{h}$ as follows:

$$\hat{\mathbf{p}}^{h \rightarrow \tilde{h}}(\mathbf{x}) = (\hat{\mathbf{M}}^{h, \tilde{h}})^T \cdot \hat{\mathbf{p}}^h(\mathbf{x}). \quad (8)$$

To ensure numerical stability and maintain better control over gradient magnitudes, we consider the computation of all the losses directly in the log domain. First, we compute the class-posterior log probabilities of classification modules $\mathbf{g}^h$ from Eq. 2:

$$\log \hat{P}^h(\omega_i^h | \mathbf{x}) = g_i^h - \text{LSE}(\mathbf{g}^h), \quad (9)$$

where $\text{LSE} : \mathbb{R}^{|\Omega^h|} \rightarrow \mathbb{R}$ is the log-sum-exp (LSE) operator:

$$\text{LSE}(\mathbf{g}^h) = \log \sum_{i=1}^{|\Omega^h|} \exp(g_i^h). \quad (10)$$

Then, we can also define the projected predictions $\hat{P}^{h \rightarrow \tilde{h}}(\omega_j^{\tilde{h}} | \mathbf{x})$ from level h to level $\tilde{h}$ in the log domain, starting from the definition in the natural domain. Applying the logarithm to both sides of Eq. 7, we obtain the following:

$$\begin{aligned} \log \left(\hat{P}^{h \rightarrow \tilde{h}}(\omega_j^{\tilde{h}} | \mathbf{x}) \right) &= \\ &= \log \left(\sum_i \hat{M}_{i,j}^{h, \tilde{h}} \cdot \hat{P}^h(\omega_i^h | \mathbf{x}) \right) \\ &= \log \left(\sum_i \hat{M}_{i,j}^{h, \tilde{h}} \cdot \frac{\exp g_i^h}{\sum_k \exp g_k^h} \right) \\ &= \log \left(\sum_i \exp \left(g_i^h + \log \hat{M}_{i,j}^{h, \tilde{h}} \right) \right) - \log \sum_k \exp g_k^h. \end{aligned} \quad (11)$$

This can finally be rewritten as the combination of two LSE operations, yielding the projected log-probability distribution at level $\tilde{h}$:

$$\begin{aligned} \log \left(P^{h \rightarrow \tilde{h}} \left(\omega_j^{\tilde{h}} | \mathbf{x} \right) \right) &= \\ &= \text{LSE} \left(\mathbf{g}^h + \log \left(\hat{\mathbf{M}}_{:,j}^{h,\tilde{h}} \right) \right) - \text{LSE} \left(\mathbf{g}^h \right). \end{aligned} \quad (12)$$

Hierarchy projection matrices can be defined directly in the log domain, *i.e.*, we learn log joint probabilities $\mathbf{L}^{h,\tilde{h}} = \log \hat{\mathbf{J}}^{h,\tilde{h}} \in \mathbb{R}^{|\Omega^h| \times |\Omega^{\tilde{h}}|}$. Then, we compute $\log \hat{\mathbf{M}}^{h,\tilde{h}}$ as follows:

$$\log \hat{\mathbf{M}}_{i,\cdot}^{h,\tilde{h}} = \mathbf{L}_{i,\cdot}^{h,\tilde{h}} - \text{LSE} \left(\mathbf{L}_{i,\cdot}^{h,\tilde{h}} \right), \quad (13)$$

which is equivalent to the row-normalization of $\hat{\mathbf{J}}^{h,\tilde{h}}$ in the log domain. The projected logits $\log \hat{\mathbf{p}}^{h \rightarrow \tilde{h}}$ can be used to calculate additional CCE losses that can contribute to the final loss [23]. However, this does not explicitly model the hierarchical consensus at the different hierarchical levels, requiring an additional step to ensure the joint classification of the classes.

2) *Semantics-Aware Hierarchical Consensus*: Given the projected prediction at each hierarchical level, we can now define a Hierarchical Consensus (HC) class probability vector $\hat{\mathbf{p}}_{HC}^{\tilde{h}}(\mathbf{x})$:

$$\begin{aligned} \hat{\mathbf{p}}_{HC}^{\tilde{h}}(\mathbf{x}) &= \\ &= \frac{1}{Z^{\tilde{h}}} \sqrt[|H|]{\hat{\mathbf{p}}^{\tilde{h}}(\mathbf{x}) \prod_{h \neq \tilde{h}} \hat{\mathbf{p}}^{h \rightarrow \tilde{h}}(\mathbf{x})} \\ &= \frac{1}{Z^{\tilde{h}}} \sqrt[|H|]{\hat{\mathbf{p}}^{\tilde{h}}(\mathbf{x}) \prod_{h \neq \tilde{h}} (\hat{\mathbf{M}}^{h,\tilde{h}})^T \cdot \hat{\mathbf{p}}^h(\mathbf{x})}, \end{aligned} \quad (14)$$

where $\hat{\mathbf{p}}_{HC}^{\tilde{h}}$ is the HC class-posterior probability defined as the geometric mean of the predictions at the target hierarchical level $\tilde{h}$ and the projected predictions of all the other h levels, and $Z^{\tilde{h}}$ is a normalization constant to ensure that $\hat{\mathbf{p}}_{HC}^{\tilde{h}}$ sums to one. This is equivalent to computing the average in the log domain [24]:

$$\begin{aligned} g_{HCj}^{\tilde{h}} &= \frac{1}{H} \left[\log \hat{P}^{\tilde{h}} \left(\omega_j^{\tilde{h}} | \mathbf{x} \right) + \sum_{h \neq \tilde{h}} \log \hat{P}^{h \rightarrow \tilde{h}} \left(\omega_j^{\tilde{h}} | \mathbf{x} \right) \right] \\ &= \frac{1}{H} \left\{ \left[g_j^{\tilde{h}} - \text{LSE} \left(\mathbf{g}^{\tilde{h}} \right) \right] + \right. \\ &\quad \left. + \sum_{h \neq \tilde{h}} \left[\text{LSE} \left(\mathbf{g}^h + \log \hat{\mathbf{M}}_{:,j}^{h,\tilde{h}} \right) - \text{LSE} \left(\mathbf{g}^h \right) \right] \right\}, \end{aligned}$$

$$\log \hat{\mathbf{p}}_{HC}^{\tilde{h}}(\mathbf{x}) = \mathbf{g}_{HC}^{\tilde{h}} - \text{LSE} \left(\mathbf{g}_{HC}^{\tilde{h}} \right), \quad (15)$$

where $\text{LSE}(\mathbf{g}_{HC}^{\tilde{h}}) = \log Z^{\tilde{h}}$. This HC represents the consensus found by the voting committee and can serve as an in-place replacement for the original classification modules during inference, providing a fully semantics-aware prediction in the downstream task.

3) *Self-Consistent Consensus Loss*: After calculating the consensus, both the prediction at the target level and the projected predictions are compared with the HC, defining an additional loss based on the divergence from the HC. To this purpose, we employ a JSD loss rather than a standard Kullback–Leibler divergence (KLD). While the KLD is widely used to measure the difference between two probability distributions, its asymmetry makes it better suited for scenarios where one distribution is a fixed “ground truth” or prior. In this case, all hierarchical levels provide dynamically trained “soft” votes, with no reference distribution. The symmetry in the JSD ensures balanced regularization between the projected predictions and the consensus, also providing stabler gradients due to its bounded nature. In the proposed framework, the JSD loss is computed as follows:

$$\begin{aligned} \text{JSD}^{h,\tilde{h}} \left(\hat{\mathbf{p}}_{HC}^{\tilde{h}} || \hat{\mathbf{p}}^{h \rightarrow \tilde{h}} \right) &= \\ &= \frac{1}{2} \left(\text{KLD} \left(\hat{\mathbf{p}}_{HC}^{\tilde{h}} || \mathbf{q} \right) + \text{KLD} \left(\hat{\mathbf{p}}^{h \rightarrow \tilde{h}} || \mathbf{q} \right) \right), \end{aligned} \quad (16)$$

where $\mathbf{q}$ is a mixture distribution of $\hat{\mathbf{p}}_{HC}^{\tilde{h}}$ and $\hat{\mathbf{p}}^{h \rightarrow \tilde{h}}$ calculated as $\mathbf{q} = \frac{1}{2}(\hat{\mathbf{p}}_{HC}^{\tilde{h}} + \hat{\mathbf{p}}^{h \rightarrow \tilde{h}})$, KLD represents the KLD, and we omit the dependency on $\mathbf{x}$ for simplicity. Please note that when $h = \tilde{h}$, we have that $\hat{\mathbf{p}}^{h \rightarrow \tilde{h}} = \hat{\mathbf{p}}^h$. Lastly, we can retrieve the JSD losses calculated at the different hierarchical levels and compute the final $\mathcal{L}_{HC}$ penalty loss:

$$\mathcal{L}_{HC} = \frac{1}{N} \sum_{n=1}^N \sum_{\tilde{h}=1}^H \frac{1}{\log |\Omega^{\tilde{h}}|} \sum_{h=1}^H \text{JSD}^{h,\tilde{h}}(\mathbf{x}_n), \quad (17)$$

where the logarithm of the number of classes at level $\tilde{h}$, *i.e.*, $|\Omega^{\tilde{h}}|$, allows the balancing of the different number of classes at each hierarchical level. This prevents skewness towards levels with larger number of classes. The $\mathcal{L}_{HC}$ measures the discrepancy between the HC prediction and the projected predictions, constraining the different layers to converge to the committee’s vote.

Finally, we also apply full supervision on the consensus predictions at the different levels considering a standard CCE loss:

$$\mathcal{L}_{CCE}^{HC} = \sum_{h=1}^H -\frac{1}{N} \sum_{n=1}^N \sum_{i=1}^{|\Omega^h|} \mathbb{1} [y_n^h = \omega_i^h] \log \hat{P}_{HC}^h \left(\omega_i^h | \mathbf{x}_n \right). \quad (18)$$

E. Overall Loss Function

Lastly, the losses are averaged and summed to $\mathcal{L}_{CCE}$ (Eq. 4):

$$\mathcal{L} = \mathcal{L}_{CCE} + \mathcal{L}_{CCE}^{HC} + \lambda_{HC} \mathcal{L}_{HC}, \quad (19)$$

where λ_{HC} is a metaparameters tuning the contribution of the HC loss. It is important to note that $\mathcal{L}_{HC}$ does not require direct supervision, as it is simply a loss representing the divergence from the HC. The HC prediction is used both within the penalty loss, and the additional supervised CCE loss during the training phase. We further consider the HC

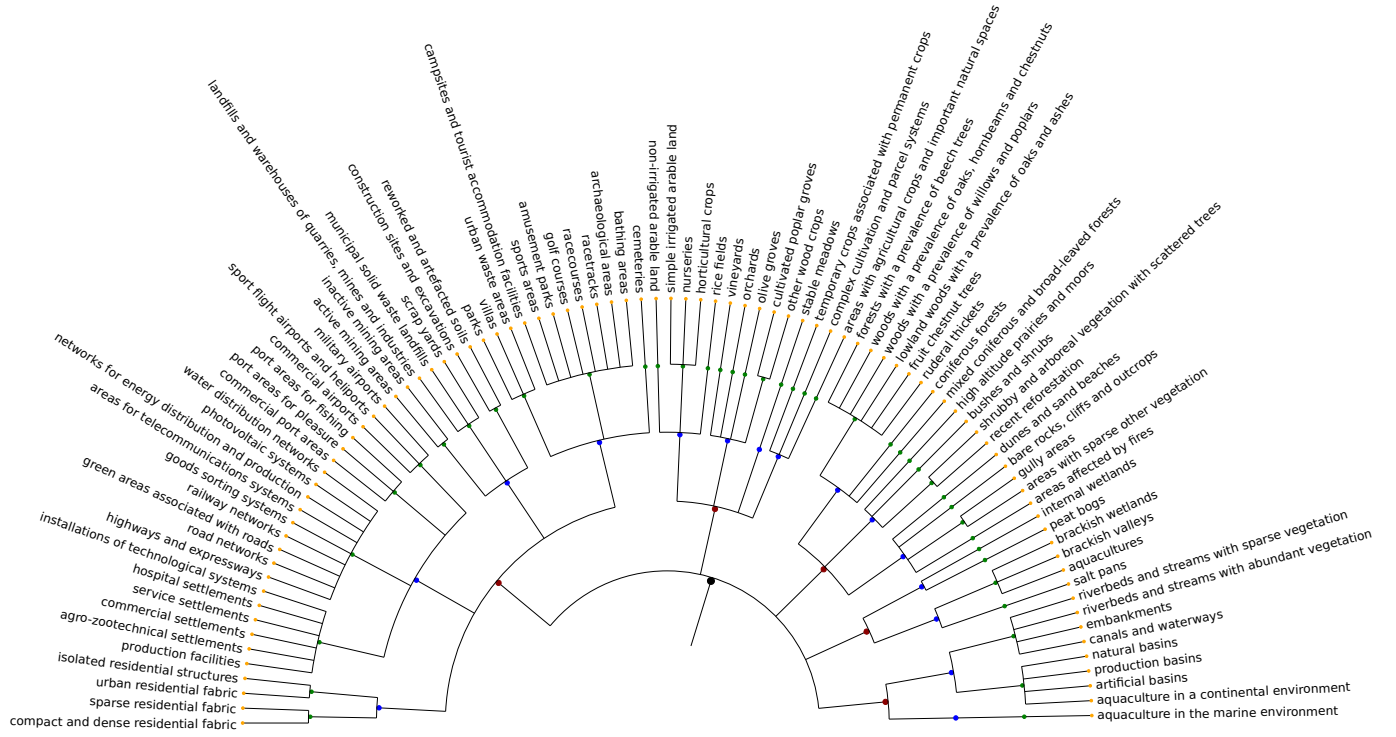


Fig. 4. A semicircular hierarchical label tree representation of the ELU dataset at the fine-grained level of the hierarchy. The leaves corresponding to the fine-grained classes are shown in orange, while the nodes for intermediate classes are depicted in green, those for coarse-grained classes are shown in blue, and the nodes for coarse in brown. The root node is represented in black.

prediction at each hierarchical level as the final predictions for inference. This allows the architecture to leverage the semantic relationships learnt during the self-consistent training phase and to perform hierarchically consistent predictions.

IV. DATASET DESCRIPTION AND EXPERIMENT SETUP

This Section provides a description of the datasets employed in this study. Then, the experimental setup, including the pre-processing phase and the benchmark methodologies adopted for evaluating the DL models, are described.

A. NWPU-RESISC45

To assess the performance of the proposed methodology on VHR images, where spatial features play a prominent role, we considered the NWPU-RESISC45 [25] benchmark as a first dataset. NWPU-RESISC45 is composed of 31,500 RGB multi-resolution images for scene classification of VHR images categorized into 45 different classes at the finest-level of detail, and exhibits a high within-class diversity and between-class similarity. The dataset is completely balanced, *i.e.*, each class has the same number of samples at the *fine-grained* level. We defined a custom-hierarchy based on the subdivision defined in MillionAID [26], resulting in 8, 24, and 45 classes, corresponding to *coarse-grained*, *intermediate*, and *fine-grained* classes, respectively.

B. ELU Dataset

To further validate the performance of the proposed methodology and demonstrate its generability on different land cover

mapping tasks, we additionally considered a multitemporal multispectral dataset differing in terms of complexity in the hierarchical structure. This dataset¹ consists of a publicly available Land-Use (LU) map of 2020 over the Emilia Romagna region, Italy. The maps were defined considering the standard CLC aggregation at four levels, showcasing a highly detailed semantic dataset. For the purpose of this analysis, we evaluated the hierarchical approach on four levels of hierarchy, consisting in the first three levels of the CLC and the additional finest level of detail defined by the LULC, here referred to as *macro*, *coarse-grained*, *intermediate* and *fine-grained*, respectively. The dataset has been built considering orthophotos and infrared images, and shows a minimum mapping unit of 0.16 hectares, with a minimum of 7m meters for linear structures. Additionally, it represents LU classes, where the same LC class could be associated with different semantic classes, such as the fine-grained classes “Commercial airports”, “Sport flight airports and heliports”, and “Military airports”. We considered the ELU dataset corresponding to the Military Grid Reference System (MGRS) Sentinel-2 (S2) granule 32TPQ (10980 × 10980 pixels, for a total area of 110 × 110 km²). In the analyzed granule, the four hierarchical levels show 5, 15, 40, and 90 classes at the macro, coarse-grained, intermediate, and fine-grained levels, respectively. Please note that such an elevated number of fine-grained classes is not commonly approached in Earth Observation (EO) classification problems, where the majority of studies focus on the intermediate or coarse-grained classification task.

¹Land Use of Emilia (2020). Accessed on April 9th 2026.

Nonetheless, this level of precision allows a better monitoring of the class of interest and the extraction of information-rich content. Due to the high number of classes and, consequently, the large size of the hierarchical tree, we refer to this hierarchy as a *complex* hierarchy. A semicircular tree representing the hierarchical structure of the dataset is shown in Fig. 4.

C. S2 Preprocessing and Dataset Preparation

1) *VHR dataset*: The training, validation, and test splits were selected accordingly to [27], considering a split-ratio of 0.6, 0.2, and 0.2, respectively. During training, we applied data augmentation techniques consisting in resize, random crops and random rotations of the images.

2) *Multitemporal MS datasets*: The input data consists of S2 multispectral (MS) satellite images provided in the Copernicus EO program. The S2 acquisitions considered have been preprocessed at Level-2A (L2A) and re-projected at 10m/pixel resolution through a nearest neighbor approach. In the experiments, all the bands except the 60m resolution band were used. Due to the high-complexity of both classification problems, we collected a time series (TS) of S2 images, fully exploiting the temporal signature and allowing the detection of seasonal targets. The S2 TS considered for the Emilia Dataset consists of acquisitions ranging from January 1st 2020 to December 31st 2020. The S2 TS has been processed to achieve a more regular TS, resulting in 12 medoid monthly composites [28].

To extract a balanced training set, we considered an approach based on the heterogeneity of the input training data. In particular, we divided the initial S2 monthly composite images into non-overlapping tiles of 1000×1000 pixels ($10km^2$). After the initial subdivision, we randomly subdivided the tiles into training, validation, and test sets by considering a split-ratio of 0.7, 0.15 and 0.15, respectively. We extracted $N = 2000$ representative points for each tile and class in the analyzed classification scheme. In the case of minoritarian classes having less than N samples, all the available samples were extracted. The training set was generated by extracting patches of 9×9 pixels around the selected points in the training tiles. To minimize spatial autocorrelation between the training patches, the corresponding reference data was extracted only for the central 3×3 pixels. In this way, we collected a training dataset representative of the entirety of the classification scheme, corresponding to $\sim 1.17\%$ ($1.413.522$ pixels) of the total number of pixels available in the MGRS S2 granule.

D. Design of Experiments

Validation has been conducted on both datasets considering different multigranularity approaches. The reported results include the values of Overall Accuracy (OA) and mean F1 Score (mF1) for both tasks, while for the multispectral dataset also Top-3 and Top-5 metrics are reported. We considered different multigranularity modules as comparison, all built on the same backbone architecture for fairness on all the datasets. We considered the following approaches:

- 1) **Baseline**: the baseline with only the *fine-grained* classification through standard CCE minimization. Selected to evaluate the improvement gained by incorporating the hierarchy.
- 2) **HAF** [13]: selected as a regularization approach adopting soft labels and geometric constraints that use auxiliary classifiers at each level to ensure consistent predictions with the hierarchy, specifically aiming to reduce the severity of mistakes.
- 3) **HRN** [15]: included as state-of-the-art benchmark that facilitates hierarchical feature interaction by forcing finer classes to inherit attributes from coarser classes through residual skip connections.
- 4) **MGM** [16]: included to assess the effectiveness of using knowledge-guided relation graphs and bi-directional interaction structures that enhance coarse-grained features through gradient truncations.
- 5) **HiCervix** [17]: chosen as benchmark for methods that model semantic correlations within an explicit tree structure using soft labels and recursive probability summation across hierarchical paths.
- 6) **ALGS** [6]: selected to evaluate the impact of a global semantic distance-based penalty term for adjusting class predictions, ensuring hierarchical consistency when classification is enforced at every level. Specifically designed for RS tasks.
- 7) **HCL** [7]: selected as comparison for representation-based learning that disentangles class-specific embeddings at multiple levels to align them with their semantic relationships. Specifically designed for RS tasks.

For the VHR task, we considered a ResNet50 backbone pretrained with ImageNet weights [29]. This choice is due to its deeply stacked convolutional layers and residual connections, making it well-suited for extracting highly localized, fine-grained spatial and morphological features necessary for VHR analysis. For the MS Satellite Image Time Series (SITS) classification task, we adopted a different backbone architecture designed to process multispectral and temporal information. This backbone consist in an adaptation to the RS scenario of a *Swin Transformer* [30] and is trained from scratch. For a detailed description of the layer topology, please refer to the cited paper [30]. The choice of this backbone is due to its shifted-window self-attention mechanism that efficiently captures both local details and long-range global contextual dependencies, making it ideal for processing complex multi-scale structural variations.

We adopted the *AdamW* optimizer for all the architectures and configurations. The metaparameters of each different configuration have been validated considering a Tree-structured Parzen Estimator (TPE) algorithm to efficiently explore the metaparameters space. The learning rate and the weight decay were sampled from log-uniform distributions $U_{\log}([10^{-6}, 10^{-3}])$. We considered 50 training epochs on both datasets, together with Stochastic Weight Averaging (SWA) to enhance model generalization [31]. Considering the proposed Semantics-Aware Hierarchical Consensus (SAHC) approach, the weight λ_{HC} associated to the HC loss has been sampled

TABLE I
COMPARISON OF THE ACCURACIES OBTAINED BY DIFFERENT HIERARCHICAL METHODOLOGIES ON THE NWPU-RESISC45 DATASET. THE BEST RESULTS ARE REPORTED IN BOLD, SECOND BEST ARE UNDERLINED

Metric	Hierarchy	Baseline [29]	HRN [15]	HAF [13]	ALGS [6]	HCL [7]	MGM [16]	HiCervix [17]	SAHC
OA (%)	Fine-grained	96.08	<u>96.48</u>	96.16	96.32	95.87	96.11	96.10	96.63
	Intermediate	—	<u>97.38</u>	97.30	97.35	97.16	97.24	97.19	97.56
	Coarse-grained	—	<u>97.86</u>	<u>97.86</u>	97.78	97.73	97.84	97.70	97.98
mF1 (%)	Fine-grained	96.08	<u>96.47</u>	96.15	96.33	95.85	96.10	96.09	96.63
	Intermediate	—	<u>96.74</u>	96.37	96.65	96.26	96.46	96.37	96.89
	Coarse-grained	—	<u>97.56</u>	97.52	97.47	97.36	97.41	97.37	97.67
Params (M)		25.8	35.5	25.9	25.9	27.0	35.5	25.9	25.9
FLOP (T)		6.22	7.03	6.22	6.22	6.51	6.88	6.22	6.22

from a log-uniform distribution $U_{\log}([10^{-2}, 10^2])$, while the weights λ^h have been set to 0.5, 0.2, and 0.3 for the VHR dataset, and to 0.4, 0.2, 0.1, and 0.3 for the MS dataset, to prioritize the fine-grained classification and reduce the influence of the intermediate, coarse-grained, and macro levels. We implemented a dynamic consistency warmup schedule for the HC loss weight, λ_{HC} . Specifically, during the training phase, λ_{HC} was initially fixed to 0 for the first 5 epochs to allow for stable feature extraction. Subsequently, the weight value was linearly ramped up from epoch 5 to 15 until reaching its sampled maximum value, where it remained constant for the duration of the training.

V. EXPERIMENTAL RESULTS

The experiments focus on validating the proposed SAHC approach over different datasets, comparing it to other state-of-the-art hierarchical approaches on different levels of the hierarchy. We first focus on the results on the specific tasks, *i.e.*, scene classification on the VHR dataset and semantic segmentation of MS SITS. Then, we analyze the importance of the hierarchical weights selection λ^h on the NWPU dataset. Finally, we discuss a visualization of the trained hierarchy projection matrices to identify the key hierarchical relationships learnt by the proposed approach.

A. Case I: VHR Scene Classification

The performance of the proposed SAHC method on the NWPU-RESISC45 dataset is summarized in Table I, providing a comparative analysis against state-of-the-art methodologies and the non-hierarchical baseline. Given the balanced class distribution characteristic of the NWPU-RESISC45 dataset, the OA and mF1 are aligned across all evaluated models. From the table, one can see that the integration of hierarchical information consistently improves classification performance over the baseline. Indeed, the majority of the evaluated hierarchical architectures outperform their non-hierarchical counterpart (with the exception of Hierarchical Contrastive Learning (HCL)), confirming that the exploitation of semantic relationships allows for a more robust distinction between fine-grained RS classes. Among the compared methodologies, the proposed SAHC achieves the highest performances at the

fine-grained level, with an OA of 96.63% and an mF1 of 96.63%. Notably, the good performance of SAHC extends across the entire taxonomic hierarchy. At the intermediate and coarse-grained levels, the proposed method achieves OAs of 97.56% and 97.98%, respectively, consistently outperforming the other frameworks. The second-best overall performance is obtained by the HRN method, which reaches a fine-grained OA of 96.48%. However, this comparison also highlights the computational overhead introduced by the different architectures. The HRN method achieves its high accuracy at the cost of architectural efficiency. Indeed, the introduction of the Granularity-Specific Block (GSB) leads to a significant increase in parameters and computational overload. In contrast, the proposed SAHC maintains the same computational footprint as the baseline architecture. This efficiency is shared by other methods such as Adaptive Label Granularity Selection (ALGS), HAF, and HiCervix, where the bottleneck remains the backbone itself. In the specific case of SAHC, the architectural overhead is due to the three auxiliary heads and their associated projection matrices, both of which are negligible relative to the dimension of the backbone. When focusing strictly on architectures that do not introduce additional complexity, the ALGS method provides the highest performance after SAHC, though it remains lower in terms of OA.

B. Case II: MS SITS Semantic Segmentation

To further evaluate the robustness of the proposed framework, an extensive analysis was conducted on the highly complex ELU dataset. Unlike the previous scenario, this dataset consists of lower-resolution multispectral images (10m/pixel) mapped to a complex taxonomy of 90 classes distributed across four hierarchical levels. The results of this analysis are reported in Table II. The complexity of the ELU dataset, combined with the potential for semantic noise or inconsistencies in the CLC LU, makes this scenario particularly challenging. Standard hierarchical models that incorporate a rigid taxonomy might struggle to adapt to these inconsistencies. The proposed SAHC mitigates this limitation by leveraging a data-driven taxonomy, allowing the architecture to dynamically adapt even when the predefined hierarchy is not perfectly aligned. This advantage can be observed in the OA and mF1 shown in

TABLE II
COMPARISON OF THE ACCURACIES OBTAINED BY DIFFERENT HIERARCHICAL METHODOLOGIES ON THE ELU DATASET. THE BEST RESULTS ARE REPORTED IN BOLD, SECOND BEST ARE UNDERLINED

Metric	Hierarchy	Baseline [29]	HRN [15]	HAF [13]	ALGS [6]	HCL [7]	MGM [16]	HiCervix [17]	SAHC
OA (%)	Fine-grained	36.39	36.41	36.63	35.90	<u>37.26</u>	35.42	36.92	38.08
	Intermediate	—	46.60	41.58	46.21	<u>48.81</u>	45.99	47.91	49.68
	Coarse-grained	—	57.13	57.89	56.43	<u>59.58</u>	56.30	58.27	59.90
	Macro	—	76.34	78.82	75.49	<u>78.36</u>	75.44	77.39	78.35
Top-3 (%)	Fine-grained	56.48	<u>57.57</u>	55.86	54.75	56.94	55.83	56.98	58.96
	Intermediate	—	71.01	63.61	69.27	<u>71.59</u>	70.13	71.31	73.48
	Coarse-grained	—	82.44	81.15	80.90	<u>82.76</u>	81.66	82.29	84.31
	Macro	—	96.77	96.36	97.10	<u>97.30</u>	96.99	97.50	97.23
Top-5 (%)	Fine-grained	64.57	<u>66.61</u>	64.21	63.11	65.54	64.89	65.48	67.84
	Intermediate	—	<u>80.68</u>	74.55	78.72	80.60	79.92	80.66	82.80
	Coarse-grained	—	90.67	88.88	90.07	90.69	90.18	<u>90.93</u>	91.98
	Macro	—	—	—	—	—	—	—	—
mF1 (%)	Fine-grained	32.02	32.38	32.12	31.63	<u>33.27</u>	31.60	32.59	34.25
	Intermediate	—	39.25	27.92	38.60	<u>41.03</u>	39.19	39.86	42.59
	Coarse-grained	—	52.64	46.42	51.44	<u>54.80</u>	51.98	53.23	55.34
	Macro	—	66.96	70.64	67.12	<u>69.47</u>	66.48	68.95	69.12
Params (M)		4.7	16.2	4.8	4.8	5.8	16.2	4.7	4.8
FLOP (T)		2.22	2.38	2.22	2.22	2.41	2.32	2.22	2.22

the Table. Indeed, SAHC achieves the best performance at the fine-grained level with an OA of 38.08% and an mF1 of 34.25%. These results are also confirmed at the intermediate (OA 49.68%, mF1 42.59%) and coarse-grained levels (OA 59.90%, mF1 55.34%), consistently better performing with respect to other state-of-the-art methodologies like HiCervix and HCL. At the macro level, SAHC demonstrates competitive performance, ranking after only to the HCL and HAF methods by a marginal gap.

To further evaluate the ability of the models to capture the semantic relationships, we analyzed the Top-3 and Top-5 accuracy metrics across all the levels of the hierarchy (see Table II). These metrics indicate whether the ground-truth class is among the highest-confidence predictions of the model, measuring the capacity of the network to maintain semantic proximity even when the absolute Top-1 prediction is incorrect. The proposed SAHC approach demonstrates the highest performance across almost all granularity levels. Specifically, SAHC achieves the highest Top-3 accuracies for the fine-grained, intermediate, and coarse-grained levels, reaching a 58.96%, 73.48%, and 84.31%, respectively. This represents a large improvement against the HRN, which ranked second in the VHR case, showing a margin of 1.39% at fine-grained level, while also outperforming the RS-oriented competitor HCL by 1.89% and 1.55% in the intermediate and coarse levels, respectively. At the macro level, Top-3 performance saturates for all evaluated models. In this aggregated space, SAHC remains robust (97.23%), performing comparably to HCL (97.30%) and HiCervix (97.50%). Because the total number of target classes at the macro level is small, Top-3 classification becomes a less discriminative metric. For the

Top-5 accuracy, SAHC maintains the overall best performance, outperforming the baseline and the compared methodologies across the evaluated granularities. At the fine-grained level, SAHC achieves 67.84%, surpassing the HRN by 1.23%. This performance gap widens at the intermediate level, where SAHC reaches 82.80% compared to HRN 80.68%. At the coarse-grained level, SAHC achieves 91.98%, outperforming the HiCervix by 1.05%. Note that Top-5 metrics are omitted at the macro level, as the limited number of classes at this highest taxonomy level renders the Top-5 metric uninformative. The improvement observed in the Top-3 and Top-5 accuracies can be attributed to the design of the SAHC architecture. When the network struggles to isolate the exact fine-grained class, the hierarchical consensus mechanism, which ensures probabilistic alignment across different hierarchical levels, guarantees that the subsequent highest-confidence predictions belong to semantically related classes within the same hierarchical branch. This minimizes the divergence between the projected predictions and the overall committee consensus, reducing severe misclassifications.

Finally, consistent with the previous analysis, SAHC achieves these state-of-the-art results without sacrificing computational efficiency. While architectures like HRN and multi-granularity module (MGM) impact on the computational efficiency, SAHC keeps the computational footprint nearly identical to the baseline model.

Lastly, Fig. 5 provides a visual comparison of the evaluated hierarchical methodologies on a small portion of the ELU dataset. From the qualitative results, one can see that the SAHC approach retrieves the most spatially-consistent land-cover maps, characterized by a significant reduction in classi-

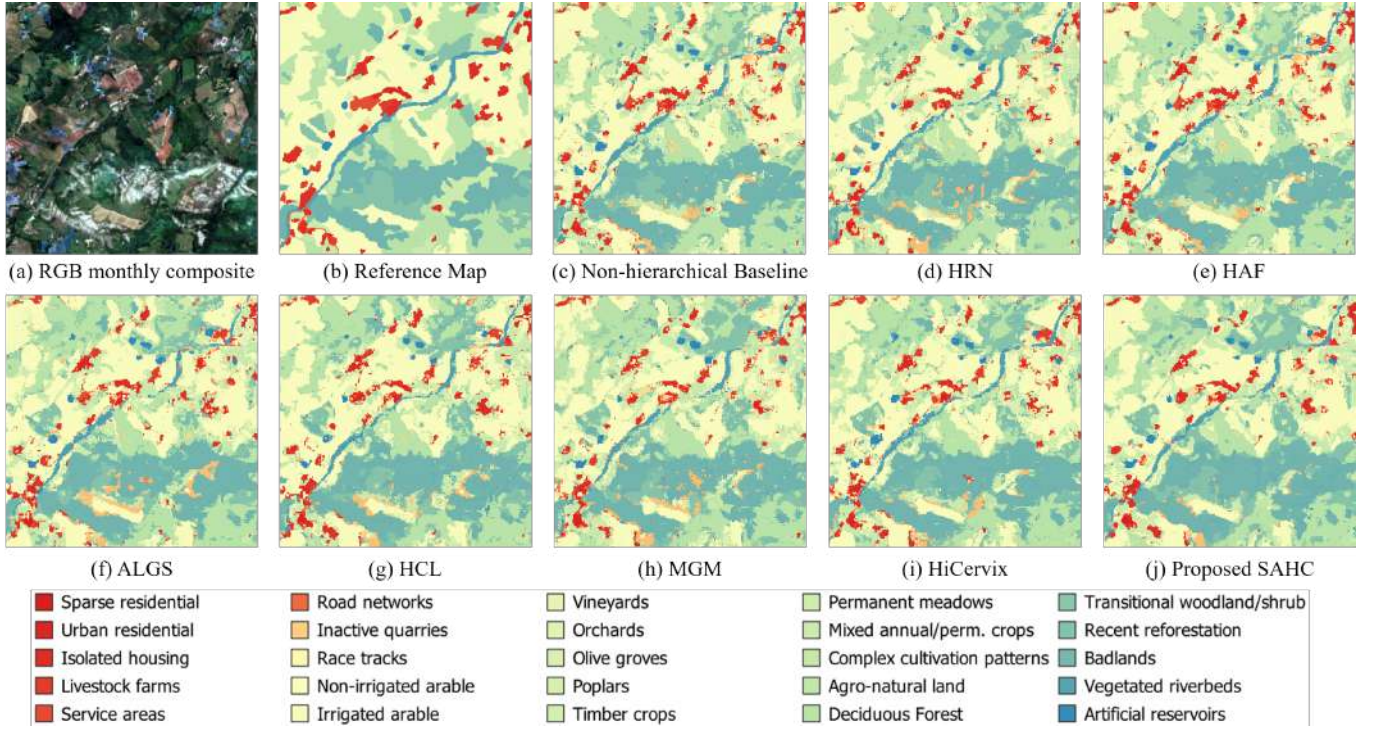


Fig. 5. Qualitative example of the predictions at the fine-grained level on the ELU dataset obtained by the considered hierarchical methodologies using the ResNet50 backbone: (a) true color (RGB) monthly composite for August 2020, (b) CLC reference map, (c) non-hierarchical baseline (d) HRN method, (e) HAF method, (f) ALGS method, (g) HCL method, (h) MGM method, (i) HiCervix method, and (j) the proposed SAHC method.

fication artifacts compared to the competing approaches. This is particularly evident in the southern sector of the study area. Indeed, while in this specific case most networks misclassify “transitional woodland/shrub” as “inactive quarries” due to sparse vegetation signatures, SAHC correctly identifies the vegetative cover, showing robustness in classification. These advantages are further evident in Fig. 6. From a broader perspective, scaling a RS classification problem to an extensive taxonomy of 90 classes inherently introduces severe spectral confusion and extreme class imbalance. In this context, the observed improvements in the mF1 score are significant. Indeed, they demonstrate that the proposed SAHC mitigates the ambiguities typical of such label space, maintaining robust and granular predictions where traditional approaches fail.

C. Sensitivity Analysis

To evaluate the robustness of the proposed hierarchical architecture, we conducted a sensitivity analysis on the VHR dataset to assess how model performance varies with the selection of different granularity weights λ^h . The search space was constrained such that the weights sum to one ($\sum_{h=1}^H \lambda^h = 1$), with each λ^h sampled using a step size of 0.1. Since the architecture adopts three hierarchical levels, the resulting parameter space can be visualized as a probability simplex that illustrates the various weight combinations across the different granularities. Fig. 7 illustrates a heatmap reporting the performance across all evaluated configurations of the three layer weights. As demonstrated in the figure, the optimal combination is achieved at $\lambda^1 = 0.5$, $\lambda^2 = 0.2$, $\lambda^3 = 0.3$. This optimal configuration justifies the hypothesis that priority

should be allocated to both the fine-grained classification task and the coarsest class. Nevertheless, the analysis also reveals the presence of several local maxima across the performance surface, which are generally skewed toward configurations in favor of the finest-level weight.

D. Ablation Study

The performance contribution of each component in the proposed hierarchical approach is analyzed in the ablation study presented in Table III. In this study, we evaluate the ResNet50 backbone across six different hierarchical configurations on the NWPU-RESISC45 dataset:

- 1) **Baseline**: the standard approach with no multi-level classifiers;
- 2) **Multi-level**: the baseline approach modified with additional classification heads for the coarse-grained and intermediate classes.
- 3) **Multi-level with HC inference $\hat{p}_{HC}$** : the multi-level approach with consensus applied solely during inference (considering *user-defined* hierarchies).
- 4) **Multi-level with $\mathcal{L}_{HC}$ training**: the multi-level approach trained with the self-consistent consensus loss, excluding consensus-based classification during inference.
- 5) **Multi-level with $\mathcal{L}_{HC}$ training and HC inference $\hat{p}_{HC}$** : the multi-level approach combining both the self-supervised consensus training and consensus-based classification during inference.
- 6) **SAHC**: the proposed full approach, which incorporates the self-supervised consensus loss, consensus inference,

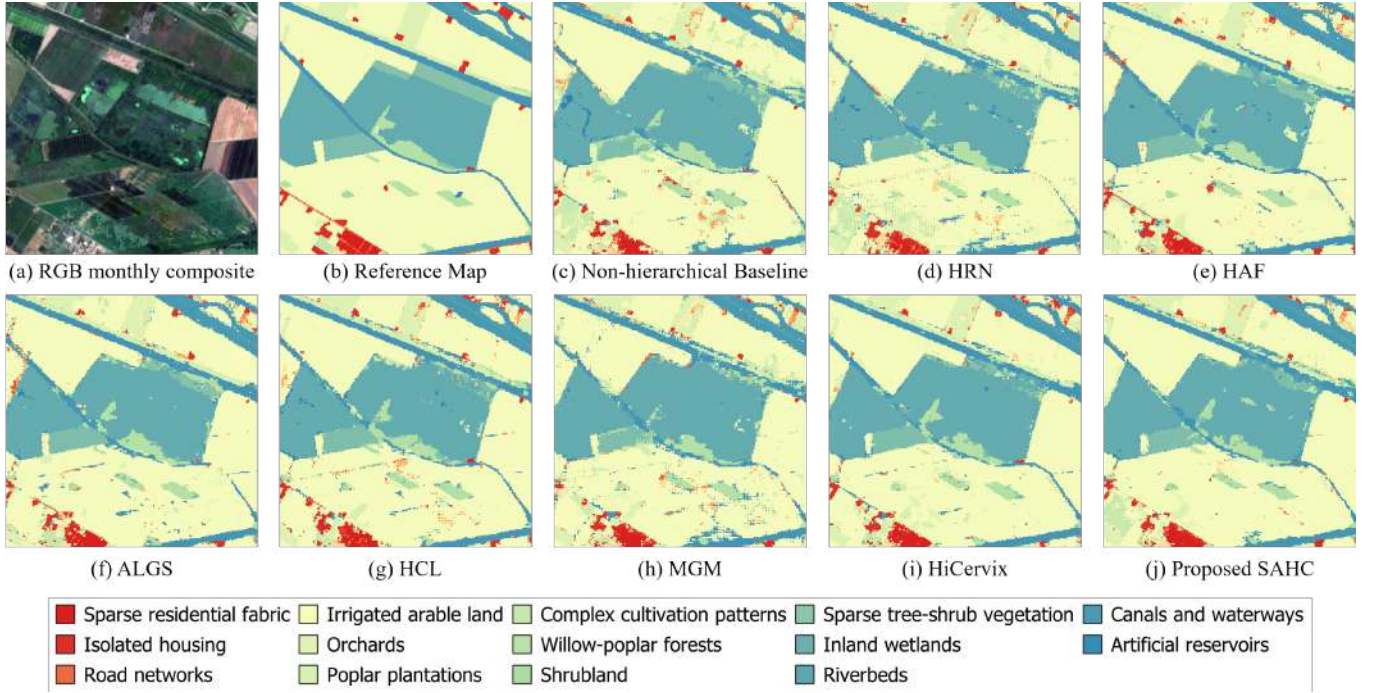


Fig. 6. Qualitative example of the predictions at the fine-grained level on the ELU dataset obtained by the considered hierarchical methodologies using the ResNet50 backbone: (a) true color (RGB) monthly composite for August 2020, (b) CLC reference map, (c) non-hierarchical baseline (d) HRN method, (e) HAF method, (f) ALGS method, (g) HCL method, (h) MGM method, (i) HiCervix method, and (j) the proposed SAHC method.

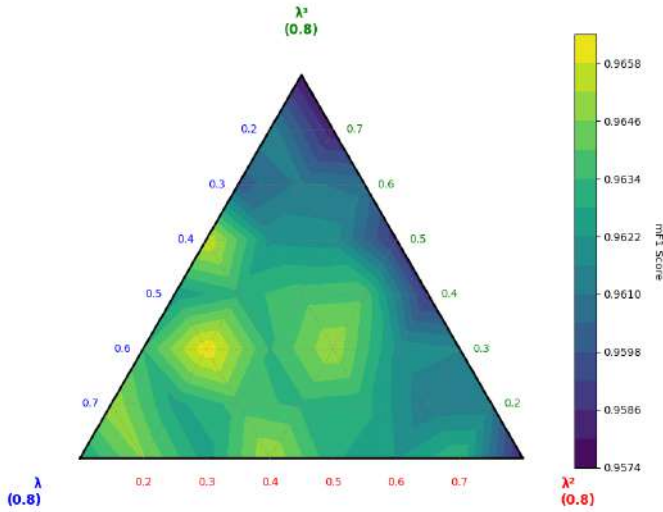


Fig. 7. Sensitivity analysis of the hierarchical weights λ^h on the VHR dataset. The performance surface is visualized as a probability simplex, where each point represents a weight combination for the three hierarchical levels sampled at a step size of 0.1. The simplex heatmap shows the optimal configuration as $\lambda^1 = 0.5$, $\lambda^2 = 0.2$, $\lambda^3 = 0.3$.

and the supervised consensus loss ($\mathcal{L}_{CCE}^{HC}$) at multiple levels.

As observed in Table III, the integration of the proposed modules consistently enhances the performance of the network. The transition from the baseline to the simple multi-level architecture shows an increase of the fine-grained OA from 96.08% to 96.44%. This demonstrates that simply forcing

TABLE III
ABLATION STUDY ON THE RESNET50 BACKBONE WITH THE
NWPU-RESISC45 DATASET. THE BEST RESULTS ARE
IN BOLD, SECOND BEST ARE UNDERLINED

Multi-Heads	L_{HC}	$\hat{p}_{HC}$	L_{CCE}^{HC}	Hierarchy	OA (%)	mF1 (%)
				Fine	96.08	96.08
				Inter.	—	—
				Coarse	—	—
✓				Fine	96.44	96.43
				Inter.	97.41	96.68
				Coarse	97.92	97.58
✓				Fine	96.49	96.48
		✓		Inter.	97.44	96.68
				Coarse	97.94	97.57
✓	✓			Fine	<u>96.52</u>	<u>96.53</u>
		✓		Inter.	97.41	96.64
				Coarse	97.76	97.33
✓	✓	✓		Fine	96.49	96.51
		✓		Inter.	<u>97.48</u>	<u>96.71</u>
				Coarse	97.90	97.48
✓	✓	✓	✓	Fine	96.63	96.63
		✓		Inter.	97.54	96.86
				Coarse	97.97	97.66

the network to learn coarse and intermediate weighted representations provides regularization for the fine-grained task. Introducing the HC inference constraint or the self-consistent loss individually provides further performance gains. Notably, using only the self-consistent consensus training leads to good fine-grained OA of 96.52% and an mF1 of 96.53%,

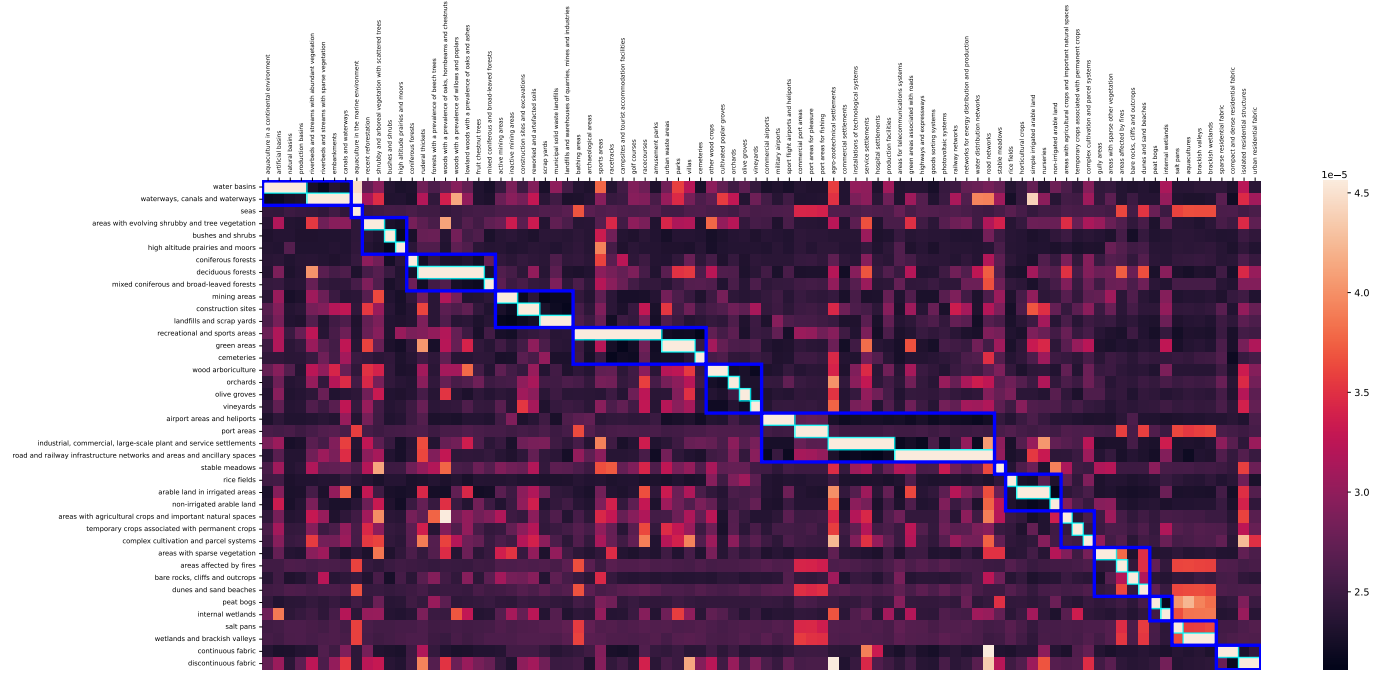


Fig. 8. Heatmap of the estimated joint probability matrix $\mathbf{J}^{h_2, h_1}$ considering the ResNet50 backbone on the ELU dataset from fine-grained classes to intermediate classes. The cyan rectangles identify the expected hierarchical aggregation of the fine-grained classes at the intermediate level.

suggesting that enforcing hierarchical consistency during the learning phase is effective. Interestingly, combining both modules creates a trade-off, reducing marginally the fine-grained accuracy while improving the intermediate OA to 97.48%. Finally, the introduction of the supervised consensus loss in the complete SAHC framework provides the best trade-off between training and inference constraints. The full SAHC configuration achieves the highest metrics across all hierarchical levels, achieving the best fine-grained OA of 96.63%. This confirms that the cooperation between self-consistent alignment, structured inference, and supervised hierarchical correction maximises the discrimination capabilities of the network.

E. Hierarchy Projection Matrices Analysis

To better visualize the impact of the hierarchical projection matrices on classification results, we analyze the estimated joint probability matrix $\mathbf{J}^{h_2, h_1}$ representing the learned hierarchy between the fine-grained and intermediate levels for the ELU dataset. The corresponding visualization on the NWPU dataset is omitted, as its simpler classification task leads to an expected result where the learned mappings perfectly align with the predefined hierarchical levels. Figure 8 presents a heatmap of this joint probability matrix, illustrating the hierarchical structure captured by the proposed SAHC methodology on the ELU dataset. The visualization confirms that the proposed approach effectively learns the hierarchy in the dataset, correctly identifying the semantic aggregations between the levels. Specifically, the cyan rectangles in the heatmap correspond to the user-defined expected mappings, from fine-grained classes to intermediate classes. The higher

weights within these regions confirm that the data-driven estimated mappings successfully respect the intended *user-defined* taxonomy. In addition to these rigid hierarchies, the network also captures important semantic relationships between different coarser groups. Indeed, the blue rectangles indicate cross-groups relationships related to coarser-grained classes. Because separate joint matrices are trained for different hierarchical transitions in the proposed architecture, the model is able to discern soft-hierarchical mappings that were strictly excluded from the initial *user-defined* hierarchy. For instance, the learned weights reveal an active semantic relationship between “water basins” and “aquaculture in the marine environment”, as well as between “continuous urban fabric” and “road networks”. These mappings highlight the importance of a data-driven approach. Indeed, rather than superimposing a rigid and potentially flawed hierarchical structure, the model dynamically adapts to the true semantic distributions present in the RS image.

VI. CONCLUSION

In scene classification and semantic segmentation RS tasks, the hierarchical relationship among the target classes is often under-exploited, resulting in the exclusion of valuable semantic information from the decision-making process. To address this limitation, this paper proposes the SAHC approach that leverages the hierarchical structure inherent in class types to improve the robustness and the accuracy of the classification. We introduced hierarchical classification branches, each designed to handle a specific hierarchical classification task, ranging from coarse-grained to fine-grained categories. Trainable hierarchy log-joint matrices are then defined to project the predictions across different hierarchical

levels, following a self-consistent data-driven approach which may not align with the rigid *user-defined* mappings between different hierarchy levels. This allows the implementation of a consensus mechanism, where predictions from multiple hierarchical levels are fused to perform the final decision. By progressively guiding the classification process from broader to specific categories, we can optimize the consensus, ensuring alignment among predictions at different hierarchical levels in a self-consistent fashion. The proposed methodology is both lightweight and scalable, making it easily adaptable to different architectures and datasets by simply reorganizing their structure in a hierarchical manner.

Extensive experimental analysis demonstrated the effectiveness of the proposed approach on two different RS datasets, representing a VHR and a MS classification tasks. Indeed, the proposed approach is particularly relevant in VHR image and RS MS SITS classification, where the semantic structure of the data and available datasets support an explicit hierarchical representation of the task. The experimental results reveal that the proposed methodology is able to exploit both complex and simpler hierarchies, demonstrating the best performances in terms of OA% and mF1% scores across experiments across all hierarchical levels. As expected, the improvements are particularly significant when complex hierarchies can be exploited within the RS data. The proposed method is general, showing top performance scores across different tasks. Furthermore, the self-consistent consensus loss effectively constrained the hierarchical levels to converge toward a unified prediction, reducing hierarchical violations in the final classification maps.

A limitation of the proposed method is the reliance on a predefined hierarchy within the labels to train the self-supervised hierarchical consensus. While the consensus itself operates in a self-supervised manner, its training remains implicitly influenced by the supervised CCE losses at the coarser levels of the hierarchy. This predefined structure may not accurately represent the true underlying relationships within the data, potentially impairing model performance, or constraining the hierarchy log-joint matrices to learn a suboptimal representation of the hierarchical structure. To address this limitation, future research will explore methods for learning the hierarchical structure directly from the data. This approach would eliminate the need for manual definition of hierarchies, thereby improving the adaptability and accuracy of the model.

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Technical Workflow & Research Progression

Stage	What is Actually Happening	What They Likely Use / Do
1. Data Collection	Collecting multi-temporal EO imagery across multiple years and ecosystems	Sentinel-2, Landsat archives, HRLC datasets
2. Preprocessing	Cleaning and harmonizing EO data before AI training	Cloud masking, atmospheric correction, temporal alignment, patch extraction
3. Historical Product Creation	Building HRLC30 historical land-cover maps (1990–2019)	Multi-year satellite processing pipeline
4. Land-Cover Label Generation	Creating labels for segmentation/classification	Existing LC products, historical maps, weak labels
5. Baseline Model Benchmarking	Comparing older segmentation/classification approaches	CNNs, U-Net-like models, classical ML baselines
6. Transformer-Based Segmentation	Using advanced DL architectures for semantic segmentation	Swin Transformer
7. Change Detection	Detecting land-cover changes between years	Multi-temporal segmentation/change analysis
8. Cross-Region Evaluation	Testing if model generalizes across ecosystems	Amazonia vs Sahel vs Siberia type evaluations
9. Operational Product Generation	Producing scalable EO products for climate applications	HRLC10 / HRLC30 / HRLCC30 outputs
10. Future Research Extensions	Improving robustness and scalability	Weak supervision, domain generalization, uncertainty estimation

MAIN RESEARCH PROBLEMS IDENTIFIED FROM ESA HRLC + D2

Problem	Why It Matters	Contextual Detail
1. Cross-region generalization	Model trained in one ecosystem may fail in another	Domain generalization across biomes
2. Temporal variability	Same class changes seasonally and yearly	Intra-annual vs inter-annual dynamics
3. Sparse historical HR data	Older years may have fewer usable images	Pre-Sentinel-2 archive constraints
4. Cloud cover & missing observations	Causes temporal gaps and inconsistent data	Cloud masking and atmospheric correction
5. Multi-sensor inconsistency	Different sensors behave differently	Harmonization of Landsat/Sentinel/HRLC
6. Semantic segmentation difficulty	Pixel-wise classification in complex environments	Advanced DL architecture performance
7. Mixed vegetation classes	Difficult spectral separation between similar classes	Spectral unmixing and class overlap
8. Spatio-temporal consistency	Products across years must remain coherent	Multi-temporal segmentation analysis
9. Large-scale operational scalability	ESA needs continental-scale robustness	Scaling HRLC10/30 production pipelines
10. Weak/noisy labels	Large-scale manual annotation impossible	Automated label generation from legacy maps

PROBLEMS YOUR FRIEND IDENTIFIED

Informant Detail	Technical Research Significance	Operational Criticality
Swin Transformer (Phase-2)	Transition to advanced transformer-based semantic segmentation	Indicates research maturity beyond standard CNN baselines
Taxonomy Mismatch	Noisy supervision resulting from external weak label inconsistency	Represents a core challenge in modern EO-AI applications
Regional Generalization	Domain shift across heterogeneous ecosystems and biomes	Fundamental for achieving global EO product deployment
Class-wise Performance	Variability in classification accuracy across land-cover types	Pervasive issue in complex EO segmentation tasks
SAR + Optical Fusion	Exploration of multi-modal Earth Observation learning	Pathway to enhanced scalability and model robustness
Medical Imaging Analogy	Cross-domain transfer of weak-supervision methodologies	Demonstrates high-level exploratory research mentality
Source Heterogeneity	Discrepancies in label definitions across various legacy maps	Induces learning instability and aleatoric uncertainty

Weakly Supervised Learning for Facial Affective Behavior Analysis: A Review

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Abstract—Recent advances in deep learning (DL) and computational capacity have enabled facial affective behavior analysis (FABA) to progress from static images captured in controlled settings to fine-grained analysis of facial expressions in real-world video data. However, training accurate DL models for FABA typically requires large-scale, expert-annotated datasets, which are costly to obtain and inherently noisy due to the ambiguity of labeling subtle facial expressions and action units (AUs). To mitigate these challenges, weakly supervised learning (WSL) has emerged as a promising paradigm for training models with weak annotations. In this paper, we present a structured taxonomy of WSL scenarios for FABA, organized according to the type of weak annotation and the specific affective task. Building on this taxonomy, we provide a critical synthesis of representative WSL methods for both classification (expression and AU recognition) and regression (expression and AU intensity estimation) tasks, focusing on their core methodological ideas, strengths, and limitations. Furthermore, we systematically summarize the comparative performance of WSL approaches along with widely adopted experimental setups and evaluation protocols. Our critical assessment identifies key challenges and future research directions, including the need for efficient adaptation of foundation models and for the development of robust, scalable FABA systems suitable for real-world applications.

Index Terms—Weakly Supervised Learning, Facial Expression Recognition, Intensity Estimation, Action Unit Detection.

I. INTRODUCTION

FACIAL affective behavior analysis (FABA) plays a vital role in computer vision and affective computing, enabling a wide range of applications such as human-computer interaction [1], medical diagnosis [2], e-education [3], and autonomous driving [4]. Although affective cues can be conveyed through multiple modalities, facial behavior manifested in the intricate movements of facial muscles provides an exceptionally rich wealth of information, making it invaluable for accurate analysis of affective states [5]. To leverage this information, three complementary annotation paradigms have emerged, ranging from coarse categorical labels to fine-grained intensity and dimensional scores. First, the categorical model defines six basic universal emotions—anger, disgust, fear, happiness, sadness, and surprise [6] (see Fig. 2). Subsequently, contempt was added as a seventh category [7]. This framework offers high-level interpretability of affective states and is widely used for classification tasks. In addition to that, micro-expressions and compound expressions are also introduced to capture the mixed emotional states [8], [9]. Second, the



Fig. 1. Examples of facial images with action units (AUs) [12].

facial action coding system (FACS) [10] provides a more granular low-level muscle-based representation by defining action units (AU), each corresponding to the movement of specific facial muscles as shown in Fig. 1. Finally, beyond these classification schemes, regression-based annotations are also explored to capture the intensity of affective states based on ordinal intensity levels or dimensional modeling. Ordinal regression represents expressions or AUs in discrete intensity levels, whereas dimensional modeling maps affect to continuous scales, often using valence (unpleasant to pleasant) and arousal (calm to excited) to capture the nuanced and fine-grained emotional states [11].

Despite their widespread use, obtaining high-quality annotations remains a significant challenge, as the ambiguity increases when transitioning from coarse categorical labels to more nuanced, intensity-based, or continuous representations of emotions¹. Categorical annotations, such as discrete emotion labels are relatively straightforward to obtain and interpret but fail to capture the fine-grained emotional states. Although FACS-based AU annotations are more granular and informative, they require substantial expertise and manual effort. Achieving certification as a FACS coder requires over 100 hours of training, and annotating even a single minute of video can take close to an hour [10]. Assessing AU intensities introduces further ambiguity, as low-intensity or subtle facial muscle activations are difficult to quantify consistently. Dimensional annotations, such as continuous valence and arousal ratings, are even more challenging, requiring sustained affective judgments over time that are cognitively taxing, prone to annotator fatigue, and often yield low inter-rater reliability.

Given these challenges in obtaining high-quality annota-

¹Though the terms “expression” and “emotion” are used interchangeably in the literature, the primary difference is that facial emotion conveys the mental state of a person, whereas facial expression is the indicator of the emotions being felt, i.e., facial expressions display a wide range of facial modulations, but facial emotions are limited.

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Fig. 2. Examples of primary universal emotions. From left to right: neutral, happy, sad, fear, angry, surprise, and disgust. Images are taken from the Compound Facial Expressions of Emotions dataset [8].

tions, training deep learning (DL) models for FABA remains difficult, as these models are both data-intensive and highly sensitive to label noise. In practice, collecting large-scale, fully annotated datasets is often infeasible, particularly for application-specific or fine-grained affective tasks. To address these limitations, two broad research paradigms have been explored: (1) unsupervised or self-supervised learning [13] and (2) weakly supervised learning [14]. While self-supervised learning offers a promising alternative, they require massive pretraining datasets and may not directly capture task-specific affective cues without additional supervision. On the other hand, WSL offers a middle ground instead of discarding supervision altogether; it leverages weak labels such as imprecise or incomplete labels (e.g., video-level emotion labels) to guide the learning process. This paradigm aligns closely with the practical realities of affective computing, where obtaining perfect labels is unrealistic, but some form of weak supervision is often available. In this paper, we examine how WSL approaches can mitigate the challenges of annotation bottlenecks to build robust FABA systems by leveraging weak annotations of expressions and AUs. Note that this survey excludes papers that rely only on self-supervised learning models without any external supervision.

Existing surveys on FABA systems focus primarily on fully supervised methods [9], [12], [15]–[17], including deep facial expression recognition (FER) [18], or specific tasks such as pain estimation [19], often assuming the availability of densely annotated datasets. Motivated by the growing importance of addressing annotation bottlenecks in real-world scenarios, this survey presents a comprehensive and critical review of state-of-the-art WSL methods developed for FABA systems. We systematically identify and organize the core sub-problems in WSL for FABA, including classification with weak categorical or discrete labels, and regression with weak ordinal or dimensional annotations. For each scenario, we analyze representative methods by summarizing their core ideas and insights, as well as their advantages and limitations. Additionally, we examine the datasets and evaluation protocols commonly used in WSL scenarios of FABA systems, including training strategies and performance trade-offs. Based on our critical analysis of existing methods, we discuss open challenges and future research directions aimed at building robust, data-efficient FABA systems, including adapting foundation models and few-shot learning.

The main contributions of this paper can be summarized as:

- (1) A structured taxonomy of WSL scenarios for FABA, highlighting their relevance across different annotation paradigms. This taxonomy formulates problem settings based on the type of weak annotation and the specific affective task, and identifies the key challenges associated with each scenario (Section III).
- (2) A comprehensive and critical review of state-of-the-art WSL methods proposed under different WSL scenarios for classification (i.e., expression and AU recognition), and for regression (i.e., expression and AU intensity estimation), along with the critical analysis of their strengths and limitations (Sections IV and V, respectively).
- (3) An overview of experimental setups and benchmarking protocols, and comparative performance analysis of these methods under different WSL scenarios (Sections IV-E and V-C).
- (4) A discussion on open challenges and future research directions, with an emphasis on scalable and reliable FABA systems under real-world weakly annotated data (Section VI).

II. DATASETS

In this section, we review widely used datasets for evaluating WSL methods under different forms of weak annotation in FABA. These datasets vary in annotation granularity, supervision level, and affective task, and are often adapted to specific WSL scenarios using standardized evaluation protocols.

UNBC-McMaster [20]: This database contains 200 video sequences of 48398 frames captured from 25 participants experiencing shoulder pain. Each frame of the video sequence is labeled with discrete intensity levels (A–E) of pain-related AUs (AU4, AU6, AU7, AU9, AU10, AU12, AU20, AU25, AU26, AU27 and AU43) by certified FACS coders. In addition, the dataset provides discrete pain intensity labels at both the frame and sequence levels, using the Prkachin and Solomon Pain Intensity Scale (PSPI; 0–15) and the Observer Pain Intensity (OPI; 0–5).

DISEFA [21]: The dataset is created using 9 short video clips from YouTube, where 27 adults are allowed to watch the short video clips about various emotions. The facial expressions of each of the participants are captured with a high-resolution video of 1024x768 pixels with a frame rate of 20fps resulting in 1,30,754 frames. Each of these frames is annotated with AUs along with the discrete intensity levels by FACS expert raters. The AUs related to the expressions in the database are AU1, AU2, AU4, AU5, AU6, AU9, AU12, AU15, AU17, AU20, AU25, and AU26, whose intensities are provided on a six-point ordinal scale (neutral < A < B < C < D < E).

BP4D [22]: The dataset was collected from 41 participants, each one asked to exhibit eight spontaneous facial expressions, resulting in the acquisition of both 2D and 3D video recordings for each task. In total, 328 video sequences were obtained. Due to the high cost and complexity of FACS-based annotation, only the most expressive temporal segments—specifically, 20-second segments with high facial activity—were annotated with facial action units (AUs). Across these sequences, 27 AUs were coded for expression analysis, with ordinal intensity annotations (0–5) provided for AU12 and AU14.

FER+ [23]: The dataset is an extension of the FER2013 dataset [24], which is labeled automatically by the Google image search engine. All the images in FER2013 have been registered and resized to 48x48 pixels. The dataset is partitioned into 28,709 training images, 3,589 validation images, and 3,589 test images with seven basic expression labels. The images are further relabeled by 10 individuals, thereby obtaining more reliable annotations.

CK+ [25] : This dataset comprises 593 video sequences captured under controlled laboratory conditions, in which 123 participants spontaneously performed facial expressions. Each sequence depicts a progression from a neutral facial state to the peak expression, with sequence lengths ranging from 10 to 60 frames. Because the original emotion labels were found to be unreliable, the annotations were subsequently refined using the FACS to assign seven basic expression categories, including contempt. Each expression category is defined by a prototypical combination of specific action units (AUs). Following this refinement process, 327 sequences were retained, with each sequence labeled at the video level with a single expression category. For static image-based approaches, the video-level label is commonly associated with the last one to three frames, corresponding to the peak expression, while treating the initial frame as neutral.

MMI [26]: The database contains 326 video sequences spontaneously captured in laboratory-controlled conditions from 32 subjects, and it includes challenging variations such as interpersonal variations, pose, etc., compared to the CK+ database. The sequences are captured as onset-apex-offset, i.e., the sequence starts with neutral expression (onset), reaches the peak (apex), and returns to neutral expression (offset). As per the standard of FACS, 213 sequences are labeled with six basic facial expressions (excluding contempt) at the video level, of which 205 sequences are captured in frontal view. They have also provided frame-level annotations for some of the sequences. For approaches based on static images, only the first frame (neutral expression) and peak frames (apex of expressions) are considered.

RAF-DB [27]: This large-scale in-the-wild dataset consists of extremely diverse facial images downloaded from the Internet. The obtained images were labeled by 315 annotators, and each image was ensured to be labeled by 40 independent annotators. The final annotations were obtained using crowdsourcing techniques for seven basic emotions (including neutral expression). The dataset is partitioned into 12,271 training samples and 3,068 test samples. Some of the images are also labeled with compound expressions of 11 classes.

AffectNet [28]: This is the largest dataset captured in the

wild, consisting of 0.4 million images, downloaded from the Internet using three search engines based on expression-related keywords. The images are labeled with seven basic expressions (excluding neutral), and partitioned to 2,80,000 samples for training and 3,500 images for validation.

FERA 2015 Challenge [29]: The dataset was constructed from the BP4D [22] and SEMAINE [30] datasets for AU occurrence detection and AU intensity estimation. For intensity estimation, only five AUs from BP4D are considered—AU6, AU10, AU12, AU14, and AU17—while AU occurrence detection is performed on a larger set of 14 AUs shared across BP4D and SEMAINE (AU1, AU2, AU4, AU6, AU7, AU10, AU12, AU14, AU15, AU17, AU23, AU25, AU28, and AU45). For BP4D, the original dataset is partitioned into subject-independent splits, with 21 subjects used for training, 20 for development, and an additional 20 subjects forming an extended test set. This results in 75,586 images for training, 71,261 for development, and 75,726 for testing. Similarly, the SEMAINE dataset comprises 48,000 training images, 45,000 development images, and 37,695 testing images. Both datasets provide frame-level annotations for AU occurrence and, where applicable, AU intensity. In the BP4D-extended set, onset and offset frames are assigned a B-level (level 2) intensity to maintain consistency with the ordinal labeling scheme. Across both datasets, most facially expressive segments are annotated, and AU intensities are encoded on an ordinal scale from 0 to 5, enabling fine-grained analysis of expressions.

MAFW [31]: This is the only data set consisting of short textual descriptions of facial expressions in two languages, English and Mandarin Chinese, along with categorical annotations of 11 categories of anger, disgust, fear, happiness, neutral, sadness, surprise, contempt, anxiety, helplessness, and disappointment in addition to 32 compound expression categories. The dataset provides a large-scale multi-modal compound affective database with 10,045 video-audio clips in the wild. Each clip is annotated with an emotional category and a couple of sentences describing the subject's affective behaviors in the clip.

III. TAXONOMY OF WEAK SUPERVISION IN FACIAL AFFECTIVE BEHAVIOR ANALYSIS

Traditionally, WSL scenarios are categorized as inexact, incomplete, and inaccurate supervision based on the availability of annotations [14]. To better understand the landscape of weak supervision in FABA, we extend this taxonomy along two intersecting dimensions: **level of supervision** and **type of affective task**. This dual-axis taxonomy captures both the granularity of supervision (ranging from global video-level labels to frame-level intensity scores) and the specific affective task (e.g., expression recognition or AU intensity estimation). This structure provides several benefits: it (1) offers a stereo-view of individual works, highlighting how different weakly supervised annotation strategies affect various FABA tasks; (2) enables researchers to efficiently locate pertinent research aligned with their objectives and annotation paradigms; and (3) consolidates key challenges and corresponding methodologies, promoting a structured and task-specific comparative analysis.

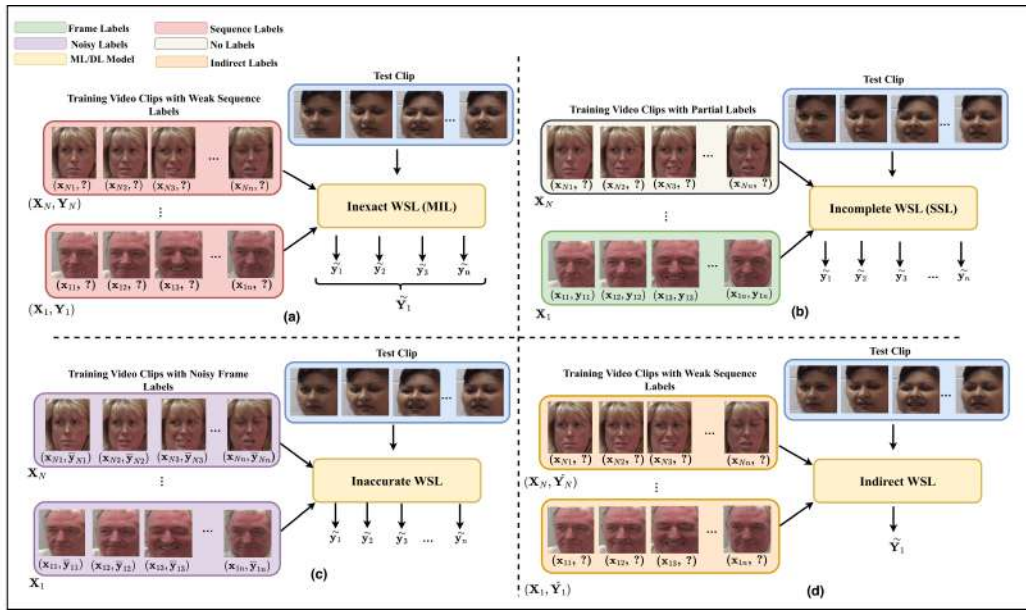


Fig. 3. An illustration of WSL scenarios for expression recognition in videos. (a) Inexact WSL: MIL with sequence-level expression labels. (b) Incomplete WSL: semi-supervised learning (SSL) using limited frame-level expression labels. (c) Inaccurate WSL: learning with noisy expression labels. (d) Indirect WSL: learning with indirect expression labels. $\mathbf{X}_1, \dots, \mathbf{X}_N$ represents N input video sequences. $\mathbf{Y}_1, \dots, \mathbf{Y}_N$ represents the direct sequence-level labels, while $\tilde{\mathbf{Y}}_1, \dots, \tilde{\mathbf{Y}}_N$ represents indirect labels. $\tilde{\mathbf{Y}}_1$ represents the sequence-level predictions of test data. $\mathbf{x}_{11}, \mathbf{x}_{12}, \dots, \mathbf{x}_{1n}$ denote n frames of sequence $\mathbf{X}_1$, and $\mathbf{y}_{11}, \mathbf{y}_{12}, \dots, \mathbf{y}_{1n}$ represents their respective frame-level expression labels. $\tilde{\mathbf{y}}_1, \tilde{\mathbf{y}}_2, \dots, \tilde{\mathbf{y}}_n$ denotes frame level predictions of test data. $\bar{\mathbf{y}}_1, \bar{\mathbf{y}}_2, \dots, \bar{\mathbf{y}}_n$ refers to noisy frame-level expression labels of sequence $\mathbf{X}_1$. “?” refers to video frames with no labels.

Specifically, we define four key categories of weak supervision scenarios: inexact (global), incomplete (sparse), inaccurate (noisy) supervision, and indirect (proxy) supervision.

A. Inexact (Global) Annotations

Annotations in this category are provided at a coarse granularity, typically at the video or image-level labels, lacking precise frame or region-level labels. The core challenge lies in associating coarse global annotations with the relevant low-level instances that reflect the target behavior, while ignoring the irrelevant ones. To tackle this, Multiple Instance Learning (MIL) [32] and attention-based models are commonly employed to identify the most discriminative frames or regions. For instance, MIL treats each sequence as a “bag” of instances (frames or clips), allowing the model to infer which instances truly contribute to the global label, with performance being influenced by bag composition, data distribution, and label ambiguity [33].

Expression Classification: This task can be formulated as static or dynamic FER, aiming to identify the spatial or temporal regions most indicative of expressions (see Fig 3a). The key challenge is to effectively model temporal dynamics to identify expression-relevant frames while simultaneously focusing on the spatial regions that convey the expression.

AU Classification: The goal is to localize AU-specific facial regions using only image-level AU labels (see Fig 4a). The challenge lies in modeling dynamic spatial variations of AUs and leveraging inter-AU relationships to resolve label ambiguity and enable precise regional attribution.

Expression or AU Intensity Estimation: This task involves capturing fine-grained temporal dynamics of expressions or

AUs to discriminate between varying intensity levels from high-level ordinal annotations (e.g., average or maximum intensity level of instance-level outputs) [34].

B. Incomplete (Sparse) Annotations

Sparse supervision arises when annotations are available for only a small subset of samples, such as selected frames in a video or a limited number of AUs, leaving most data unlabeled. The central challenge is to leverage this sparse supervision to guide learning from abundant unlabeled data without amplifying errors introduced by incorrect inductive assumptions or noisy pseudo-labels. Models trained under partial annotation are also prone to overfitting observed categories while ignoring unannotated ones, which can result in biased representations.

Expression Classification: This scenario includes (1) sparsely annotated images or video frames, requiring generalization from labeled to unlabeled data, and (2) fully labeled source datasets combined with unlabeled target data (see Fig. 3b), requiring adaptation across domains. Semi-supervised methods such as pseudo-labeling and consistency regularization exploit the structure of unlabeled data to recover missing annotations [35]. However, pseudo-labeling can amplify errors when generated labels are unreliable. Consistency regularization may encourage representation collapse, i.e., diverse inputs may have similar embeddings, leading to uninformative features.

AU Classification: The problem can be formulated as (1) limited annotations, where only a few samples have complete AU labels (see Fig. 4b), or (2) partial annotations, where only some AUs are labeled per image (missing-label setting). Challenges include modeling context-aware AU relationships

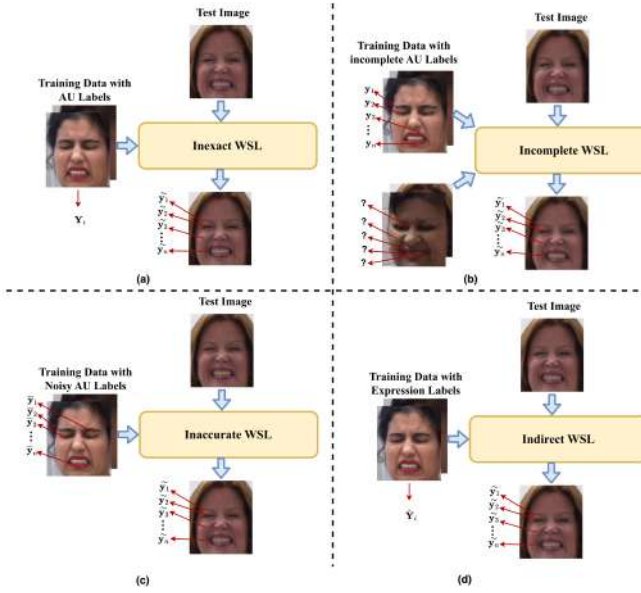


Fig. 4. An illustration of WSL scenarios for AU recognition in images. (a) Inexact WSL: MIL with image-level annotations. Y_i represents the image-level expression label for image i (b) Incomplete WSL: SSL with limited AU annotations, (c) Inaccurate WSL with noisy AU annotations. $y_1, y_2, \dots, y_n$ denote the AU labels. $\hat{y}_1, \hat{y}_2, \dots, \hat{y}_n$ represent the AU predictions of test data, and $\bar{y}_1, \bar{y}_2, \dots, \bar{y}_n$ denote the noisy AU labels. (d) Indirect WSL: WSL from implicit expression labels $\bar{Y}_i$ represents the image-level expression label for image i . Finally, "?" refers to the case with no annotations.

for reliable label propagation and maintaining flexibility to accommodate subject-specific dynamics of different AUs.

Expression or AU Intensity Estimation: This involves estimating intensity trajectories when only sparse annotations (e.g., peak or key frames) are available. Models must generalize intensity predictions across frames using temporal coherence and pseudo-label propagation.

C. Inaccurate (Noisy) Annotations

Noisy supervision requires learning robust representations that capture genuine affective cues while suppressing spurious labels introduced by subjective or ambiguous interpretation.

Expression Classification: Label noise in FER frequently arises from semantic ambiguity, where similar facial expressions may correspond to different emotions (e.g., surprise versus fear), leading to inconsistent annotations (see Fig. 3c). This ambiguity complicates the separation of overlapping affective cues from noisy supervision. Approaches such as confidence-based sample reweighting and label distribution learning mitigate these effects by modeling uncertainty rather than enforcing hard labels. Key challenges remain in obtaining reliable confidence estimates, learning uncertainty-aware representations, and handling class-specific ambiguity.

AU Classification: For AU classification, inaccurate supervision primarily stems from mislabeling of AU presence, annotator disagreement on subtle or short-lived activations, and systematic class imbalance, where less frequent AUs are under-annotated (see Fig. 4c). Such noise biases decision boundaries and weakens the learning of inter-AU dependencies, leading to implausible co-activations or missed exclusivity constraints.

As a result, models trained under noisy AU supervision may overemphasize dominant or visually salient AUs while failing to reliably detect subtle or infrequent ones. This highlights the need for uncertainty-aware learning and structured relational constraints.

Expression or AU Intensity Estimation: This task introduces the additional challenges of subjective ratings, temporal imprecision in onset–apex–offset annotation, and ambiguity at low intensity levels. These factors violate smooth temporal progression and ordinal consistency, making direct regression unstable.

D. Indirect (Proxy) Annotations

Proxy supervision refers to learning from related but indirect signals when explicit task labels are unavailable. These proxy cues often include expressions for AU recognition or textual descriptions for emotion recognition, leveraging cross-modal or inter-task correlations to transfer supervision.

Expression Classification: In FER, indirect supervision often utilizes dialogue transcripts or linguistic descriptions as auxiliary sources (see Fig. 3d). The major challenges are modeling weak cross-modal correlations, handling noisy proxy labels, ensuring robust alignment, and addressing domain variability between text and vision.

AU Classification: Facial expressions are widely used as proxy signals to guide AU detection (see Fig. 4d). This paradigm leverages the inherent correlations between expressions and localized AU activations. The main challenges include modeling inter-AU dependencies (co-occurrence and mutual exclusivity), capturing nuanced expression–AU relationships, and accounting for temporal dynamics to predict temporally coherent AUs.

Expression or AU Intensity Estimation: Estimating intensity from indirect cues such as expressions or linguistic descriptions adds another layer of complexity. The challenges include learning indirect correlations between modalities, managing semantic inconsistencies, preserving fine-grained affective nuances, and maintaining temporal coherence across sequences.

IV. METHODS FOR CLASSIFICATION

A. Inexact Annotations

1) *Expression Recognition:* Expression recognition under inexact supervision can be formulated either as spatial localization of expressions from image-level labels or as temporal localization from video-level labels, where the model must infer which spatial regions or frames actually display the target expression. Most of the existing approaches for spatial localization from image-level labels often relied on auxiliary signals (facial landmarks [36], [37], facial attributes [38], or AUs [39]), which fall outside the framework of inexact supervision. We therefore focus on temporal localization in videos from video-level labels.

Instance Level Approaches: Early works on temporal expression localization cast the problem as an instance-selection task. Each video is treated as a bag of short segments (instances), and a positive video is assumed to have at least one video segment, exhibiting the expression label. Therefore, models assign

a score to each segment and aggregate these scores to predict the video label. Sikka et al. [40] adopted this formulation using a boosting-based classifier [41], where segment scores are combined via max pooling, so that the most salient segment dominates the prediction. This design is effective when expressions are sparse and intense, and was shown to effectively capture short, high-intensity pain episodes better than treating each frame independently. Wu et al. [42] extended this framework by coupling segment-level scores with a Hidden Markov Model [43], which constrains predictions to follow a plausible temporal progression (e.g., neutral-onset-apex-offset) rather than fluctuating independently across segments. By enforcing temporal consistency, this extension improves robustness to noisy or irrelevant frames. Despite these advantages, instance-centric MIL formulations are inherently biased toward peak expressions. They perform well when a single brief expression dominates the label, but tend to underperform when multiple expression cues are distributed over time, as weaker yet persistent segments may be suppressed by max-based aggregation.

Bag Level Approaches: To better handle co-occurring or distributed expression cues, subsequent works shifted from selecting one key segment to learning richer bag representations that capture multiple affective cues and their evolution in time. Ruiz et al. [44] proposed a multi-concept approach where each video is scored by several expression-specific classifiers, rather than forcing a single expression to explain the entire video. These expression-specific classifiers allow the model to recognize multiple emotions co-occurring in the same video. This improves multi-label accuracy compared to single-expression models, but fails to capture the temporal progression of emotions across time (neutral $\rightarrow$ onset $\rightarrow$ peak $\rightarrow$ offset). Sikka et al. [45] addressed this problem by introducing temporal ordering constraints: instead of treating segments independently, it learns “templates” that enforce realistic progression of expressions by leveraging the gradual intensity changes and phase transitions of emotions (rather than just peak moments). This leads to better performance, but requires more complex training to learn these structured templates. Huang et al. [46] observed that universal bag representations often fail to generalize across individuals, as individuals express emotions differently. Their personalized approach clusters video segments according to subject-specific expression patterns, then learns customized bag-level representations for each individual. This strategy improves performance on datasets with high inter-subject variability but risks overfitting when per-subject data is limited. Finally, Wang et al. [47] noted that expressions show strong short-term correlations (nearby frames are similar) but also need long-term context to understand full dynamics. They replaced hand-crafted pooling rules with a hierarchical encoder: 3D CNNs [48] capture motion in short clips, while LSTMs/Transformers model dependencies across the full sequence. This end-to-end design better handles subtle, distributed expressions spread across long videos, but demands more computation and can be sensitive to noisy video-level labels.

Discussion: Overall, FER under inexact supervision has evolved from peak-evidence MIL, which explains the video label with a single expression [40], [42] to bag-level and tempo-

rally structured models that can integrate evidence across time and accommodate multiple, co-occurring expressions [44], [45], [47]. Methodologically, performance gains are largely driven by two complementary ideas. First, explicit temporal modeling, typically implemented through hierarchical short or long-term modeling [47], enables the model to capture both local motion patterns and long-term context, making it more effective when expression cues are weak but persistent over multiple segments [45]. Second, progression constraints, such as ordinal or phase-based modeling, impose an interpretable ordering on expression intensity or temporal stages (e.g., onset-apex-offset), helping disambiguate gradual transitions from noise [45]. Empirically, these design choices consistently improve robustness when expressions are temporally distributed rather than concentrated in a single peak. In practice, instance-centric MIL remains attractive when labels are reliable, and expression events are sparse and well-localized, whereas bag-level and hierarchical approaches [49] are better suited to complex affective scenarios involving co-occurrence, subject variability, and gradual temporal evolution.

2) AU Recognition: To reduce dependency on dense spatial annotations or predefined AU regions, recent WSL approaches seek to jointly infer where AUs occur and how they relate, using attention mechanisms and relational reasoning. The core idea is to let the model discover informative facial regions from image-level labels while simultaneously modeling inter-AU dependencies, rather than relying on fixed regions or external priors. Shao et al. [50] introduced attention and relation learning (ARL), a dual-branch architecture in which one branch learns region-level attention to localize discriminative facial areas, while the other branch explicitly models relationships among AUs to capture common co-occurrence and relational patterns. This synergy allows spatial localization and semantic dependency modeling to reinforce each other under weak supervision. However, focusing solely on spatially diverse regions may fail to capture AU-specific characteristics when multiple AUs share overlapping muscle activations, often leading to suboptimal performance. To address this problem, Li et al. [51] shifted the focus from where the model attends to how it encodes AU-specific patterns within shared spatial regions. They introduced a Self-diversified Multichannel Attention Network (SMA-Net) by employing a diversity loss that maximizes the cosine distance between attention maps of different heads, encouraging them to extract complementary features even when operating within the same region of interest. This mechanism reduces representational redundancy and mitigates attention collapse, where multiple attention heads converge on dominant AUs without capturing AU-specific features.

Discussion: Collectively, these approaches reflect a broader shift in weakly supervised AU recognition away from fixed spatial priors and toward flexible, data-driven attention mechanisms. Early works such as ARL [50] laid the groundwork by explicitly coupling spatial attention with inter-AU relational reasoning. This dual-branch architecture performs well when AU interactions follow consistent relational patterns across individuals. However, in scenarios involving overlapping muscle activations or subtle AUs, representation redundancy, and

attention collapse can hinder performance. Addressing this, more recent methods like SMA-Net [51] introduce internal attention diversification to improve feature discrimination within shared spatial regions. By enforcing disagreement among attention heads, the model learns complementary representations without requiring additional labels. Together, these strategies signal a broader trend toward context-aware and representation-diverse modeling, improving robustness in AU recognition under inexact supervision.

B. Incomplete Annotations

1) *Expression Recognition*: In this category, three dominant strategies have emerged to exploit unlabeled or sparsely labeled data: consistency regularization [52], pseudo-labeling [53], and hybrid frameworks that integrate both.

Consistency Regularization (CR): These strategies avoid explicit pseudo-label generation and instead encourage the model to produce stable predictions under input perturbations or alternative views of the same data. The core idea is that reliable semantic structure should be invariant to noise, or augmentation, allowing unlabeled data to contribute supervision without committing to potentially incorrect hard labels. Progressive Teacher (PT) [54] adopted a teacher–student paradigm, where two networks initialized differently exchange supervision only on confident, low-loss samples. By progressively refining the teacher and restricting supervision to stable predictions, this strategy improves robustness while reducing the risk of confirmation bias and preserving representation diversity. Cho et al. [55] integrate contrastive learning, transferring relational information—such as similarity and dissimilarity between samples—from labeled to unlabeled data. By selecting reliable positive and negative pairs based on learned attention patterns, they implicitly refine latent labels and promote more discriminative feature representations. In practice, consistency-based approaches are particularly effective when pseudo-labels are unreliable, but they require careful regularization to prevent representation collapse or over-smoothing when invariance constraints are enforced too aggressively.

Pseudo-Labeling: These strategies exploit high-confidence model predictions as surrogate supervision for unlabeled data, but their effectiveness is often limited by label noise, spurious attention to non-discriminative regions, and severe class imbalance. To tackle the problem of attending non-discriminatory regions, Contrastive Feature Refinement Network (CFRN) [56] explicitly refined class activation maps using a feature dropout module, which is used in a complementary fashion along with a feature emphasis module. In the feature dropout module, spurious attended regions (e.g., background or identity-specific cues) are identified using a threshold-based mask, while the feature emphasis module is used to strengthen the true key regions. By anchoring the predicted labels to more stable and meaningful visual regions, this approach helps prevent mistakes from compounding as the model continues to learn from its own predictions. Another line of approach focuses on identifying unreliable pseudo-labels through prediction disagreement. Regularized Mixture of Predictions (ReMoP) [57] compared the outputs of two classifiers that learn in different

ways—a center-loss-trained model [58], which encourages tightly grouped class features, and a simple linear classifier. When the two models disagree on a prediction, the sample is flagged as uncertain and either down-weighted or discarded. This helps the model avoid learning from noisy labels and reduces class imbalance by preventing dominant classes from dominating the training process. Complementarily, Dynamic Direction Learning (DDL) [59] adopted a teacher–student framework in which pseudo-labels are generated by a teacher model calibrated using a small, balanced validation set. Rather than enforcing strict label matching, DDL aligns the feature distributions of teacher and student using Maximum Mean Discrepancy, encouraging the student to follow the teacher’s learning direction while remaining robust to noisy supervision. In practice, such distribution-level regularization stabilizes training, but introduces additional complexity and sensitivity to the quality of the validation set.

Hybrid Approaches: These methods combine the strengths of pseudo-labeling and consistency-based learning by treating unlabeled samples differently based on how reliable their predictions appear. The key idea is to use confident predictions as supervision signals, while handling uncertain ones more cautiously guiding the model through training objectives that promote stability and separation between classes, without enforcing potentially incorrect labels. MarginMix [60] exemplified this strategy by stabilizing pseudo-labels through both margin-based class separation and prediction aggregation. By extending the center-loss objective [58], MarginMix enforces large inter-class margins in the feature space, making pseudo-labels less sensitive to small perturbations. At the same time, predictions are averaged across multiple data augmentations, so only samples with consistently stable outputs are reinforced. This combination improves pseudo-label reliability while preserving clear class boundaries in the learned representations. Building on this idea, AdaCM [61] addressed class imbalance and varying class difficulty by introducing class-specific confidence margins. Instead of using a fixed threshold for all classes, it applies higher confidence thresholds to dominant or ambiguous classes, where incorrect predictions are more likely and lower thresholds to rare or hard-to-classify classes. Confident samples are used for supervision, while uncertain ones guide representation learning through contrastive learning, improving robustness without forcing the model to rely on unreliable labels. In practice, such hybrid designs offer more stable training under weak supervision by balancing confident learning with cautious use of uncertain predictions.

Discussion: Early semi-supervised approaches largely treated high-confidence model predictions as surrogate labels for unlabeled data. While effective when decision boundaries were well separated, this strategy proved fragile in affective recognition settings, where dataset bias, class imbalance, and semantic ambiguity frequently cause confident but incorrect predictions. As a result, performance often degraded as errors accumulated over iterations. To overcome these challenges, recent pseudo labeling methods have moved beyond simple confidence thresholding toward strategies that explicitly improve pseudo-label reliability. These approaches refine where supervision is drawn from (e.g., focusing on spatially reliable

regions [56]), when pseudo-labels should be trusted (e.g., discrepancy-based filtering [57]), and how predictions are calibrated across domains or classes (e.g., distribution alignment [59]). In contrast to pseudo-labeling, CR methods [54], [55] encourage stable predictions across augmented or perturbed versions of the same input by maintaining coherence in how inputs are represented and classified under different conditions. This makes CR especially effective when temporal structure, cross-sample relationships, or augmentation-invariant features are available. However, their success often depends on the quality of augmentations and the reliability of the consistency signals they enforce. By leveraging the benefits of both pseudo labeling and CR strategies, hybrid frameworks [60], [61] enable more stable optimization and better use of unlabeled data under severe label sparsity. Overall, the field is moving away from deterministic label assignment toward uncertainty-aware representation learning, where confidence estimation governs how supervision is applied. Across methods, performance gains increasingly stem from calibrated supervision, adaptive regularization, and improved pseudo-labels.

2) *AU Recognition*: The central challenge in AU recognition with incomplete supervision is to exploit partially available annotations while modeling inter-AU dependencies that guide reliable label propagation.

Multi-label Learning with Missing Labels: Early approaches to AU recognition under incomplete supervision relied on structured priors and hand-crafted representations to compensate for missing labels. A common strategy was to explicitly encode prior knowledge about AU structure and co-occurrence to guide predictions when annotations were sparse. For instance, Song et al. [62] formulated AU recognition based on the assumption that only a small subset of AUs is active in each frame, and related AUs tend to co-occur. Their Bayesian Graphical Compressed Sensing (BGCS) framework encodes known AU co-occurrence patterns and encourages groups of related AUs to activate together, enabling effective prediction even when supervision is limited. Complementary efforts focused on propagating known labels across related samples. Wu et al. [63] introduced a multi-label learning framework with missing labels (MLML) that encourages consistency between predicted and observed labels and enforces smoothness across visually or semantically similar instances. This formulation assumes that samples, which are closer in feature space, should share similar AU activations, allowing partial annotations to influence unlabeled examples. Despite their effectiveness under structured assumptions, these methods were fundamentally limited by their reliance on hand-crafted features, which made it difficult to disentangle expression-specific facial movements from identity-related variations. To alleviate this limitation, Wu et al. [64] proposed a deep AU recognition framework that integrates Restricted Boltzmann Machines (RBM) [65] to learn latent representations capturing AU co-occurrence and mutual exclusivity directly from data. By combining unsupervised pretraining with supervised fine-tuning on partially labeled samples, this approach marked an early transition from hand-crafted features to data-driven representation learning, while still retaining explicit modeling of AU relationships.

Incomplete Supervision with Partially Labeled Samples:

Recent works have increasingly adopted end-to-end learning frameworks that integrate relational reasoning, contrastive objectives, and temporal cues, with the goal of learning AU representations that are both discriminative and robust to partial supervision. A central theme across these approaches is to replace fixed structural priors with learned dependency modeling, allowing relationships between AUs to be inferred directly from data. Multi-Label Co-Regularization (MLCR) [66] exemplified this shift by combining co-training with relational learning. Two networks are encouraged to produce consistent predictions on partially labeled data, using agreement as a regularizing signal, while graph convolutional networks are employed to learn AU dependencies dynamically rather than relying on predefined co-occurrence rules. This co-regularization stabilizes learning under sparse supervision, but introduces additional model complexity and sensitivity to graph construction. Complementary approaches leverage contrastive learning to compensate for partial supervision by directly learning discriminative features to promote class separation and improve representations. Liu et al. [67] improved performance by enforcing discrimination at both the instance (sample) level and the prototype (class) level, encouraging samples that share known AUs to cluster in the feature space while separating samples with conflicting annotations. To ensure that contrastive learning focuses on meaningful facial evidence, anatomical priors are used to guide attention toward plausible AU-related regions. This design helps to reduce the influence of spurious cues under noisy supervision. However, because attention masks are fixed and AU relationships are not modeled explicitly, the approach may generalize poorly across subjects, poses, or expression variations that deviate from the assumed spatial patterns. The Knowledge-Spreader (KS) framework [68] adopted a teacher–student architecture in which a transformer learns AU dependencies from sparsely annotated keyframes and propagates this knowledge to unlabeled frames via pseudo-labeling and temporal modeling. By unifying spatial attention, relational reasoning, and temporal consistency within a single framework, KS achieves strong performance under partial supervision, although its effectiveness depends on the representation quality of the annotated keyframes. Finally, Yan et al. [69] integrate AU recognition with auxiliary supervisory signals, combining region-of-interest inpainting to enforce localized AU representations, optical flow to inject motion cues, and a transformer that encodes AU relationships. When coupled with MixMatch [70], this multi-signal supervision yields robust features despite sparse labels, but at the cost of increased computational complexity and reliance on the accuracy of inpainting and motion estimation.

Discussion: Approaches like BGCS [62] and MLML [63] compensate for missing AU labels by imposing static structural priors such as sparsity, co-occurrence, or label smoothness on hand-crafted features. While effective under controlled conditions, these assumptions limit their ability to adapt to complex, context-dependent AU interactions and to generalize across subjects, poses, and uncontrolled environments. DL-based methods [64] alleviate these limitations by jointly learning feature representations and label structure from data,

enabling flexible modeling of AU relationships. This shift paved the way for adaptive relation modeling using graph neural networks and transformers [66], [68], which infer AU co-occurrence and mutual exclusivity patterns directly from observations rather than relying on predefined rules. By updating relational structure during training, these models can better capture dataset-specific dependencies, albeit at the cost of increased computational complexity and sensitivity to noisy supervision. To further exploit partially labeled data, several methods introduce auxiliary learning signals that do not rely solely on explicit labels. Co-training and contrastive objectives [66], [67] extract supervisory signal by encouraging consistency between models or by shaping the feature space so that samples sharing known AUs are closer than those with conflicting annotations. These strategies are particularly effective when anatomical priors or reliable similarity relationships are available, but their performance can degrade when such priors are inaccurate or incomplete. Temporal teacher–student frameworks [68] learn spatial and relational AU representations from a small set of reliable key-frames and enables efficient use of sparse annotations, but its success is influenced by the representation quality of keyframes and the calibration of confidence thresholds. The field has transitioned from rigid priors and hand-crafted features toward adaptive, end-to-end frameworks that learn inter-AU relations from data, propagate supervision from partial or missing labels, and exploit temporal context to stabilize learning.

C. Inaccurate Annotations

1) *Expression Recognition*: Though expression recognition with noisy annotations has also been addressed due to pose and occlusion challenges, we focus specifically on methods that explicitly model and mitigate noisy labels.

Label Refinement from Clean Sample Selection These approaches refine supervision by explicitly separating reliable samples from uncertain ones, enabling the model to learn from cleaner signals while reducing the impact of noisy labels. Wang et al. [71] proposed Self-Cure Network (SCN), which estimates sample reliability within each mini-batch using a self-attention–based confidence score. The key intuition is that samples consistently receiving strong, focused attention are more likely to be correctly labeled. SCN therefore prioritizes these high-confidence samples for supervision, while low-confidence samples are softly relabeled using prediction agreement, preventing early overfitting to noise. Zhang et al. [72] introduced Relative Uncertainty Learning (RUL), which measures uncertainty through relative sample difficulty rather than relying on raw loss magnitudes or static confidence thresholds. By applying feature-level mixup, RUL exposes inconsistencies in facial representations and down-weights unreliable features and labels. This design allows the model to retain informative but hard samples, instead of discarding them as noise. Building on this idea, Nie et al. [73] proposed the IntraClass-Divide framework, which models loss distributions separately for each expression class using Gaussian Mixture Models (GMMs). Unlike global GMM approaches that often confuse inherently difficult expressions with mislabeled samples, this class-specific modeling better captures heterogeneous

noise patterns and enables more precise separation of clean and noisy data. Together, these methods demonstrate how uncertainty estimation and class-aware sample selection can stabilize learning under noisy annotations without requiring additional supervision.

Label Distribution Learning (LDL): LDL addresses annotation noise and ambiguity by replacing hard labels with soft label distributions [74] that encode uncertainty over multiple affective states. This formulation is particularly well suited to facial expression analysis, where annotators often disagree or where expressions lie between discrete categories. Early LDL-based approaches focused on estimating reliable label distributions from noisy supervision. She et al. [75] inferred soft labels by aggregating predictions from multiple auxiliary classifiers and weighting them using uncertainty estimates. By encouraging samples with similar feature representations to share similar label distributions, the method smoothens inconsistent annotations while preserving inter-class separability in the feature space. Subsequent works incorporated teacher-guided refinement to stabilize distribution estimation. Lukov et al. [76] adopted a mean-teacher framework, where averaged teacher predictions are used to generate soft label distributions for the student network. High-confidence logits are converted into soft probability distributions, while low-confidence classes receive small but non-zero probabilities. This discourages the model from making overly confident predictions on uncertain inputs. To further reduce the impact of noisy or unreliable outputs, the process is regularized using uncertainty estimation techniques [71]. More recent methods addressed the dynamic nature of label noise during training. Self-Paced Label Distribution Learning (SPLDL) [77] introduced a curriculum strategy that progressively learns hard and noisy samples. Using a predefined label distribution generator [78], the model first learns from low-loss, reliable instances and gradually adapts the label distributions as training progresses, improving robustness to severe and class-dependent noise. Overall, LDL has evolved from static, heuristically defined soft labels to adaptive, uncertainty-aware, and curriculum-driven distributions. This progression improves robustness to ambiguous supervision but introduces additional complexity and sensitivity to distribution initialization and uncertainty estimation quality.

Ensembling for Noise Mitigation: Multi-view aggregation methods mitigate annotation noise by combining predictions or feature representations obtained from multiple annotations, model instances, or data perturbations, thereby reducing the influence of any single unreliable view. Zeng et al. [79] proposed a framework that explicitly treats noisy labels as multiple inconsistent observations of an underlying latent ground truth. Their method trains models on these conflicting annotations and then infers latent labels by aggregating pseudo-annotations through log-likelihood maximization, effectively filtering noise by favoring label configurations that are globally consistent with the data. Subsequent works shifted attention from label aggregation to feature- and attention-level consistency. Zhang et al. [80] observed that, under noisy supervision, deep models tend to overfit spurious local regions that correlate with incorrect labels. To counter this, they

enforce consistency between attention maps generated from augmented versions of the same image, encouraging the model to rely on stable, globally informative facial regions rather than noisy cues. More recently, Zheng et al. [81] proposed a geometry-aware framework that improves robustness to label noise and class imbalance. Instead of treating all samples equally, the method detects mislabeled data by measuring how sensitive a sample's features are to small adversarial changes in a geometry-informed space. Samples that react abnormally are flagged as likely noisy. The model then applies a divide-and-conquer strategy, treating clean and noisy data separately and rebalancing training to avoid bias toward dominant classes. By explicitly incorporating facial geometric structure, the approach provides more reliable training even in the presence of annotation errors. Overall, these approaches reflect a progression from label-level voting schemes toward representation- and geometry-aware consistency mechanisms, with increasing emphasis on identifying unreliable supervision and enforcing stability across views to guide learning under noisy annotations.

Discussion: Clean-sample selection methods have evolved from global, loss-based heuristics to class-aware and uncertainty-weighted filtering, explicitly accounting for the fact that annotation noise often varies across expressions [71]–[73]. By estimating sample reliability in a class-conditional manner and weighting supervision accordingly, these approaches improve robustness to class-specific noise patterns. However, their effectiveness depends critically on accurate confidence estimation: when confidence scores are poorly calibrated, clean and noisy samples become difficult to separate, limiting their benefit. LDL addresses noise from a different perspective by replacing hard labels with soft distributions that encode ambiguity and annotator disagreement. Early LDL approaches relied on auxiliary prediction heads to estimate these distributions [75], while more recent methods incorporate teacher-guided smoothing and self-paced curriculum that progressively refine soft labels as training stabilizes [76], [77]. This formulation is particularly effective when expressions are inherently ambiguous or subjectively annotated, but its performance hinges on well-calibrated soft labels and reliable teacher signals. Ensembling and consistency-based methods take yet another route, stabilizing learning by enforcing agreement across multiple views, model instances, or feature geometries [79]–[81]. Rather than correcting labels explicitly, these approaches constrain representations to remain invariant under perturbations, which is especially effective when noise manifests as spurious saliency or fragmented supervision. Across these methods, a unifying trend is a shift away from deterministic relabeling toward probabilistic, representation-centric noise handling, emphasizing uncertainty awareness, agreement-based regularization, and calibration. Persistent challenges remain, including structured label noise, underrepresented expressions, and biases correlated with identity or context. These limitations increasingly motivate hybrid pipelines that combine soft-label modeling with consistency constraints, and confidence calibration with class- or geometry-aware sample selection.

2) *AU Recognition:* Recent approaches mitigate label noise in weakly supervised AU recognition by exploiting inherent data structure, using representation refinement and consistency constraints instead of explicit label correction. Zhao et al. [82] addressed AU learning by collecting large-scale web images through expression-based keyword searches, treating the retrieved expression labels as noisy AU annotations. Their method employs a scalable clustering to refine these weak labels and learns a spectral embedding that pulls together visually and semantically similar faces, followed by rank-order clustering to form confident AU-specific groups. These clusters are re-labeled via majority vote and used to train AU detectors. This weakly supervised pipeline effectively denoises noisy labels without manual annotation, achieving high agreement with expert labels. The approach relies on the accuracy of initial tags and models each AU independently, but shows that embedding-guided relabeling can generate high-quality supervision from large, noisy web image collections. Fabian et al. [83] introduced a global–local learning objective that strengthens region-level AU cues while enforcing structural consistency within images. The key intuition is that the spatial organization of facial movements remains stable even with noisy annotations. Encouraging agreement between local AU evidence and global facial structure helps prevent overfitting to annotation errors. More recently, hybrid co-training frameworks have combined clean-sample selection with noise-aware regularization. ReCoT [84] separated training into two complementary streams: a clean network trained on class-balanced, high-confidence samples, and a noisy network regularized using all available data. Cross-view consistency aligns predictions between these networks, while temporal smoothing and fusion progressively refine noisy labels, mitigating class imbalance and reducing error accumulated over time.

Discussion: Embedding- and clustering-based approaches [82] are most effective when large weakly labeled datasets exhibit coherent latent structure, providing lightweight denoising with modest computational cost, but remaining sensitive to clustering granularity and representation quality. Global–local consistency methods [83] primarily improve spatial reliability under coarse supervision, helping suppress spurious attention but offering limited correction for systematic class-level noise. Co-training frameworks such as ReCoT [84] is found to be promising in handling severe noise and class imbalance by explicitly separating trusted from unreliable supervision and enforcing cross-view agreement, albeit with higher complexity and greater sensitivity to hyperparameter choices (e.g., confidence thresholds and agreement criteria). Overall, these approaches illustrate a shift from confidence-based filtering toward structure- and agreement-driven noise handling, emphasizing robustness through data organization, representation constraints, and training dynamics rather than direct label correction.

D. Indirect (Proxy) Annotations

Indirect supervision leverages auxiliary signals such as captions, dialogue, or language-model outputs—that are semantically related to facial expressions but do not provide explicit

emotion annotations. The central intuition is that language carries rich affective information that can guide representation learning when direct labels are unavailable, allowing models to scale to large, in-the-wild datasets.

1) *Expression Recognition*: Recent vision–language approaches replace categorical emotion labels with language priors learned through cross-modal alignment. EmotionCLIP [85] aligned facial features with dialogue-based sentiment using contrastive learning within a CLIP-style video–text framework [86]. To improve grounding in affective signals, it uses subject-aware context encoding to help the model attend to nonverbal cues (like facial expressions and posture) and employs sentiment-guided contrastive learning to align visual features with verbal emotion cues. Trained on multimodal data from films and vlogs, it can learn affect-rich representations without relying on manual emotion labels. However, since speech and facial expressions may not always align temporally or semantically (e.g., sarcasm, off-screen dialogue), the weak supervision signal can introduce noise—particularly at the frame level. EmoCLIP [87] addressed this limitation by replacing coarse emotion categories with sample-level descriptive captions, which provide richer and more context-aware supervision. They fine-tuned CLIP to directly align video frames with their associated textual descriptions using contrastive objectives, leading to richer supervision and better generalization to subtle expressions. This benefit, however, comes at the cost of relying on high-quality captions, which often require expert annotation or careful curation. More recently, Exp-CLIP [88] removes human annotation entirely by aligning facial representations with LLM-derived expression embeddings, enabling label-efficient and fine-grained zero-shot FER. By training a lightweight projection head to match facial features with LLM-derived descriptions through contrastive learning, the model learns expression-specific semantics without manual labels, achieving fine-grained, zero-shot FER with strong generalization across diverse datasets. However, its effectiveness depends on the quality of prompts, the semantic coverage of the generated embeddings, and the diversity of the unlabeled visual data used for alignment.

Discussion: Vision–language weak supervision for FER replaces explicit emotion labels with language priors, introducing a spectrum of trade-offs between scalability, semantic precision, and robustness. Dialogue-driven supervision [85] scales naturally with large volumes of multimodal video, making it attractive for data-hungry models, but is prone to modality drift when speech content is weakly aligned with facial expressions, particularly in the absence of frame-level grounding. Caption-driven supervision [87] offers more targeted and interpretable affective cues, improving semantic specificity, but its effectiveness depends on caption quality and limits throughput due to annotation or curation costs. LLM-derived embedding supervision [88] is the most label-efficient alternative, enabling fine-grained zero-shot recognition, yet it remains sensitive to prompt design, embedding calibration, and the semantic coverage of the unlabeled corpus. Across these variants, empirical evidence suggests that temporal grounding (e.g., frame- or segment-level alignment), face-centric constraints to reduce off-screen or contextual leakage,

and normalization of linguistic supervision (prompt or caption style and vocabulary) are critical for stabilizing training and preserving fine-grained facial cues. Collectively, these approaches highlight a core trade-off in indirect supervision for FER: reducing annotation cost and improving scalability comes at the expense of increased sensitivity to language priors and cross-modal alignment quality.

2) *AU Recognition*: Motivated by the lower cost and wider availability of expression-level annotations, early approaches relied on fixed expression–AU mappings derived from statistical correlations (see Table I). Ruiz et al. [89] introduced a hidden task learning framework that transfers supervision from expressions to AUs using predefined expression–AU associations. During training, the model predicts both expressions and AUs, but only expression labels are supervised directly. A consistency loss is explored ensures that the predicted AUs align with the expected AU pattern of the predicted expression, allowing gradients to train the AU branch without AU labels. While conceptually simple, this method assumed static and independent mappings, limiting its ability to model joint AU dependencies or adapt to individual or contextual variability. To address these limitations, subsequent probabilistic methods introduced probabilistic priors to model uncertainty in expression–AU relationships. These methods typically generate pseudo AU labels from soft priors (e.g., expression–AU statistics) and then refine these labels using models that encode structural dependencies among AUs. For instance, Wang et al. [90] used RBM to learn the joint distribution of AUs by modeling co-activation and mutual exclusivity between AUs as a latent binary representation, where visible units correspond to AU labels and hidden units capture high-level dependencies. During training, pseudo AU labels (inferred from expression priors) are input as constraints, and the RBM learns to adjust the joint AU predictions to reflect plausible co-occurrence patterns. This allows the model to correct noisy pseudo-labels by enforcing biologically and contextually realistic AU combinations. In contrast, Peng et al. [91] explored adversarial framework by guiding the AU predictor using a discriminator that distinguishes between the predicted AU distributions and a prior distribution derived from expression–AU associations. The AU predictor is trained to “fool” the discriminator by producing AU distributions that resemble the prior, while the discriminator learns to detect mismatches. This adversarial objective ensures that AU outputs align with known priors while still allowing flexibility to deviate when needed — thus balancing guidance and adaptability. These strategies improve flexibility over fixed mappings, but their performance remains sensitive to the quality and transferability of handcrafted priors. To tackle this problem, ordinal and fusion-based frameworks aimed to improve flexibility and robustness by moving beyond reliance on a single prior. Wang et al. [92] incorporated domain knowledge in the form of partial ordering constraints between AUs. For example, if AU12 is typically stronger than AU14 in expressions of happiness, the model enforces this ordering during learning by introducing a ranking loss or ordinal margin. This guides the model to predict relative AU activations even when absolute labels are unavailable, helping it focus on relational structure rather than precise

intensity values. On the other hand, LP-SM [93] addressed the unreliability of individual priors by combining multiple weak cues such as expression-to-AU mappings, region-based saliency, and statistical priors—into a single model in a probabilistic fusion framework. By learning to weigh each prior dynamically based on the consistency of its predictions with the data, LP-SM can attenuate the influence of misleading sources while emphasizing more reliable ones. This fusion improves robustness within a dataset, though generalization across datasets remains difficult when expression–AU associations vary significantly. To further enhance weak supervision, dual-task and dual-direction frameworks introduce stronger inductive constraints by coupling AU recognition with face synthesis, thereby enforcing consistency between prediction and visual appearance. In WSDL [94], the model is trained in a two-stage setup: first, an AU recognition branch predicts AU activations from facial images, and then a face generation module (a decoder or generator network) attempts to reconstruct the original face image conditioned on the predicted AUs. Based on the reconstruction error, the self-reconstruction loop allows the model to verify the plausibility of AU predictions by checking whether they can explain the visual input. Peng et al. [95] extend this idea with a GAN-based framework where a generator synthesizes face images from predicted AUs, and a discriminator checks whether the output matches both the real distribution and AU configuration. In parallel, an AU recognition branch is trained on both real and generated faces, promoting consistency between AU predictions and facial appearance. This bidirectional setup strengthens learning under weak supervision by ensuring that predicted AUs are both discriminative and visually plausible.

TABLE I
LIST OF AUs OBSERVED IN EXPRESSIONS [8].

Expression	AUs
Anger	4, 5, 7, 10, 17, 22-26
Disgust	9, 10, 16, 17, 25, 26
Fear	1, 2, 4, 5, 20, 25, 26, 27
Happiness	6, 12, 25
Sadness	1, 4, 6, 11, 15, 17
Surprise	1, 2, 5, 26, 27
Pain	4, 6, 7, 9, 10, 12, 20, 25, 26, 27, 43

Discussion: AU recognition from expression supervision has progressed from fixed expression–AU mappings toward probabilistic and bidirectional formulations that better accommodate ambiguity, subject variability, and inter-AU dependencies. Fixed mappings [89] are effective when expression–AU relationships are reliable, and variation is limited, but they break down under cross-subject differences and context-dependent expression realizations. Probabilistic pseudo-AU approaches [89], [91] fuse multiple weak cues such as expression tags, co-occurrence statistics, or region scores, and are preferable when supervision is heterogeneous; however, they remain sensitive to bias or mismatch in the underlying priors. Ordinal and unified-prior models [92], [93] provide a more stable alternative when absolute AU probabilities are unreliable but relative relationships between AUs remain consistent, allowing

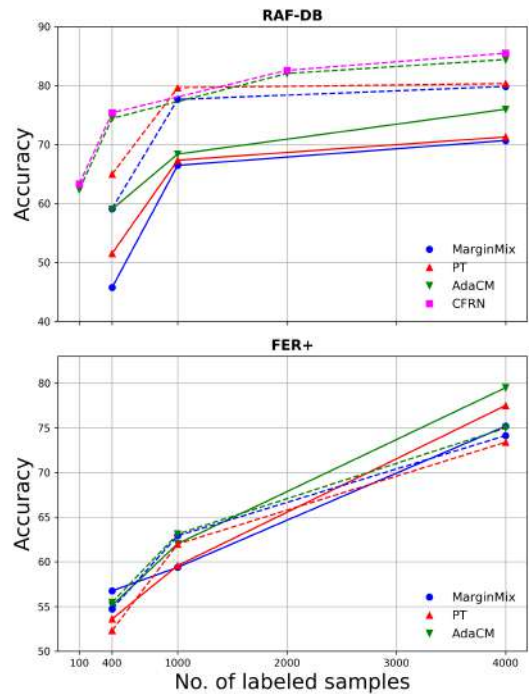


Fig. 5. Accuracy of expression classification methods with incomplete annotations on RAF-DB (top) and FER+ (bottom) datasets. Dotted and dashed lines denote performance using Resnet-18 and WideResnet-28-2, respectively.

learning to proceed from partial yet robust structural information. Dual-task or dual-direction frameworks [94], [95] impose the strongest inductive constraints by requiring predicted AUs to both explain and reconstruct facial configurations, yielding improved robustness under weak supervision at the cost of increased computational burden and tuning complexity. Across these paradigms, recurring challenges—including biased priors, cross-dataset drift, and optimization overhead—highlight the need for uncertainty-aware weighting of pseudo-labels, deformation-sensitive localization to handle non-rigid facial motion, and lightweight temporal grounding to stabilize learning without relying on dense AU annotations.

E. Model Evaluation Protocols

To compare methods, the annotations of datasets are further modified to match the corresponding task. For the classification task, the performance of expressions and AUs are expressed in terms of accuracy and F1-score, respectively. For a fair comparison, we have chosen methods that follow the same experimental protocol for expressions and AUs and presented in Tables II and III, respectively. Unless specified, all the results are shown in a within-database setting.

Inexact Annotations: Although the UNBC-McMaster dataset is annotated with pain intensity levels, it has been used for classification by converting its ordinal labels (OPI ratings) to binary labels based on a threshold, i.e., $OPI \geq 3$ is considered as pain and $OPI=0$ no pain, resulting in a total of 149 sequences with 57 positive and 92 negative bags. In all these methods, Leave-One-Subject-Out (LOSO) cross-validation is followed. For AU classification, BP4D and DISFA are widely

TABLE II

PERFORMANCE OF STATE-OF-THE-ART FACIAL EXPRESSION CLASSIFICATION APPROACHES UNDER VARIOUS WSL SETTINGS ON THE MOST WIDELY EVALUATED DATASETS. 'CONVENTIONAL' REFERS TO THE TRAINING AND TESTING PARTITIONS PROVIDED BY THE DATASET ORGANIZERS. † DENOTES THAT THE MODEL IS TRAINED ON BOTH RAF-DB AND AFFECTNET TRAINING SETS. FOR INCOMPLETE SUPERVISION, AFFECTNET IS USED AS UNLABELED DATA.

WSL Setting	Dataset	Method	Task	Features	Learning Model	Validation	Accuracy
Inexact	UNBC - McMaster	Sikka et al. [40] (FG 2013)	Pain [2 classes]	BoW	MILBOOST	LOSO	83.70
		Wu et al. [42] (FG 2015)	Pain [2 classes]	geometric	HMM	LOSO	85.23
		Adria et al. [44] (BMVC 2014)	Pain [2 classes]	3D-SIFT	RMC-MIL	LOSO	85.70
		Sikka et al. [45] (CVPR 2016)	Pain [2 classes]	SIFT	LOMo	LOSO	87.00
		Huang et al. [46] (TAFFC 2016)	Pain [2 classes]	geometric	P-MIL	LOSO	84.40
Incomplete	RAF-DB	Li et al. [61] (CVPR 2022)	Expression [7 classes]	ResNet-18	Ada-CM	Conventional	89.28
		Jiang et al. [54] (TAFFC 2023)	Expression [7 classes]	ResNet-18	PT	Conventional	88.69
		Li et al. [57] (IJCAI 2025)	Expression [7 classes]	ResNet-18	ReMoP	Conventional	90.29
		Li et al. [59] (TAFFC 2025)	Expression [7 classes]	ResNet-18	DDL	Conventional	89.58
	FER+	Li et al. [61] (CVPR 2022)	Expression [8 classes]	ResNet-18	Ada-CM	Conventional	87.80
		Jiang et al. [54] (TAFFC 2023)	Expression [8 classes]	ResNet-18	PT	Conventional	86.60
		Li et al. [57] (IJCAI 2025)	Expression [8 classes]	ResNet-18	ReMoP	Conventional	89.06
		Li et al. [59] (TAFFC 2025)	Expression [8 classes]	ResNet-18	DDL	Conventional	89.16
Inaccurate	RAF-DB	Zeng et al. [79]† (ECCV 2018)	Expression [7 classes]	ResNet [96]	IPA2LT	Conventional	86.77
		Wang et al. [71] (CVPR 2020)	Expression [7 classes]	ResNet-18	SCN	Conventional	87.03
		She et al. [75] (CVPR 2021)	Expression [7 classes]	ResNet-18	DMUE	Conventional	88.76
		Zhang et al. [72] (NeurIPS 2021)	Expression [7 classes]	ResNet-18	RUL	Conventional	88.98
		Lukov et al. [76] (ECCV 2022)	Expression [7 classes]	ResNet-18	SLS	Conventional	90.42
		Zhang et al. [80] (ECCV 2022)	Expression [7 classes]	ResNet-18	EAC	Conventional	89.99
		Shao et al. [77] (ACM MM 2022)	Expression [7 classes]	ResNet-18	SPLDL	Conventional	88.59
		Zheng et al. [81] (AAAI 2023)	Expression [7 classes]	ResNet-18	GAAVE	Conventional	91.53
	AffectNet	Zeng et al. [79]† (ECCV 2018)	Expression [8 classes]	ResNet [96]	IPA2LT	Conventional	55.71
		Wang et al. [71] (CVPR 2020)	Expression [8 classes]	Resnet-18	SCN	Conventional	60.23
		Shao et al. [77] (ACM MM 2022)	Expression [7 classes]	ResNet-18	SPLDL	Conventional	59.76
		She et al. [75] (CVPR 2021)	Expression [8 classes]	ResNet-18	DMUE	Conventional	62.84
		Zhang et al. [72] (NeurIPS 2021)	Expression [8 classes]	ResNet-18	RUL	Conventional	61.43
		Lukov et al. [76] (ECCV 2022)	Expression [8 classes]	ResNet-18	SLS	Conventional	62.69
		Zhang et al. [80] (ECCV 2022)	Expression [7 classes]	ResNet-18	EAC	Conventional	65.32
		Zheng et al. [81] (AAAI 2023)	Expression [8 classes]	ResNet-18	GAAVE	Conventional	63.25
	FER+	Wang et al. [71] (CVPR 2020)	Expression [8 classes]	ResNet-18	SCN	Conventional	89.35
		She et al. [75] (CVPR 2021)	Expression [8 classes]	ResNet-18	DMUE	Conventional	88.64
		Zhang et al. [72] (NeurIPS 2021)	Expression [8 classes]	ResNet-18	RUL	Conventional	88.75
		Zhang et al. [80] (ECCV 2022)	Expression [8 classes]	ResNet-18	EAC	Conventional	89.64
		Lukov et al. [76] (ECCV 2022)	Expression [8 classes]	ResNet-18	SLS	Conventional	88.60
		Zheng et al. [81] (AAAI 2023)	Expression [8 classes]	ResNet-18	GAAVE	Conventional	89.29
Indirect	MAFW	Zhang et al. [85] (CVPR 2023)	Expression [43 classes]	ViT	EmotionCLIP	5-fold	11.65
		Foteinopoulou et al. [87] (FG 2024)	Expression [43 classes]	ViT	EmoCLIP	5-fold	33.49
		Zhang et al. [88] (WACV 2025)	Expression [43 classes]	ViT	ExpCLIP	5-fold	26.98

used, where 8 AUs are considered for DISFA and 12 AUs for BP4D. For the DISFA dataset, AU intensities equal or greater than 2 are considered as occurrences, otherwise considered as non-occurrences. The DISFA dataset has a more significant data imbalance problem than BP4D, in which most of the AUs have very low occurrence rates.

Incomplete Annotations: RAF-DB and FER+ are widely used datasets for expression classification with SSL. SSL methods for FER have been evaluated using two experimental setups. First, the labeled images are randomly chosen at growing levels of supervision at 100, 400, 1000, 2000 and 4000 within RAF-DB and FER+, where the remaining images are considered to be unlabeled. Since most approaches [54], [60], [61] use both Resnet-18 [96] and WideResNet-28-2 [98]

as backbones pre-trained on MS-Celeb-1M [99], we have compared methods on both backbones (see Figure 5). Second, some methods [54], [57], [59], [61] explored training with the complete labeled images of RAF-DB and FER+, while AffectNet images are used as unlabeled images. For AU detection, BP4D and DISFA also widely rely on two experimental setups. Some of the methods [62], [64], [68] explored within-dataset evaluation, where the number of available labels is randomly dropped with a growing proportion of 50, 60, 70, 80 and 90 as shown in Figure 6. In contrast, some of the methods [66], [69] used the EmotionNet dataset as unlabeled images, while the complete set of labels of within-data are used.

Inaccurate Annotations: RAF-DB, FER+, and AffectNet have been widely used to deal with noisy annotations for

TABLE III
PERFORMANCE OF STATE-OF-THE-ART AU CLASSIFICATION APPROACHES UNDER WSL SETTINGS ON MOST WIDELY EVALUATED DATASETS. FOR INCOMPLETE SUPERVISION, EMOTIONET [97] IS USED AS UNLABELED DATA.

WSL Setting	Dataset	Method	Task	Features	Learning Model	Validation	F1-Score
Inexact	DISFA	Shao et al. [50] (TAFFC 2022)	AU [8 AUs]	CNN	ARL	3-fold	58.7
		Li et al. [51] (TAFFC 2024)	AU [8 AUs]	CNN	SMA-Net	3-fold	62.4
	BP4D	Shao et al. [50] (TAFFC 2022)	AU [12 AUs]	CNN	ARL	3-fold	61.1
		Li et al. [51] (TAFFC 2024)	AU [12 AUs]	ViT	SMA-Net	3-fold	65.2
Incomplete	DISFA	Niu et al. [66] (NeurIPS 2019)	AU [8 AUs]	ResNet-34	MLCR	3-fold	59.2
		Yan et al. [69] (TMM 2023)	AU [8 AUs]	ResNet-18	WSRTL	3-fold	64.6
	BP4D	Niu et al. [66] (NeurIPS 2019)	AU [12 AUs]	ResNet-34	MLCR	3-fold	59.8
		Yan et al. [69] (TMM 2023)	AU [12 AUs]	ResNet-18	WSRTL	3-fold	65.9
Indirect	UNBC - McMaster	Ruiz et al. [89] (ICCV 2015)	AU [6 AUs]	SIFT	HTL	5-fold	23.5
		Wang et al. [90] (TAFFC 2020)	AU [6 AUs]	geometric	RBM-P	5-fold	35.1
		Wang et al. [94] (TMM 2019)	AU [6 AUs]	geometric	WSDL	5-fold	40.0
		Peng et al. [91] (CVPR 2018)	AU [6 AUs]	geometric	RAN	5-fold	37.6
	CK+	Ruiz et al. [89] (ICCV 2015)	AU [12 AUs]	SIFT	HTL	5-fold	46.9
		Wang et al. [90] (TAFFC 2020)	AU [12 AUs]	geometric	RBM-P	5-fold	70.5
		Wang et al. [94] (TMM 2019)	AU [12 AUs]	geometric	WSDL	5-fold	74.0
		Peng et al. [91] (CVPR 2018)	AU [12 AUs]	geometric	RAN	5-fold	71.5
		Zhang et al. [93] (CVPR 2018)	AU [8 AUs]	LBP	LP-SM	5-fold	73.2
	MMI	Ruiz et al. [89] (ICCV 2015)	AU [14 AUs]	SIFT	HTL	5-fold	43.1
		Wang et al. [90] (TAFFC 2020)	AU [13 AUs]	geometric	RBM-P	5-fold	51.6
		Wang et al. [94] (TMM 2019)	AU [13 AUs]	geometric	WSDL	5-fold	53.0
		Peng et al. [91] (CVPR 2018)	AU [13 AUs]	geometric	RAN	5-fold	52.0
		Zhang et al. [93] (CVPR 2018)	AU [8 AUs]	LBP	LP-SM	5-fold	48.1
	DISFA	Ruiz et al. [89] (ICCV 2015)	AU [12 AUs]	SIFT	HTL	5-fold	37.1
		Wang et al. [90] (TAFFC 2020)	AU [12 AUs]	geometric	RBM-P	5-fold	42.4

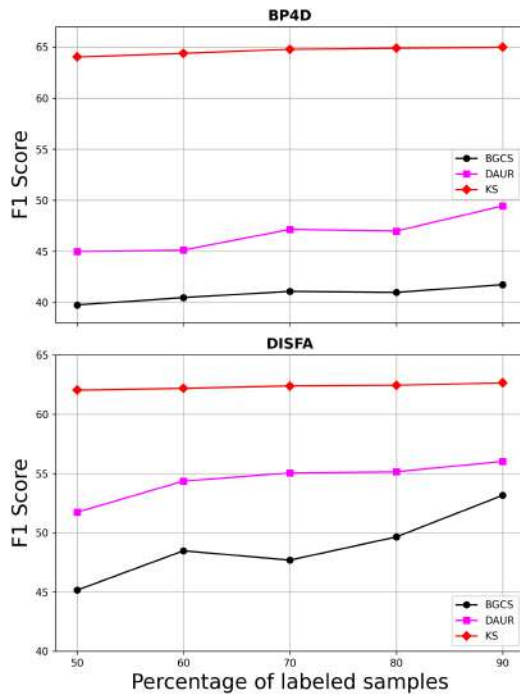


Fig. 6. F1 Score of AU detection methods with incomplete annotations on BP4D and DISFA datasets.

expression recognition. For the AffectNet dataset, 33,803 facial images are considered clean data, and 1,71,005 facial images are considered noisy data for training. 3500 facial images are further used for validation, with 500 images per expression. In the case of RAF-DB, 12271 training samples are used for training, and the results are reported on the test set of 3068 samples. For the FER+ dataset, 28,709 images are used for training, 3,589 for validation, and 3,589 for testing. With all these datasets, the results are shown based on the training and testing partitions provided by the dataset organizers.

Indirect Annotations: For expression classification, the MAFW dataset has been widely explored for indirect supervision, as it is the only dataset with textual descriptions for the facial expressions. For AU classification, 7319 frames are chosen from 30 video sequences of 17 subjects that exhibit the expression of pain with $PSPI \geq 5$. Six AU labels are associated with the chosen frames, i.e., AU4, AU6, AU7, AU9, AU10, and AU43, which are related to pain expressions. In all these methods, 5-fold cross-validation is used. For CK+, MMI, and DISFA datasets, AUs, which are available for more than 10% of all frames, are chosen. Based on this criterion, 309 sequences of 106 subjects are chosen from 593 sequences of 123 subjects, resulting in 12 AUs for the CK+ dataset. For the MMI dataset, AUs are available for more than 10% of all samples, resulting in 171 sequences from 27 subjects with 13

labels. For the DISFA dataset, 482 apex frames are chosen based on AU intensity levels, for which expression labels are obtained by FACS. Similar to CK+ and MMI, 9 AUs are considered. 5-fold cross-validation is deployed, where 20% of the whole database is used as a validation set according to subjects. All the experiments are conducted in a subject-independent protocol.

F. Critical Analysis and Results

1) *Temporal Modeling*: For FER, temporal modeling has been explored primarily under inexact and indirect supervision. Under inexact supervision, early approaches aggregated frame evidence via max-pooling [40] or leveraged facial landmark displacements [42]. Subsequent work of LOMo [45] encoded temporal progression with ordinal constraints, yielding clear gains over conventional aggregates [40], [42]. More recently, [47] jointly captured short- and long-term dynamics using R3D features with multi-head attention, further improving robustness. Under indirect supervision, EmoCLIP [87] enhanced CLIP image embeddings with transformer-based temporal modeling and reported a substantial performance margin (about 6%).

For AU recognition under WSL, most prior work emphasizes modeling inter-AU relationships, while temporal cues remain comparatively underexplored. Notable exceptions under incomplete annotations include KS [68], which models temporal evolution in the student with transformers and propagates temporal context to refine the teacher, exhibiting strong label-efficiency (minimal degradation as the labeled fraction decreases) and reporting an absolute improvement of roughly 15% on BP4D (see Fig 6). Similarly, WSRTL [69] incorporates optical flow as an auxiliary objective to inject motion information, yielding consistent gains of about 5% across both datasets of BP4D and DISFA (see Table III). These results indicate that explicit temporal modeling complements relational AU reasoning and is particularly beneficial when annotations are sparse.

2) *Mining AU relationships*: Effective modeling of intricate inter-AU dependencies is pivotal for AU recognition under weak supervision, as AUs exhibit structured co-occurrence and mutual exclusivity patterns. Early approaches [64], [90] explored RBMs for modeling inter-AU relationships due to the inherent ability of capturing higher dependencies across multiple AUs, regularizing missing supervision. Recently, SMA-Net [51] addressed attention collapse by encouraging each head to encode distinct AU-related features, promoting representational diversity within the same spatial regions. This internal diversification reduces redundancy among heads and improves the model's ability to capture co-occurring or subtle AUs under weak supervision, thereby outperforming prior baselines under inexact supervision (about 4%). In the case of incomplete supervision, a subset of methods [66], [68], [69] explicitly models AU relationships, while others focus on learning more discriminative AU features [63], [67] (implicitly capturing dependencies). MLCR [66] uses GCNs to encode dynamic AU co-activation graphs, propagating information among related AUs. WSRTL [69] adopts a transformer encoder-decoder design in which the encoder refines AU-aware

context and the decoder retrieves AU-specific cues via AU queries. Complementary to these, KS [68] exploits key frames to encode AU relationships using transformers along with temporal context to guide the student model, demonstrating that relational priors and temporal cues are synergistic when labels are sparse. Overall, transformer-based relational encoders provide a flexible and empirically strong mechanism for encoding inter-AU structure under WSL, offering significant improvement over prior baselines. Another promising direction is to jointly model expression-AU relations along with inter-AU dependencies for indirect supervision. [89] explored expression-AU mappings with individual AUs, whereas subsequent works [90], [91], [93] emphasized modeling both expression-dependent and expression-independent AUs and showed consistent performance improvement on multiple datasets. Recent methods employed dual/auxiliary-task objectives to implicitly strengthen consistency of expression-AU relationships [94], achieving state-of-the-art results under indirect supervision.

3) *Pseudo-label Refinement*: Ensuring pseudo-label reliability is key to achieving robust performance when annotations are sparse. CFRN [56] exploits CAMs to detect spurious pseudo-labels under the premise that incorrect labels induce fake saliency, achieving state-of-the-art results on RAF-DB (see Fig. 5). Beyond CAM-based refinement, recent methods [57], [59] target pseudo-label noise from the perspectives of class imbalance and domain shift. ReMoP [57] explored consistency with a mixture of model predictions to identify unreliable pseudo-labels and mitigate confirmation bias. DDL [59] learns adaptive, class-aware attention constraints calibrated by a balanced validation set to restore minority-class recall. Both approaches deliver comparable gains and achieve state-of-the-art performance on both RAF-DB and FER+, underscoring that reliability mechanisms, whether saliency, ensemble, or calibration-driven are crucial to standard semi-supervised pipelines.

4) *Uncertainty Estimation*: Discriminating clean from noisy labels and quantifying label uncertainty are central challenges for inaccurate supervision. Confidence-based ranking (SCN) [71] prioritizes putatively clean samples, while relative (sample-wise) uncertainty estimation (RUL) [72] improves the detection of ambiguous labels. Complementary, class-specific uncertainty modeling [73] accounts for varying uncertainty levels across classes. Beyond hard one-hot targets, assigning soft label distributions mitigates memorization of noisy labels. DMUE [75] estimated reliable label distributions by aggregating diverse predictors using multi-auxiliary classifiers. Confidence-aware label smoothing (SLS) [76] tempers targets using prediction confidence, yielding a modest improvement over [75] on RAF-DB and comparable performance on Affect-Net and FER+. Another line of research to deal with uncertain labels is ensemble-based approaches. Model ensembling (IPA2LT) [79] aggregates predictions across multiple networks to mitigate the influence of unstable labels, while GAAVE [81] explored adversarial-sensitivity analysis to flag uncertain samples by measuring prediction inconsistencies under small, targeted perturbations. Similar to the idea of exploiting CAMs [56] to detect fake pseudo-labels, EAC [80] enforced attention

consistency across strong augmentations to deal with uncertain labels, achieving the best performance across AffectNet and FER+ datasets.

V. METHODS FOR REGRESSION

Since regression in FABA under weak supervision has primarily been studied in the context of inexact and incomplete supervision, we organize this section around expression and AU intensity estimation under these two scenarios.

A. Inexact Annotations

1) *Expression Intensity Estimation*: Early works on weakly supervised expression intensity estimation framed the problem as mapping a global ordinal label (e.g., low–medium–high pain) to latent frame-level intensities. Ruiz et al. [100] introduced Multi-instance Dynamic Ordinal Random Fields (MI-DORF), which treats frame-level pain intensity as an unobserved ordinal variable constrained by the video-level label. The model represents frame-level intensities as latent ordinal variables linked in a temporal sequence, and uses a structured probabilistic framework with high-order potentials to capture both temporal dynamics and ensure consistency with the global video-level label. This formulation enables instance-level inference without dense labels, but relies on hand-crafted features and strong distributional assumptions. Subsequently, Praveen et al. [101] proposed a weakly supervised domain adaptation (WSDA) framework that combines MIL with adversarial domain adaptation. Using deep spatiotemporal features, the model transfers intensity knowledge from a fully annotated source dataset to a sparsely labeled target dataset. This approach shows that deep models benefit more from additional cross-dataset supervision than earlier feature-based methods, improving generalization across domains with different subjects and recording conditions.

2) *AU Intensity Estimation*: The MI-DORF framework [102] was later extended to tackle AU intensity estimation, which introduced added challenge of modeling multiple AUs simultaneously, each with its own temporal dynamics and co-occurrence structure. The model treats frame-level AU intensities as latent ordinal variables and enforces consistency with the sequence-level label using maximum or relative aggregation functions. However, relying on global labels can lead to under-representation of more subtle or distributed intensity variations, especially when different AUs evolve asynchronously within a sequence. To address this limitation, Zhang et al. [103] introduced Bilateral Ordinal Relevance Multiple Instance Regression (BORMIR), which injects limited domain knowledge by identifying both peak and valley frames. By jointly optimizing ordinal relevance, temporal smoothness, and relevance consistency, BORMIR captures the progressive nature of facial behavior more accurately than peak-only formulations. This highlights the importance of modeling intensity evolution, not just extremes, under weak supervision.

3) *Discussion*: Overall, ordinal MIL formulations provide a principled way to propagate global intensity annotations to instance-level predictions by exploiting ordering

constraints [100]. However, early approaches are constrained by hand-crafted representations and limited adaptability across datasets [102], [103]. Incorporating domain knowledge, such as peak–valley structure, improves discrimination between neighboring intensity levels [103], while leveraging deep representations and cross-dataset supervision substantially enhances robustness and generalization [101]. These trends indicate a shift from static, feature-driven ordinal models toward data-driven, domain-adaptive frameworks that combine ordinal structure with learned representations to mitigate dataset bias and sparse annotation.

B. Incomplete Annotations

1) *Expression Intensity Estimation*: Directly addressing incomplete expression intensity annotations remains relatively underexplored. Rui et al. [104] formulated the problem using ordinal support vector regression (OSVR) by explicitly modeling intensity levels as ordered categories rather than arbitrary numerical values. This design choice reflects the key intuition that relative ordering of intensities is more reliable than absolute intensity values under sparse supervision. By optimizing a max–margin objective that enforces ordinal consistency, OSVR enables stable learning even when annotations are irregular or missing. While the optimization is solved efficiently, the main contribution lies in preserving ordinal structure, which proves crucial for robust intensity estimation under weak supervision.

2) *AU Intensity Estimation*: Most early works on AU intensity estimation under partial supervision rely on supervision based on keyframes, where a small number of peak and valley intensity frames serve as anchors for learning. The core assumption is that these frames provide reliable intensity cues, which can then be propagated to unlabeled frames by exploiting temporal continuity and facial structure. KBSS [105] exemplified this paradigm by transferring supervision from peak and valley frames using domain priors such as facial symmetry, monotonic intensity evolution over time, and appearance similarity between neighboring frames. These priors constrain the learning process, allowing the model to infer smooth intensity trajectories despite sparse annotations. Building on this idea, KJRE [106] improved the performance by jointly learning both the facial representations and the intensity regression model, rather than treating the feature extractor as fixed. To compensate for the absence of dense labels, it incorporated various forms of weak supervision derived from human knowledge (feature smoothness, label consistency, etc) as soft or hard constraints. This hybrid supervision strategy reduces sensitivity to keyframe selection errors and improves generalization when annotations are irregularly distributed. Recognizing that different AUs exhibit distinct activation patterns and learning speeds, Zhang et al. [107] proposed a context-aware fusion framework for weakly supervised AU intensity estimation that operates on facial patches instead of full-face images. A feature fusion module models spatial interactions by adapting each patch's contribution based on the target AU, while a label fusion module captures AU-specific temporal dynamics. These attention-based modules enable the

model to handle diverse activation patterns and support training with partially labeled data, improving robustness under limited supervision. More recently, Self-SL [108] demonstrates that self-supervised pretraining on large unlabeled video corpora can provide strong spatiotemporal representations that already encode facial dynamics and temporal consistency. As a result, fine-tuning on sparsely labeled AU intensities requires fewer handcrafted priors or explicit keyframe annotations, shifting the burden from manual supervision design to data-driven representation learning.

3) *Discussion:* For expression intensity estimation, ordinal formulations such as OSVR [104] highlight the importance of respecting label ordering when supervision is incomplete, offering stable learning across heterogeneous annotation densities. In AU intensity estimation, early reliance on peak and valley frames provides strong but sparse supervisory signals [105], while domain priors—facial symmetry, temporal smoothness, and relative intensity ordering—play a central role in propagating reliable predictions to unlabeled frames [105], [106]. Later work emphasizes AU-specific learning dynamics and inter-AU structure, improving calibration and plausibility under partial labels [107], [109]. Self-supervised video pretraining [108] emerges as a scalable alternative that shifts reliance away from handcrafted priors toward data-driven representations. The field is moving from tightly constrained, keyframe-driven supervision toward representation-centric and pretraining-based approaches that better accommodate sparse, irregular, and incomplete intensity annotations.

C. Model Evaluation Protocol

The intensities of pain and AUs are evaluated in terms of Mean Absolute Error (MAE), Pearson Correlation Coefficient (PCC), and Intra-class Correlation Coefficient (ICC). The comparison of performances of state-of-the-art methods, that follow the same experimental protocol for each category of WSL for regression is shown in Table IV.

Inexact Annotations: In case of regression on the UNBC-McMaster dataset, PSPI labels of the frames are converted to 6 ordinal levels: 0(0), 1(1), 2(2), 3(3), 4-5(4), 6-15(5). The bag label of each pain sequence is considered the maximum of frame labels. Out of 25 subjects, 15 are used for training, 9 for validation, and 1 for testing. LOSO cross-validation is followed. For the DISFA and FERA datasets, bags are considered sequences with monotonically increasing or decreasing offset and onset frames. Similar to UNBC, the bag label is considered the maximum of the frame labels of AU intensities. 5-fold cross-validation is deployed for DISFA and official training/validation split for FERA. In order to compare the work of [102] with the conventional approach of pain detection [40], MILBOOST is deployed and the output probabilities of pain detection are normalized between 0 and 5 to have a fair comparison with that of [102].

Incomplete Annotations: For the UNBC-McMaster dataset, only 10% of annotations are considered in each sequence for the task of pain regression. For the task of AU regression in DISFA datasets, only 10% of annotated frames are considered in [102], [109], and [104] in order to incorporate the setting

of incomplete annotations, whereas [106], [107], [112] and [105] considered only the annotations of peak and valley frames, [108] uses only 2% of randomly annotated frames. In the case of the FERA 2015 dataset, official training and development sets provided by the FERA 2015 challenge [29] are deployed. Similar to UNBC-McMaster and DISFA, [109], [113] considers only 10% of annotated frames while [104], [103], [105] considers annotations of peak and valley.

D. Critical Analysis and Results

Estimating the intensity of expressions or AUs is intrinsically harder than classification, because models must capture fine-grained, temporally evolving appearance changes and cope with label uncertainty. So, these regressions tasks are relatively under-explored. Under inexact supervision (sequence-level labels only), effective designs impose ordinal/MIL structure to map global labels to frame/segment intensities. Classical models encode temporal ordinal relations and smoothness constraints often via peak/valley anchors [102], [103]. Subsequent methods showed superior performance by employing deep spatiotemporal backbones that pair MIL with domain adaptation and ordinal priors [101], [111].

For incomplete annotations, [105] and [106] explored temporal, feature, and label smoothness in addition to the temporal relevance, where the former shows better performance by leveraging more constraints. [107] explored attention mechanisms at both the feature and label levels, showing an improvement over prior approaches [105], [106]. [108] and [109] further improved performance by leveraging large-scale pre-trained models based on self-supervised learning and RBMs to model AU relationships, where the latter has shown better performance. Compared to classical ML models [102], [103], DL models [108], [109] have shown significant improvements for both inexact and incomplete annotations. However, a challenge in using DL models for estimating the intensity of facial expressions or AUs is the requirement of a large number of representative samples with intensity annotations, which is costly and requires domain expertise. Estimation of intensity levels using DL models is therefore explored in the context of incomplete annotations by leveraging large-scale pre-trained models [108] or large-scale unlabeled data [112].

VI. CHALLENGES AND OPPORTUNITIES

This section highlights key open challenges for FABA under weak supervision and outlines promising research opportunities to address them. While robust FER systems face many general hurdles (e.g., identity bias, data sparsity), we focus here on challenges unique to weakly annotated scenarios and how they may be addressed in future work.

A. Few-Shot Learning

FSL offers a principled way to adapt models to new or rare expression categories from only a handful of labeled examples [114]. Recently, FSL has also gained attention in FER as a solution to the problem of novel classes for compound emotion recognition [115]–[117]. Xinyi et al. [115]

TABLE IV
PERFORMANCE OF STATE-OF-ART METHODS FOR REGRESSION OF EXPRESSIONS AND AUS UNDER VARIOUS MODES OF WSL SETTING ON THE MOST WIDELY EVALUATED DATASETS.

WSL Setting	Dataset	Method	Task	Features	Learning Model	Validation	MAE	PCC	ICC
Inexact	UNBC-McMaster	Ruiz et al. [102] (TIP 2018)	Pain [6 levels]	geometric	DORF	LOSO	0.710	0.360	0.340
		Praveen et al. [101] (FG 2020)	Pain [6 levels]	I3D [110]	WSDA	LOSO	0.714	0.630	0.567
		Praveen et al. [111] (IMAVIS 2020)	Pain [6 levels]	I3D [110]	WSDA-OR	LOSO	0.530	0.705	0.696
		Zhang et al. [103] (CVPR 2018)	Pain [6 levels]	geometric	BORMIR	LOSO	0.821	0.605	0.531
	DISFA	Ruiz et al. [102] (TIP 2018)	12 AUs [6 levels]	geometric	DORF	5-fold	1.130	0.400	0.260
		Zhao et al. [104] (CVPR 2016)	12 AUs [6 levels]	LBP	OSVR	5-fold	1.380	0.350	0.150
		Zhang et al. [103] (CVPR 2018)	12 AUs [6 levels]	geometric	BORMIR	5-fold	0.789	0.353	0.283
	FERA 2015	Zhang et al. [103] (CVPR 2018)	5 AUs [6 levels]	geometric	BORMIR	Valstar et al. [29]	0.852	0.635	0.620
Incomplete	UNBC-McMaster	Ruiz et al. [102] (TIP 2018)	Pain [6 levels]	geometric	DORF	LOSO	0.510	0.460	0.460
		Zhao et al. [104] (CVPR 2016)	Pain [6 levels]	LBP	OSVR	LOSO	0.951	0.544	0.495
	DISFA	Ruiz et al. [102] (TIP 2018)	12 AUs [6 levels]	geometric	DORF	5-fold	0.480	0.420	0.380
		Zhao et al. [104] (CVPR 2016)	12 AUs [6 levels]	LBP	OSVR	5-fold	0.800	0.370	0.290
		Zhang et al. [105] (CVPR 2018)	12 AUs [6 levels]	CNN	KBSS	3-fold	0.330	-	0.360
		Wang et al. [109] (TAFFC 2019)	12 AUs[6 levels]	RBM	RBM-P	3-fold	0.431	0.592	0.549
		Zhang et al. [106] (CVPR 2019)	12 AUs [6 levels]	geometric	KJRE	5-fold	0.910	0.370	0.350
		Zhang et al. [107] (ICCV 2019)	12 AUs [6 levels]	Resnet-18	CFLF	3-fold	0.329	-	0.408
		Sanchez et al. [108] (ACCV 2020)	12 AUs [6 levels]	Resnet-18	Self-SL	3-fold	0.376	-	0.413
	FERA 2015	Wang et al. [109] (TAFFC 2019)	5 AUs [6 levels]	RBM	RBM-P	Valstar et al. [29]	0.728	0.605	0.585
		Zhao et al. [104] (CVPR 2016)	5 AUs [6 levels]	LBP	OSVR	Valstar et al. [29]	1.077	0.545	0.544
		Zhang et al. [105] (CVPR 2018)	5 AUs [6 levels]	CNN	KBSS	Valstar et al. [29]	0.660	-	0.670
		Zhang et al. [106] (CVPR 2019)	5 AUs [6 levels]	geometric	KJRE	Valstar et al. [29]	0.870	0.620	0.600
		Sanchez et al. [108] (ACCV 2020)	12 AUs [6 levels]	Resnet-18	Self-SL	3-fold	0.798	-	0.680
		Zhang et al. [107] (ICCV 2019)	5 AUs [6 levels]	Resnet-18	CFLF	Valstar et al. [29]	0.741	-	0.661

proposed a novel emotion-guided similarity network to deal with the limited annotations of compound expressions, where the pre-trained model is generalized to limited samples of novel classes (support set) of compound emotions. Zou et al. [117] also addressed the problem of compound FER in a cross-domain setting using a cascaded decomposition network for FSL. Chen et al. [116] explored self-supervised vision transformers by jointly pertaining with multiple pretext tasks, and leveraged FSL to train the deep model with fewer labeled samples. Despite these advances in fully supervised settings, FSL remains an under-explored problem for weakly annotated FABA. For instance, incomplete supervision with severe class imbalance can be reframed as a few-shot problem by treating minority classes as new categories. By training on the majority classes and unlabeled data using semi-supervised strategies, a model could then adapt to the new minority classes, expanding the space for modeling mixed emotional states. This approach offers a novel way to mitigate confirmation bias and reduce spurious pseudo-labels. More broadly, coupling FSL with WSL provides a unified framework to address both limited/weak annotations and limited target-domain data.

B. Multimodal Learning

Exploring multiple modalities can reduce annotation cost and error by providing complementary cues [118]. To foster

the progress, several multimodal datasets (audio, visual, text) have been released [119]–[122], and many supervised works study audio–visual fusion for emotion recognition [123]–[126]. Text has recently gained traction as a surrogate for human annotations [88], yet other modalities—such as physiology and speech—remain underexplored in the WSL setting. Thermal imagery offers invariance to illumination and can complement RGB for FER [127], [128]. Using 3D/4D facial data can help address pose and occlusion, capturing fine-grained AU changes (e.g., AU18 vs. AU10+AU17+AU24) via depth cues, see [129]. Optical flow provides motion-sensitive descriptors of facial muscle activity by converting temporal changes into static textures [130], [131], with efficacy shown for both macro- and micro-expressions [132]. Therefore, modalities such as text, audio, optical flow, depth, and thermal images are promising auxiliary signals to supplement weak supervision. They can supply proxy labels, stabilize localization, and improve robustness under missing, inexact, or noisy annotations, laying a path toward resilient multimodal FABA under WSL.

C. Fairness in Weakly Supervised Learning

Fairness and weak annotation are intrinsically coupled in FABA. Recent studies demonstrated that many FER datasets

contain significant annotation biases between genders, especially between happy and angry expressions [133], [134]. Though the issue of fairness has been explored with supervised learning settings [134], [135], it remains under-explored in the context of WSL. Weak labels (incomplete, inexact, or noisy) are rarely distributed uniformly across demographic groups. Source biases arising from annotator perspectives, cultural differences, varying lighting and camera domains, and demographic imbalances frequently result in group-dependent label noise. A model can inadvertently amplify this noise, especially when employing confidence-driven pseudo-labeling or MIL strategies that disproportionately favor already well-represented groups. This critical linkage motivates several promising directions, such as (i) uncertainty-aware, fairness-calibrated pseudo-labeling with group-normalized thresholds, (ii) design fairness-aware MIL/localization that balances attention across skin tones, ages, and genders, and (iii) incorporates causal/counterfactual augmentation (e.g., edit pose/illumination/skin tone while holding expression constant) to probe and reduce spurious correlations. Therefore, this line of research holds significant promise for advancing both the robustness and ethical integrity of FABA under WSL.

D. Micro-Expression Recognition Under Weak Supervision

Micro-expressions are brief, involuntary facial movements that reflect concealed or unconscious emotions, typically lasting less than 500 milliseconds [9]. Their subtlety and sparsity make them particularly challenging to annotate at frame-level granularity, especially in unconstrained or real-world settings. Consequently, WSL presents a compelling alternative by reducing dependence on dense annotations and leveraging indirect or auxiliary signals. Recent studies have explored the close correlation between action units (AUs) and micro-expressions [136], suggesting that AU activations can serve as effective implicit supervision signals for micro-expression recognition. Robust temporal modeling is critical for capturing the fine-grained dynamics of micro-expressions in videos, particularly under weak supervision [137]. Techniques such as spatiotemporal contrastive learning have demonstrated the ability to distinguish subtle muscle activations from background noise or overlapping macro-expressions [138]. Furthermore, cross-domain transfer from macro-expression datasets or AU activation patterns may enhance generalization, especially when guided by domain adaptation or multi-task objectives. The integration of auxiliary tasks such as macro-expression recognition offers additional supervision pathways to compensate for weak visual signals [139]. Emerging efforts also leverage multimodal large language models (MLLMs) to fuse human-like perception of subtle facial movements with powerful semantic reasoning capabilities [140]. To advance the field, there is a need for new benchmark datasets incorporating coarse, probabilistic, or confidence-weighted annotations, as well as evaluation protocols tailored to the weak supervision regime. Progress in this direction holds the potential to enable robust and fine-grained micro-expression recognition systems, particularly for real-time applications.

E. Leveraging Large Language Models

LLMs have strong potential to enhance FABA by compensating for weak or noisy supervision, injecting linguistic structure, expert priors, and explanatory rationales into the training pipeline. Acting as label refiners, LLMs transform low-quality annotations, such as clip-level tags, coarse timestamps, or ASR transcripts, into candidate expression or AU labels, complete with textual rationales and uncertainty scores that can serve as soft targets or consistency constraints for vision models. This process can be informed by encoded expert knowledge, like display rules and AU-expression taxonomies, to probabilistically fuse multiple weak signals. Recently, LLM-derived annotations have been successfully explored for indirect supervision for FER, circumventing the need for human annotations [88]. In multimodal settings, LLM-derived summaries provide temporal priors that guide attention mechanisms toward relevant frames, while also supporting data-centric curation efforts by generating descriptive prompts for diffusion models to synthesize diverse, label-coherent faces, and rewriting crowd-sourced tags into calibrated, subgroup-aware pseudo-labels. Furthermore, LLMs enable explanation-aware supervision by verifying whether model saliency aligns with natural language descriptions of AUs, triggering reweighting or active learning queries when rationales misalign, thus regularizing MIL and enabling text–video contrastive training. Future directions and evaluation must consider not only accuracy but also rationale faithfulness, localization, calibration, and subgroup performance gaps, while mitigating cultural bias and hallucination via self-consistency checks, verifier prompts, and robust human-in-the-loop audits.

F. Towards Unified and Interpretable Models

Most current methods tackle weakly supervised classification, regression, or AU detection in isolation. We argue for a unified framework that natively accommodates heterogeneous supervision—categorical, ordinal, and continuous—under incomplete, inexact, or noisy labels. Such a system would eliminate isolated pipelines, improve robustness, and generalize better across datasets and tasks. Concretely, the framework should enable multi-task learning (expressions, AUs, valence–arousal), cross-modal reasoning (video, audio, text/ASR), and interpretable outputs (region/temporal rationales and concept bottlenecks). Building on strong pretrained backbones, it can use modular adapters such as LoRA for rapid task/domain personalization, and integrate uncertainty estimation to gate pseudo-labels, calibrate decisions, and handle label noise. Training this architecture at scale by pooling datasets spanning multiple weak-supervision regimes (MIL, sparse/clip-level tags, noisy multi-annotator labels) would yield a foundation model for affective behavior, later adapted with lightweight heads/adapters to diverse downstream settings—reducing annotation cost while delivering consistent, explainable performance across tasks, subjects, and domains.

VII. CONCLUSION

This survey provides a structured review of WSL methods for FABA. To better understand the field, we categorize the reviewed literature based on affective tasks and WSL scenarios.

We systematically analyze state-of-the-art methods for recognizing expressions and AUs under various WSL conditions, discussing their challenges and strengths. By comparing results from widely used experimental protocols, we provide new insights into the benefits and limitations of these approaches. This review highlights the diverse WSL techniques available for developing robust deep learning models for real-world FABA applications. We also present open challenges and suggest future research directions to advance the field. This survey serves as a comprehensive reference for researchers and practitioners, fostering progress in FABA under WSL settings.

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